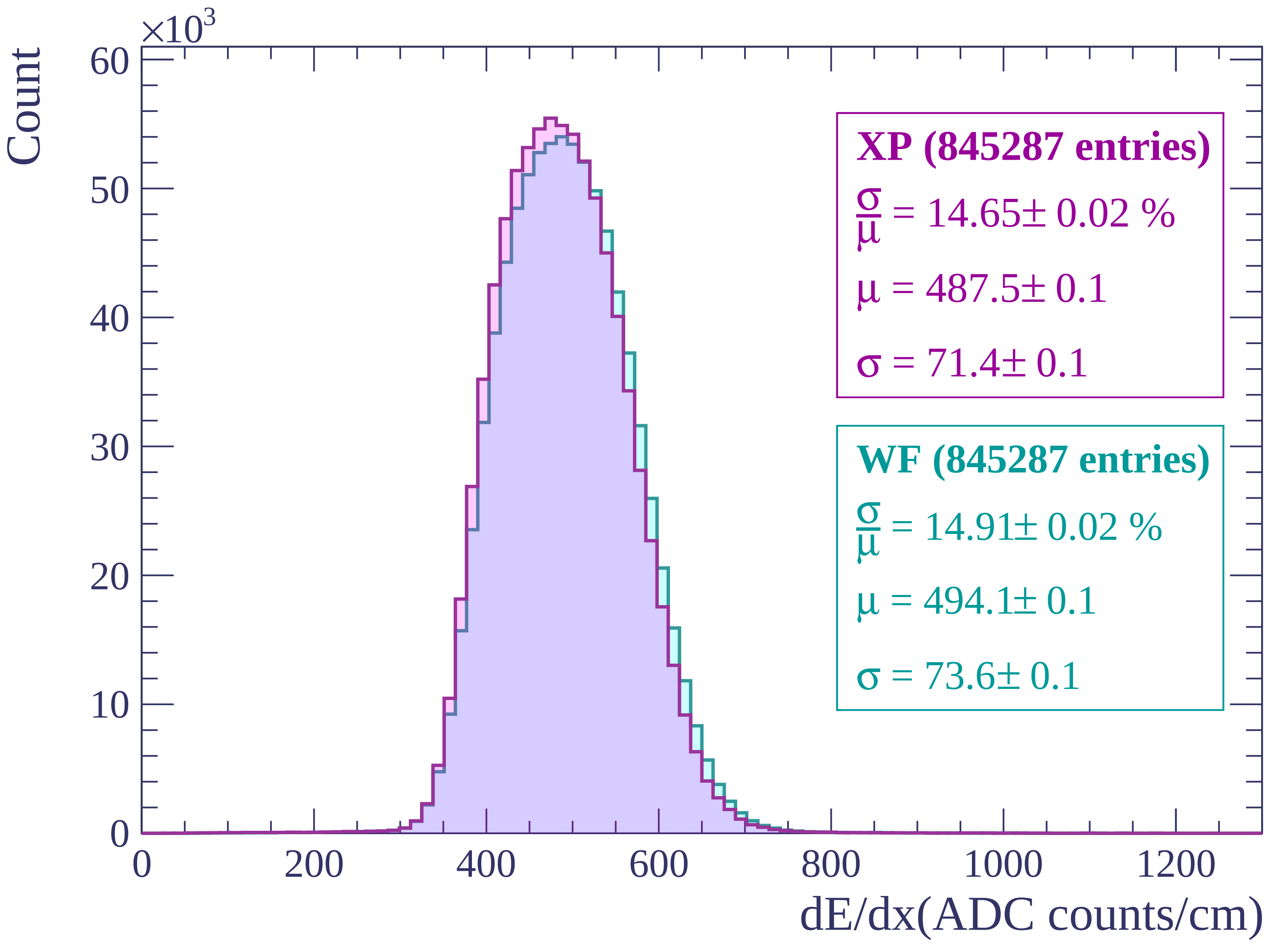
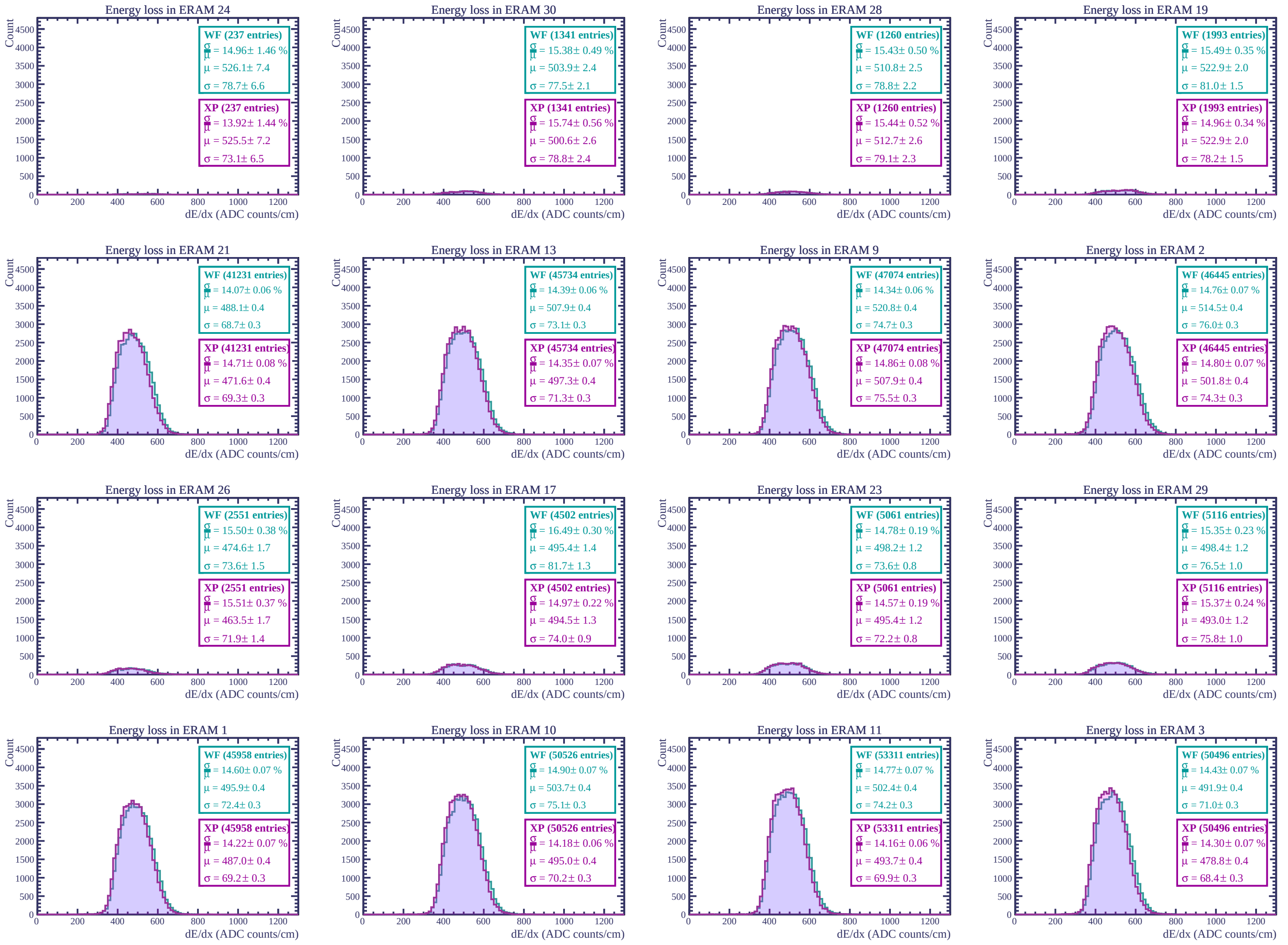
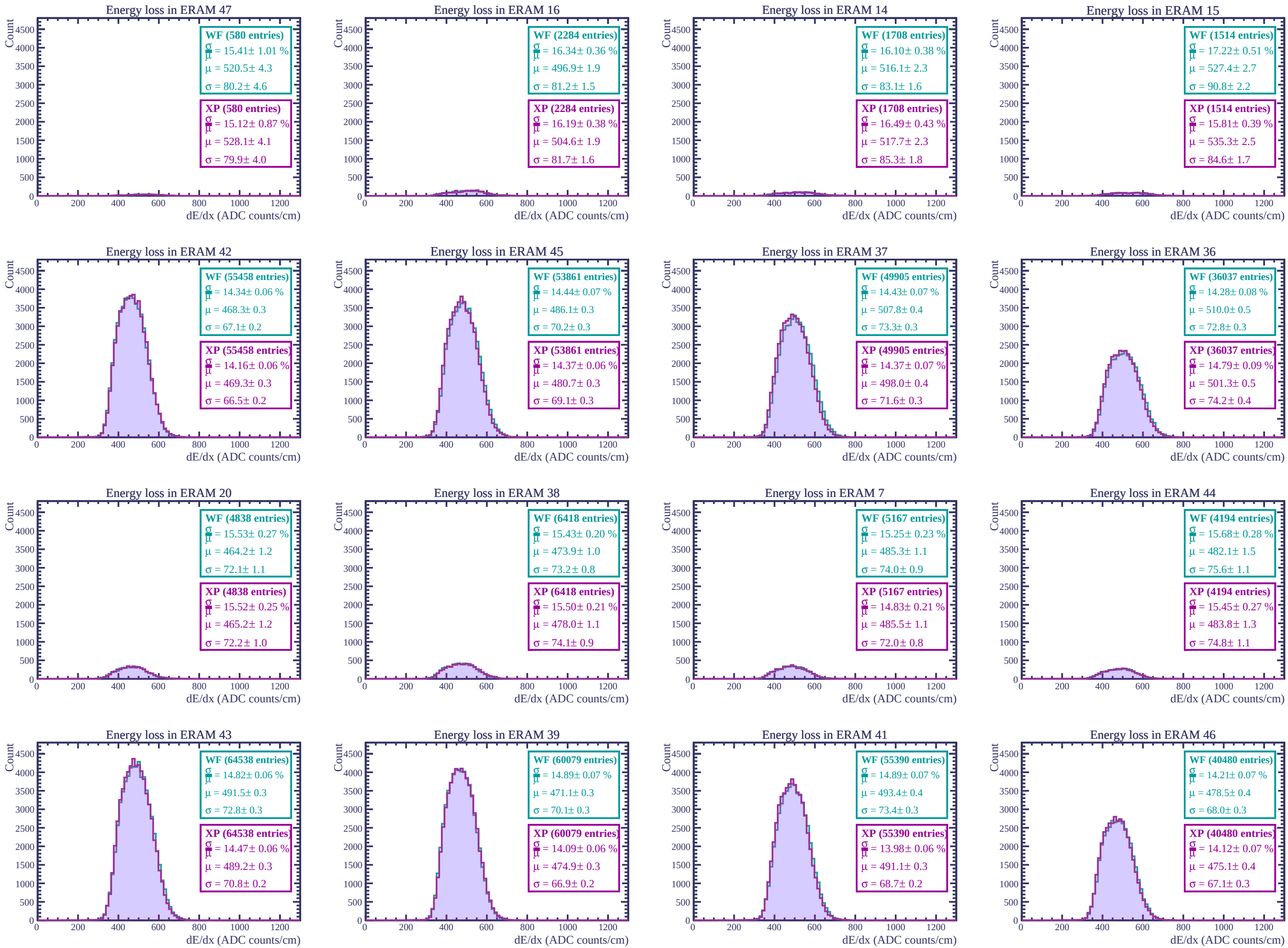
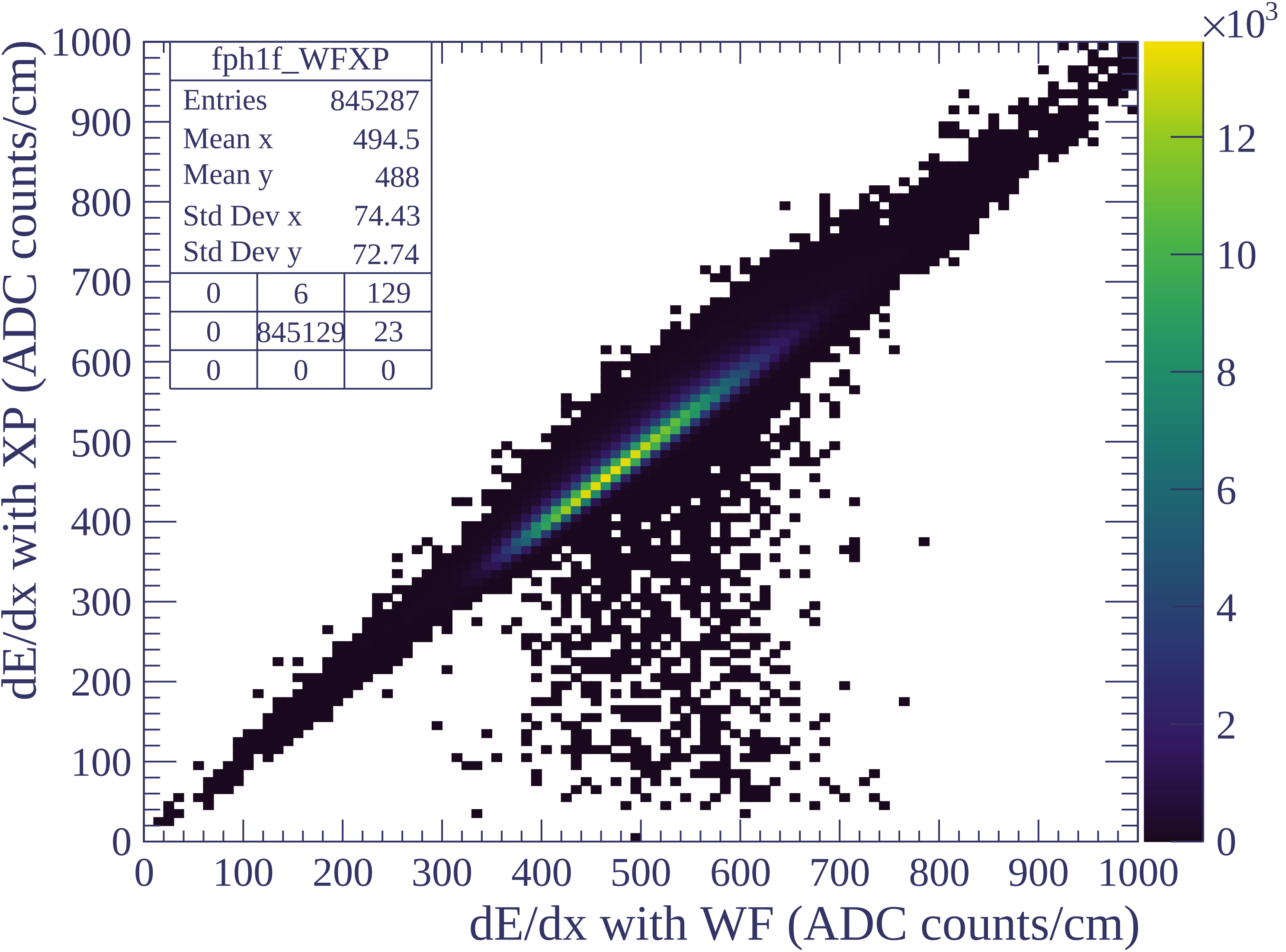
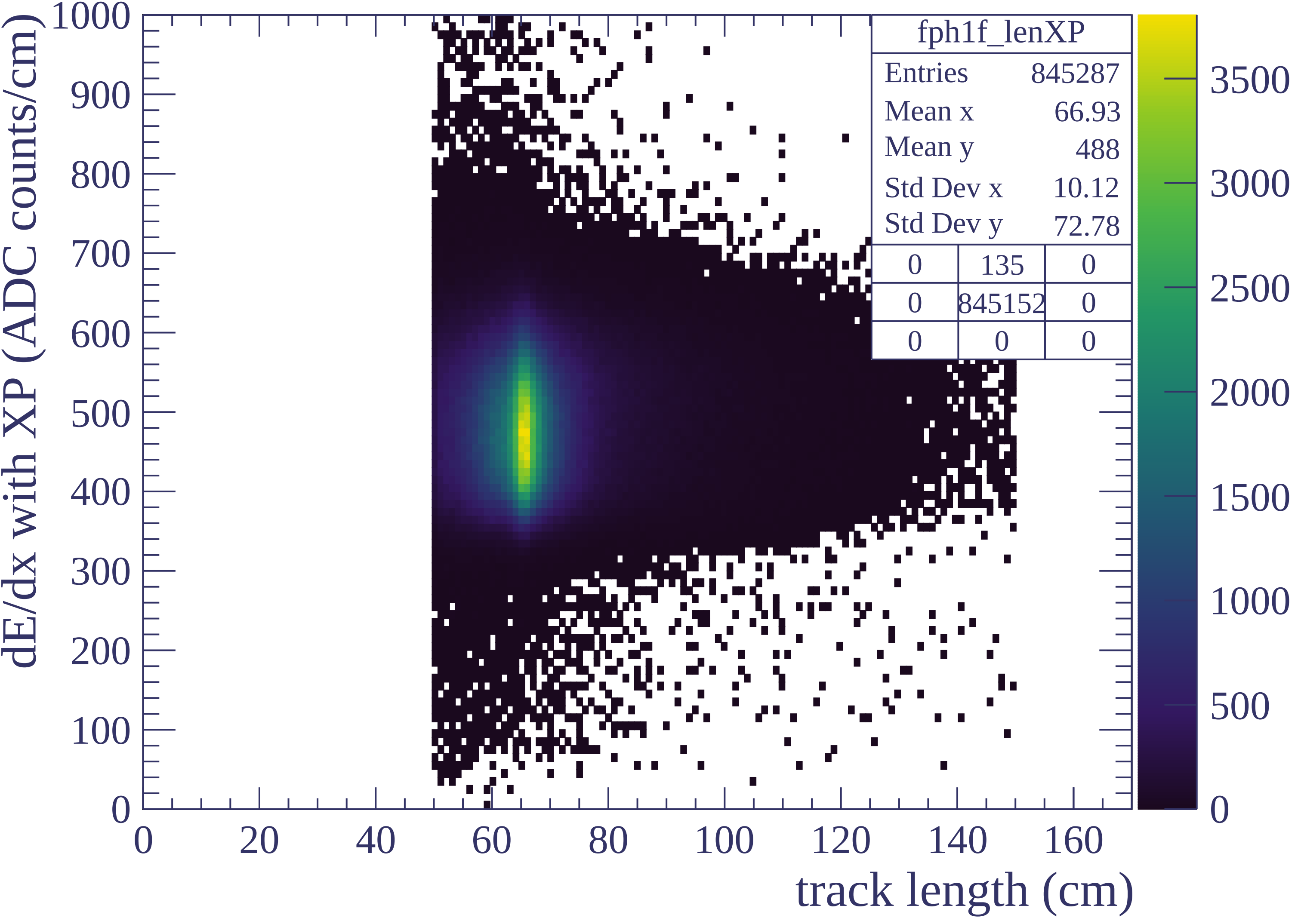


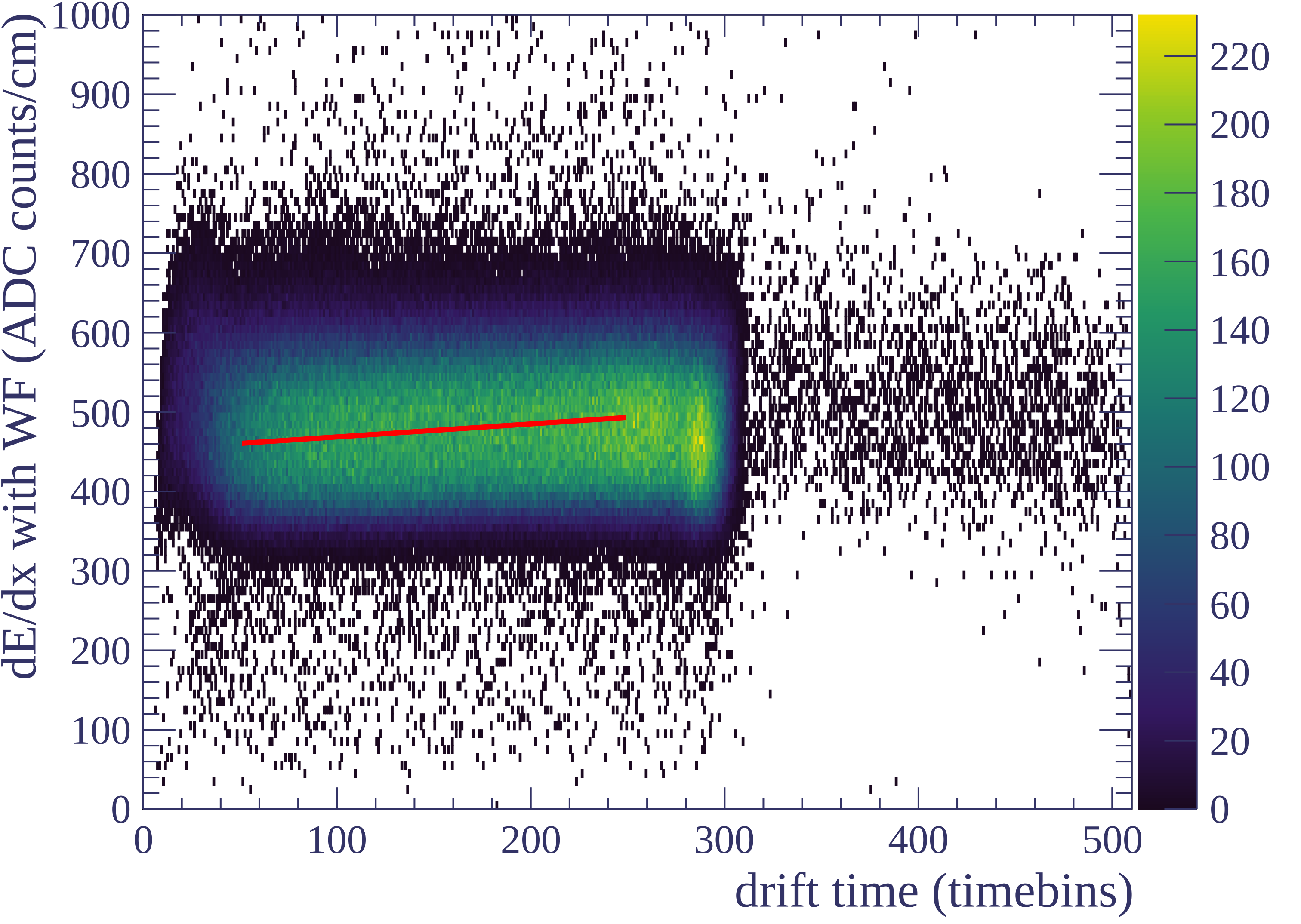
$\times 10^3$
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (845287 entries)
$\frac{\sigma}{\mu}$ = 14.65± 0.02 %
μ = 487.5± 0.1
σ = 71.4± 0.1
WF (845287 entries)
$\frac{\sigma}{\mu}$ = 14.91± 0.02 %
μ = 494.1± 0.1
σ = 73.6± 0.1


Energy loss in ERAM 24
WF (237 entries)
σ/μ = 14.96± 1.46 %
μ = 526.1± 7.4
σ = 78.7± 6.6
XP (237 entries)
σ/μ = 13.92± 1.44 %
μ = 525.5± 7.2
σ = 73.1± 6.5
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 30
WF (1341 entries)
σ/μ = 15.38± 0.49 %
μ = 503.9± 2.4
σ = 77.5± 2.1
XP (1341 entries)
σ/μ = 15.74± 0.56 %
μ = 500.6± 2.6
σ = 78.8± 2.4
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 28
WF (1260 entries)
σ/μ = 15.43± 0.50 %
μ = 510.8± 2.5
σ = 78.8± 2.2
XP (1260 entries)
σ/μ = 15.44± 0.52 %
μ = 512.7± 2.6
σ = 79.1± 2.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 19
WF (1993 entries)
σ/μ = 15.49± 0.35 %
μ = 522.9± 2.0
σ = 81.0± 1.5
XP (1993 entries)
σ/μ = 14.96± 0.34 %
μ = 522.9± 2.0
σ = 78.2± 1.5
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 21
WF (41231 entries)
σ/μ = 14.07± 0.06 %
μ = 488.1± 0.4
σ = 68.7± 0.3
XP (41231 entries)
σ/μ = 14.71± 0.08 %
μ = 471.6± 0.4
σ = 69.3± 0.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 13
WF (45734 entries)
σ/μ = 14.39± 0.06 %
μ = 507.9± 0.4
σ = 73.1± 0.3
XP (45734 entries)
σ/μ = 14.35± 0.07 %
μ = 497.3± 0.4
σ = 71.3± 0.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 9
WF (47074 entries)
σ/μ = 14.34± 0.06 %
μ = 520.8± 0.4
σ = 74.7± 0.3
XP (47074 entries)
σ/μ = 14.86± 0.08 %
μ = 507.9± 0.4
σ = 75.5± 0.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 2
WF (46445 entries)
σ/μ = 14.76± 0.07 %
μ = 514.5± 0.4
σ = 76.0± 0.3
XP (46445 entries)
σ/μ = 14.80± 0.07 %
μ = 501.8± 0.4
σ = 74.3± 0.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 26
WF (2551 entries)
σ/μ = 15.50± 0.38 %
μ = 474.6± 1.7
σ = 73.6± 1.5
XP (2551 entries)
σ/μ = 15.51± 0.37 %
μ = 463.5± 1.7
σ = 71.9± 1.4
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 17
WF (4502 entries)
σ/μ = 16.49± 0.30 %
μ = 495.4± 1.4
σ = 81.7± 1.3
XP (4502 entries)
σ/μ = 14.97± 0.22 %
μ = 494.5± 1.3
σ = 74.0± 0.9
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 23
WF (5061 entries)
σ/μ = 14.78± 0.19 %
μ = 498.2± 1.2
σ = 73.6± 0.8
XP (5061 entries)
σ/μ = 14.57± 0.19 %
μ = 495.4± 1.2
σ = 72.2± 0.8
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 29
WF (5116 entries)
σ/μ = 15.35± 0.23 %
μ = 498.4± 1.2
σ = 76.5± 1.0
XP (5116 entries)
σ/μ = 15.37± 0.24 %
μ = 493.0± 1.2
σ = 75.8± 1.0
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 1
WF (45958 entries)
σ/μ = 14.60± 0.07 %
μ = 495.9± 0.4
σ = 72.4± 0.3
XP (45958 entries)
σ/μ = 14.22± 0.07 %
μ = 487.0± 0.4
σ = 69.2± 0.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 10
WF (50526 entries)
σ/μ = 14.90± 0.07 %
μ = 503.7± 0.4
σ = 75.1± 0.3
XP (50526 entries)
σ/μ = 14.18± 0.06 %
μ = 495.0± 0.4
σ = 70.2± 0.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 11
WF (53311 entries)
σ/μ = 14.77± 0.07 %
μ = 502.4± 0.4
σ = 74.2± 0.3
XP (53311 entries)
σ/μ = 14.16± 0.06 %
μ = 493.7± 0.4
σ = 69.9± 0.3
Count
dE/dx (ADC counts/cm)
Energy loss in ERAM 3
WF (50496 entries)
σ/μ = 14.43± 0.07 %
μ = 491.9± 0.4
σ = 71.0± 0.3
XP (50496 entries)
σ/μ = 14.30± 0.07 %
μ = 478.8± 0.4
σ = 68.4± 0.3
Count
dE/dx (ADC counts/cm)

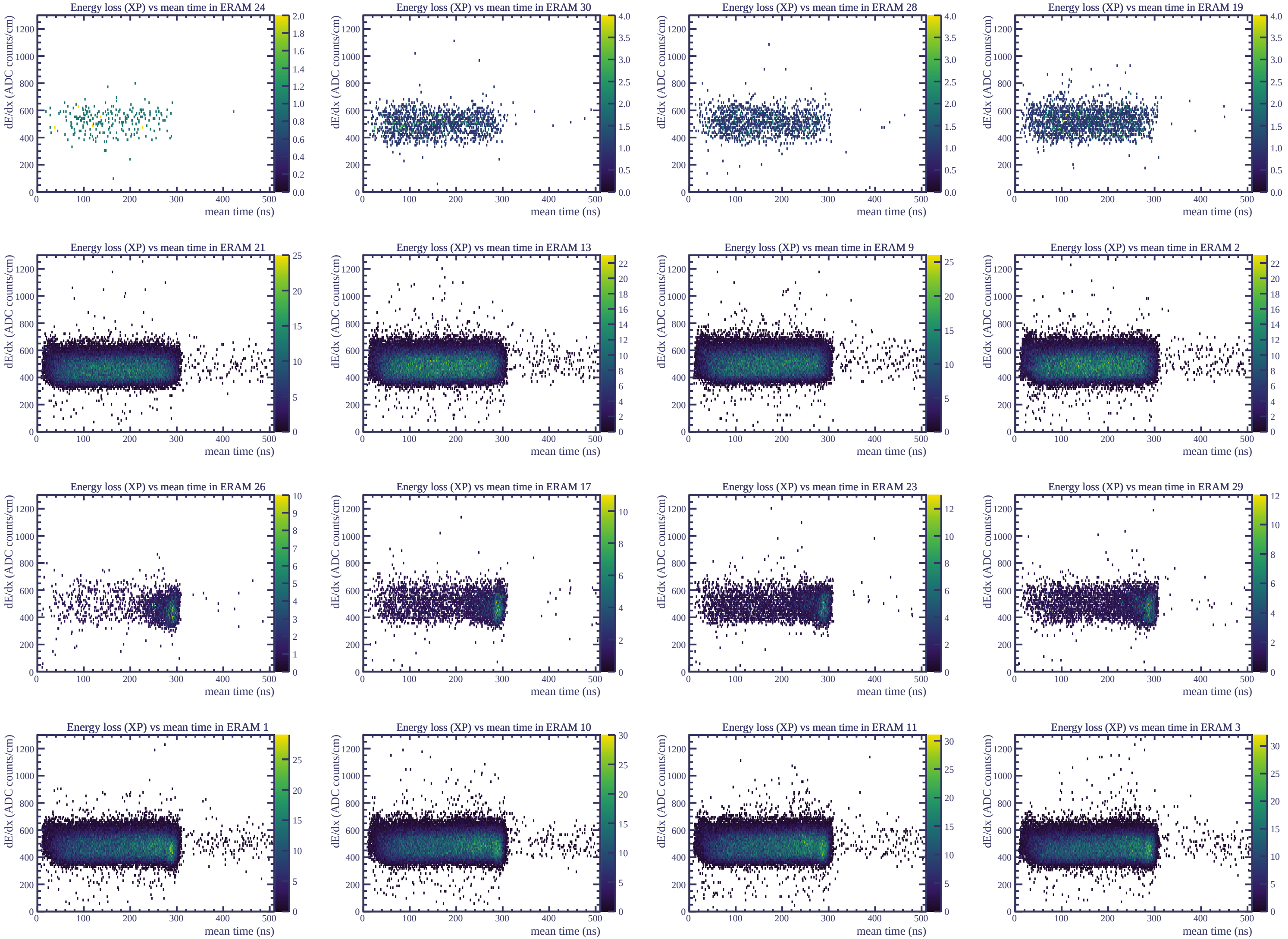

Energy loss in ERAM 47
WF (580 entries)
σ/μ = 15.41± 1.01 %
μ = 520.5± 4.3
σ = 80.2± 4.6
XP (580 entries)
σ/μ = 15.12± 0.87 %
μ = 528.1± 4.1
σ = 79.9± 4.0
Energy loss in ERAM 16
WF (2284 entries)
σ/μ = 16.34± 0.36 %
μ = 496.9± 1.9
σ = 81.2± 1.5
XP (2284 entries)
σ/μ = 16.19± 0.38 %
μ = 504.6± 1.9
σ = 81.7± 1.6
Energy loss in ERAM 14
WF (1708 entries)
σ/μ = 16.10± 0.38 %
μ = 516.1± 2.3
σ = 83.1± 1.6
XP (1708 entries)
σ/μ = 16.49± 0.43 %
μ = 517.7± 2.3
σ = 85.3± 1.8
Energy loss in ERAM 15
WF (1514 entries)
σ/μ = 17.22± 0.51 %
μ = 527.4± 2.7
σ = 90.8± 2.2
XP (1514 entries)
σ/μ = 15.81± 0.39 %
μ = 535.3± 2.5
σ = 84.6± 1.7
Energy loss in ERAM 42
WF (55458 entries)
σ/μ = 14.34± 0.06 %
μ = 468.3± 0.3
σ = 67.1± 0.2
XP (55458 entries)
σ/μ = 14.16± 0.06 %
μ = 469.3± 0.3
σ = 66.5± 0.2
Energy loss in ERAM 45
WF (53861 entries)
σ/μ = 14.44± 0.07 %
μ = 486.1± 0.3
σ = 70.2± 0.3
XP (53861 entries)
σ/μ = 14.37± 0.06 %
μ = 480.7± 0.3
σ = 69.1± 0.3
Energy loss in ERAM 37
WF (49905 entries)
σ/μ = 14.43± 0.07 %
μ = 507.8± 0.4
σ = 73.3± 0.3
XP (49905 entries)
σ/μ = 14.37± 0.07 %
μ = 498.0± 0.4
σ = 71.6± 0.3
Energy loss in ERAM 36
WF (36037 entries)
σ/μ = 14.28± 0.08 %
μ = 510.0± 0.5
σ = 72.8± 0.3
XP (36037 entries)
σ/μ = 14.79± 0.09 %
μ = 501.3± 0.5
σ = 74.2± 0.4
Energy loss in ERAM 20
WF (4838 entries)
σ/μ = 15.53± 0.27 %
μ = 464.2± 1.2
σ = 72.1± 1.1
XP (4838 entries)
σ/μ = 15.52± 0.25 %
μ = 465.2± 1.2
σ = 72.2± 1.0
Energy loss in ERAM 38
WF (6418 entries)
σ/μ = 15.43± 0.20 %
μ = 473.9± 1.0
σ = 73.2± 0.8
XP (6418 entries)
σ/μ = 15.50± 0.21 %
μ = 478.0± 1.1
σ = 74.1± 0.9
Energy loss in ERAM 7
WF (5167 entries)
σ/μ = 15.25± 0.23 %
μ = 485.3± 1.1
σ = 74.0± 0.9
XP (5167 entries)
σ/μ = 14.83± 0.21 %
μ = 485.5± 1.1
σ = 72.0± 0.8
Energy loss in ERAM 44
WF (4194 entries)
σ/μ = 15.68± 0.28 %
μ = 482.1± 1.5
σ = 75.6± 1.1
XP (4194 entries)
σ/μ = 15.45± 0.27 %
μ = 483.8± 1.3
σ = 74.8± 1.1
Energy loss in ERAM 43
WF (64538 entries)
σ/μ = 14.82± 0.06 %
μ = 491.5± 0.3
σ = 72.8± 0.3
XP (64538 entries)
σ/μ = 14.47± 0.06 %
μ = 489.2± 0.3
σ = 70.8± 0.2
Energy loss in ERAM 39
WF (60079 entries)
σ/μ = 14.89± 0.07 %
μ = 471.1± 0.3
σ = 70.1± 0.3
XP (60079 entries)
σ/μ = 14.09± 0.06 %
μ = 474.9± 0.3
σ = 66.9± 0.2
Energy loss in ERAM 41
WF (55390 entries)
σ/μ = 14.89± 0.07 %
μ = 493.4± 0.4
σ = 73.4± 0.3
XP (55390 entries)
σ/μ = 13.98± 0.06 %
μ = 491.1± 0.3
σ = 68.7± 0.2
Energy loss in ERAM 46
WF (40480 entries)
σ/μ = 14.21± 0.07 %
μ = 478.5± 0.4
σ = 68.0± 0.3
XP (40480 entries)
σ/μ = 14.12± 0.07 %
μ = 475.1± 0.4
σ = 67.1± 0.3
Count
dE/dx (ADC counts/cm)

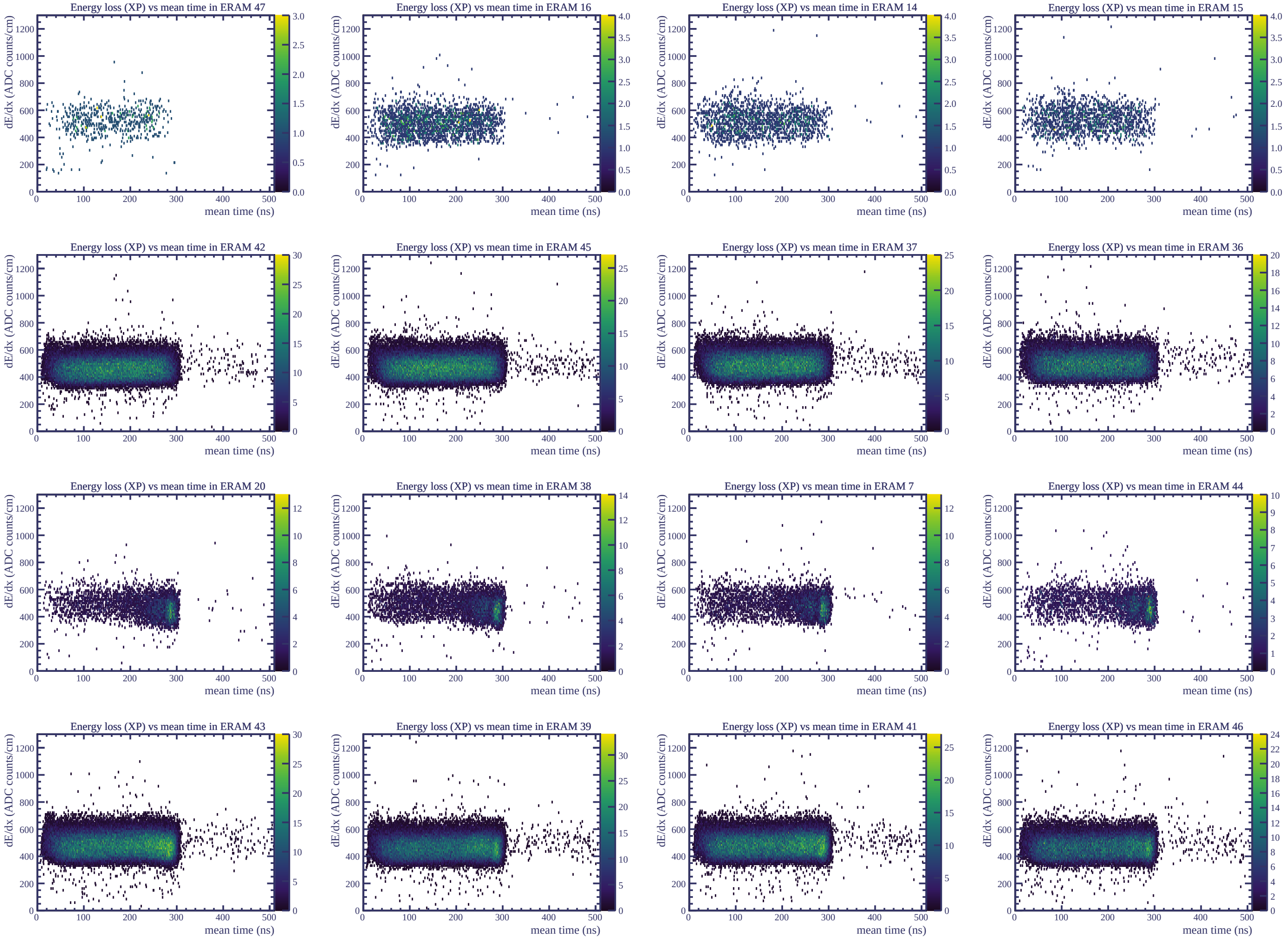

fph1f_WFXP
Entries 845287
Mean x 494.5
Mean y 488
Std Dev x 74.43
Std Dev y 72.74
0 6 129
0 845129 23
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3

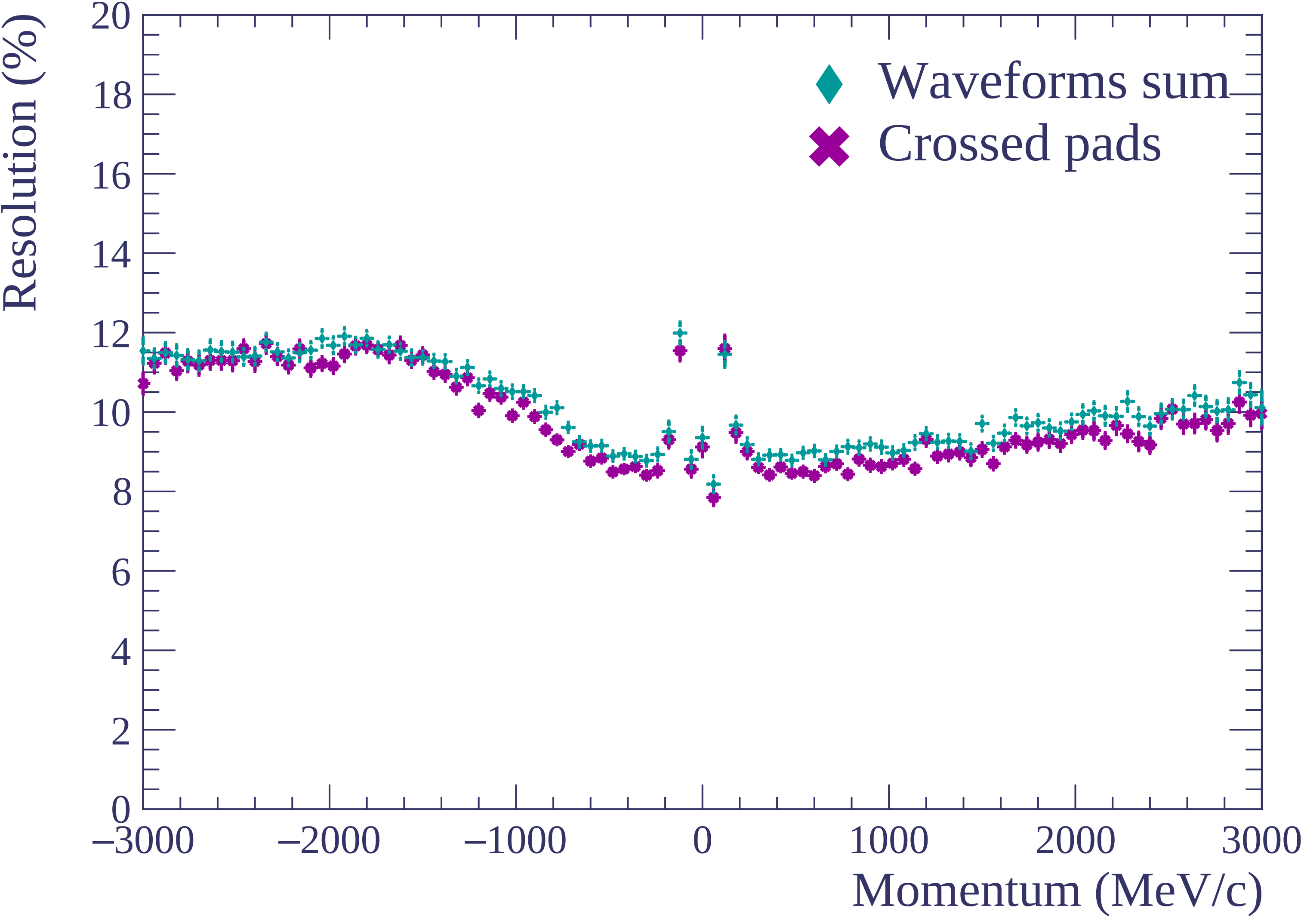

fph1f_lenXP
Entries 845287
Mean x 66.93
Mean y 488
Std Dev x 10.12
Std Dev y 72.78
0 135 0
0 845152 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
500
1000
1500
2000
2500
3000
3500



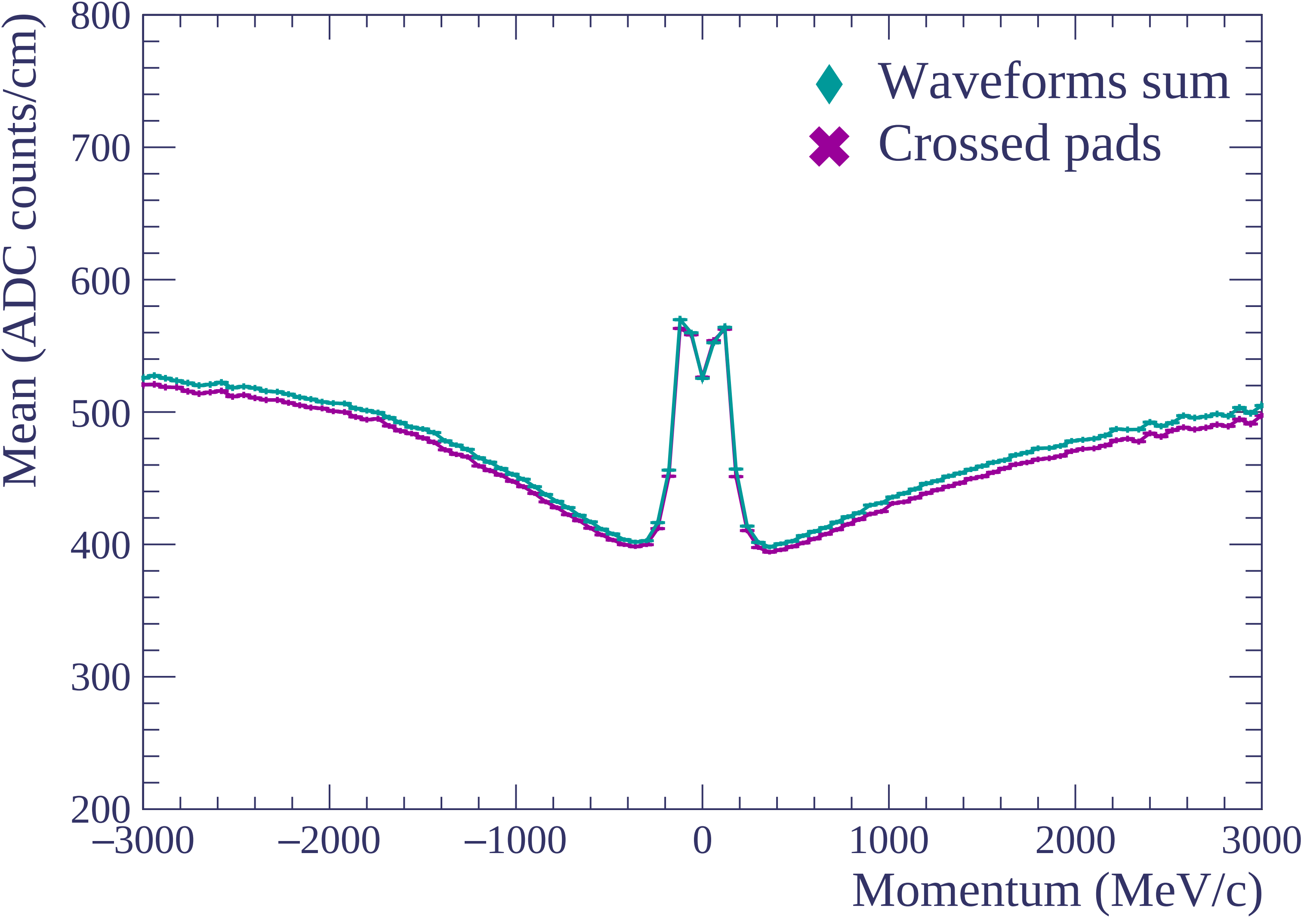
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
20
40
60
80
100
120
140
160
180
200
220


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

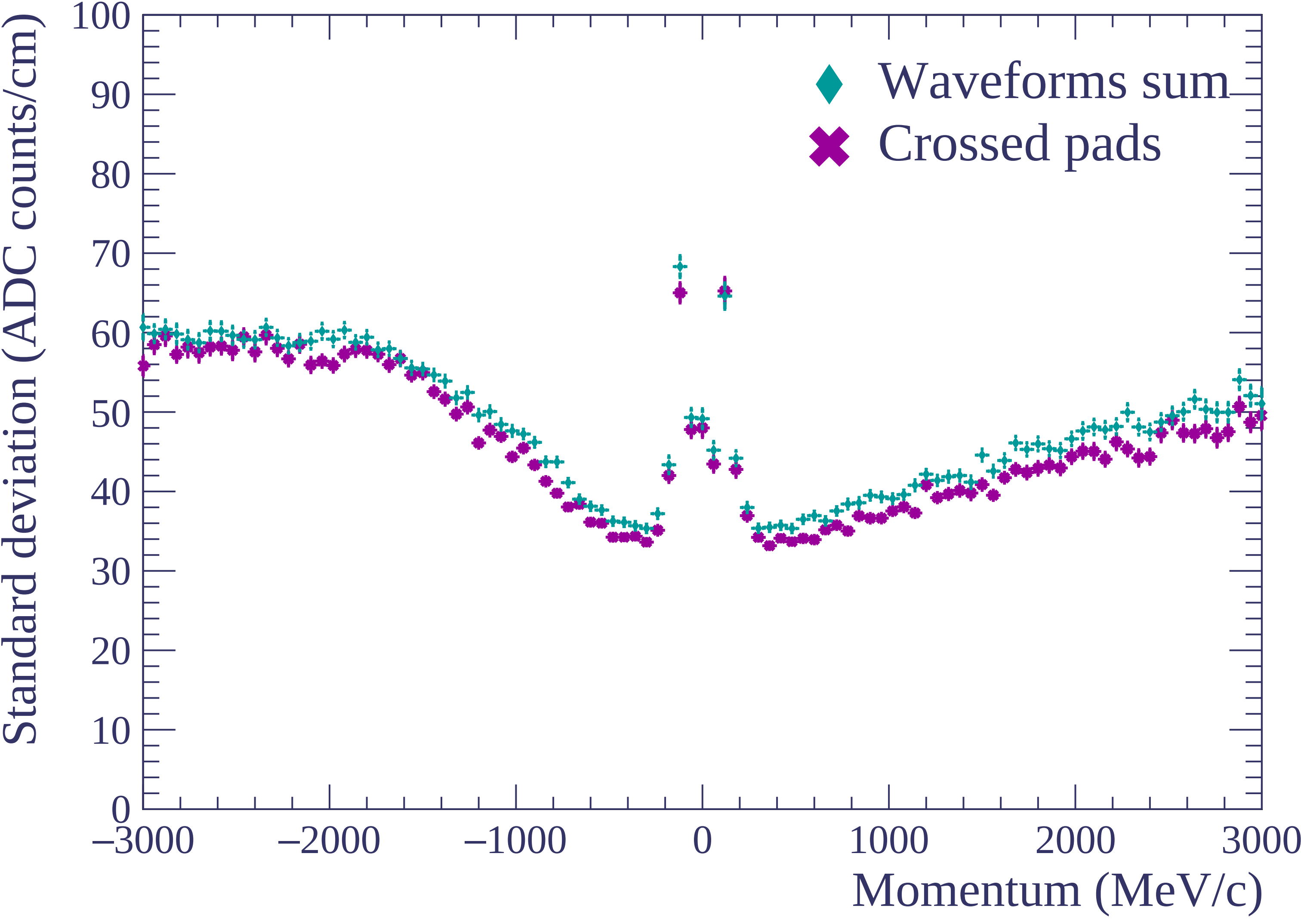

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

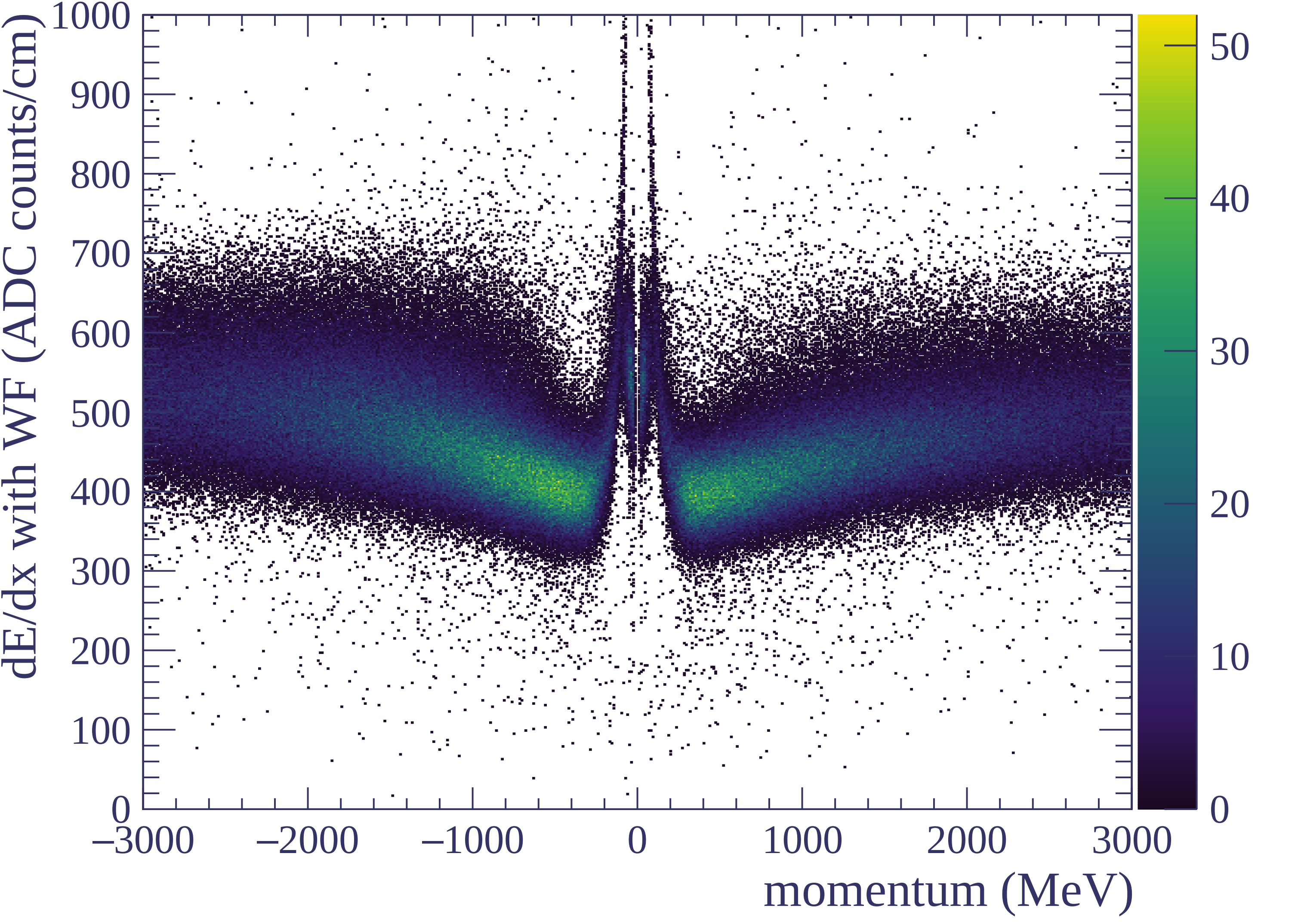


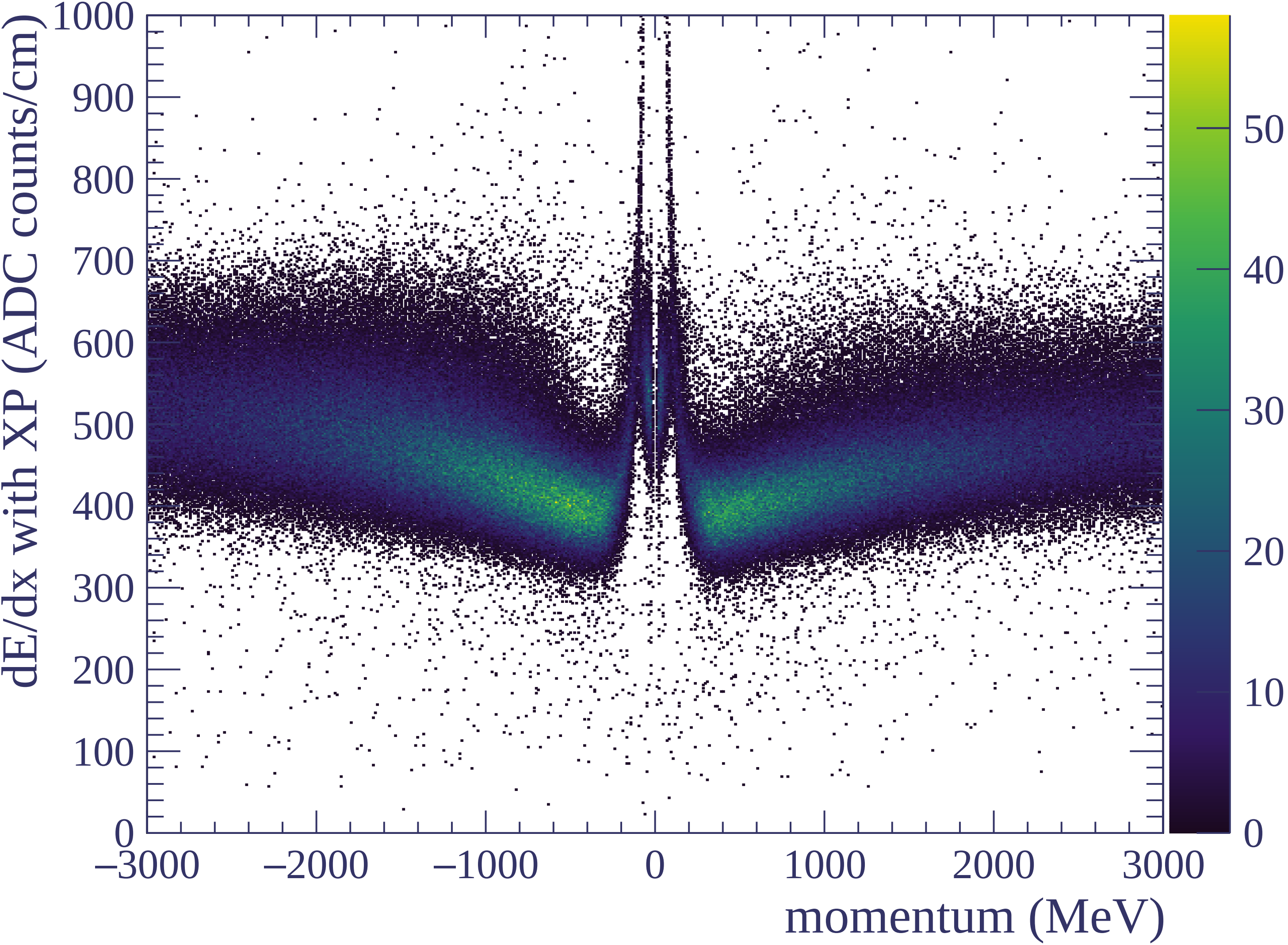
Resolution (%)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000



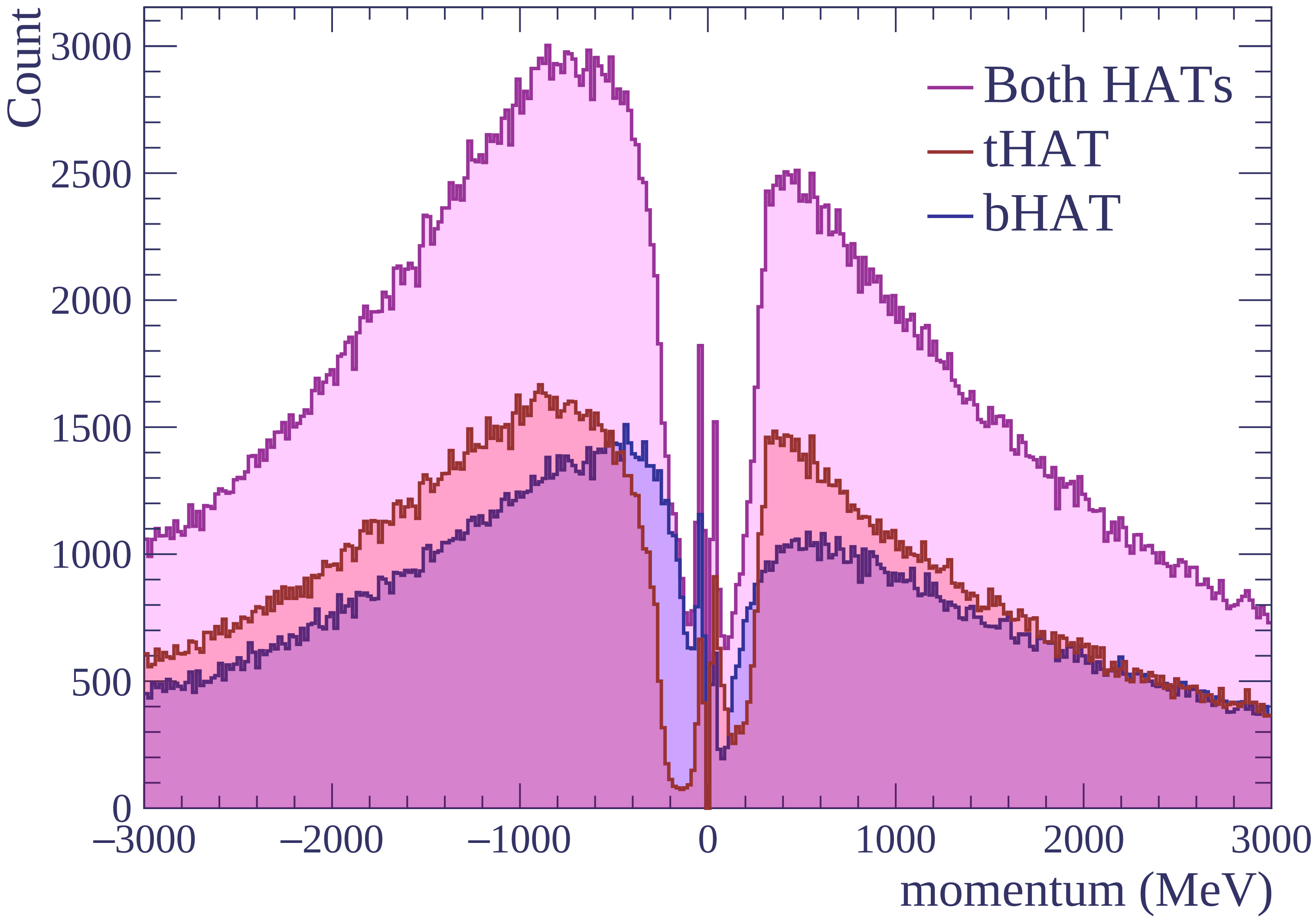
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000

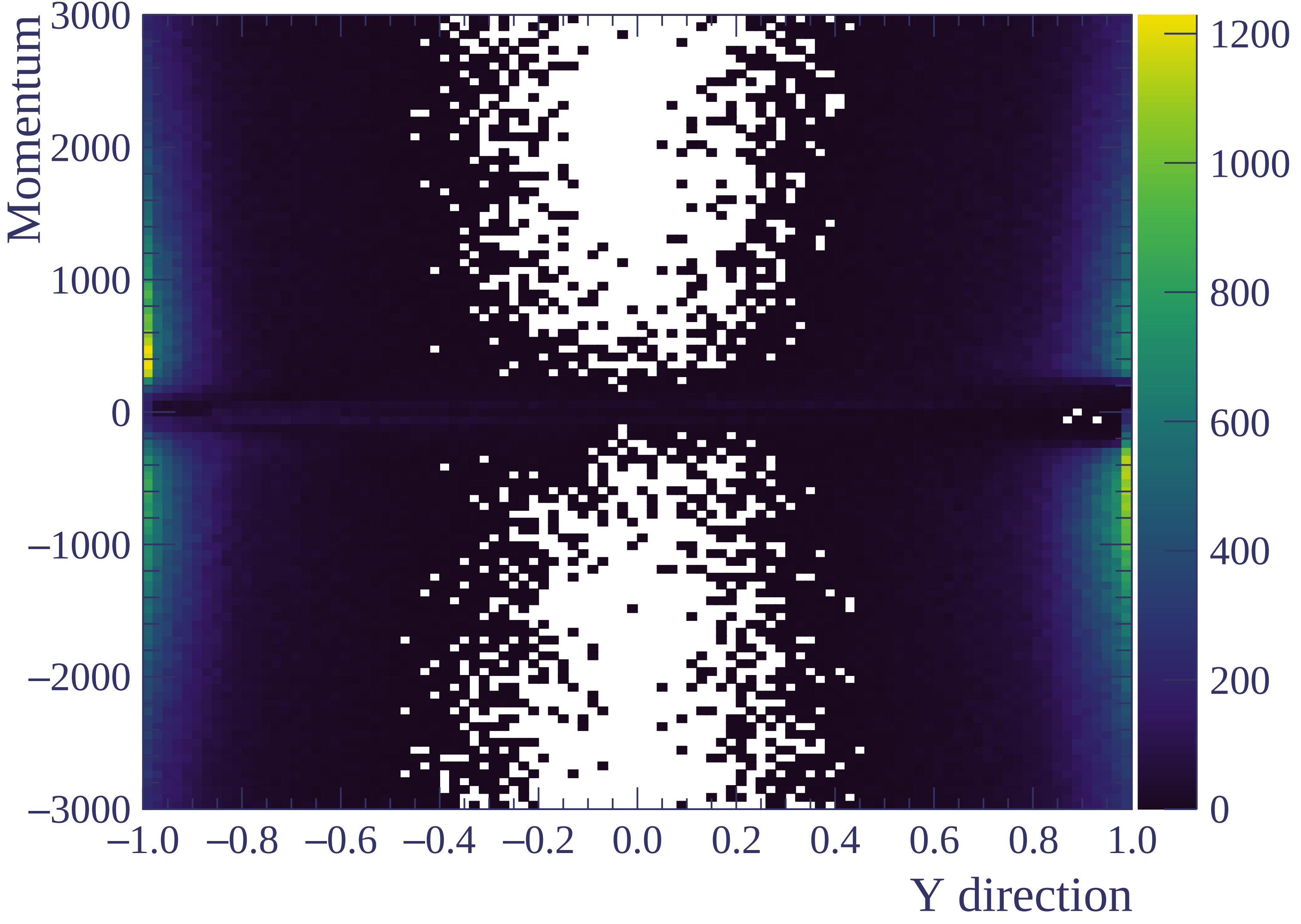

Waveforms sum
Crossed pads
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
0
10
20
30
40
50
60
70
80
90
100
−3000
−2000
−1000
0
1000
2000
3000

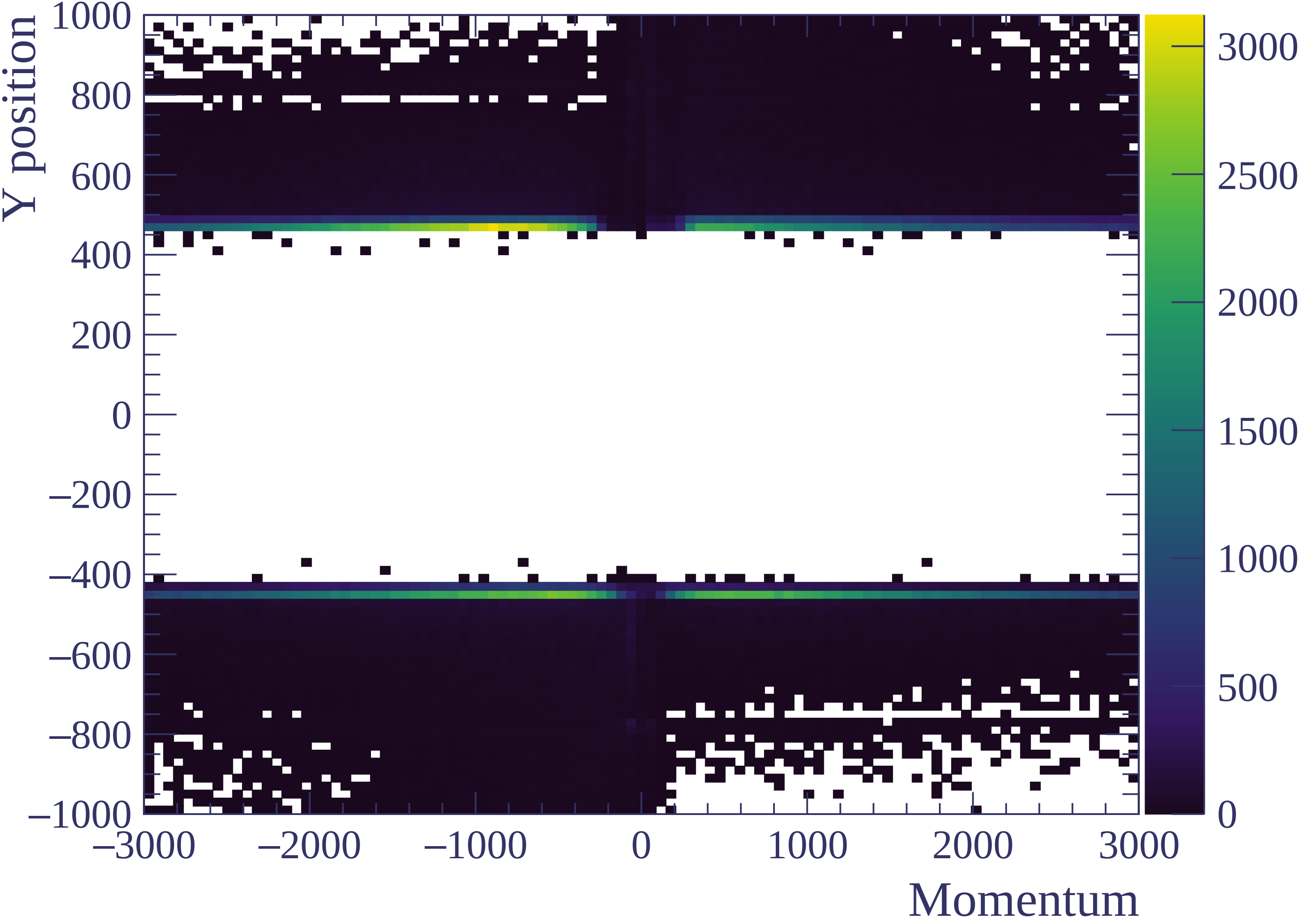

dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
0
10
20
30
40
50



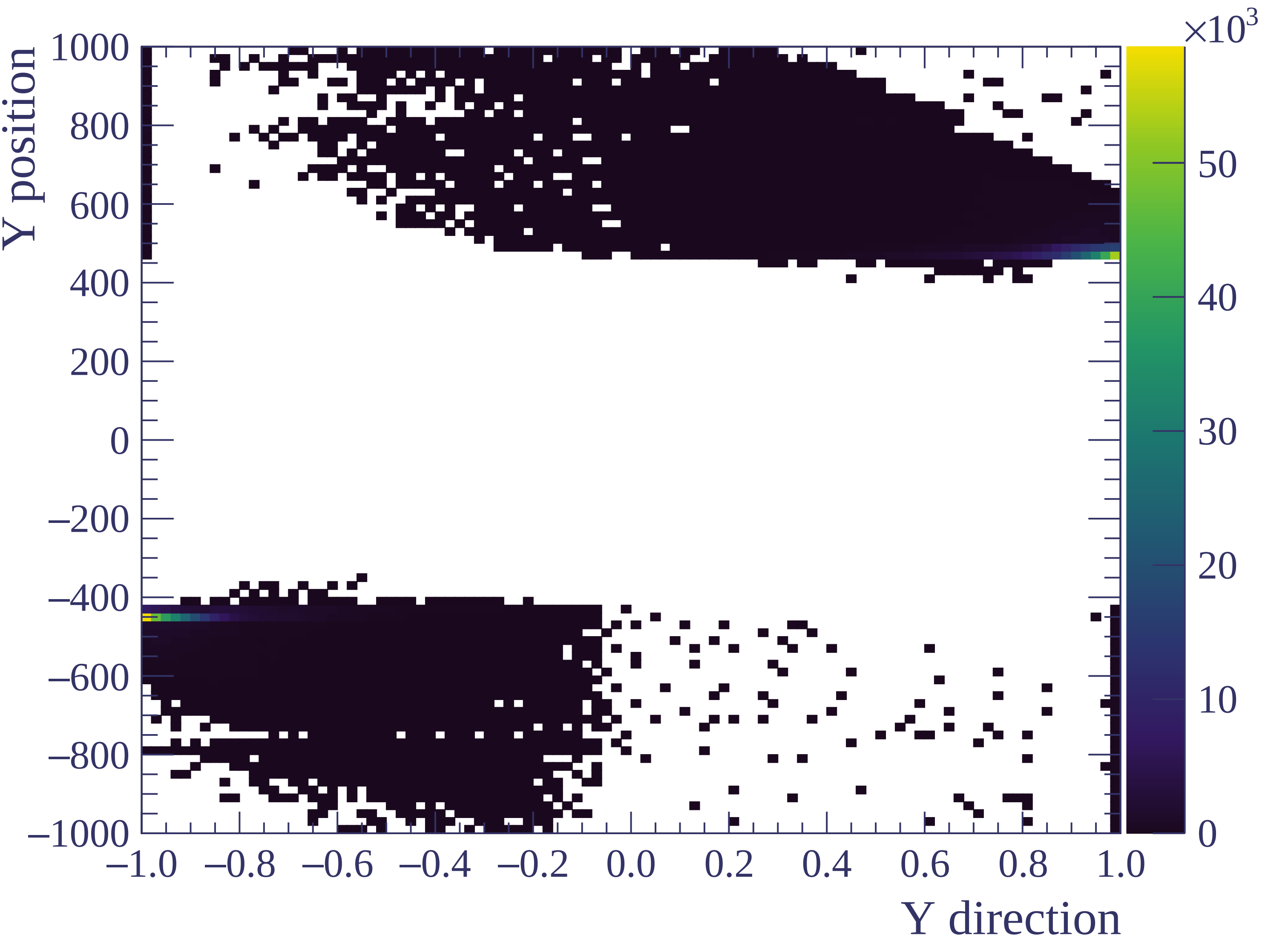
dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
10
20
30
40
50


Count
3000
2500
2000
1500
1000
500
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

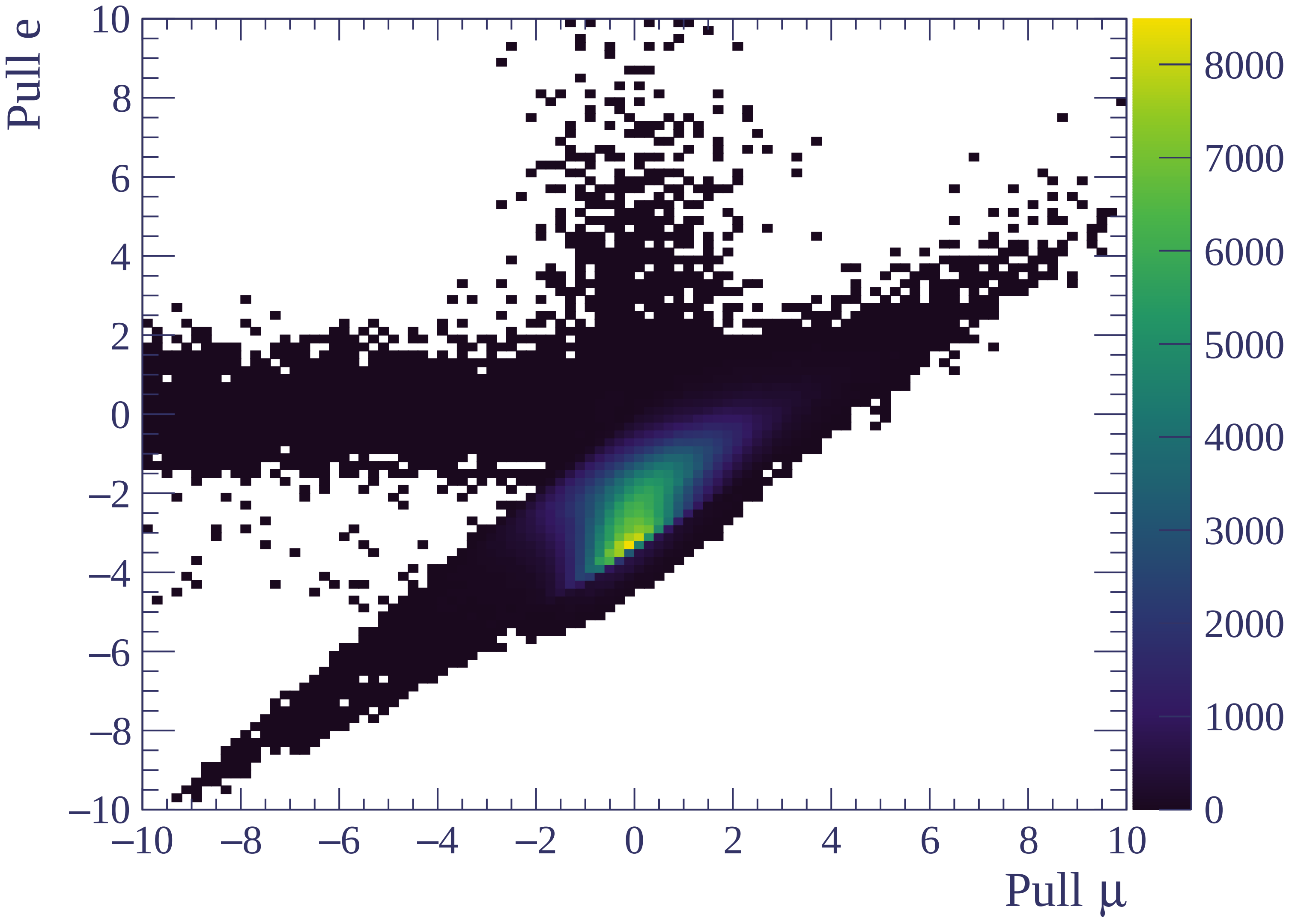

Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
1200
1000
800
600
400
200
0



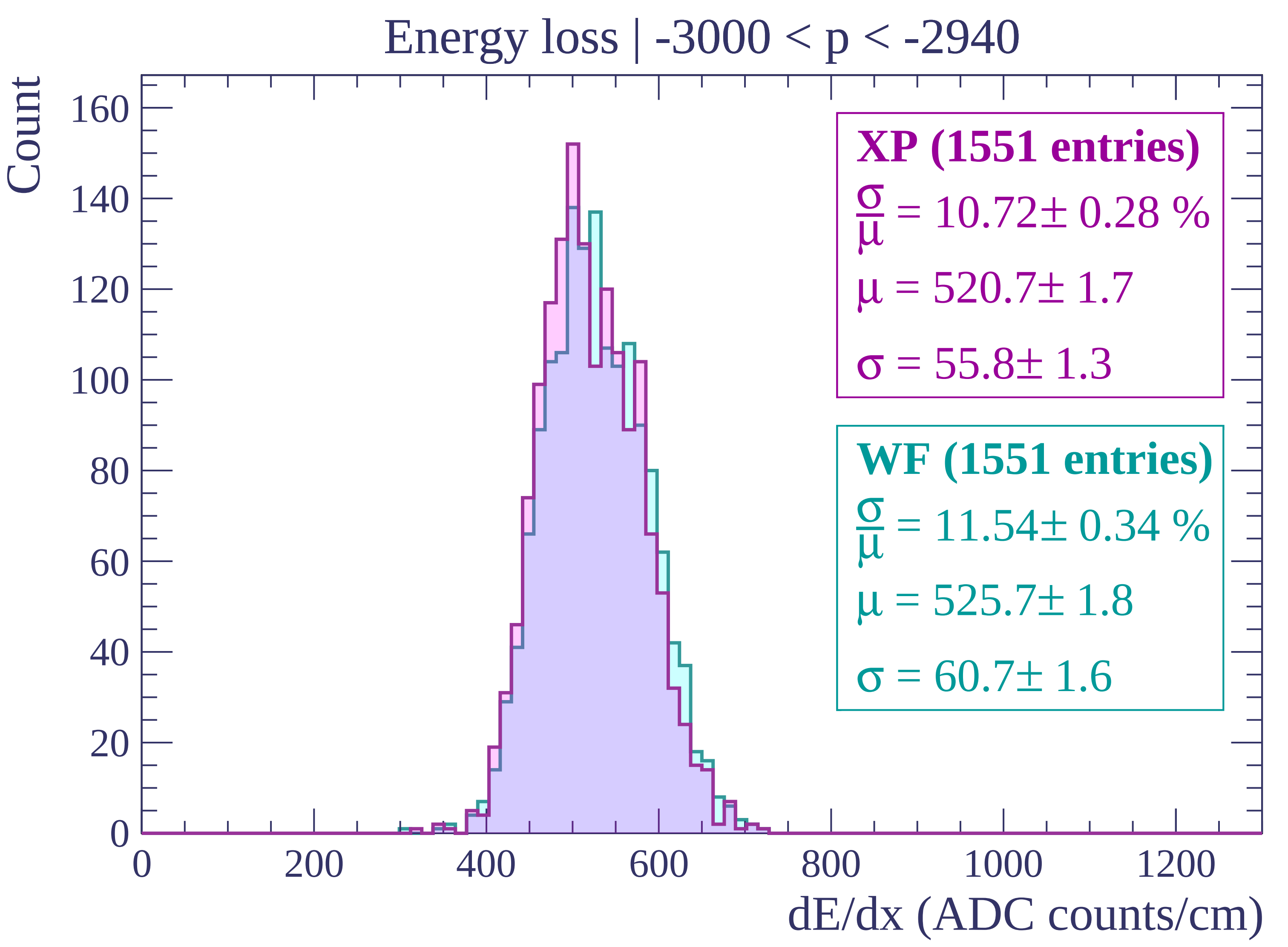
Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
Momentum
3000
2500
2000
1500
1000
500
0



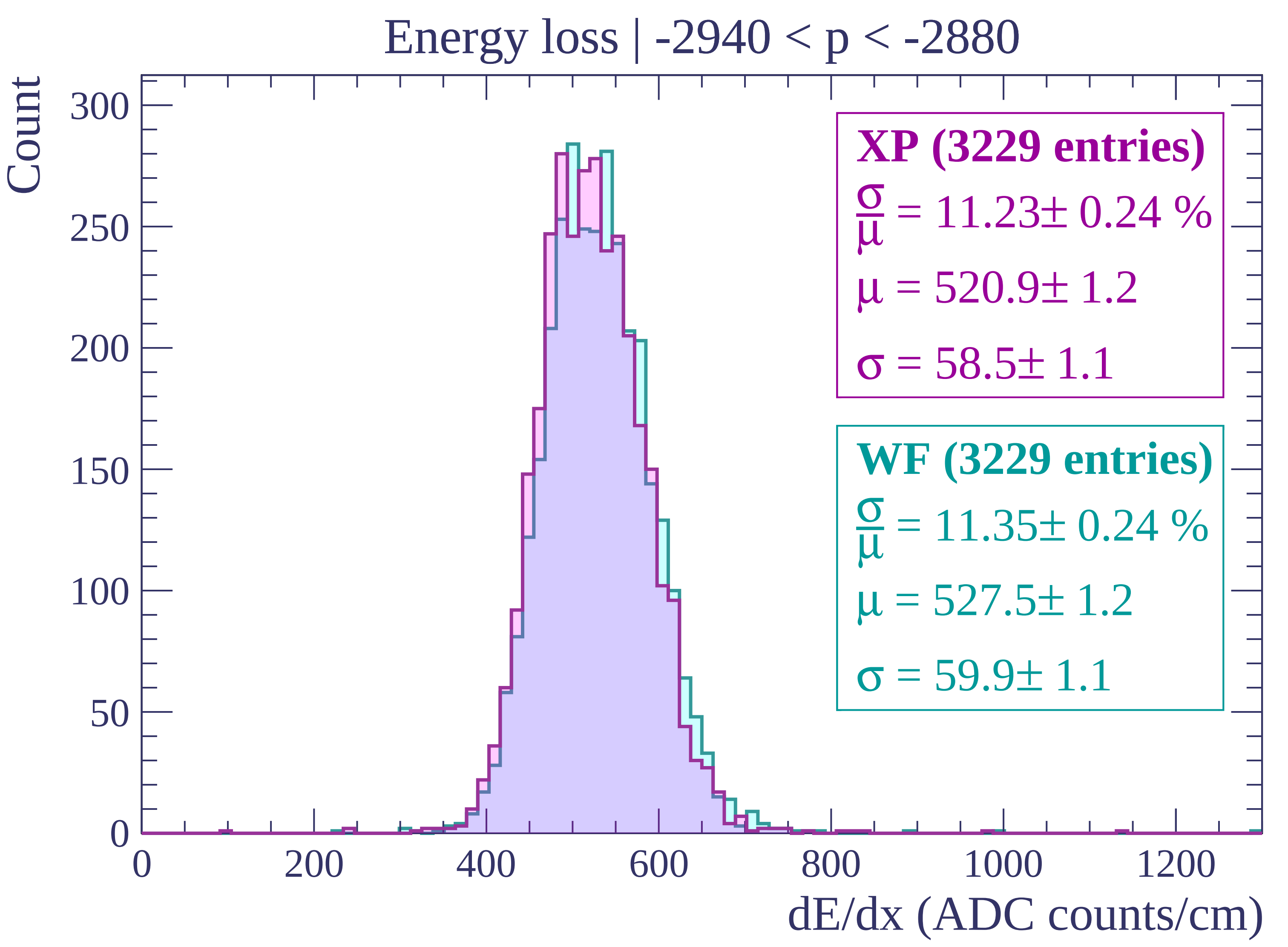
Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
×10³
50
40
30
20
10
0



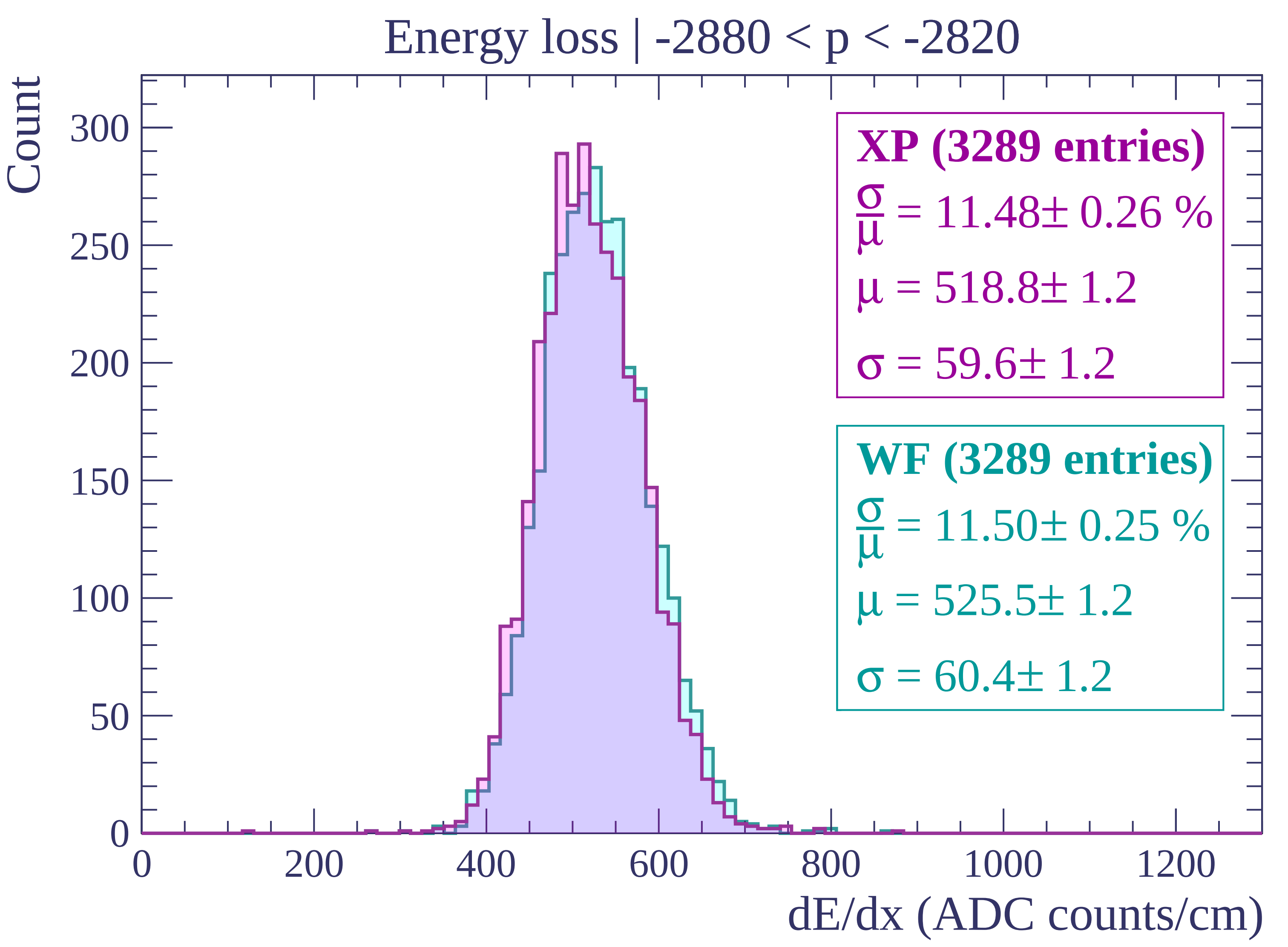
Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0
2
4
6
8
10
8000
7000
6000
5000
4000
3000
2000
1000
0



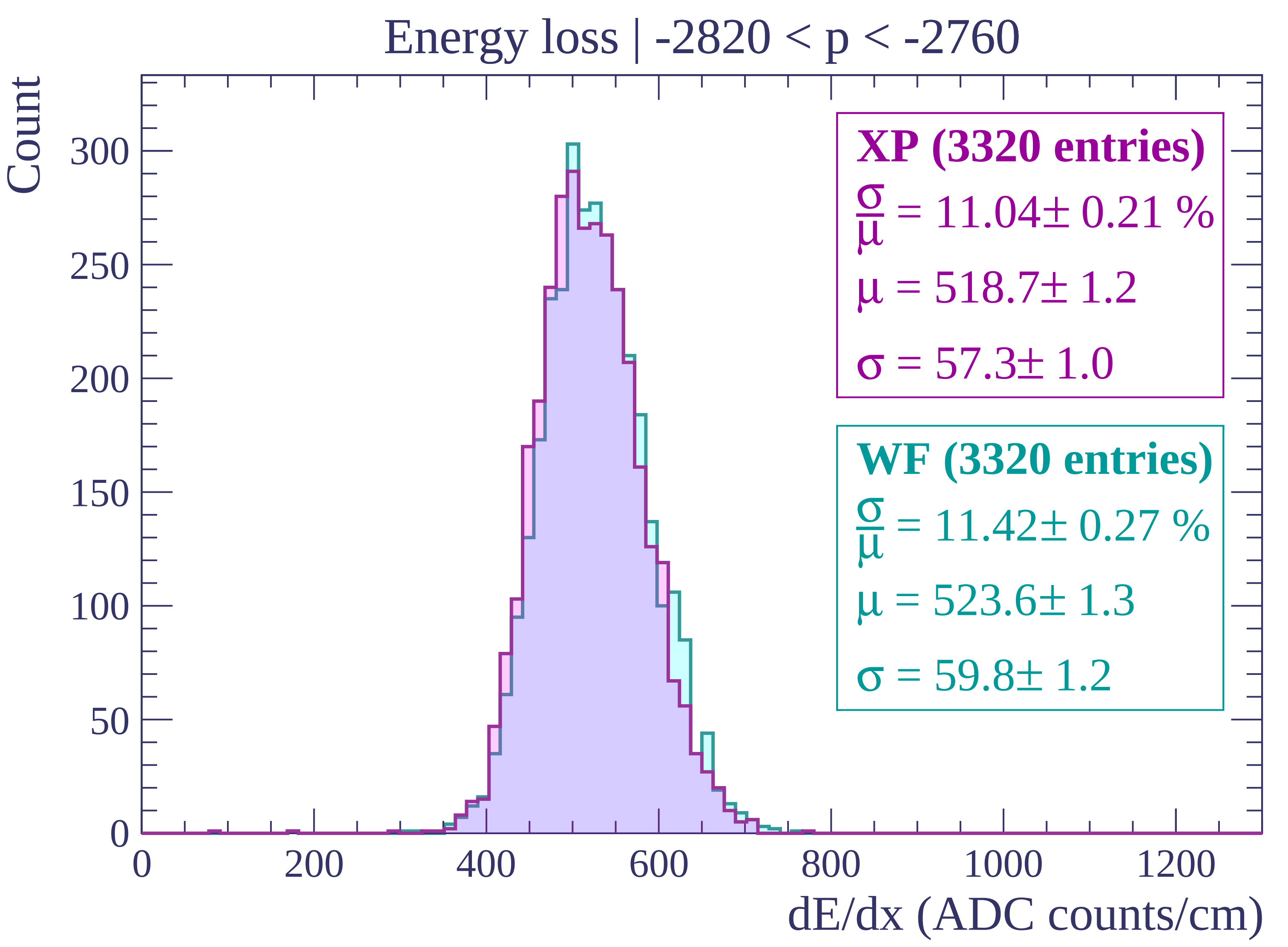
Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
XP (1551 entries)
$\frac{\sigma}{\mu}$ = 10.72± 0.28 %
μ = 520.7± 1.7
σ = 55.8± 1.3
WF (1551 entries)
$\frac{\sigma}{\mu}$ = 11.54± 0.34 %
μ = 525.7± 1.8
σ = 60.7± 1.6
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



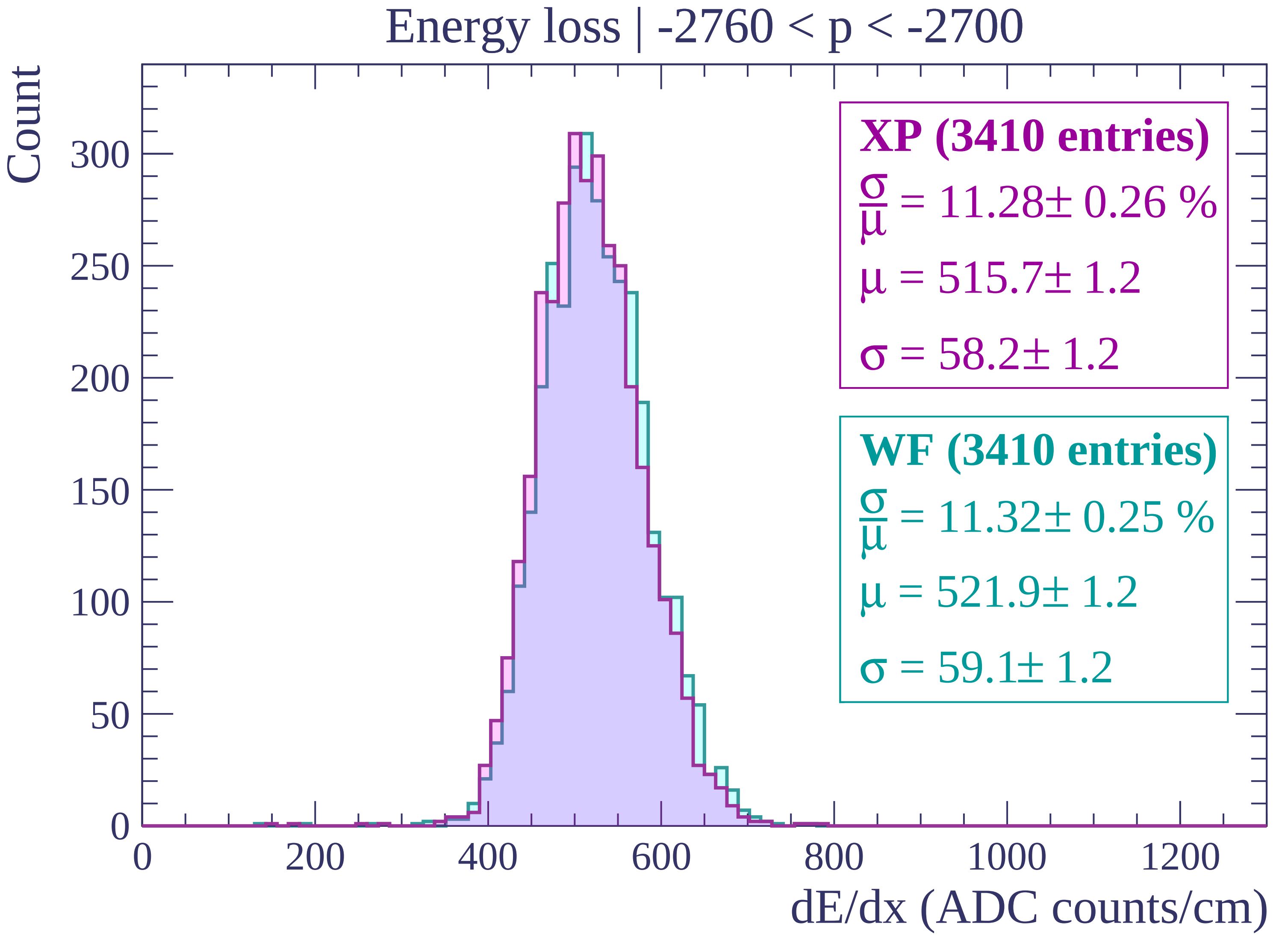
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP (3229 entries)
σ/μ = 11.23± 0.24 %
μ = 520.9± 1.2
σ = 58.5± 1.1
WF (3229 entries)
σ/μ = 11.35± 0.24 %
μ = 527.5± 1.2
σ = 59.9± 1.1

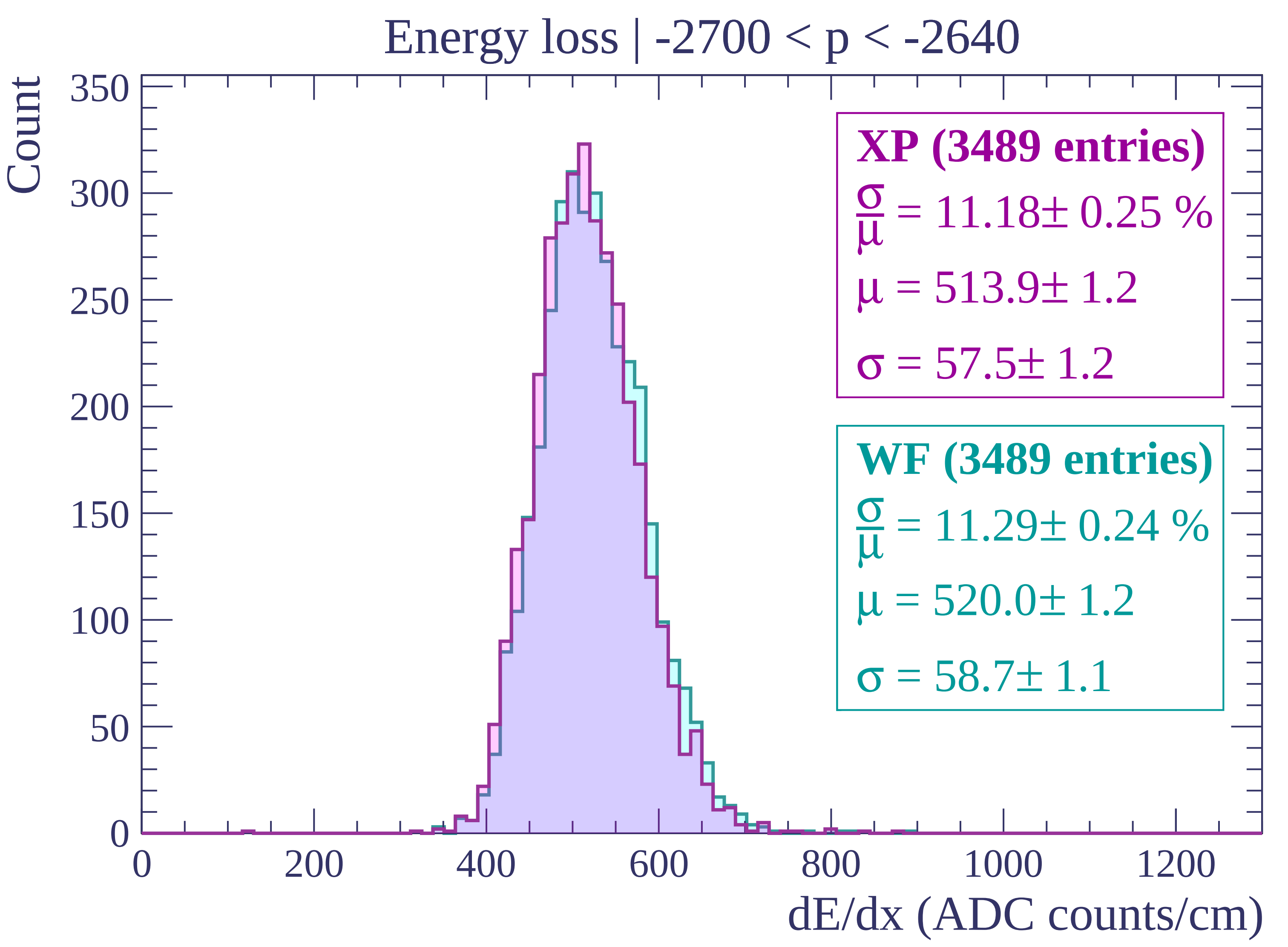


Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3289 entries)
σ/μ = 11.48± 0.26 %
μ = 518.8± 1.2
σ = 59.6± 1.2
WF (3289 entries)
σ/μ = 11.50± 0.25 %
μ = 525.5± 1.2
σ = 60.4± 1.2

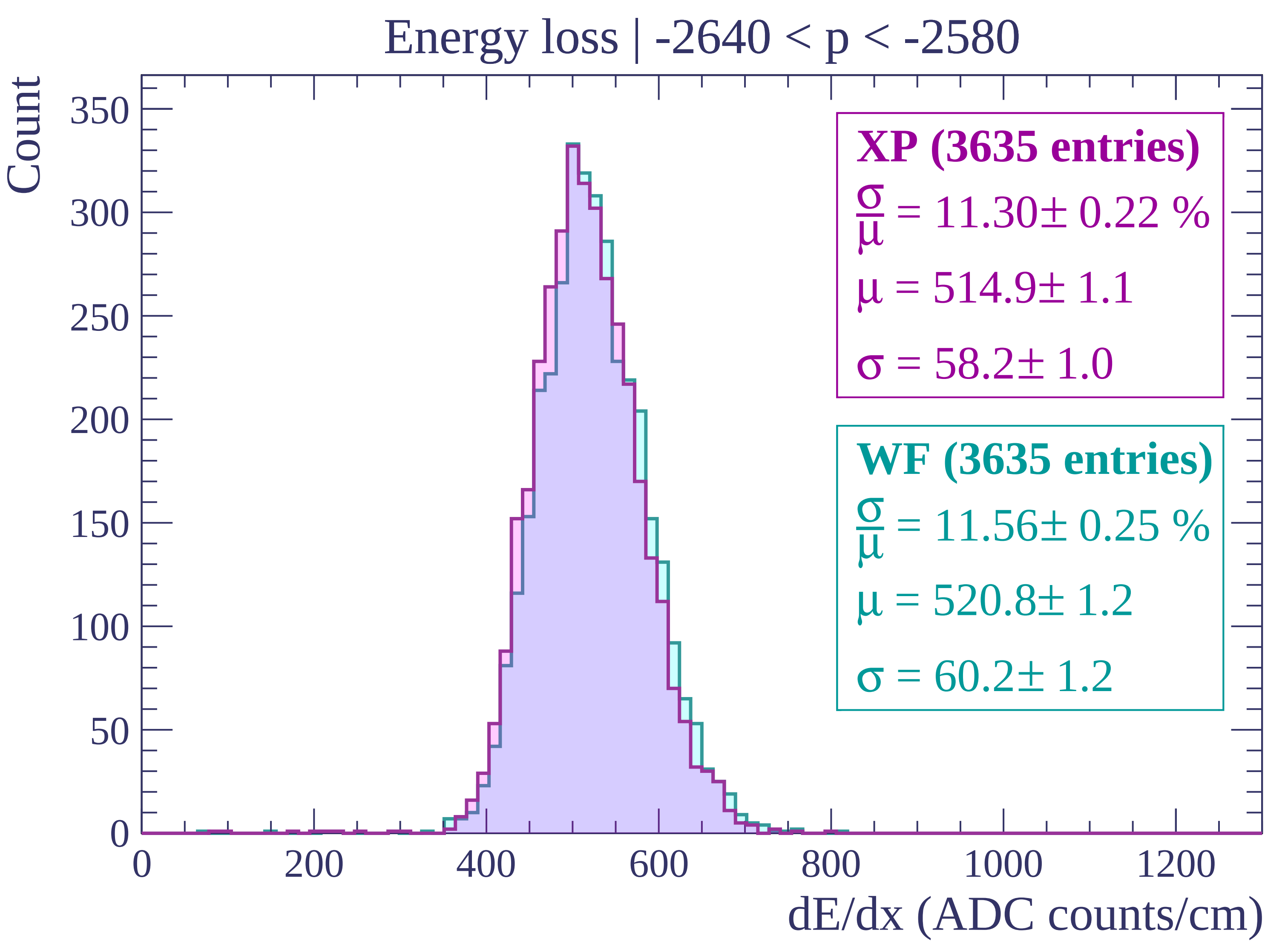


Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3320 entries)
σ/μ = 11.04± 0.21 %
μ = 518.7± 1.2
σ = 57.3± 1.0
WF (3320 entries)
σ/μ = 11.42± 0.27 %
μ = 523.6± 1.3
σ = 59.8± 1.2

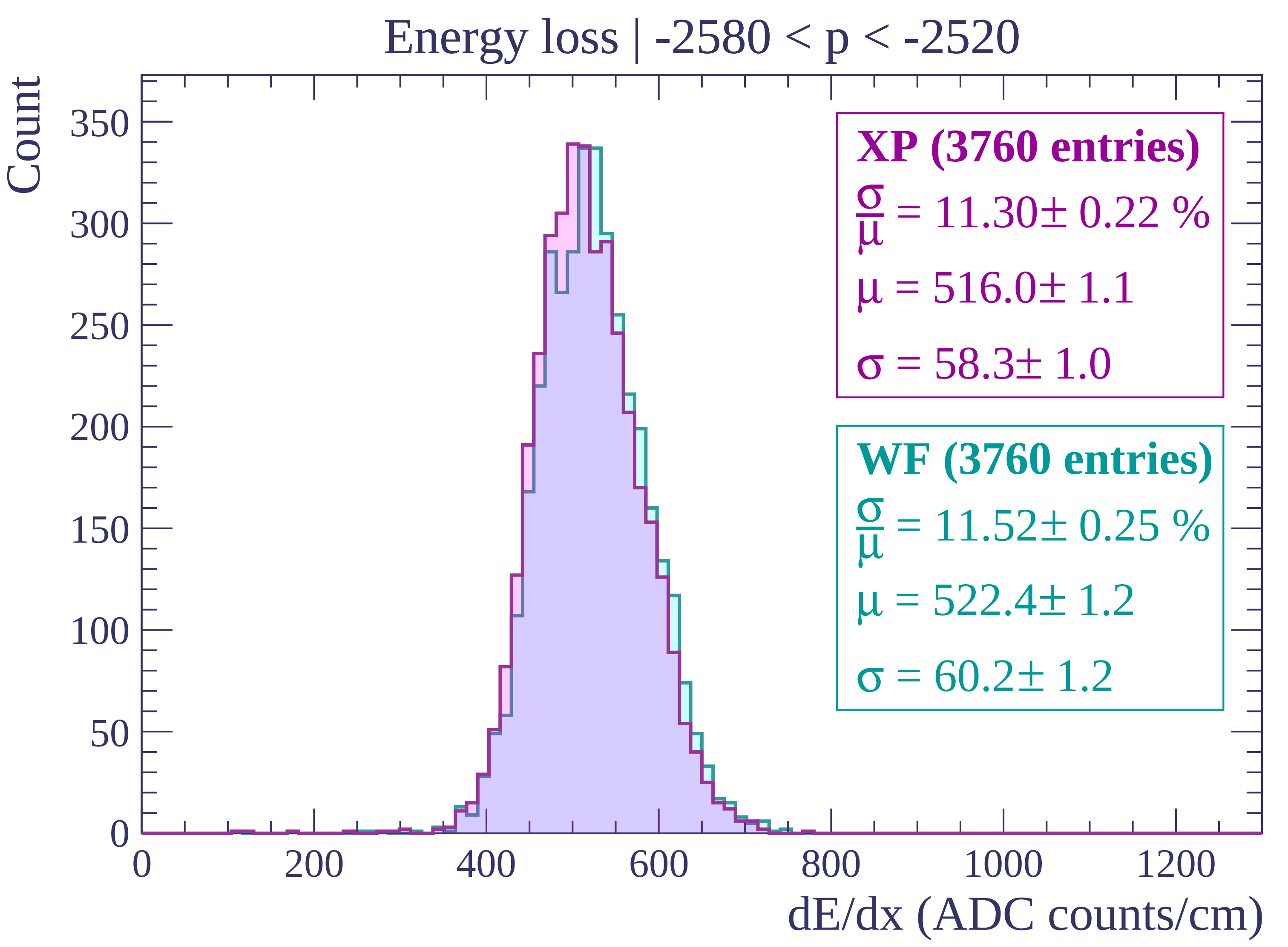

Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
XP (3410 entries)
σ/μ = 11.28± 0.26 %
μ = 515.7± 1.2
σ = 58.2± 1.2
WF (3410 entries)
σ/μ = 11.32± 0.25 %
μ = 521.9± 1.2
σ = 59.1± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



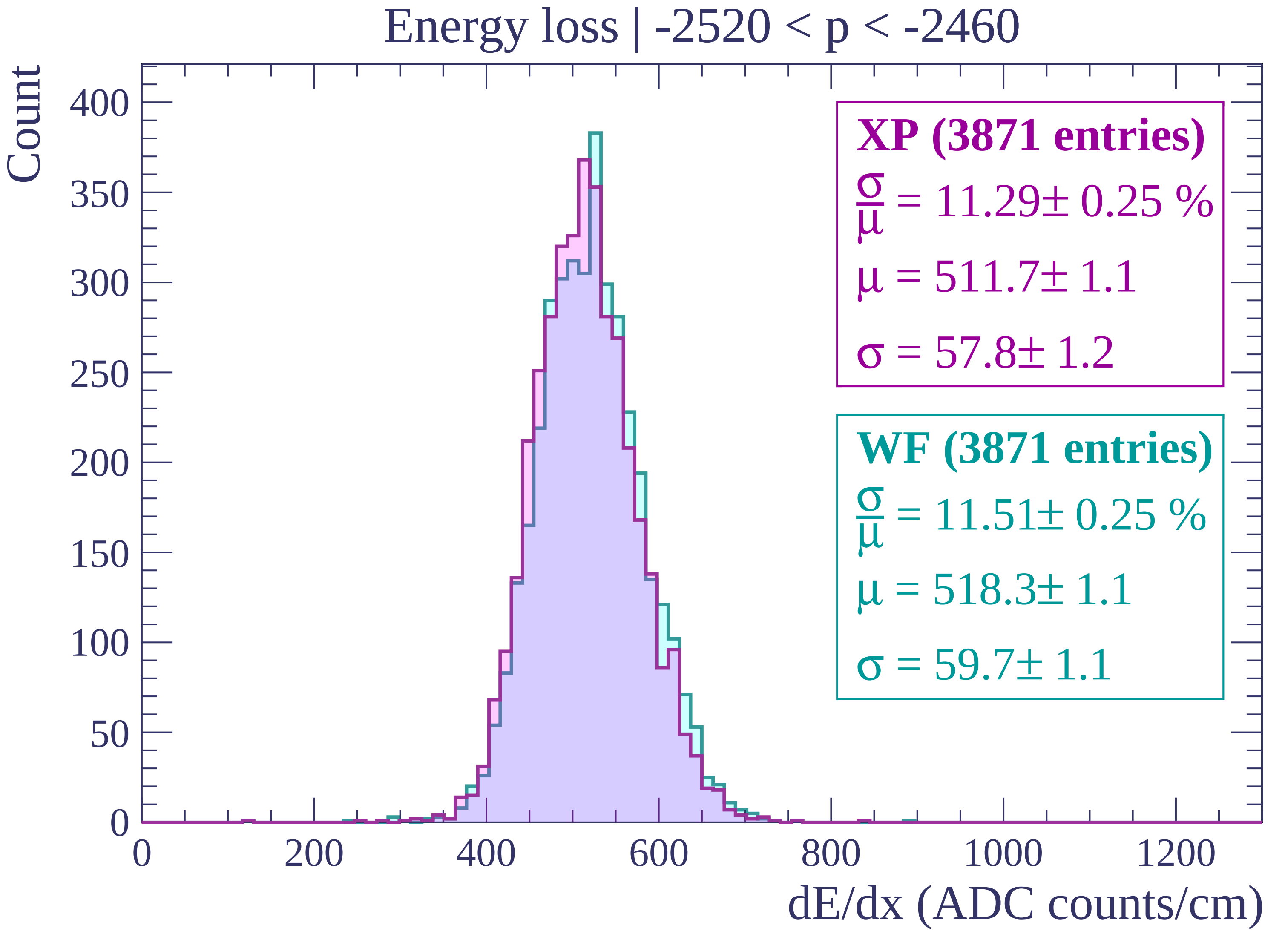
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3489 entries)
$\frac{\sigma}{\mu}$ = 11.18± 0.25 %
μ = 513.9± 1.2
σ = 57.5± 1.2
WF (3489 entries)
$\frac{\sigma}{\mu}$ = 11.29± 0.24 %
μ = 520.0± 1.2
σ = 58.7± 1.1



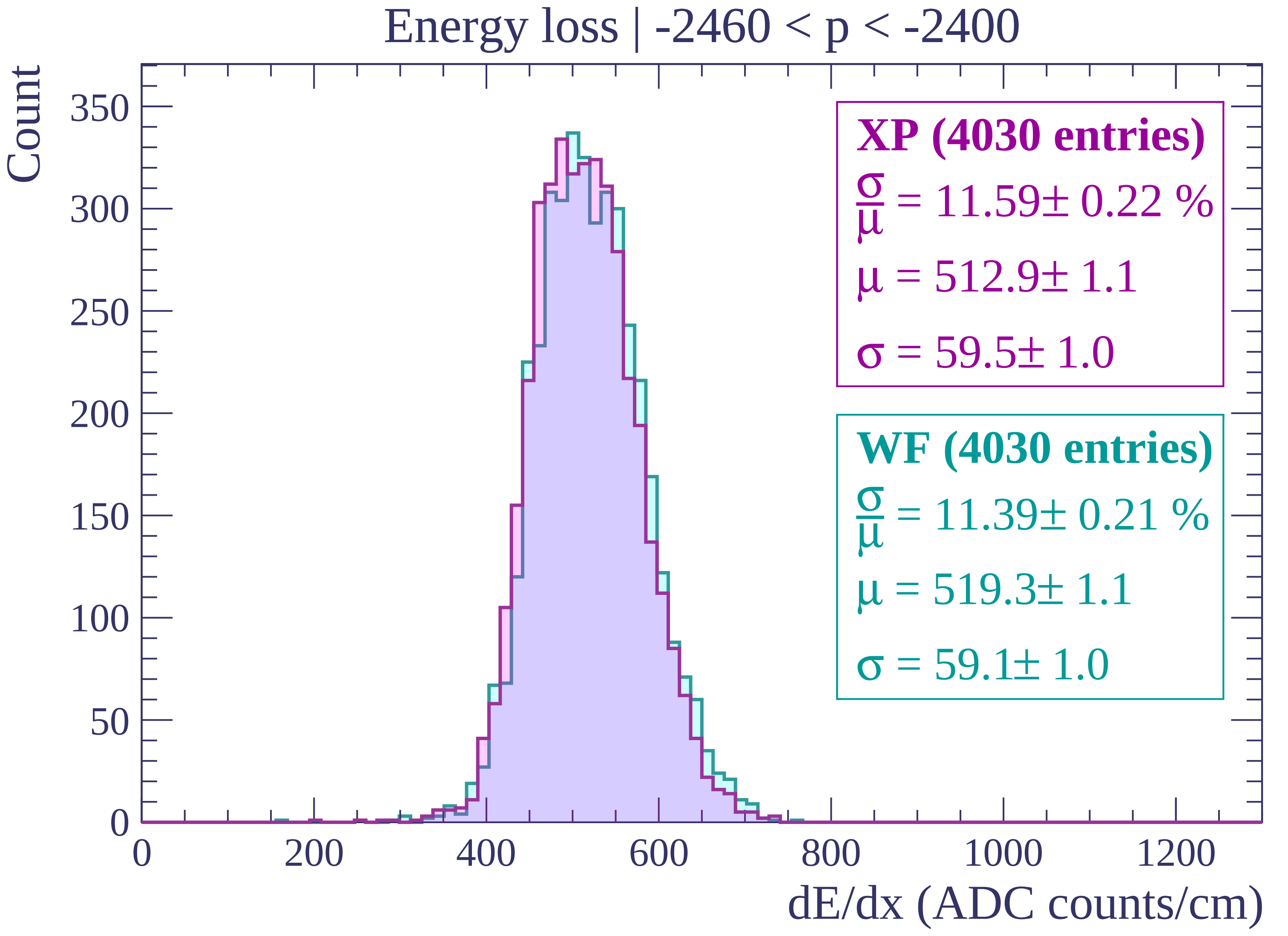
Energy loss | -2640 < p < -2580
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3635 entries)
σ/μ = 11.30± 0.22 %
μ = 514.9± 1.1
σ = 58.2± 1.0
WF (3635 entries)
σ/μ = 11.56± 0.25 %
μ = 520.8± 1.2
σ = 60.2± 1.2



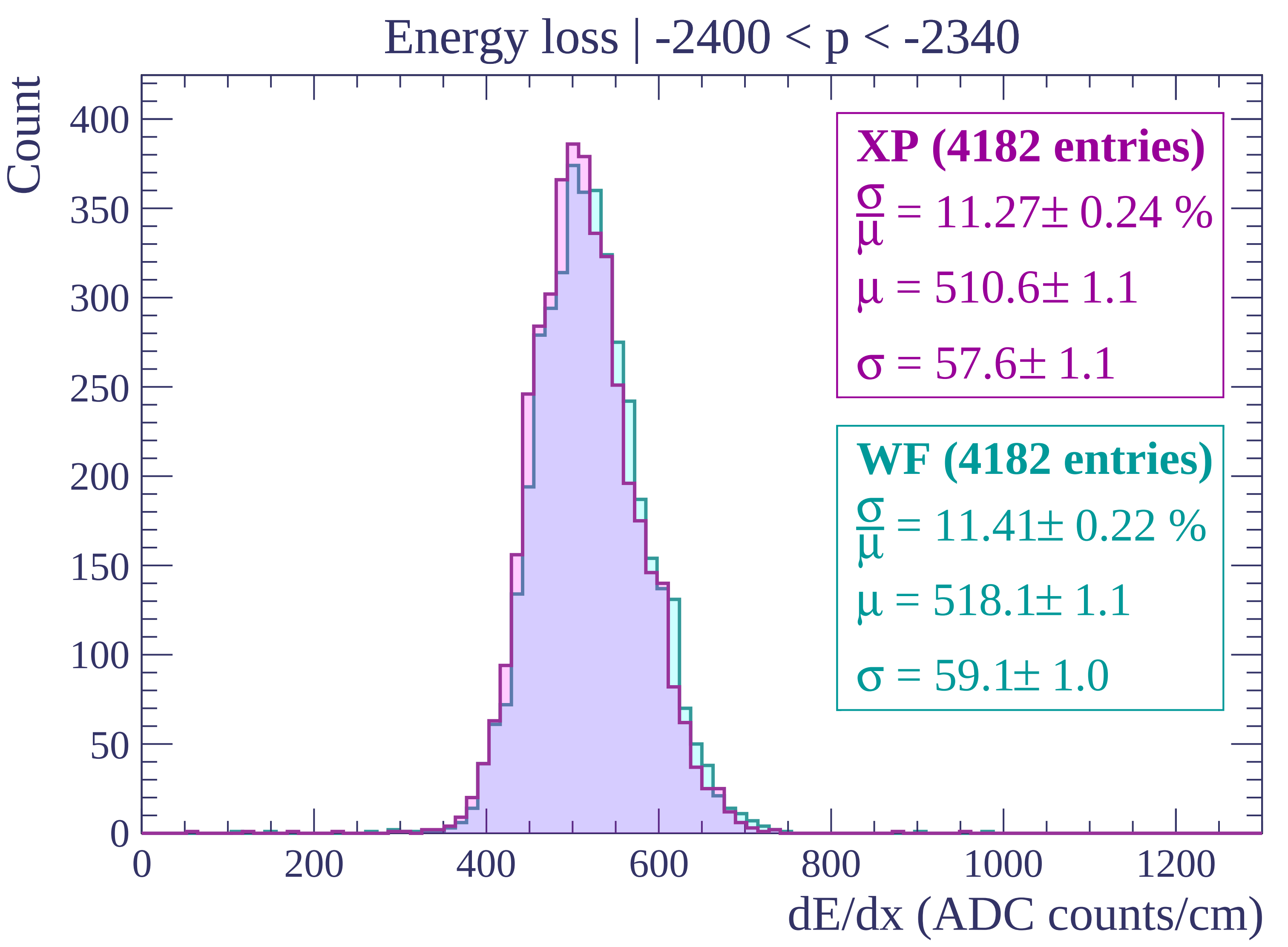
Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3760 entries)
$\frac{\sigma}{\mu}$ = 11.30± 0.22 %
μ = 516.0± 1.1
σ = 58.3± 1.0
WF (3760 entries)
$\frac{\sigma}{\mu}$ = 11.52± 0.25 %
μ = 522.4± 1.2
σ = 60.2± 1.2

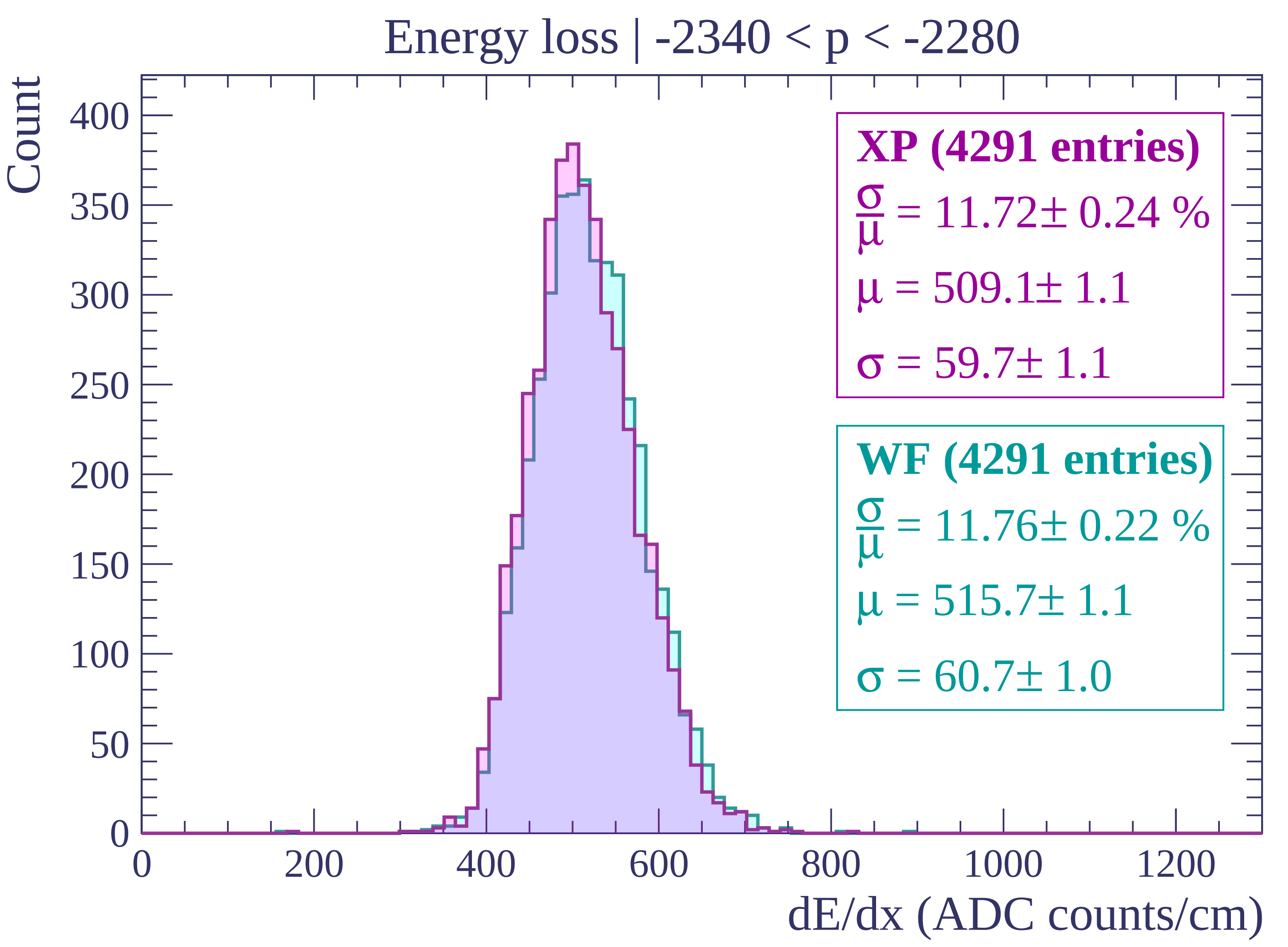


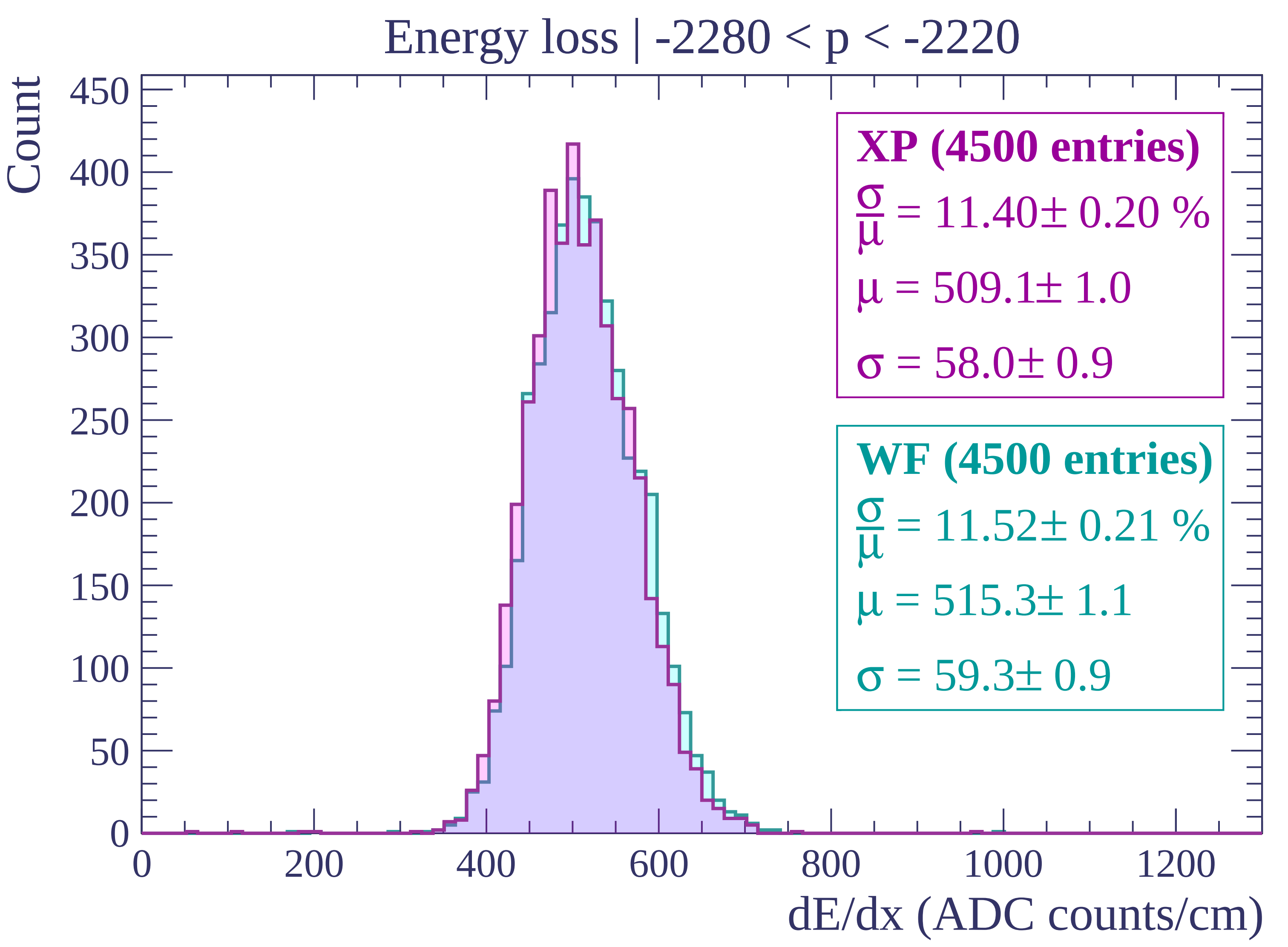
Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
XP (3871 entries)
$\frac{\sigma}{\mu}$ = 11.29± 0.25 %
μ = 511.7± 1.1
σ = 57.8± 1.2
WF (3871 entries)
$\frac{\sigma}{\mu}$ = 11.51± 0.25 %
μ = 518.3± 1.1
σ = 59.7± 1.1



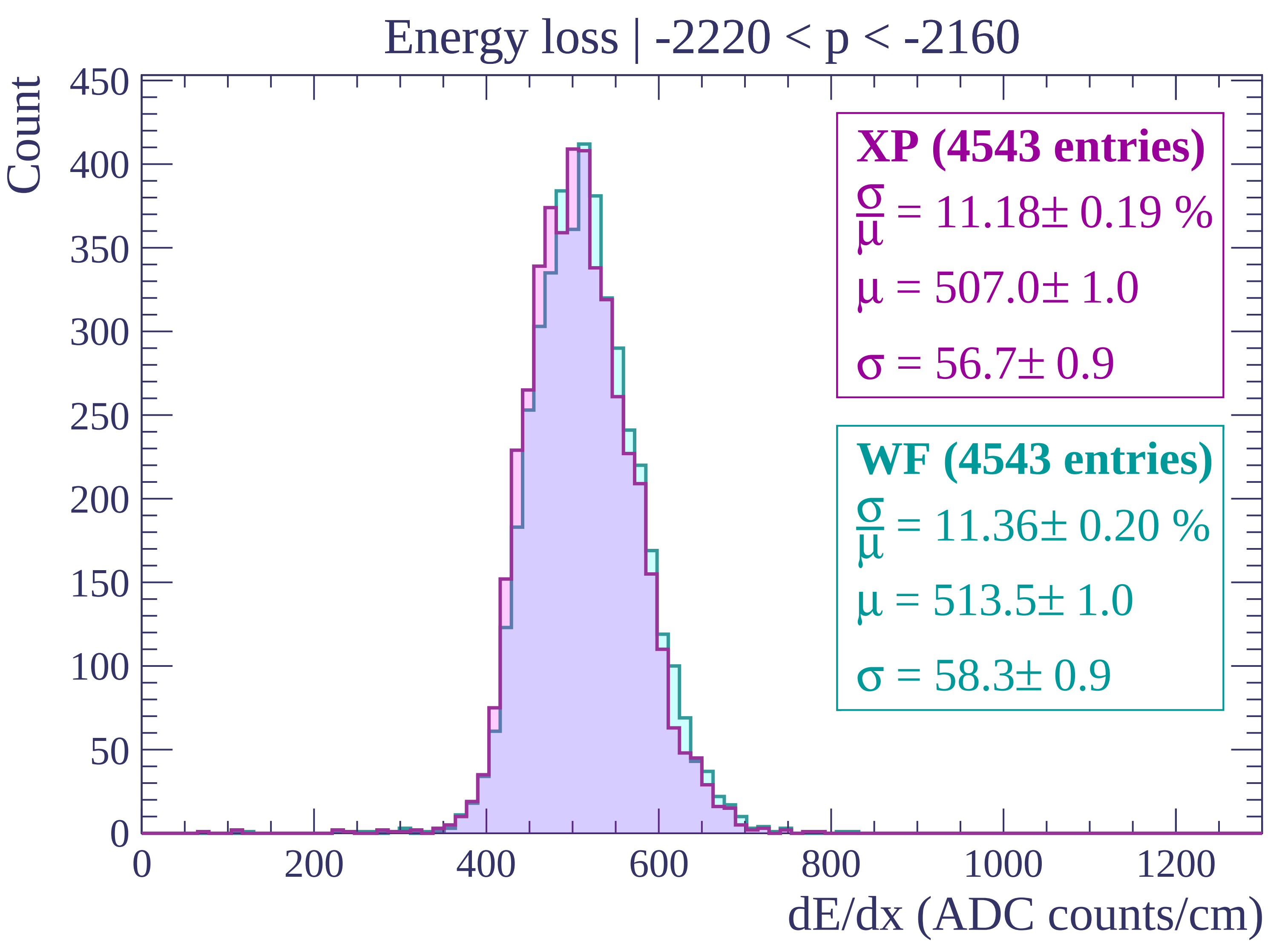
Energy loss | -2460 < p < -2400
Count
XP (4030 entries)
$\frac{\sigma}{\mu}$ = 11.59± 0.22 %
μ = 512.9± 1.1
σ = 59.5± 1.0
WF (4030 entries)
$\frac{\sigma}{\mu}$ = 11.39± 0.21 %
μ = 519.3± 1.1
σ = 59.1± 1.0
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
XP (4182 entries)
$\frac{\sigma}{\mu}$ = 11.27± 0.24 %
μ = 510.6± 1.1
σ = 57.6± 1.1
WF (4182 entries)
$\frac{\sigma}{\mu}$ = 11.41± 0.22 %
μ = 518.1± 1.1
σ = 59.1± 1.0
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

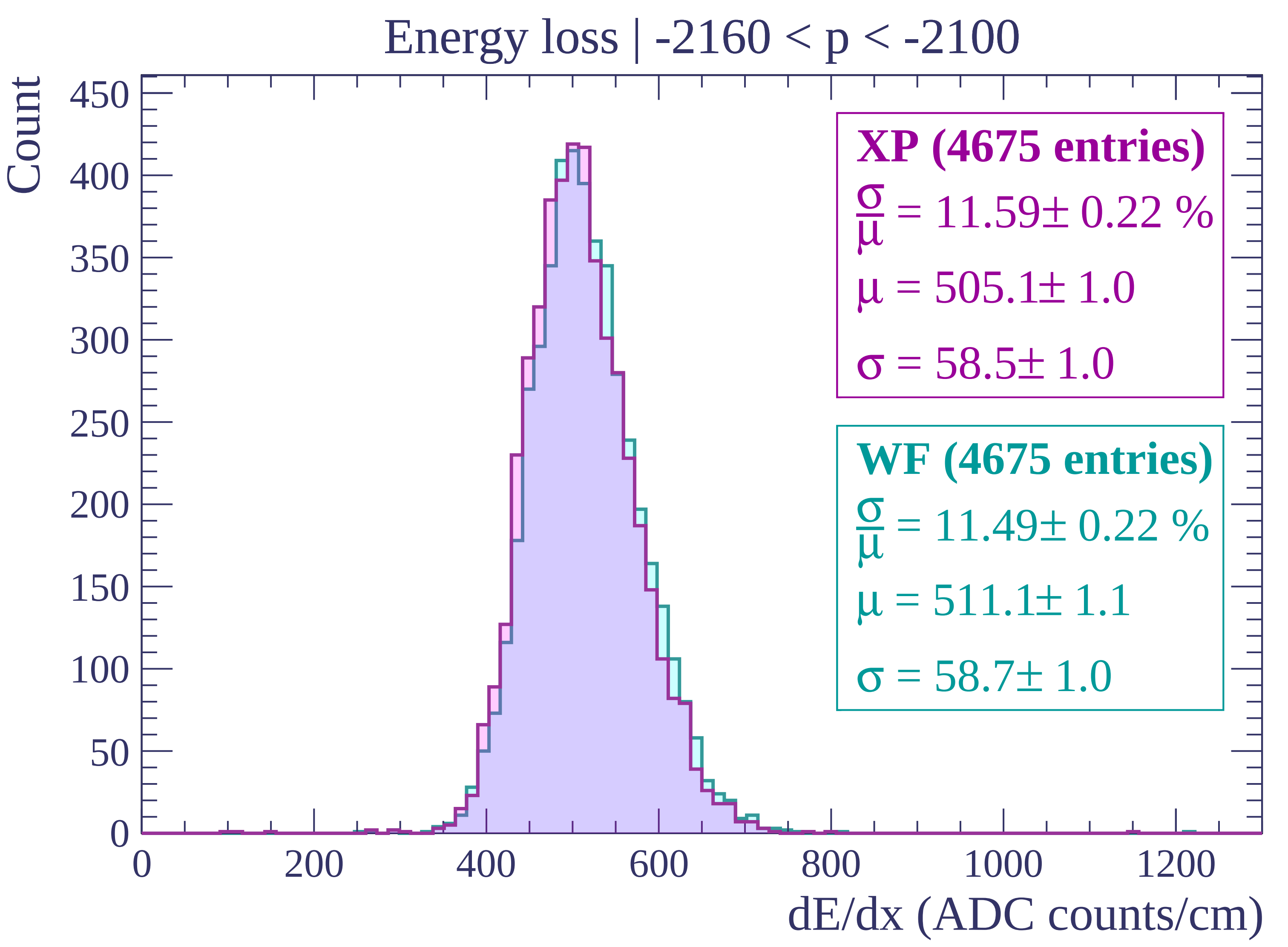

Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4291 entries)
$\frac{\sigma}{\mu}$ = 11.72± 0.24 %
μ = 509.1± 1.1
σ = 59.7± 1.1
WF (4291 entries)
$\frac{\sigma}{\mu}$ = 11.76± 0.22 %
μ = 515.7± 1.1
σ = 60.7± 1.0



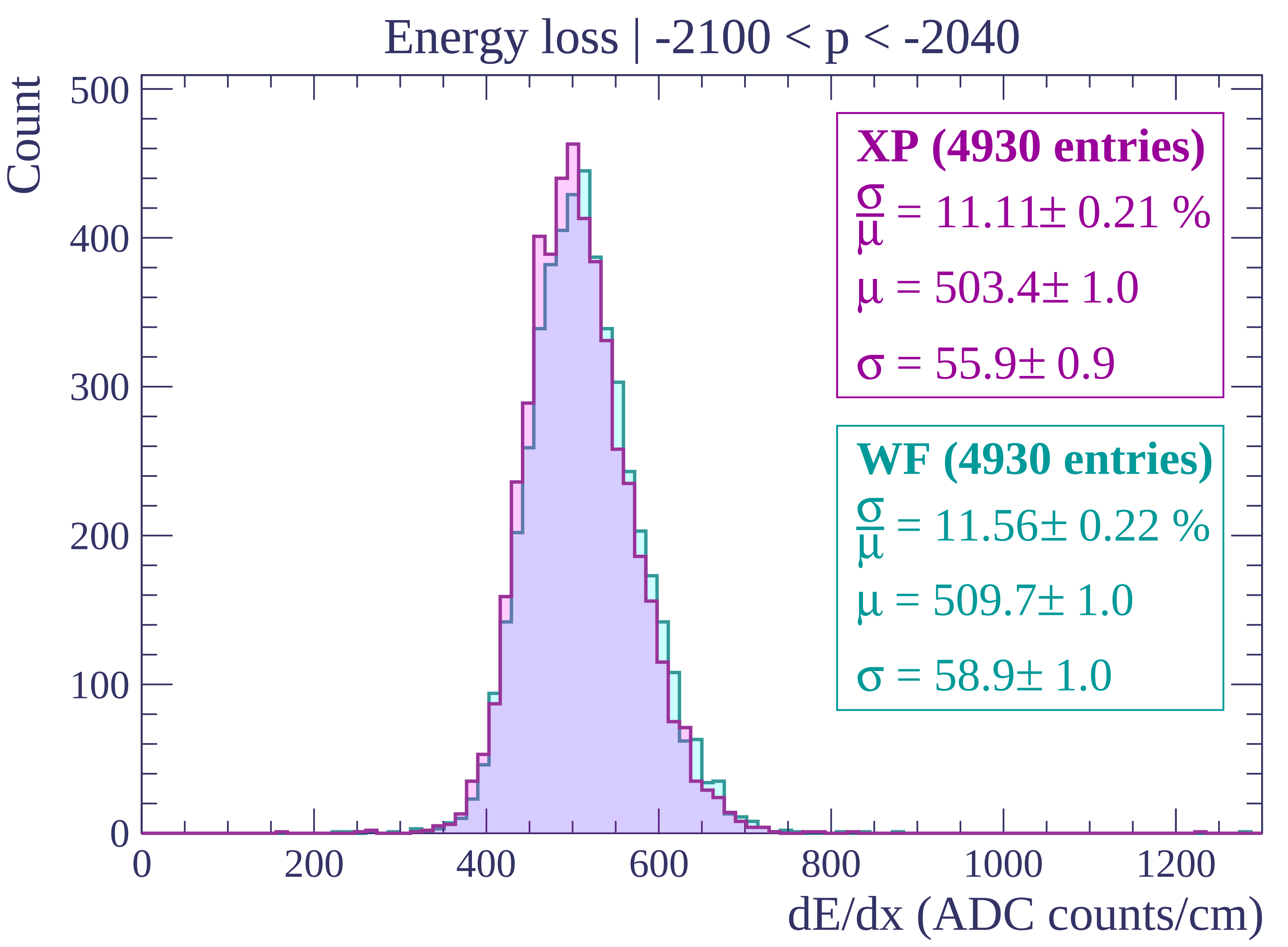
Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4500 entries)
σ/μ = 11.40± 0.20 %
μ = 509.1± 1.0
σ = 58.0± 0.9
WF (4500 entries)
σ/μ = 11.52± 0.21 %
μ = 515.3± 1.1
σ = 59.3± 0.9



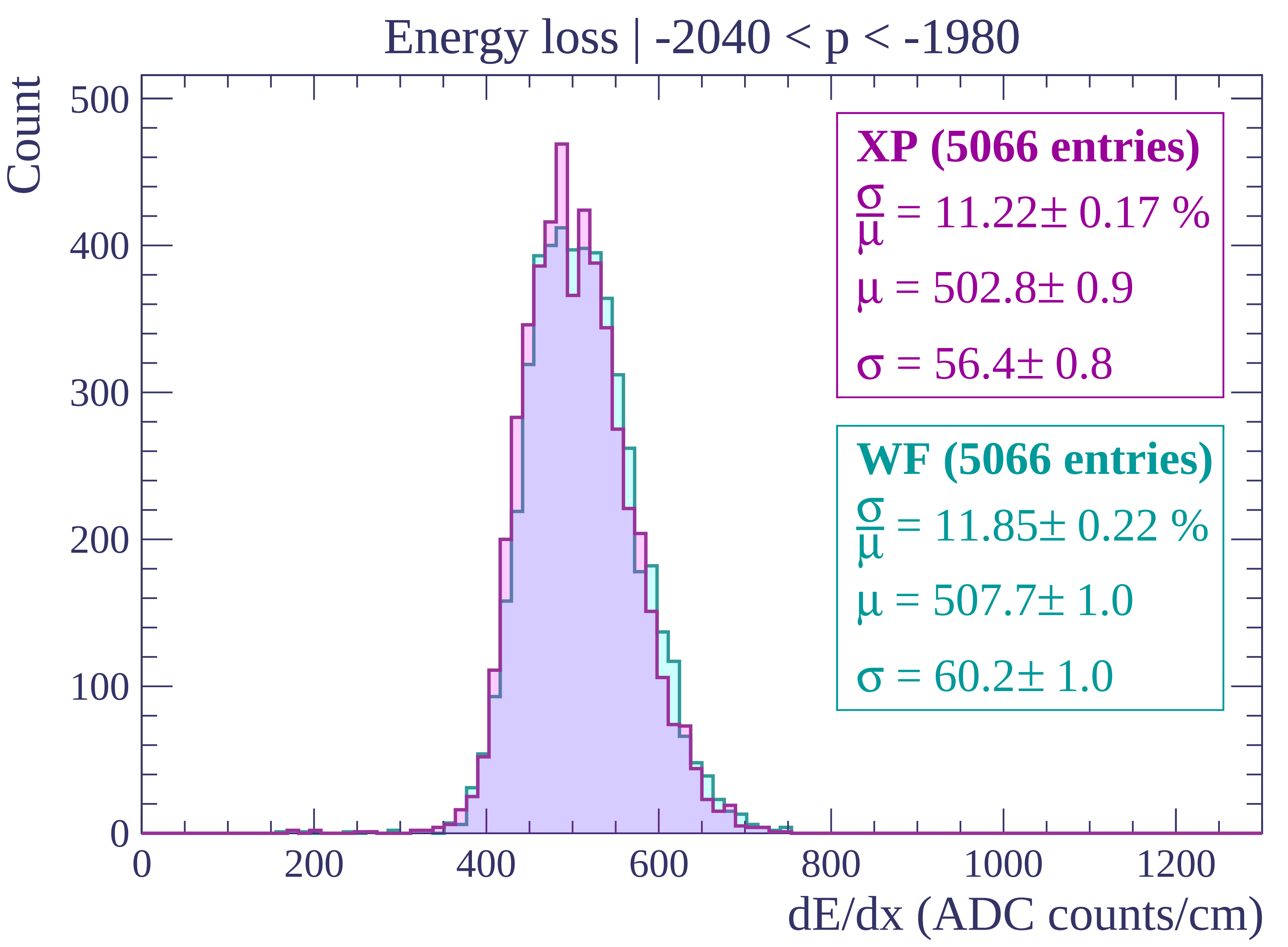
Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4543 entries)
$\frac{\sigma}{\mu}$ = 11.18± 0.19 %
μ = 507.0± 1.0
σ = 56.7± 0.9
WF (4543 entries)
$\frac{\sigma}{\mu}$ = 11.36± 0.20 %
μ = 513.5± 1.0
σ = 58.3± 0.9

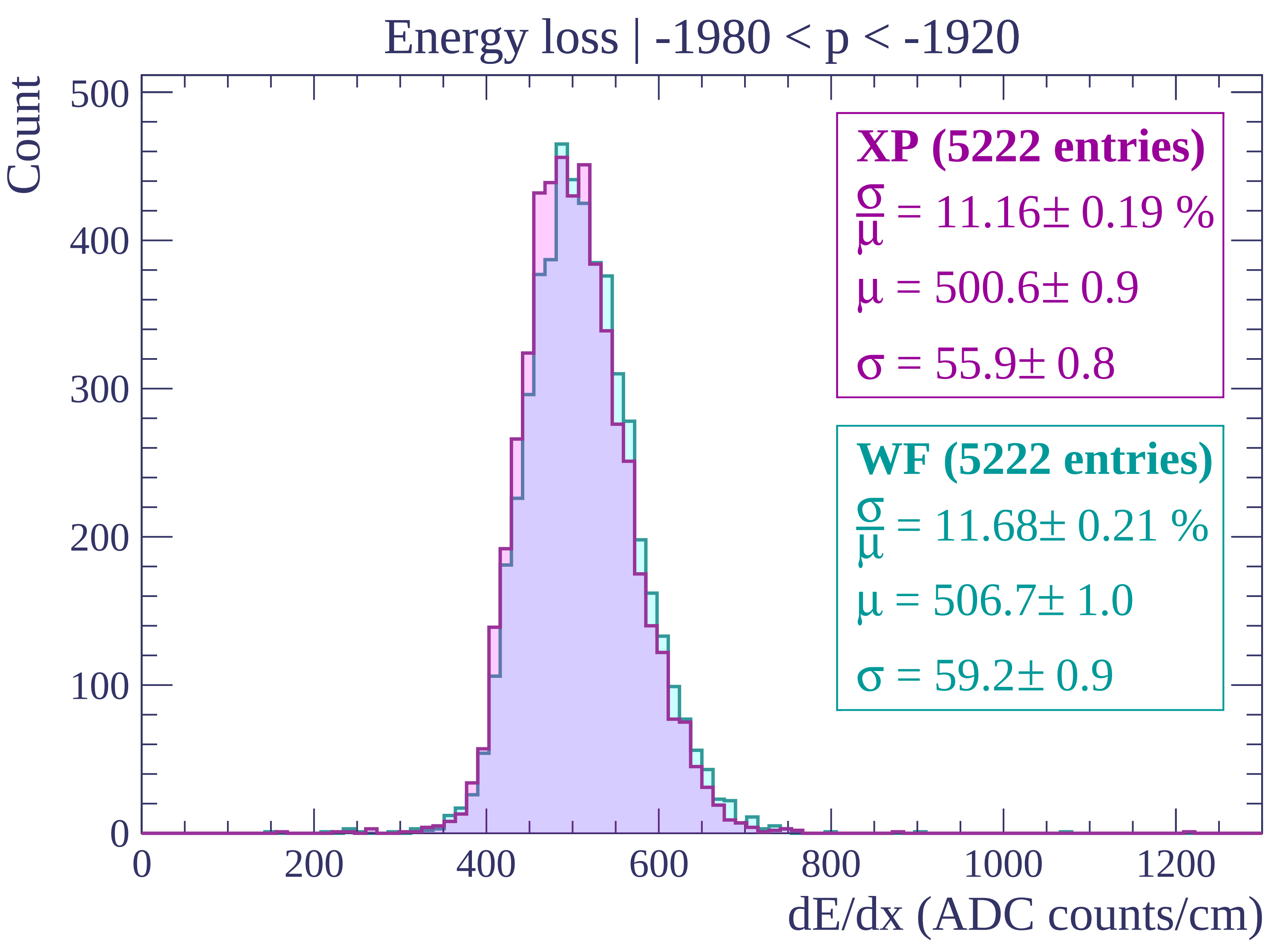


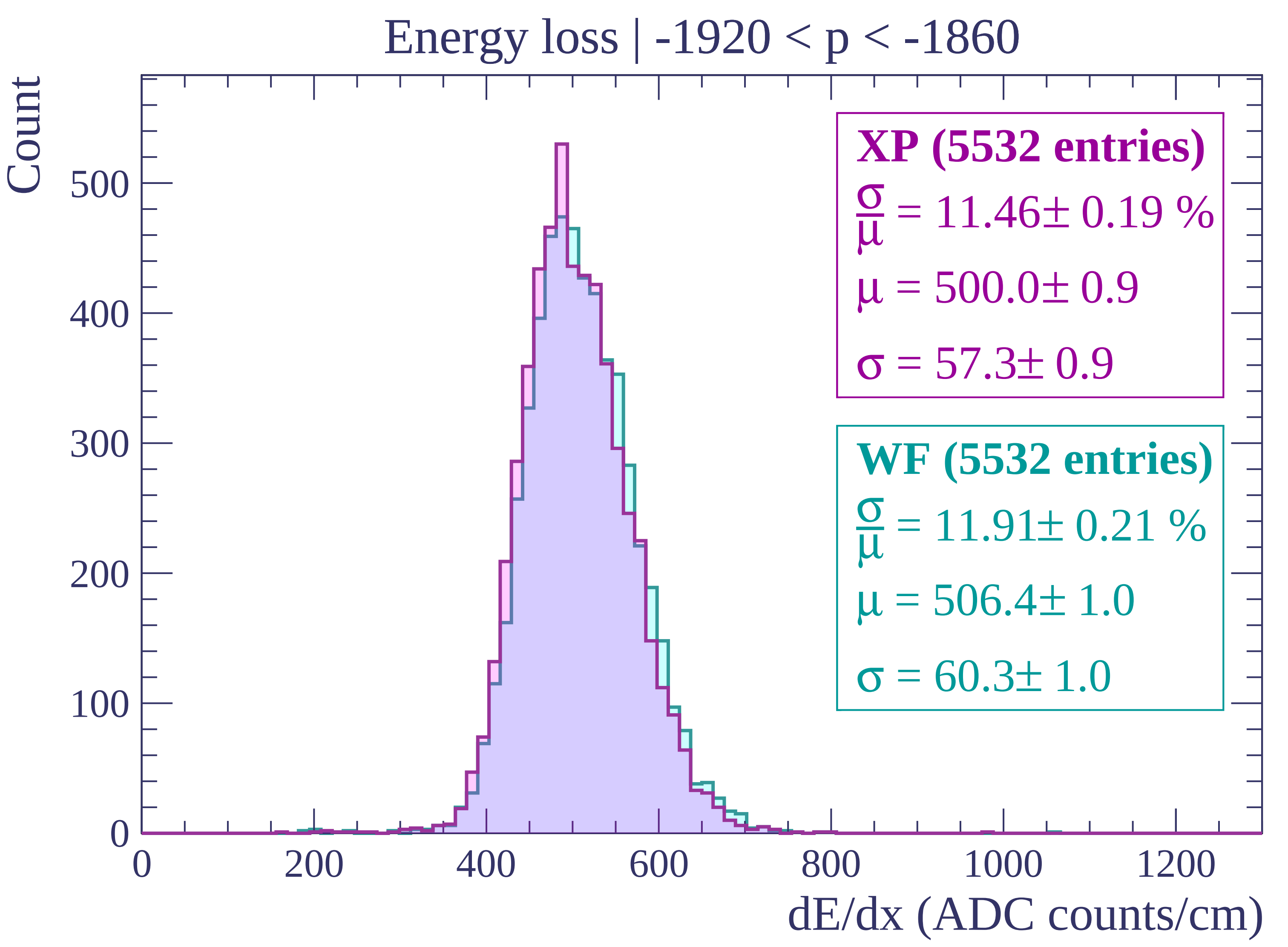
Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4675 entries)
σ/μ = 11.59± 0.22 %
μ = 505.1± 1.0
σ = 58.5± 1.0
WF (4675 entries)
σ/μ = 11.49± 0.22 %
μ = 511.1± 1.1
σ = 58.7± 1.0



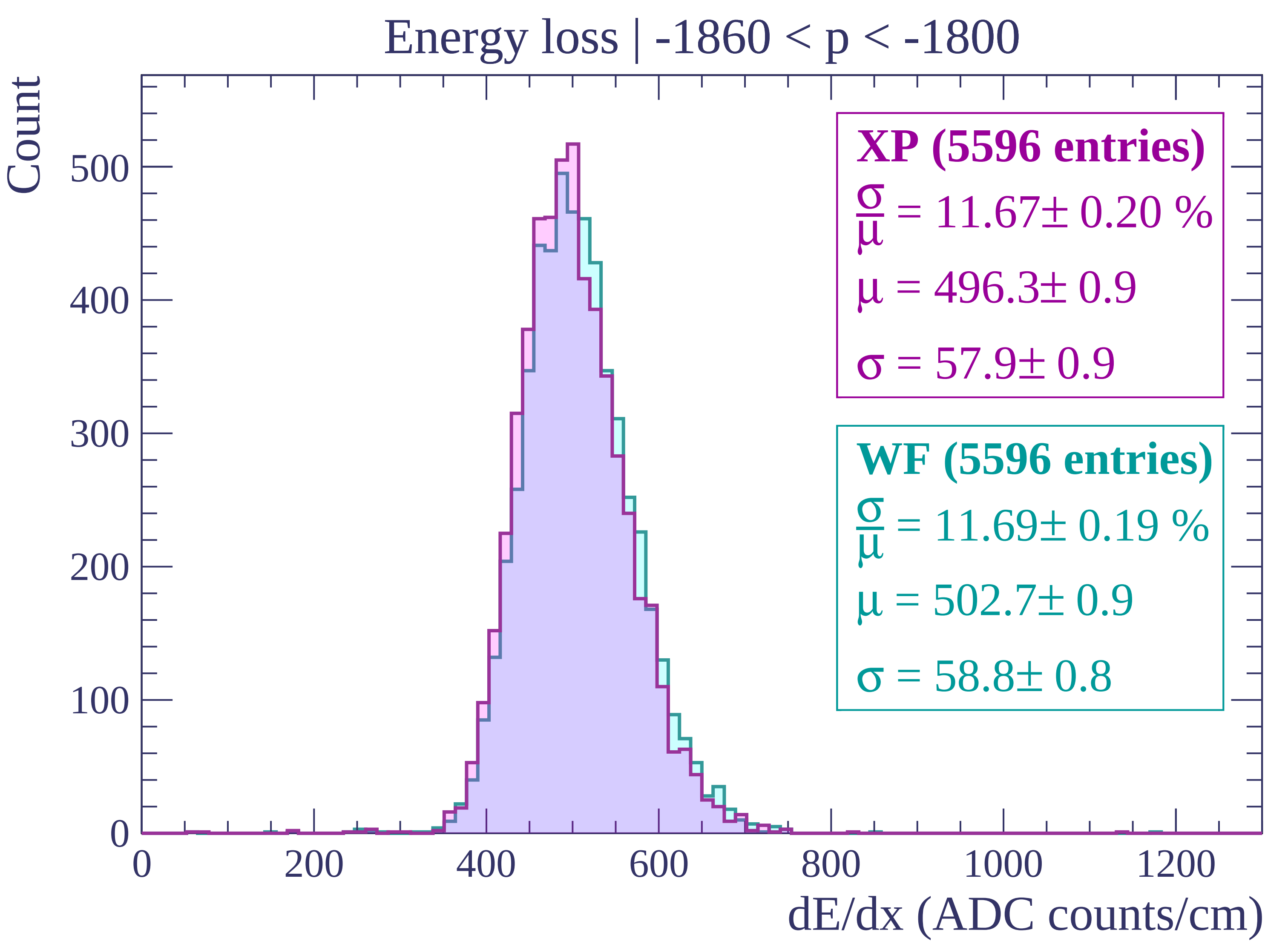
Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4930 entries)
$\frac{\sigma}{\mu}$ = 11.11± 0.21 %
μ = 503.4± 1.0
σ = 55.9± 0.9
WF (4930 entries)
$\frac{\sigma}{\mu}$ = 11.56± 0.22 %
μ = 509.7± 1.0
σ = 58.9± 1.0


Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
XP (5066 entries)
σ/μ = 11.22± 0.17 %
μ = 502.8± 0.9
σ = 56.4± 0.8
WF (5066 entries)
σ/μ = 11.85± 0.22 %
μ = 507.7± 1.0
σ = 60.2± 1.0
0
100
200
300
400
500
0
200
400
600
800
1000
1200

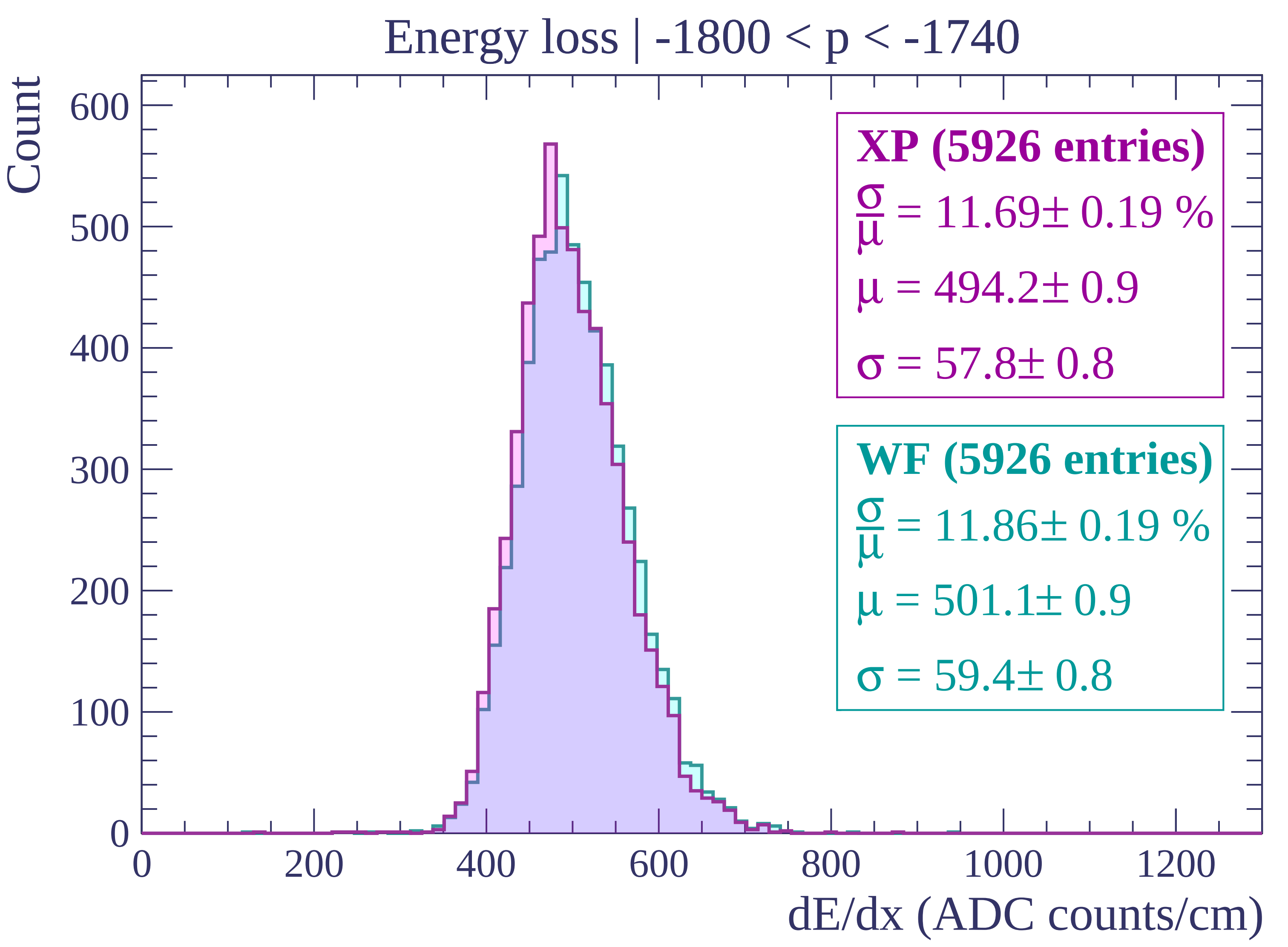

Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
100
200
300
400
500
XP (5222 entries)
σ/μ = 11.16± 0.19 %
μ = 500.6± 0.9
σ = 55.9± 0.8
WF (5222 entries)
σ/μ = 11.68± 0.21 %
μ = 506.7± 1.0
σ = 59.2± 0.9

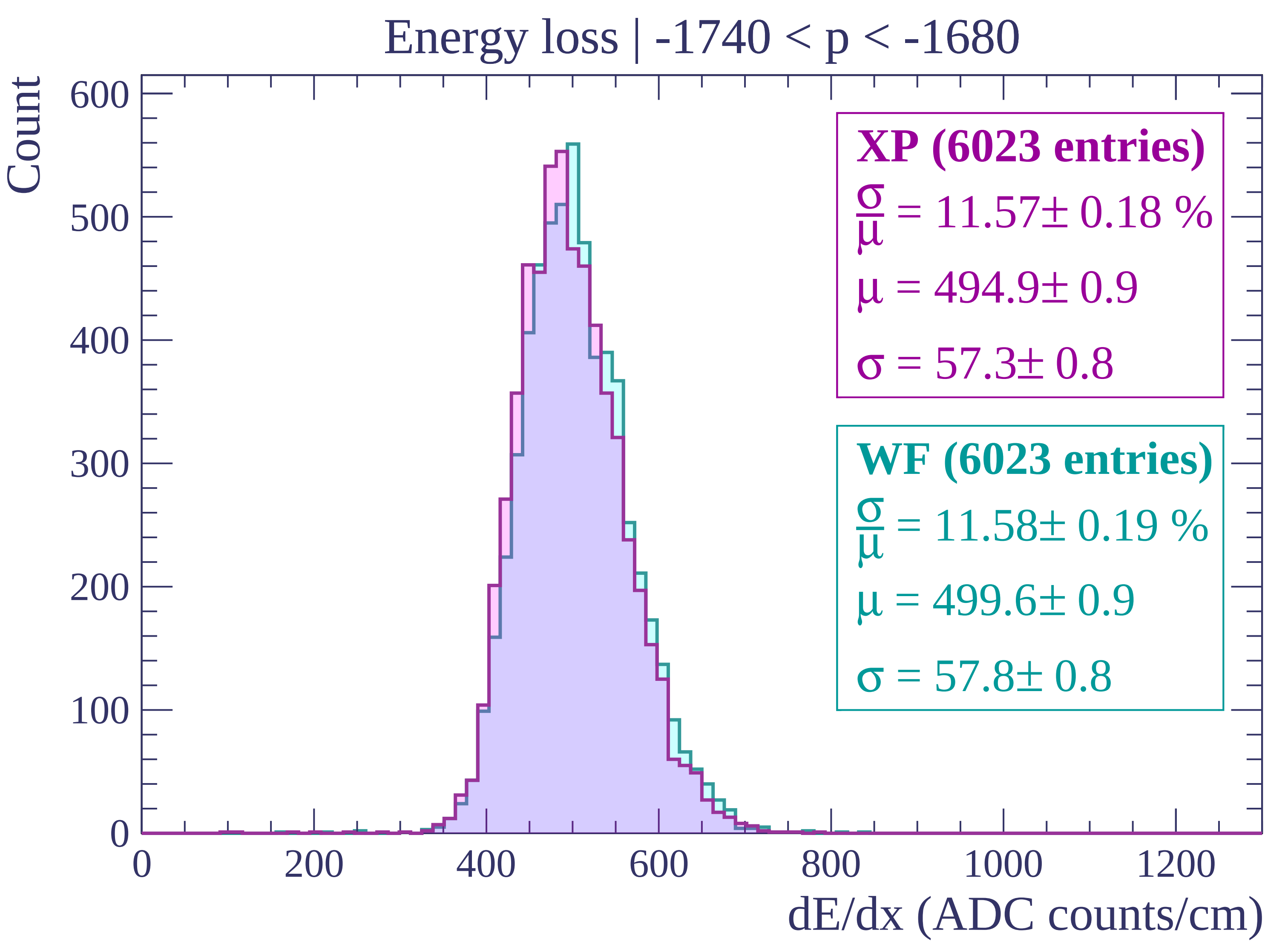


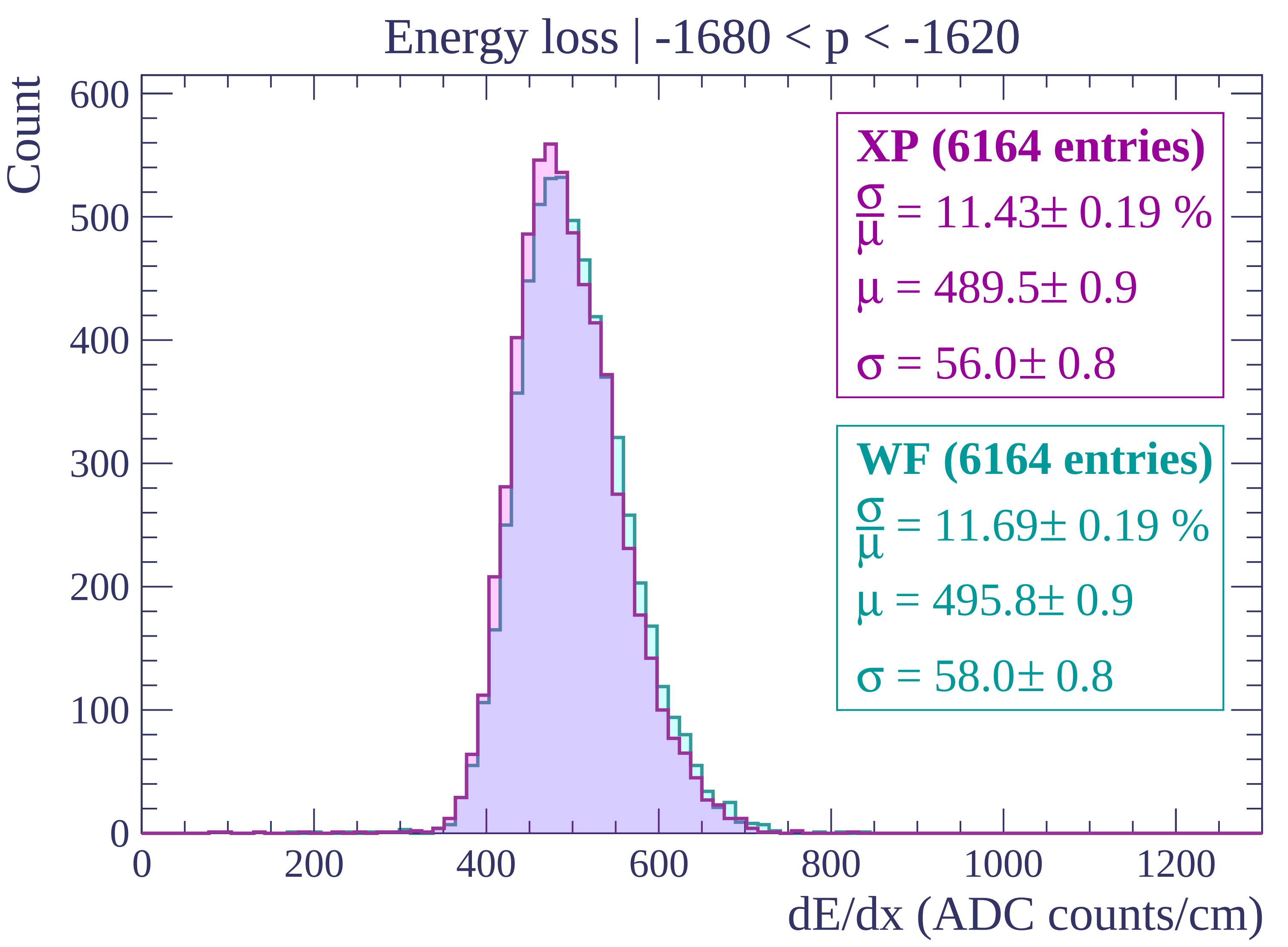
Energy loss | -1920 < p < -1860
Count
XP (5532 entries)
σ/μ = 11.46± 0.19 %
μ = 500.0± 0.9
σ = 57.3± 0.9
WF (5532 entries)
σ/μ = 11.91± 0.21 %
μ = 506.4± 1.0
σ = 60.3± 1.0
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



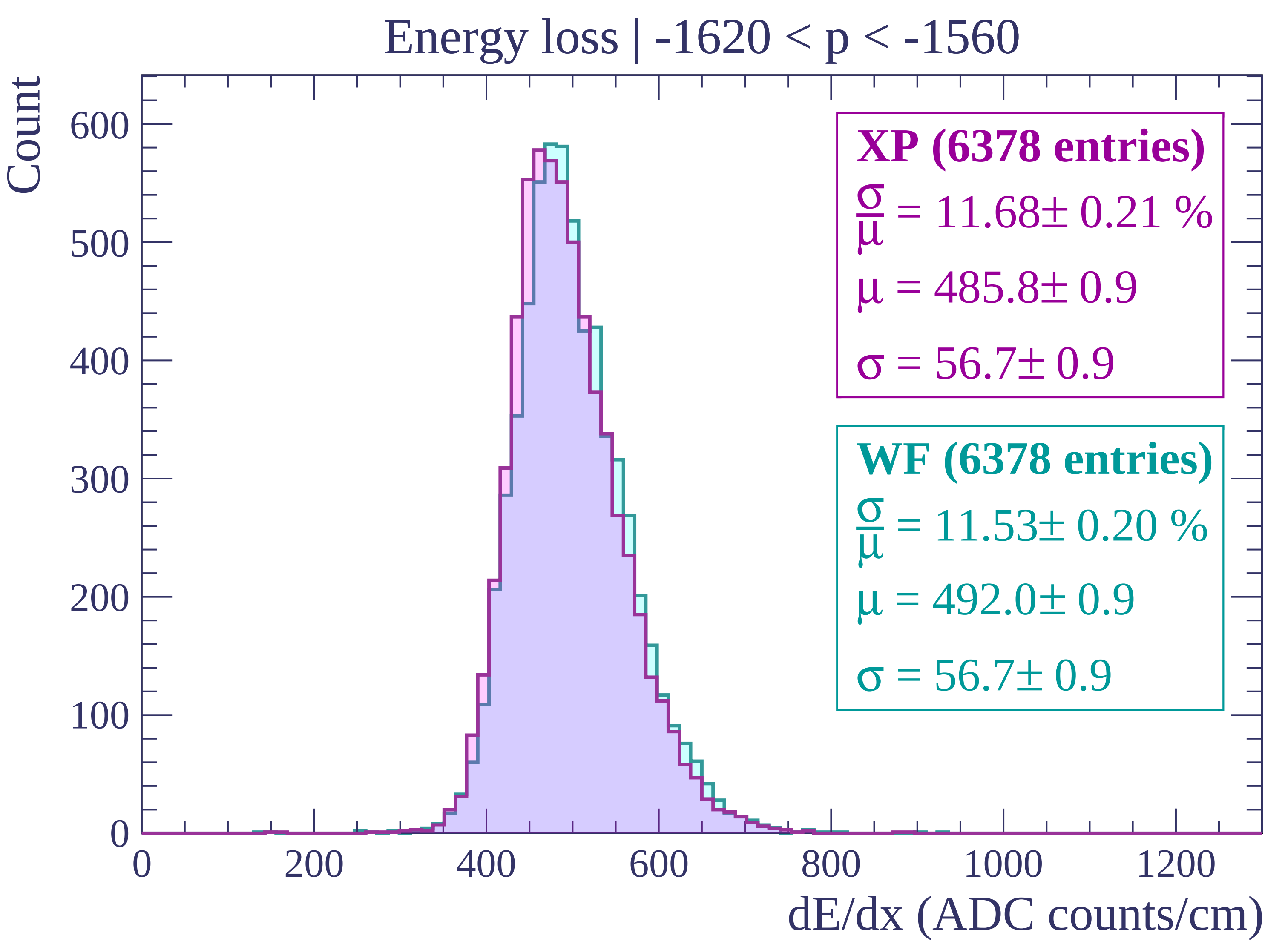
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5596 entries)
$\frac{\sigma}{\mu}$ = 11.67± 0.20 %
μ = 496.3± 0.9
σ = 57.9± 0.9
WF (5596 entries)
$\frac{\sigma}{\mu}$ = 11.69± 0.19 %
μ = 502.7± 0.9
σ = 58.8± 0.8


Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5926 entries)
$\frac{\sigma}{\mu}$ = 11.69± 0.19 %
μ = 494.2± 0.9
σ = 57.8± 0.8
WF (5926 entries)
$\frac{\sigma}{\mu}$ = 11.86± 0.19 %
μ = 501.1± 0.9
σ = 59.4± 0.8

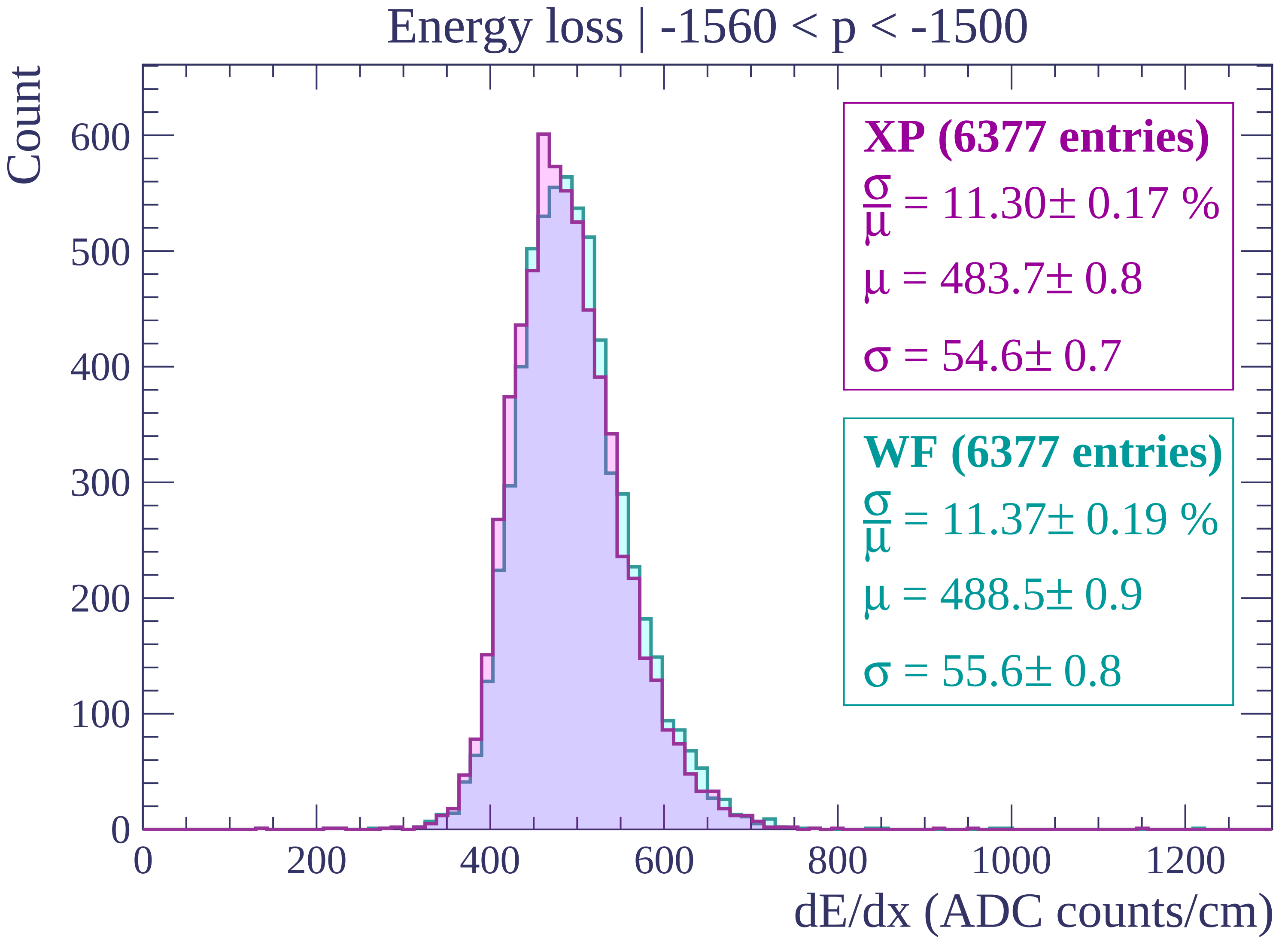

Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (6023 entries)
σ/μ = 11.57± 0.18 %
μ = 494.9± 0.9
σ = 57.3± 0.8
WF (6023 entries)
σ/μ = 11.58± 0.19 %
μ = 499.6± 0.9
σ = 57.8± 0.8



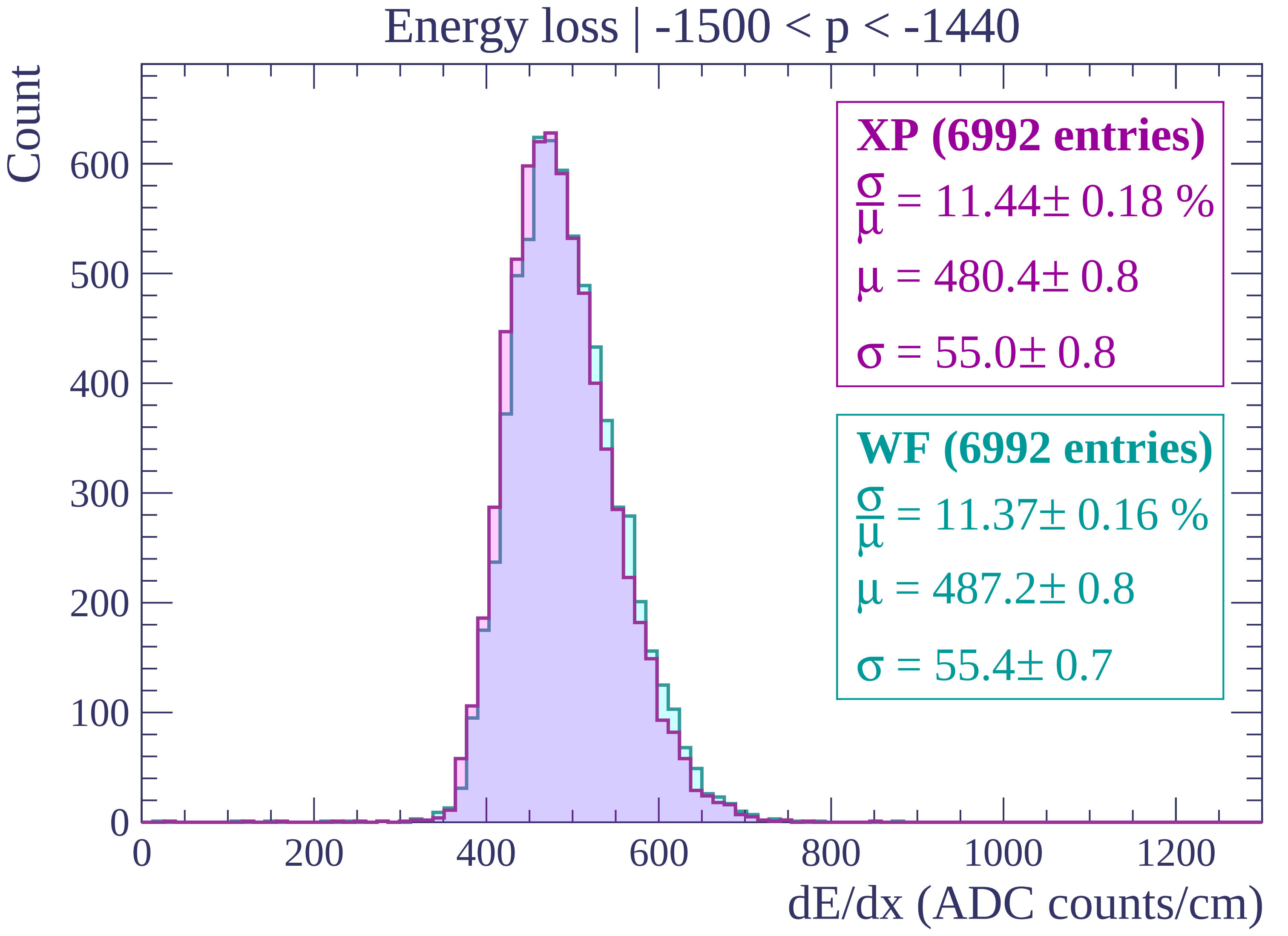
Energy loss | -1680 < p < -1620
Count
XP (6164 entries)
σ/μ = 11.43± 0.19 %
μ = 489.5± 0.9
σ = 56.0± 0.8
WF (6164 entries)
σ/μ = 11.69± 0.19 %
μ = 495.8± 0.9
σ = 58.0± 0.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



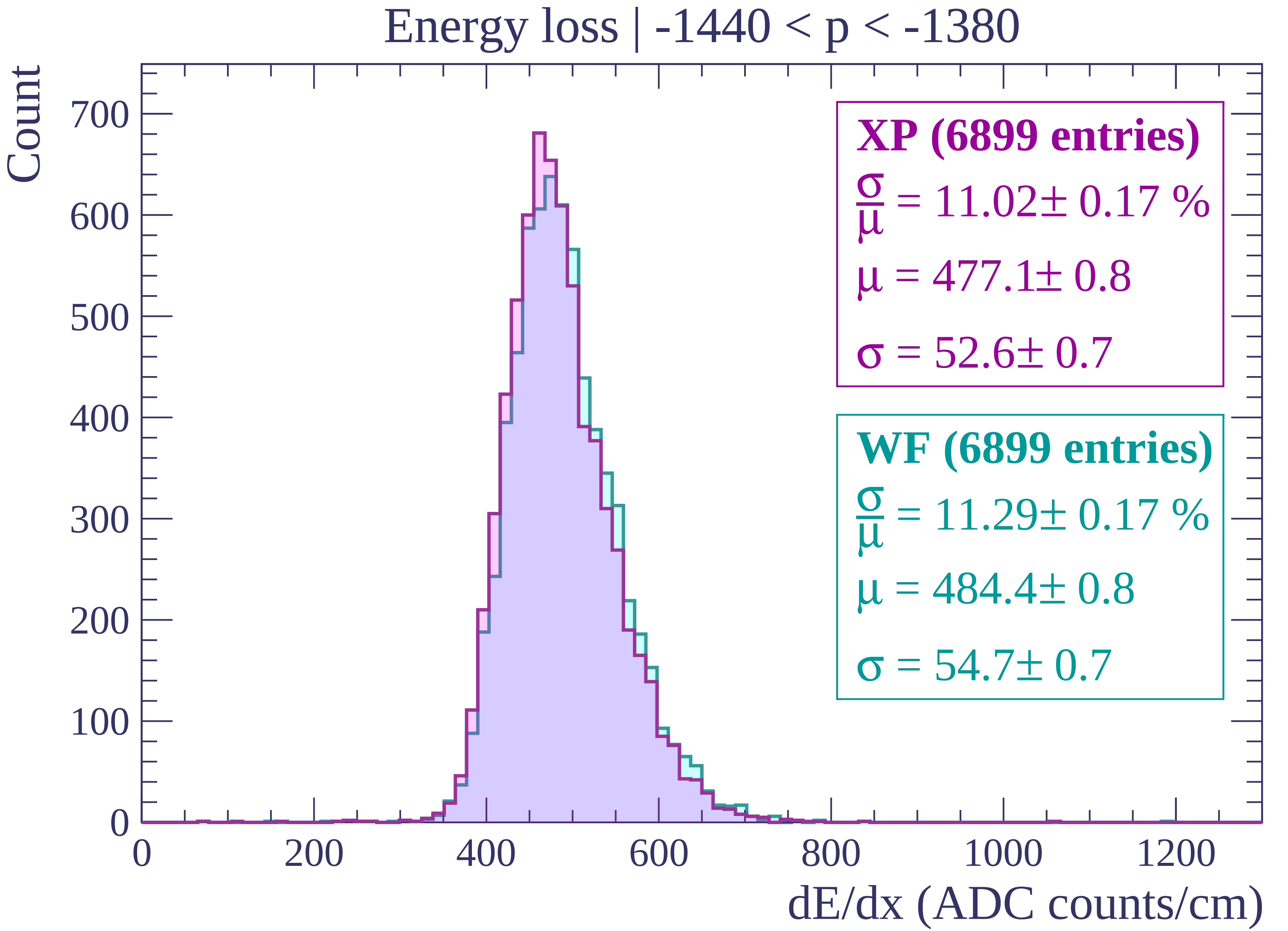
Energy loss | -1620 < p < -1560
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6378 entries)
σ/μ = 11.68± 0.21 %
μ = 485.8± 0.9
σ = 56.7± 0.9
WF (6378 entries)
σ/μ = 11.53± 0.20 %
μ = 492.0± 0.9
σ = 56.7± 0.9

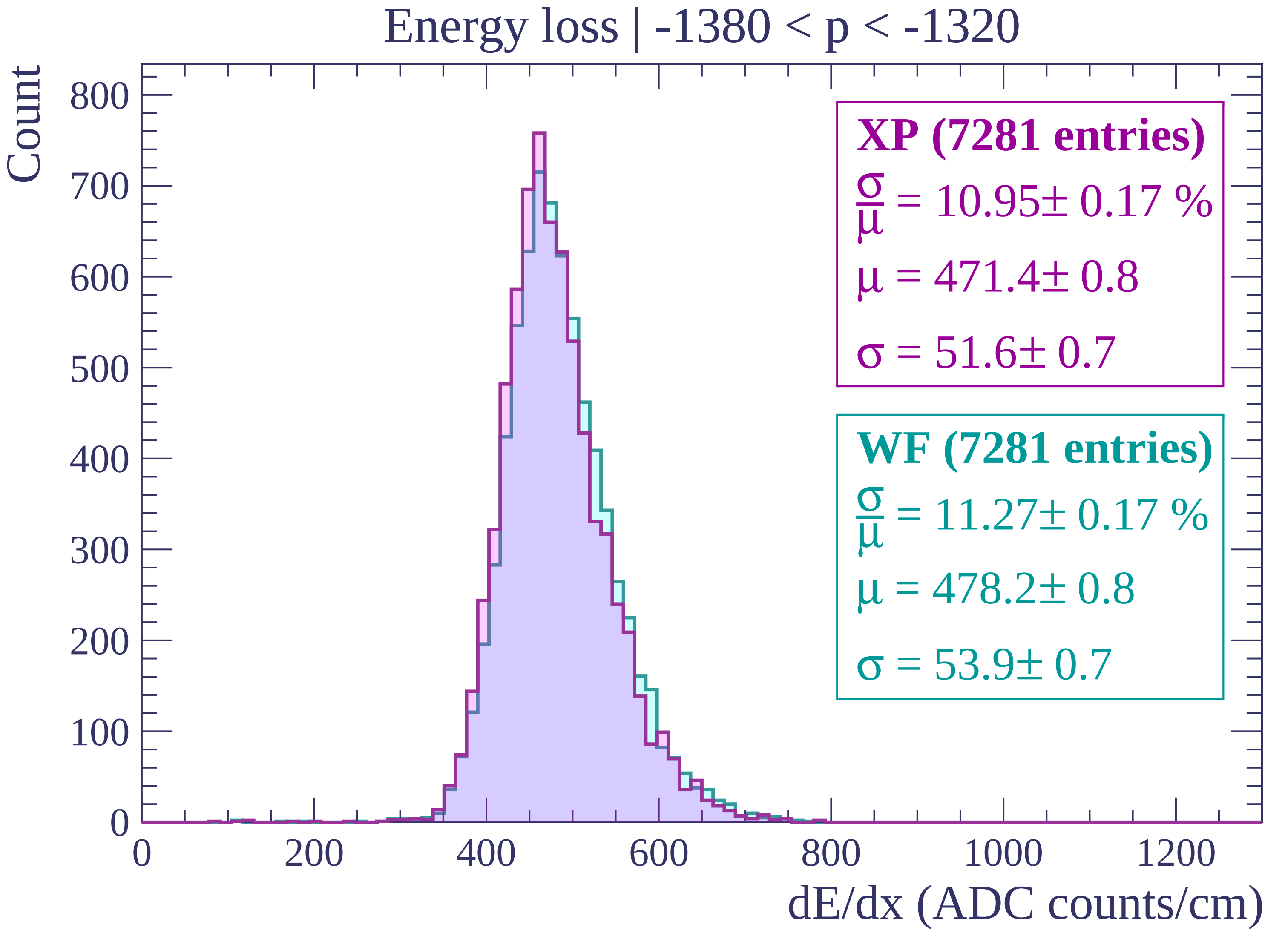


Energy loss | -1560 < p < -1500
Count
XP (6377 entries)
$\frac{\sigma}{\mu}$ = 11.30± 0.17 %
μ = 483.7± 0.8
σ = 54.6± 0.7
WF (6377 entries)
$\frac{\sigma}{\mu}$ = 11.37± 0.19 %
μ = 488.5± 0.9
σ = 55.6± 0.8
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

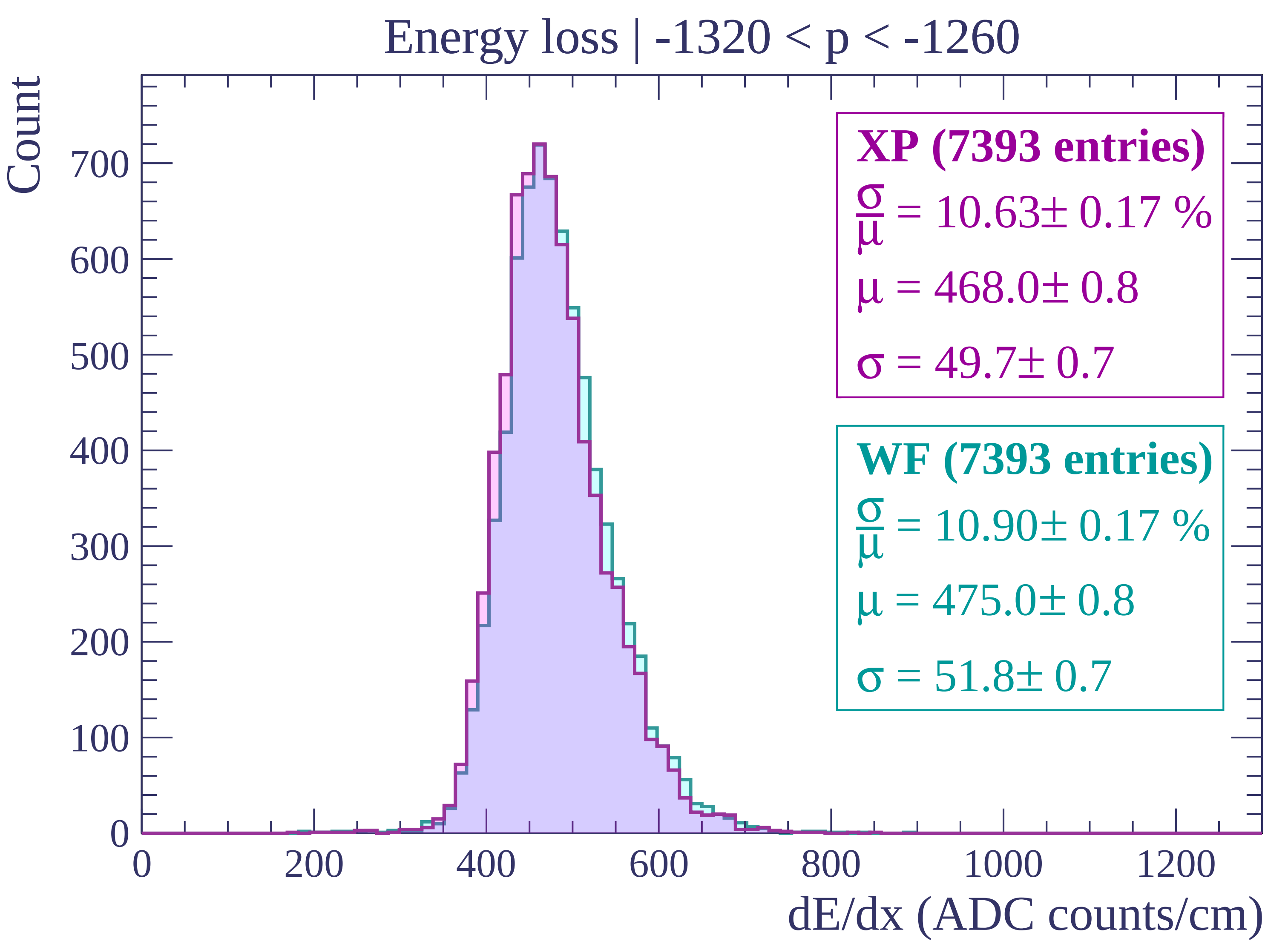


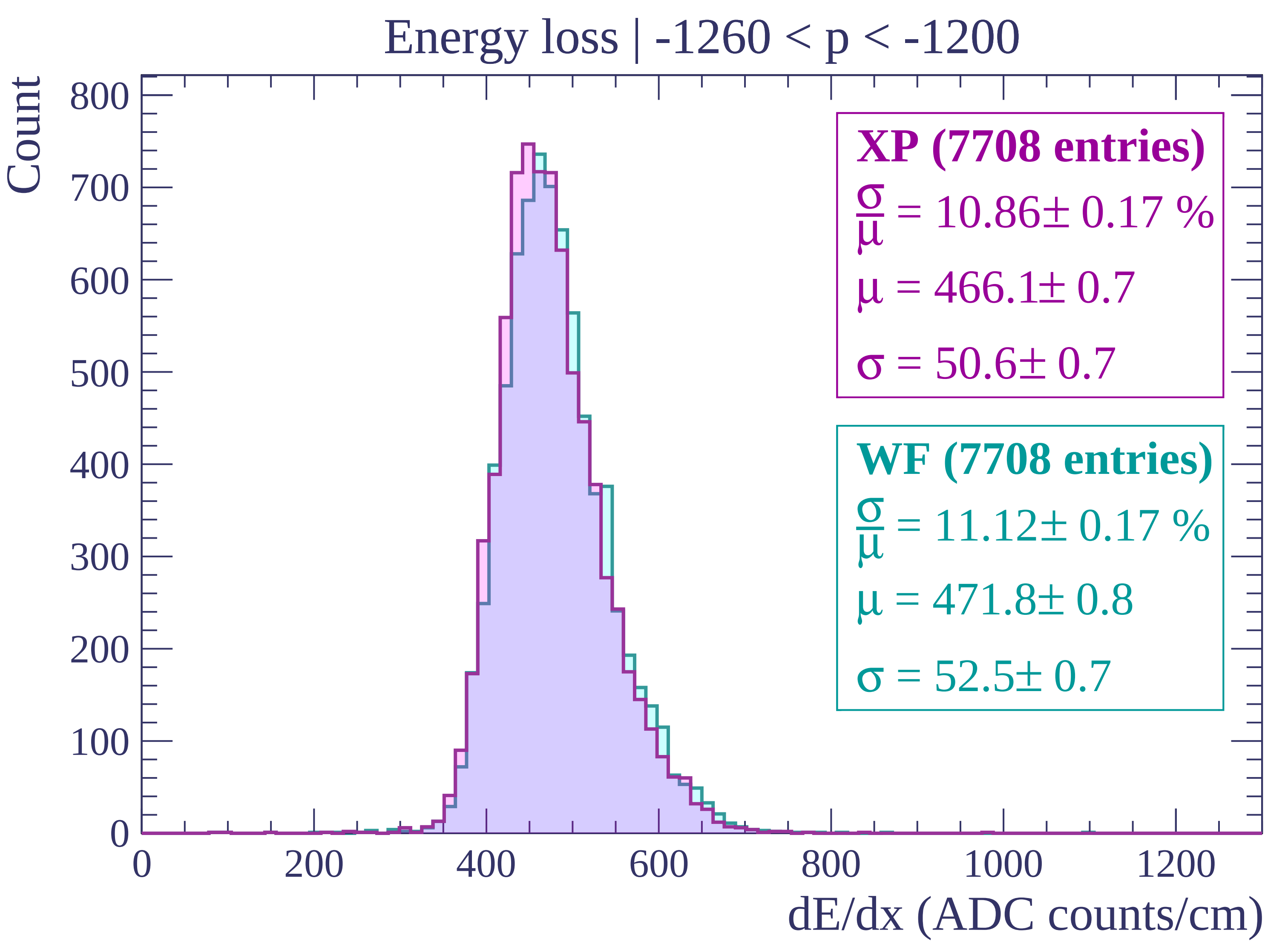
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (6992 entries)
σ/μ = 11.44± 0.18 %
μ = 480.4± 0.8
σ = 55.0± 0.8
WF (6992 entries)
σ/μ = 11.37± 0.16 %
μ = 487.2± 0.8
σ = 55.4± 0.7


Energy loss | -1440 < p < -1380
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6899 entries)
σ/μ = 11.02± 0.17 %
μ = 477.1± 0.8
σ = 52.6± 0.7
WF (6899 entries)
σ/μ = 11.29± 0.17 %
μ = 484.4± 0.8
σ = 54.7± 0.7

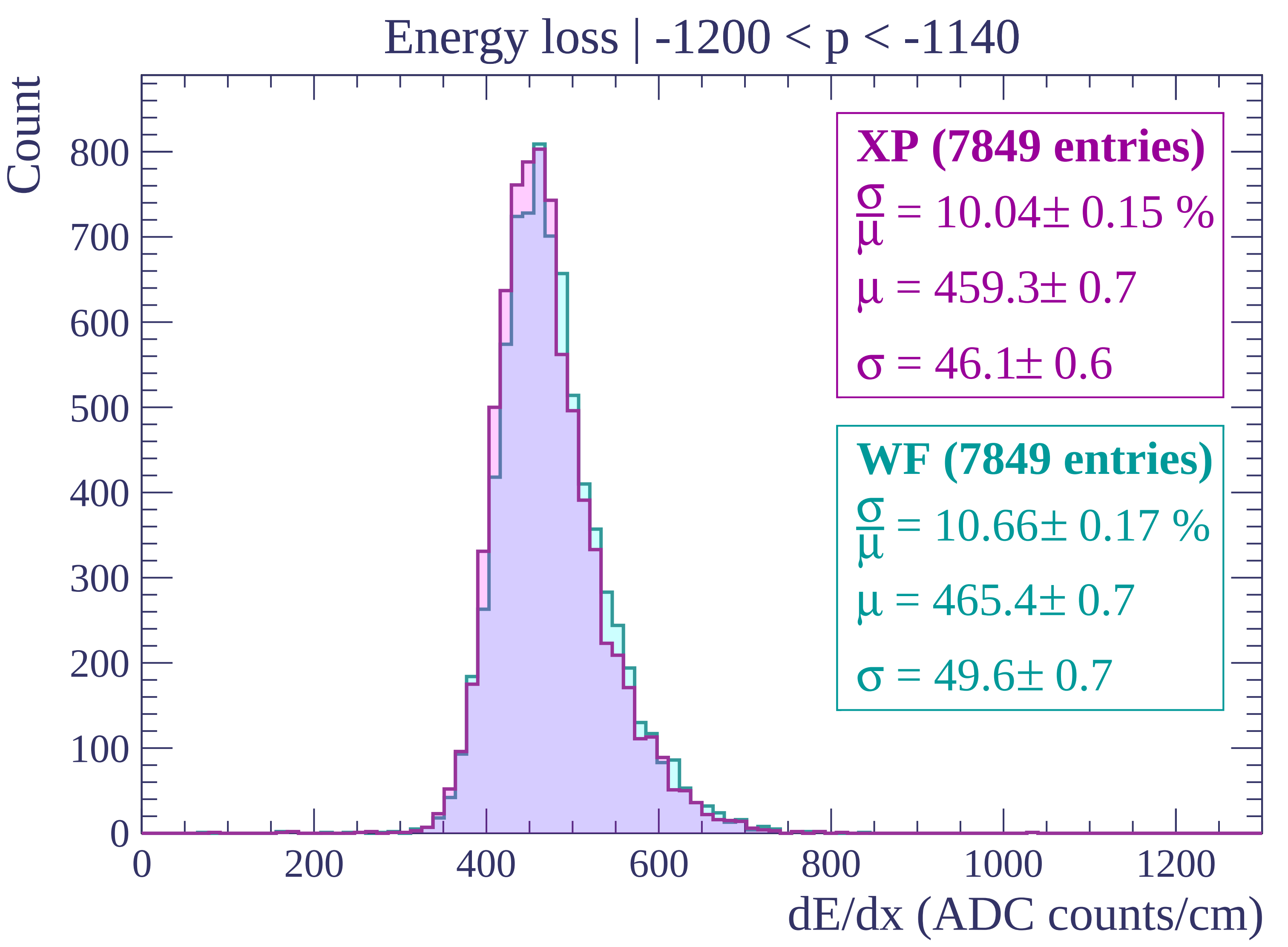


Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (7281 entries)
$\frac{\sigma}{\mu}$ = 10.95± 0.17 %
μ = 471.4± 0.8
σ = 51.6± 0.7
WF (7281 entries)
$\frac{\sigma}{\mu}$ = 11.27± 0.17 %
μ = 478.2± 0.8
σ = 53.9± 0.7

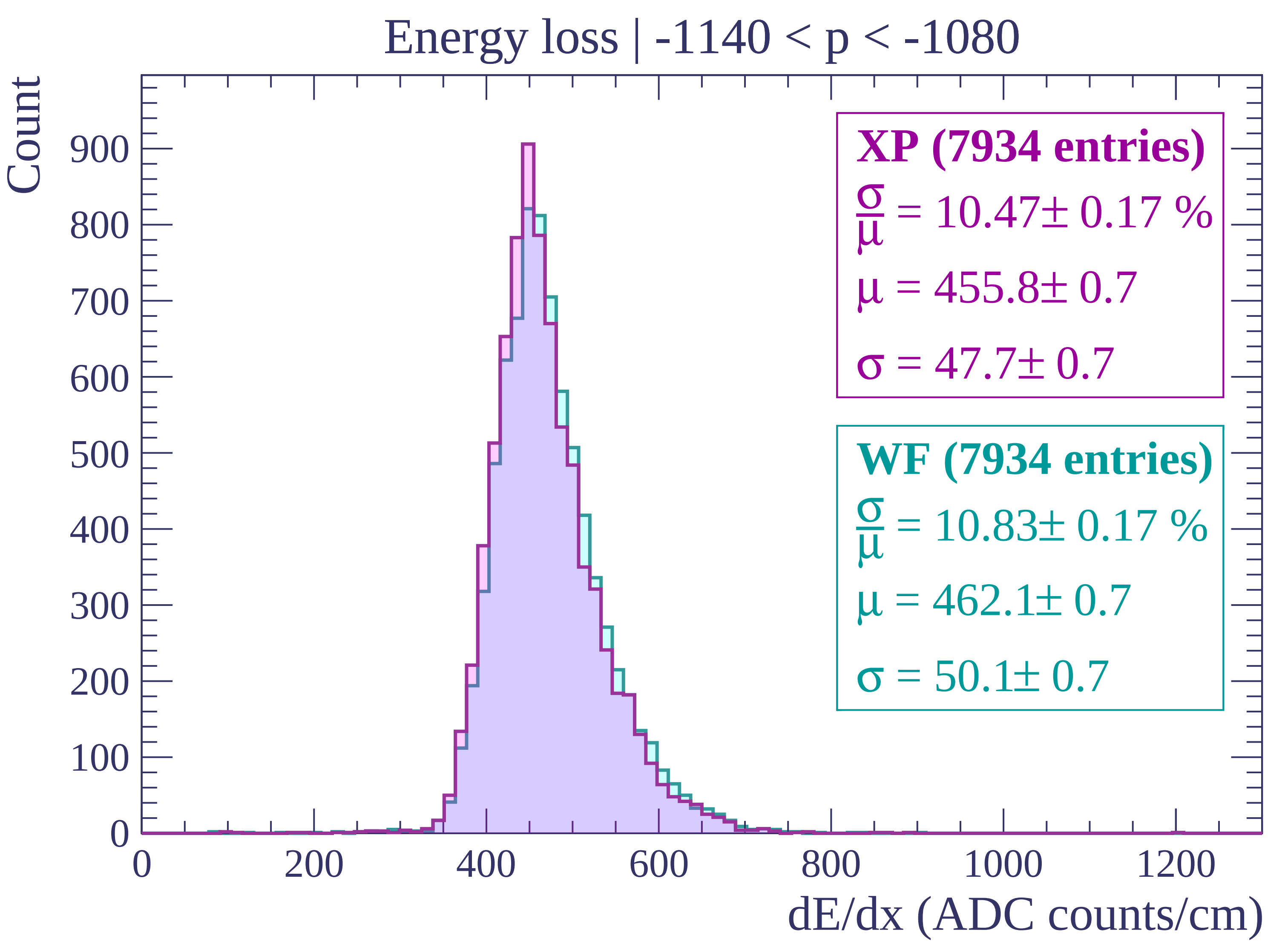

Energy loss | -1320 < p < -1260
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7393 entries)
σ/μ = 10.63± 0.17 %
μ = 468.0± 0.8
σ = 49.7± 0.7
WF (7393 entries)
σ/μ = 10.90± 0.17 %
μ = 475.0± 0.8
σ = 51.8± 0.7



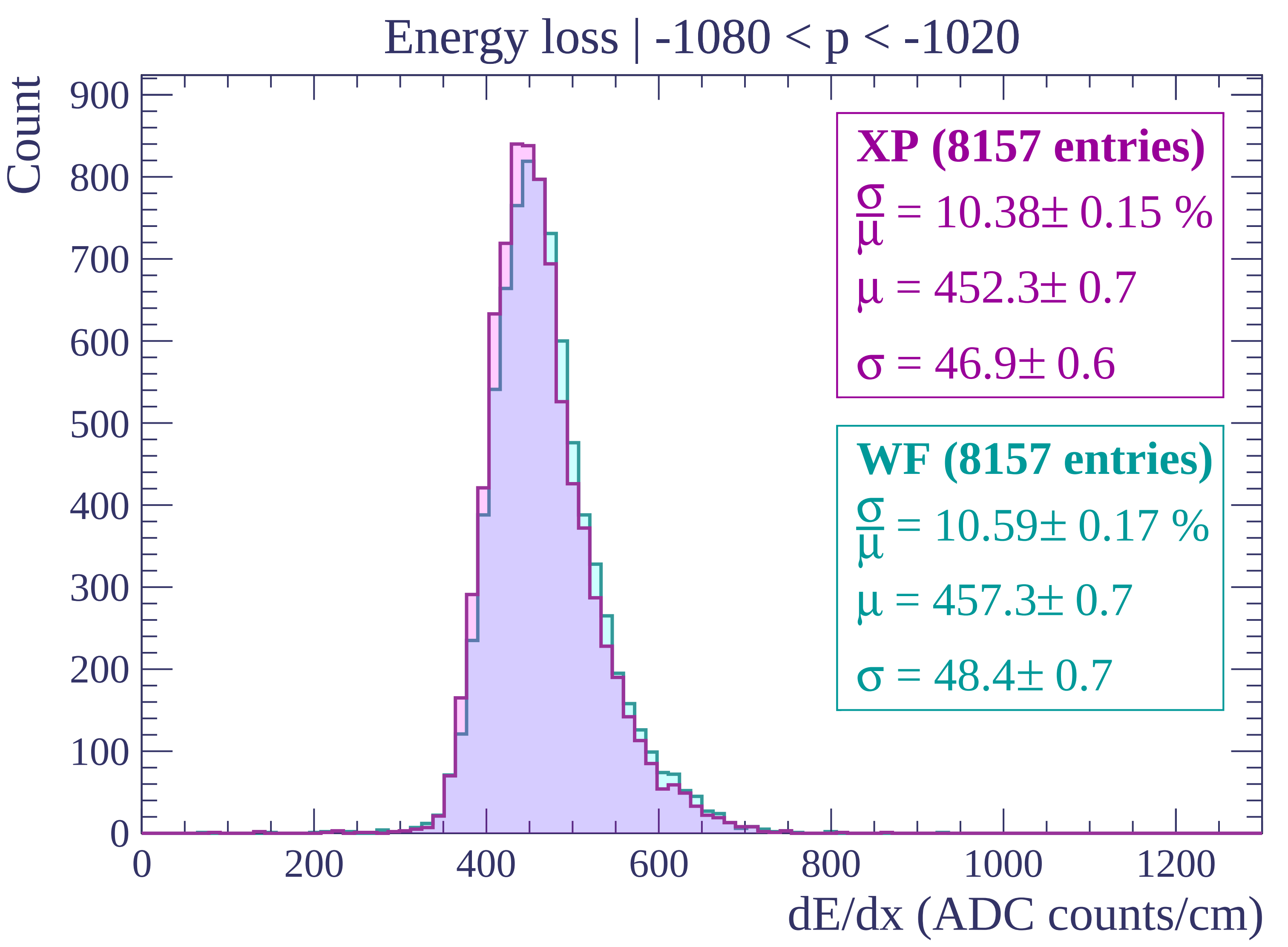
Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (7708 entries)
σ/μ = 10.86± 0.17 %
μ = 466.1± 0.7
σ = 50.6± 0.7
WF (7708 entries)
σ/μ = 11.12± 0.17 %
μ = 471.8± 0.8
σ = 52.5± 0.7



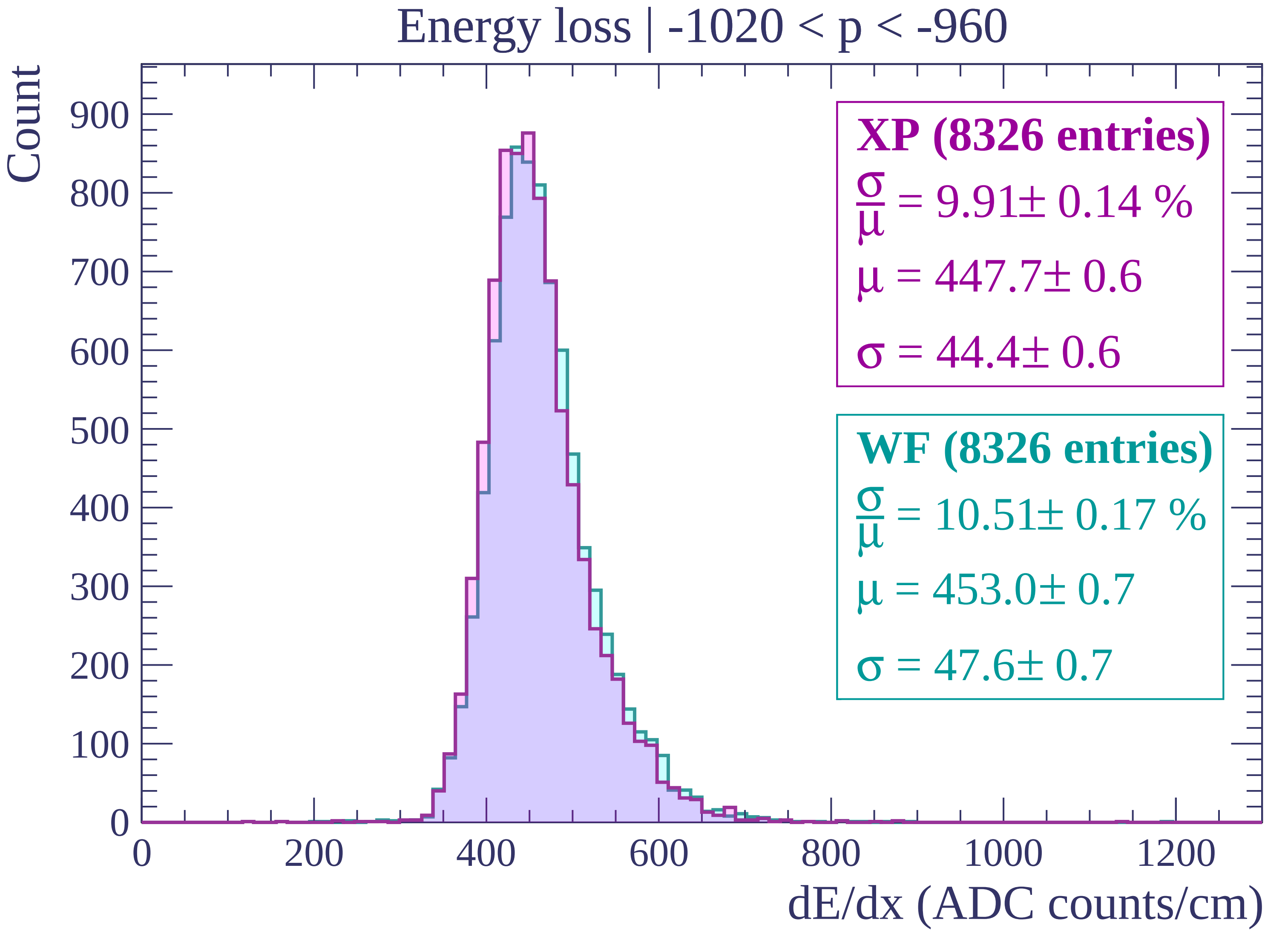
Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
XP (7849 entries)
$\frac{\sigma}{\mu}$ = 10.04± 0.15 %
μ = 459.3± 0.7
σ = 46.1± 0.6
WF (7849 entries)
$\frac{\sigma}{\mu}$ = 10.66± 0.17 %
μ = 465.4± 0.7
σ = 49.6± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

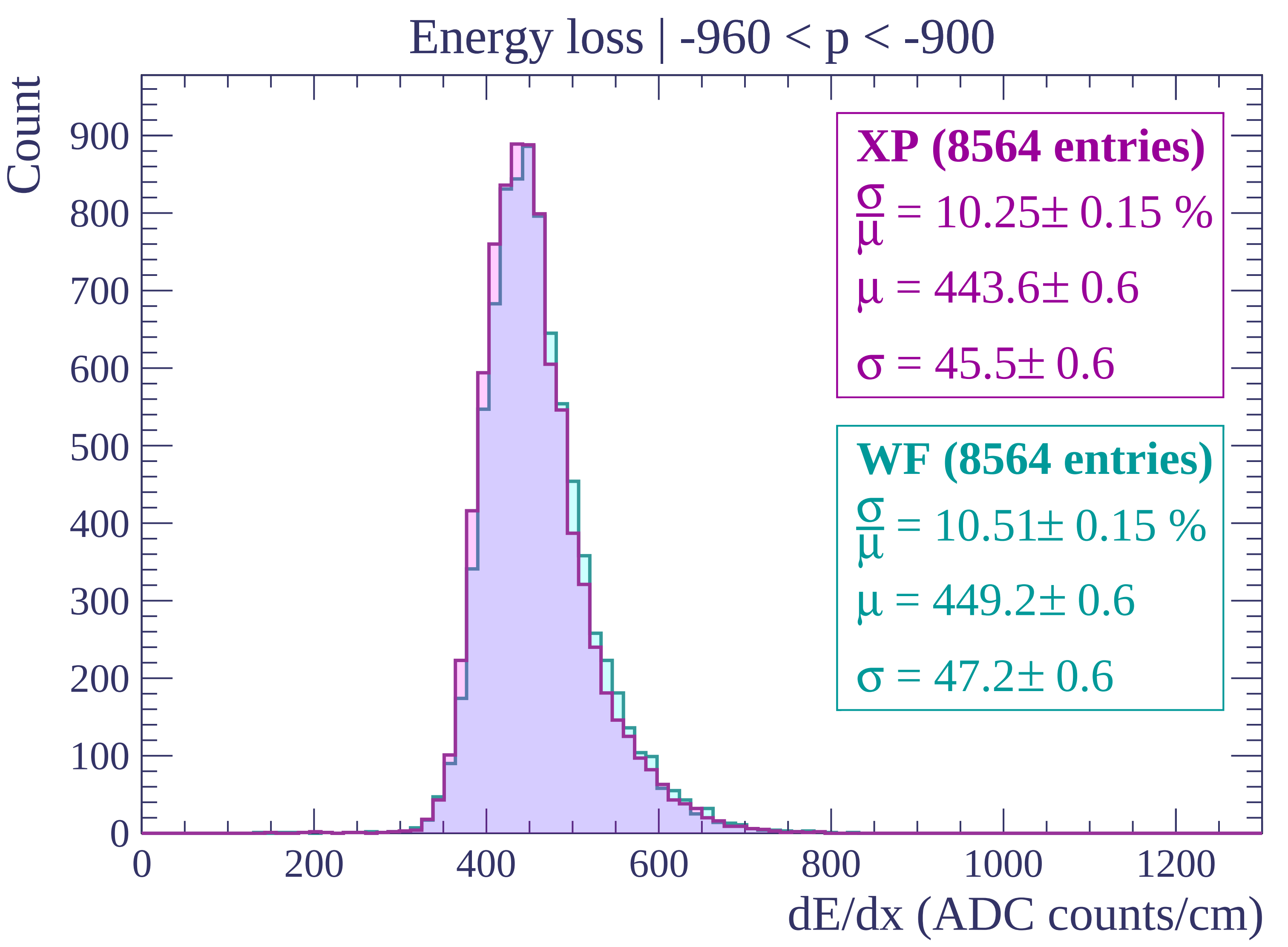


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
XP (7934 entries)
$\frac{\sigma}{\mu}$ = 10.47± 0.17 %
μ = 455.8± 0.7
σ = 47.7± 0.7
WF (7934 entries)
$\frac{\sigma}{\mu}$ = 10.83± 0.17 %
μ = 462.1± 0.7
σ = 50.1± 0.7
0
200
400
600
800
1000
1200
100
300
500
700
900

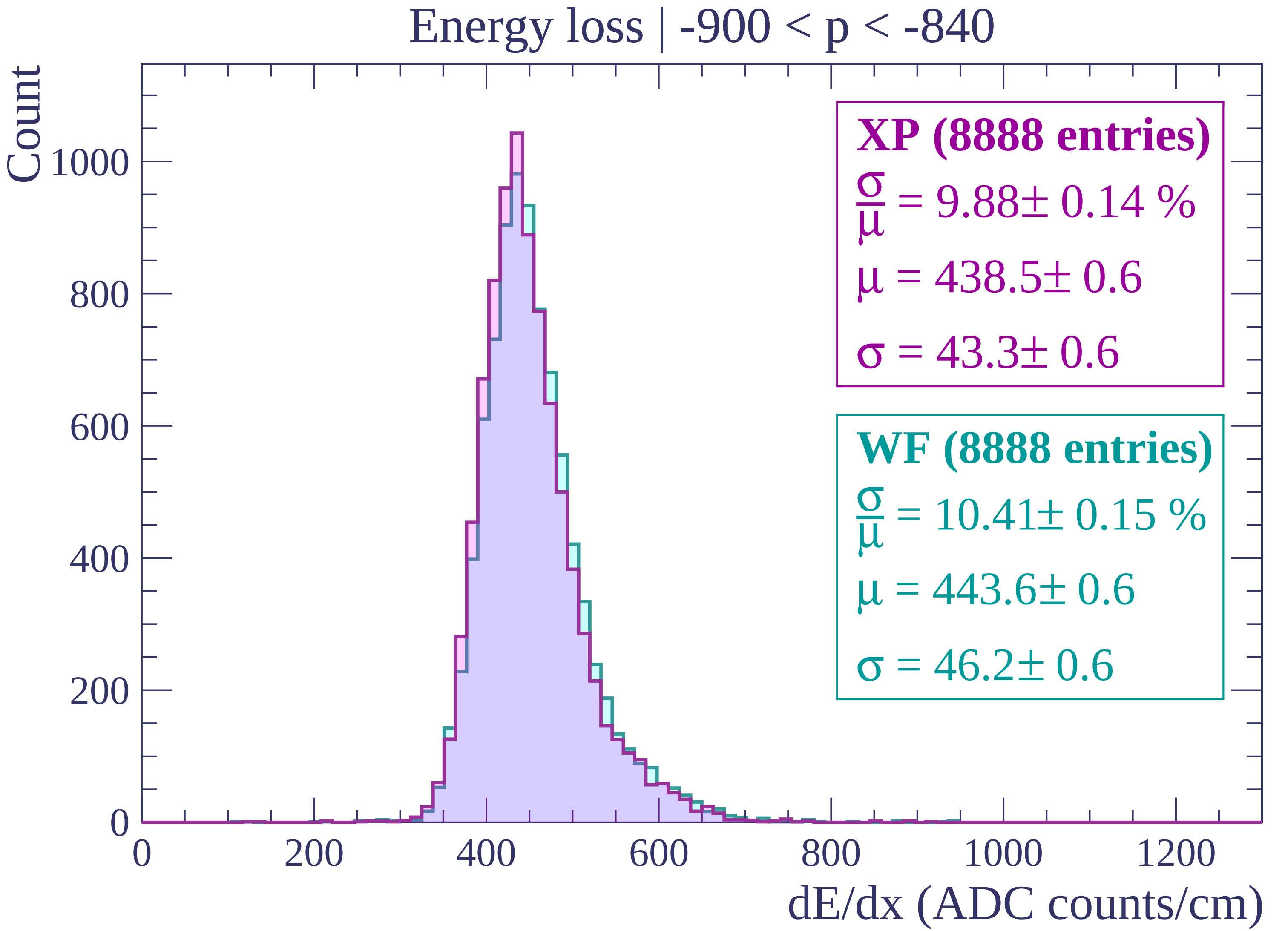


Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (8157 entries)
σ/μ = 10.38± 0.15 %
μ = 452.3± 0.7
σ = 46.9± 0.6
WF (8157 entries)
σ/μ = 10.59± 0.17 %
μ = 457.3± 0.7
σ = 48.4± 0.7

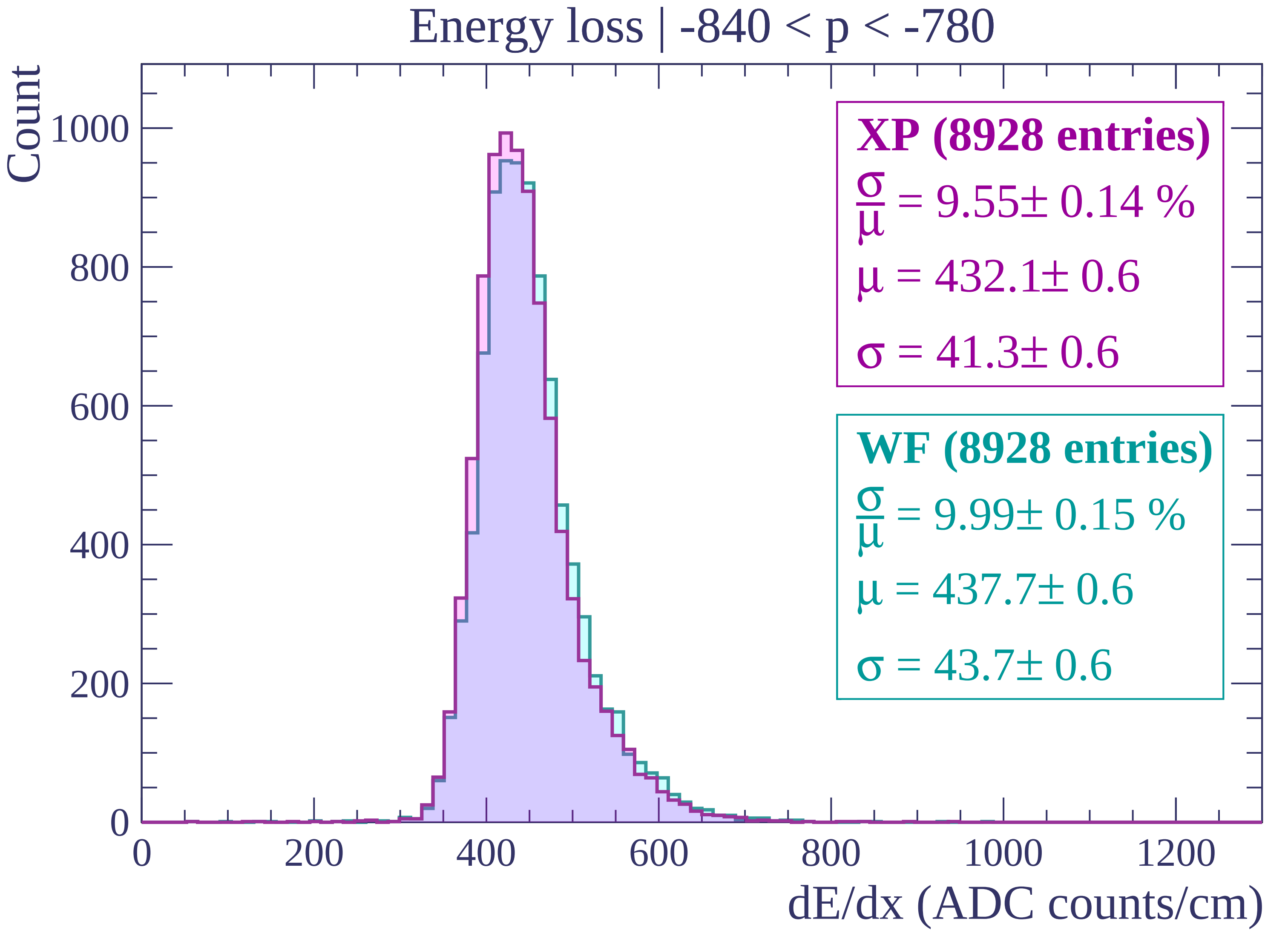

Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
900
XP (8326 entries)
σ/μ = 9.91± 0.14 %
μ = 447.7± 0.6
σ = 44.4± 0.6
WF (8326 entries)
σ/μ = 10.51± 0.17 %
μ = 453.0± 0.7
σ = 47.6± 0.7



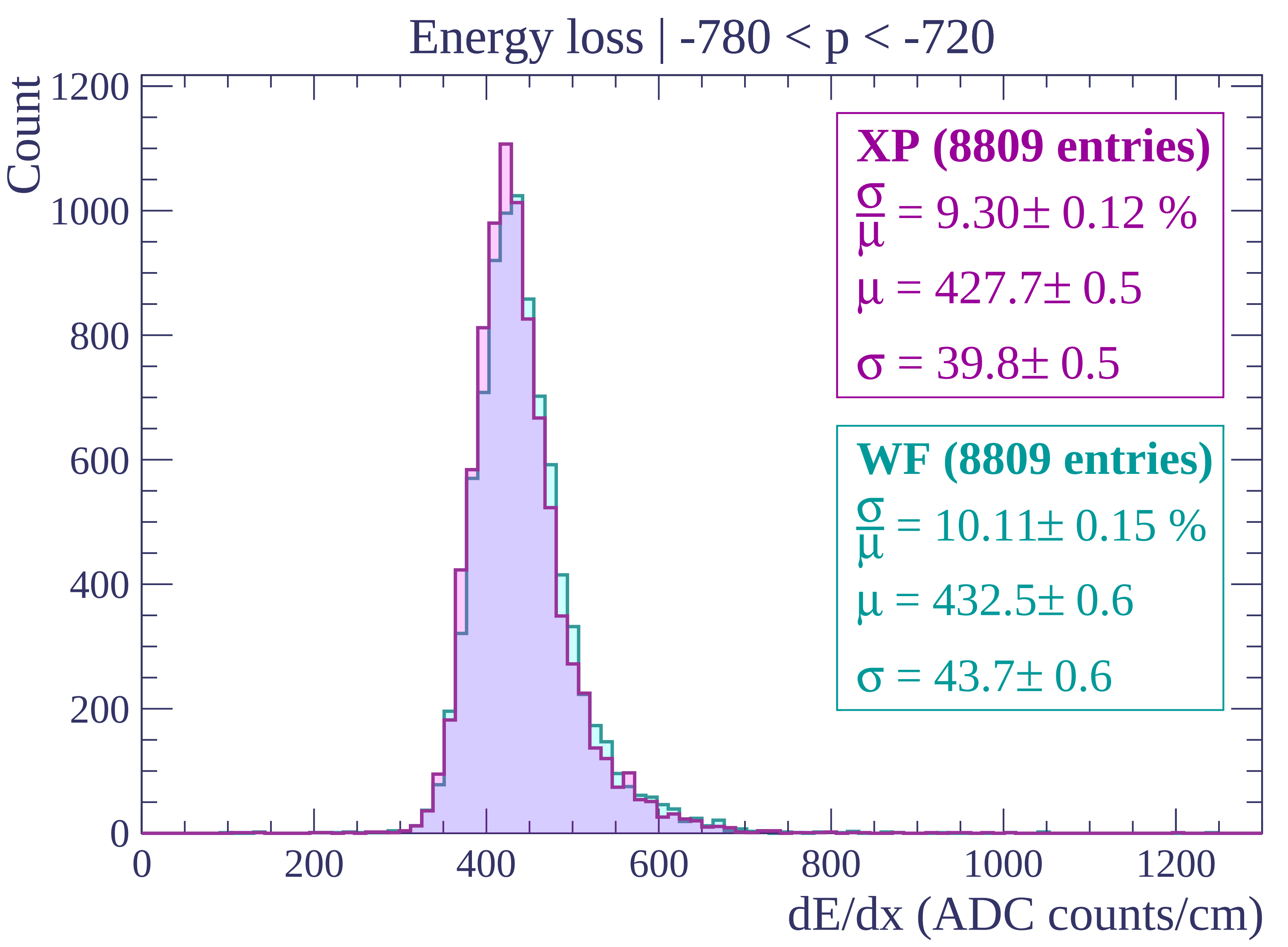
Energy loss | -960 < p < -900
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8564 entries)
σ/μ = 10.25± 0.15 %
μ = 443.6± 0.6
σ = 45.5± 0.6
WF (8564 entries)
σ/μ = 10.51± 0.15 %
μ = 449.2± 0.6
σ = 47.2± 0.6

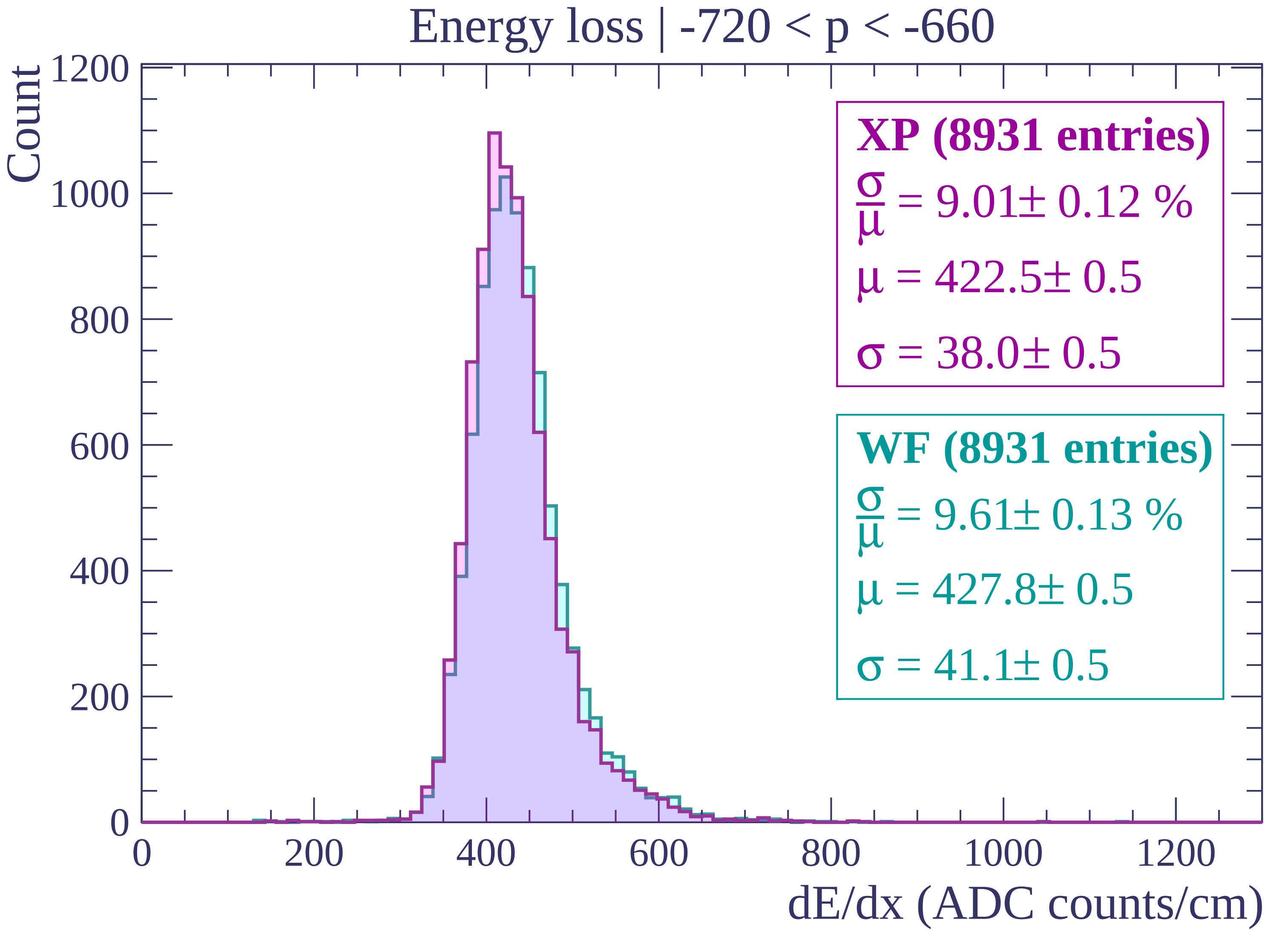


Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
200
400
600
800
1000
XP (8888 entries)
$\frac{\sigma}{\mu}$ = 9.88± 0.14 %
μ = 438.5± 0.6
σ = 43.3± 0.6
WF (8888 entries)
$\frac{\sigma}{\mu}$ = 10.41± 0.15 %
μ = 443.6± 0.6
σ = 46.2± 0.6

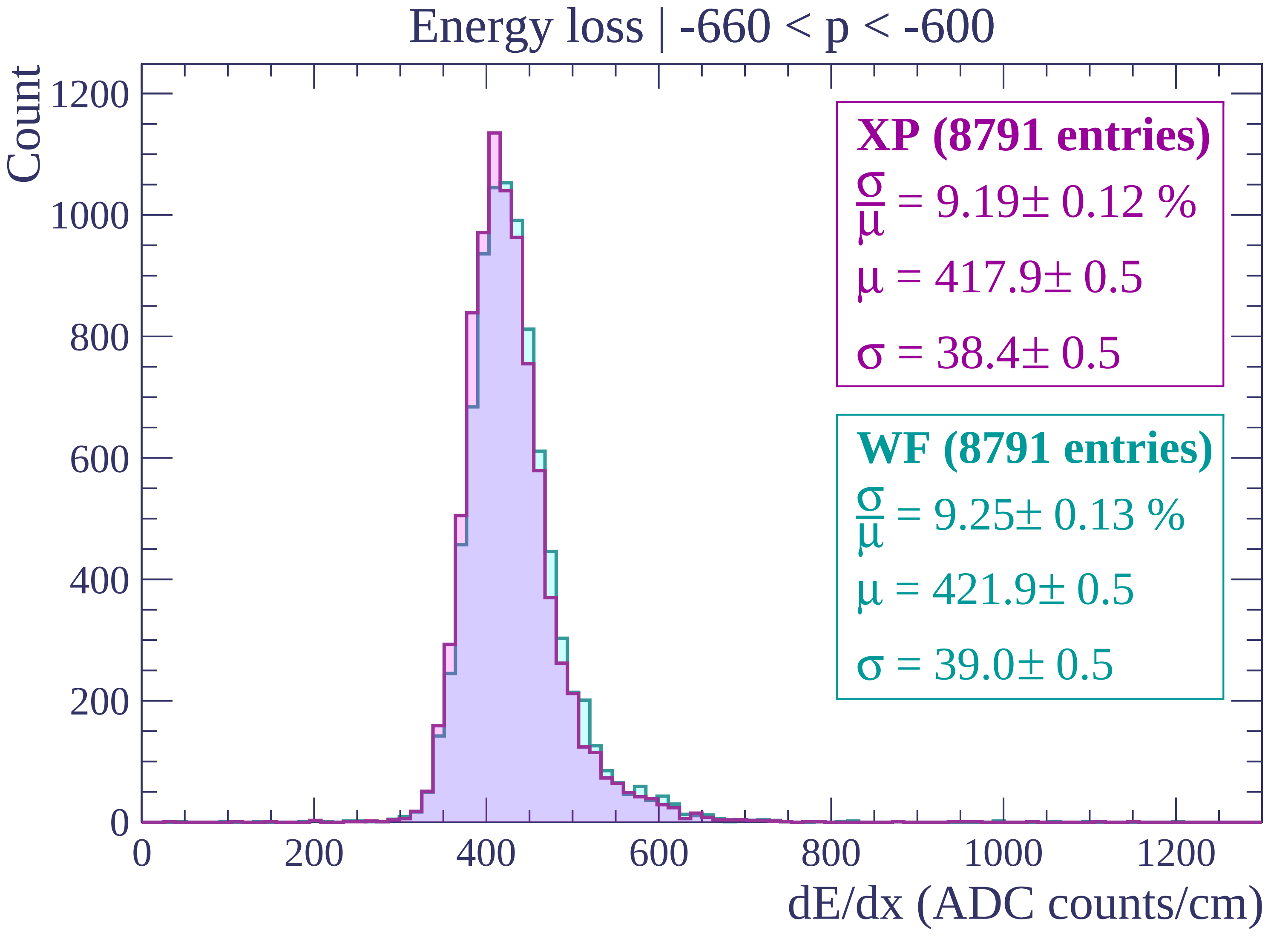


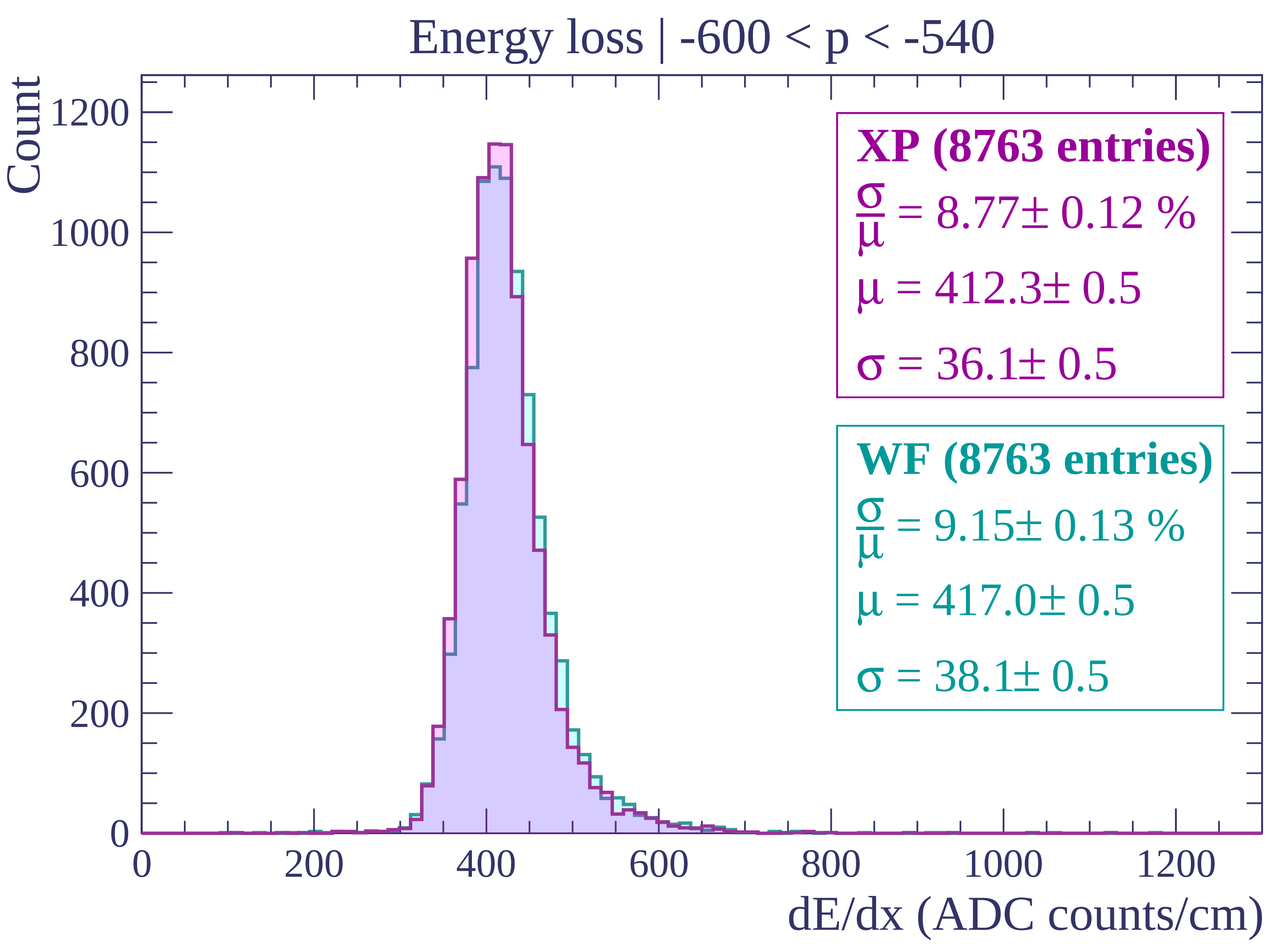
Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
XP (8928 entries)
$\frac{\sigma}{\mu}$ = 9.55± 0.14 %
μ = 432.1± 0.6
σ = 41.3± 0.6
WF (8928 entries)
$\frac{\sigma}{\mu}$ = 9.99± 0.15 %
μ = 437.7± 0.6
σ = 43.7± 0.6
0
200
400
600
800
1000
1200

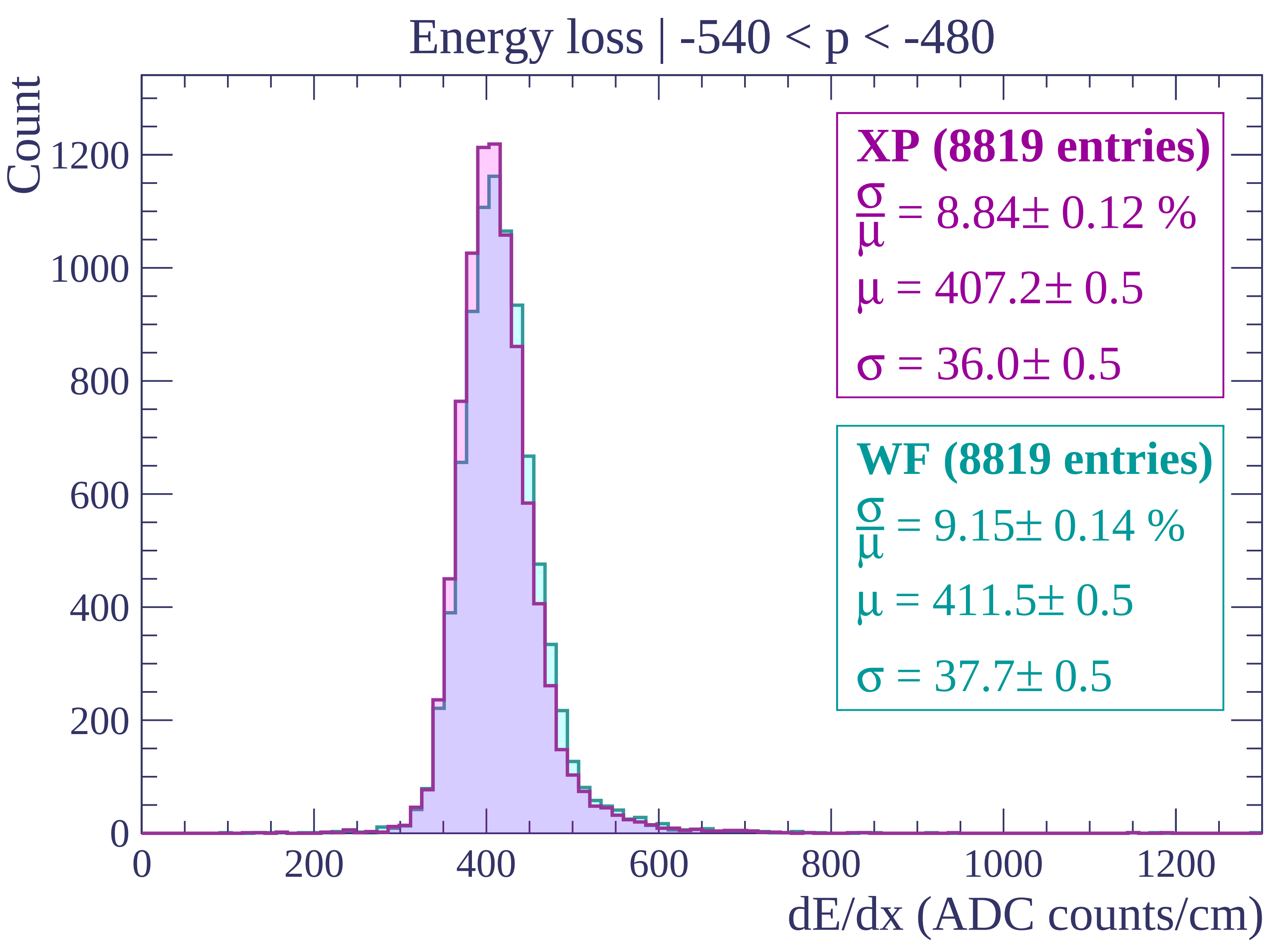

Energy loss | -780 < p < -720
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8809 entries)
σ/μ = 9.30± 0.12 %
μ = 427.7± 0.5
σ = 39.8± 0.5
WF (8809 entries)
σ/μ = 10.11± 0.15 %
μ = 432.5± 0.6
σ = 43.7± 0.6



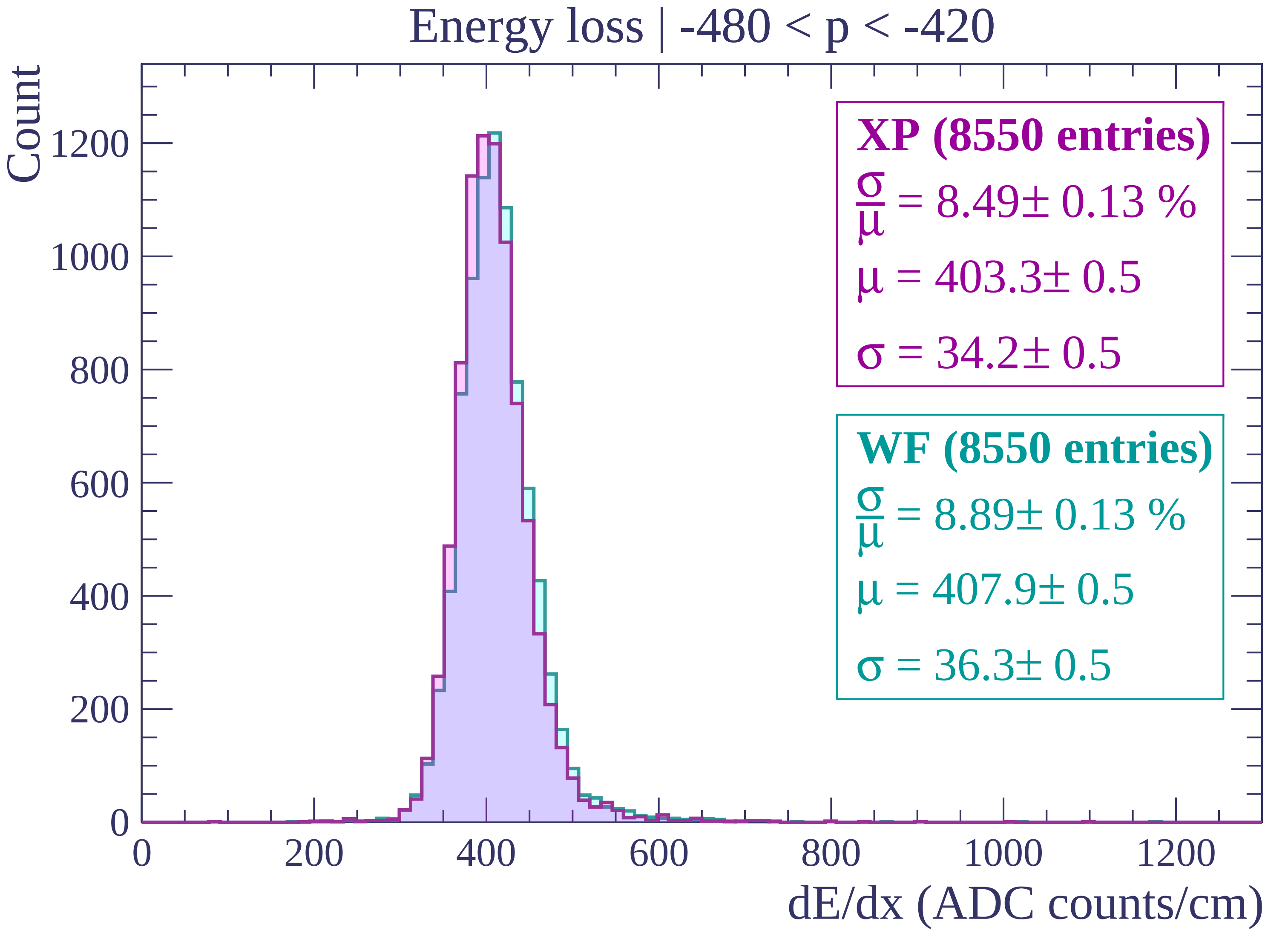
Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8931 entries)
σ/μ = 9.01± 0.12 %
μ = 422.5± 0.5
σ = 38.0± 0.5
WF (8931 entries)
σ/μ = 9.61± 0.13 %
μ = 427.8± 0.5
σ = 41.1± 0.5


Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (8791 entries)
$\frac{\sigma}{\mu}$ = 9.19± 0.12 %
μ = 417.9± 0.5
σ = 38.4± 0.5
WF (8791 entries)
$\frac{\sigma}{\mu}$ = 9.25± 0.13 %
μ = 421.9± 0.5
σ = 39.0± 0.5
0
200
400
600
800
1000
1200

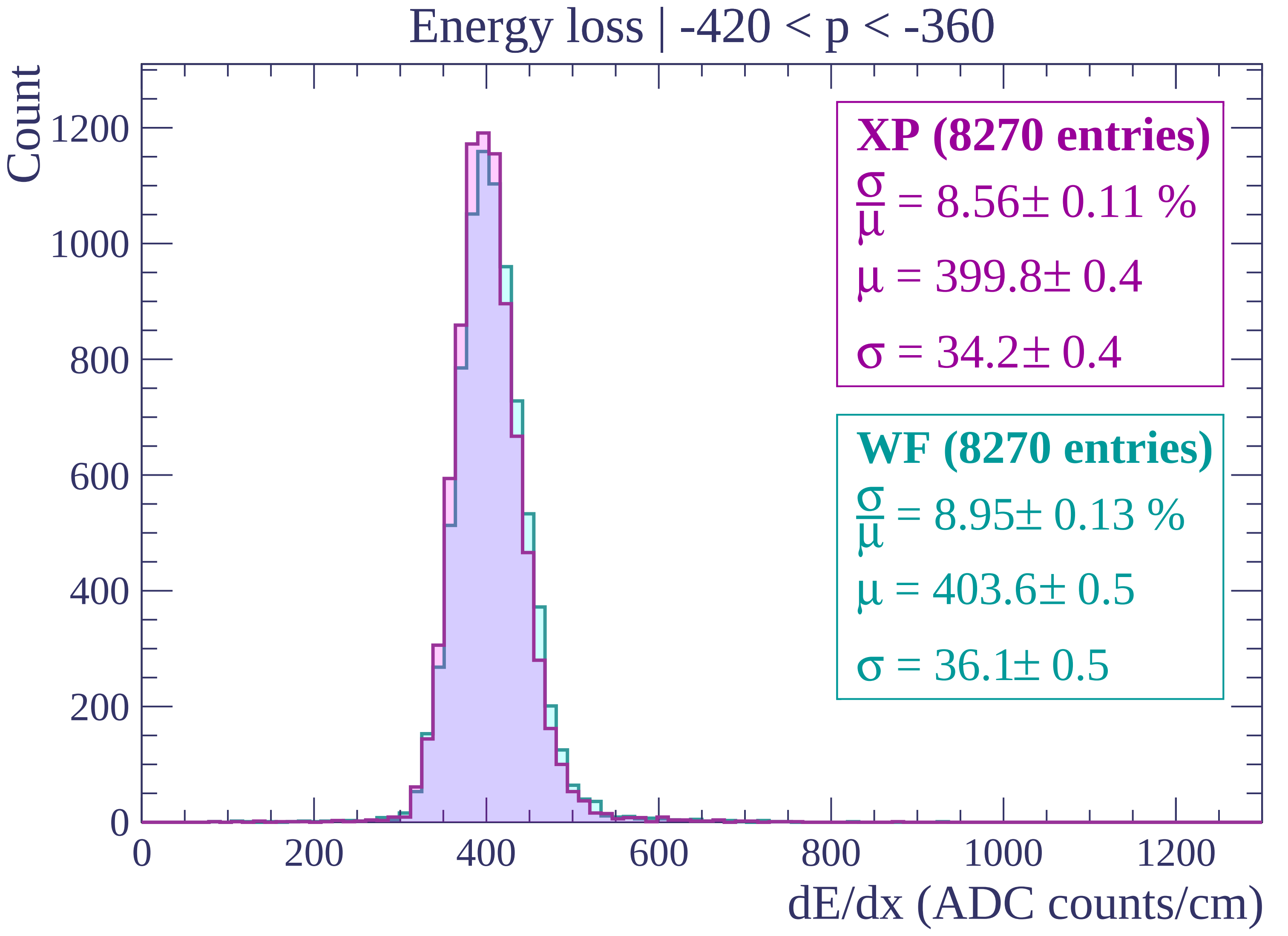

Energy loss | -600 < p < -540
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8763 entries)
σ/μ = 8.77± 0.12 %
μ = 412.3± 0.5
σ = 36.1± 0.5
WF (8763 entries)
σ/μ = 9.15± 0.13 %
μ = 417.0± 0.5
σ = 38.1± 0.5

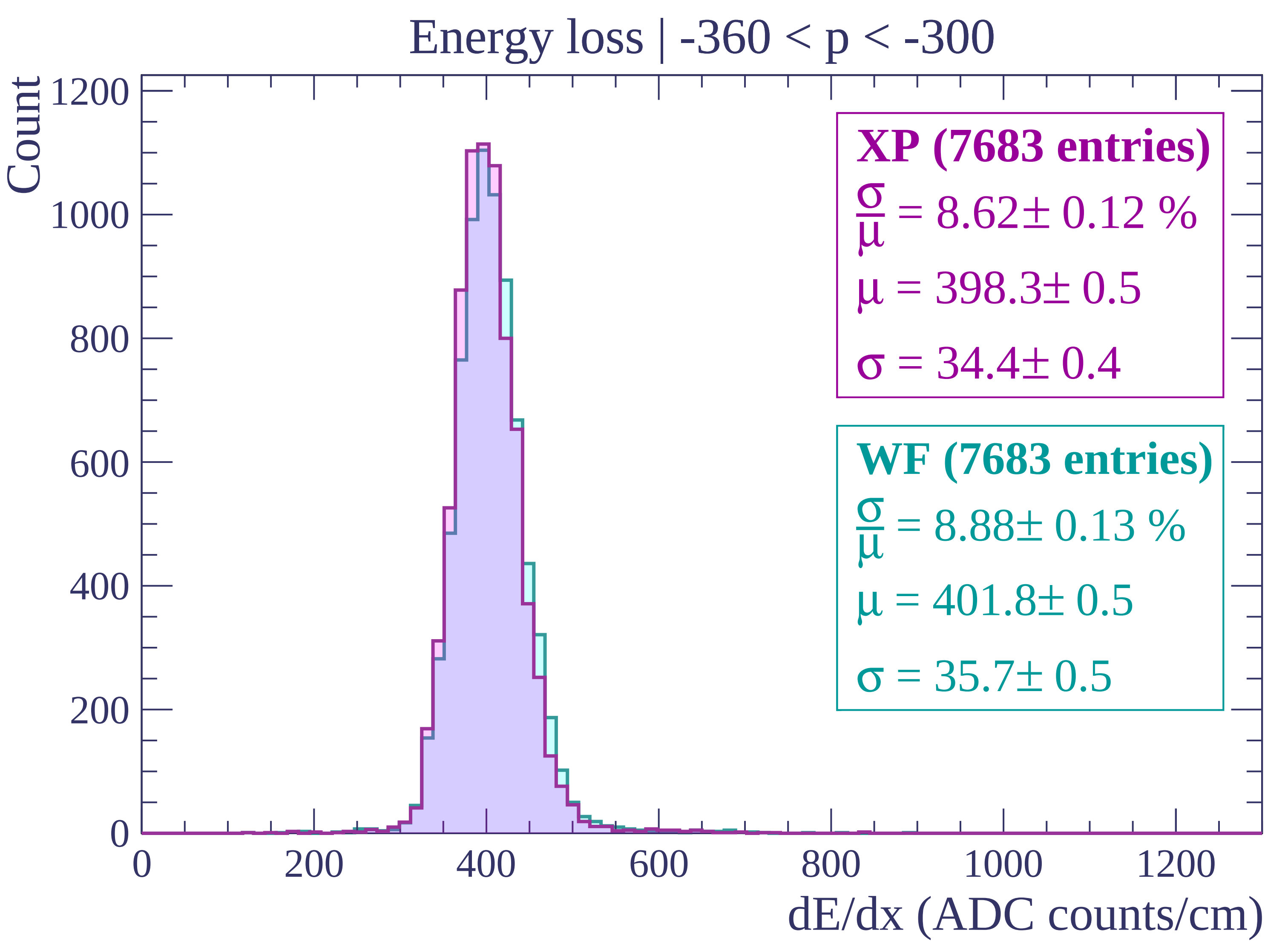


Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
XP (8819 entries)
$\frac{\sigma}{\mu}$ = 8.84± 0.12 %
μ = 407.2± 0.5
σ = 36.0± 0.5
WF (8819 entries)
$\frac{\sigma}{\mu}$ = 9.15± 0.14 %
μ = 411.5± 0.5
σ = 37.7± 0.5

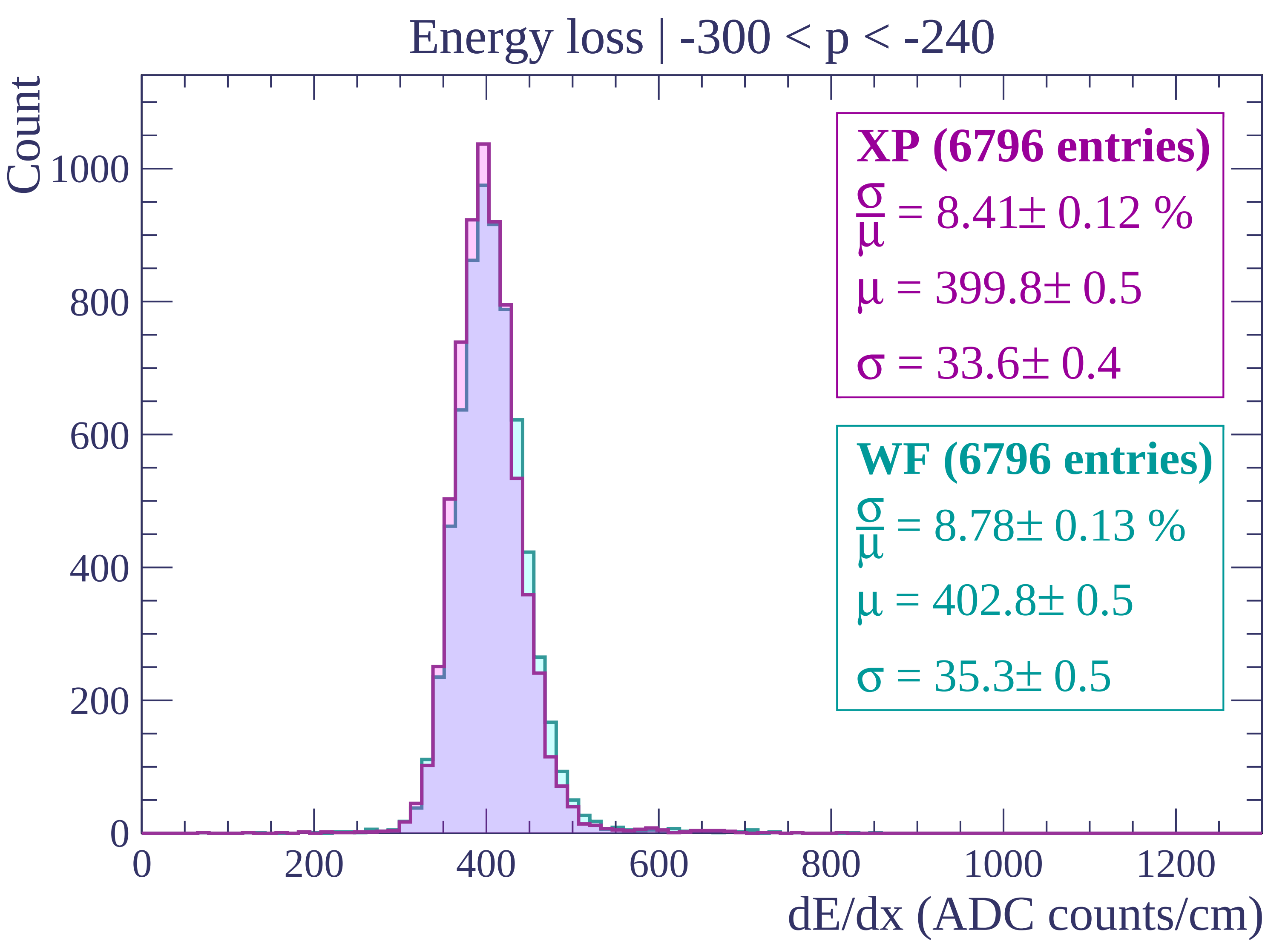


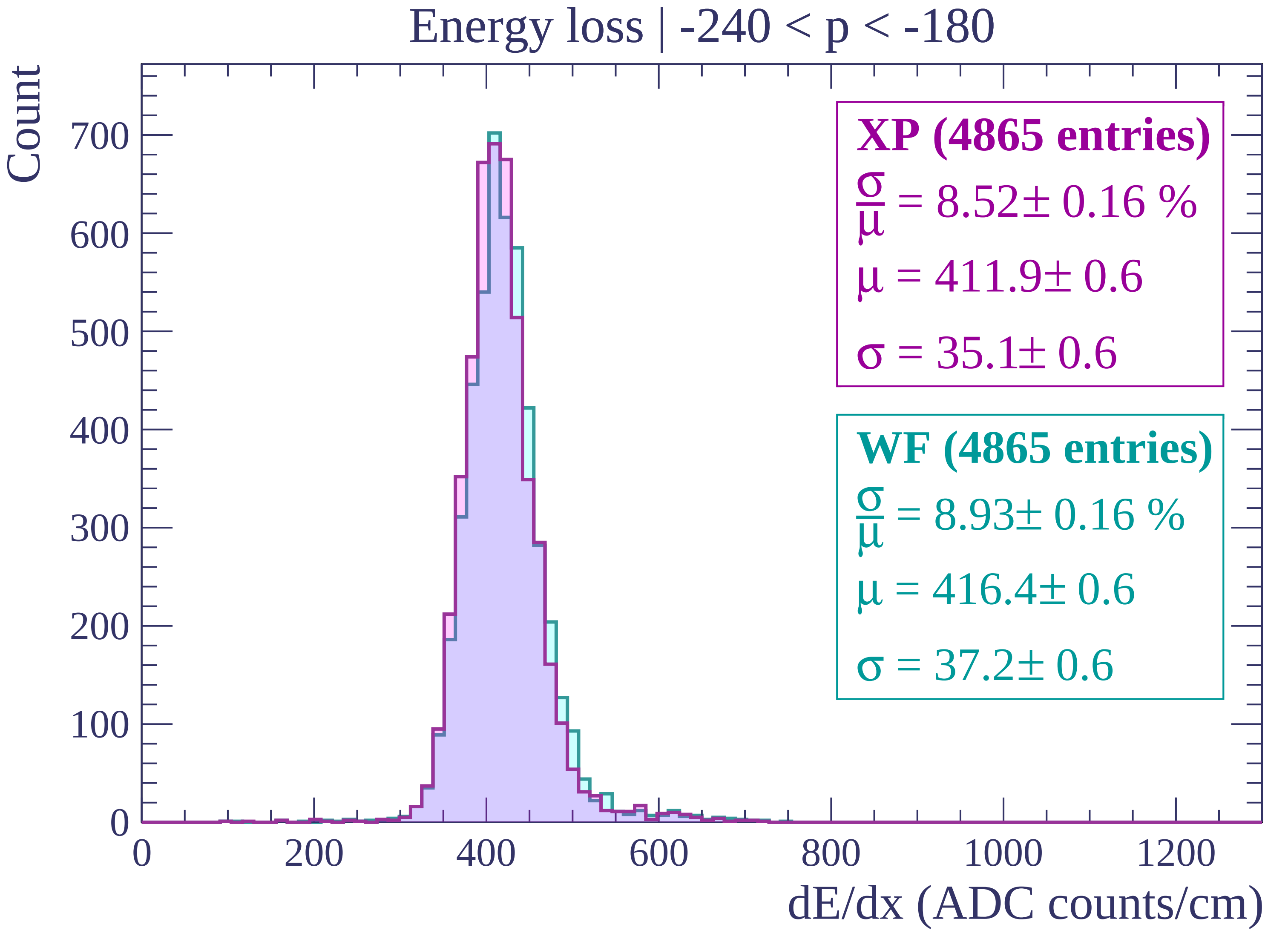
Energy loss | -480 < p < -420
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8550 entries)
σ/μ = 8.49± 0.13 %
μ = 403.3± 0.5
σ = 34.2± 0.5
WF (8550 entries)
σ/μ = 8.89± 0.13 %
μ = 407.9± 0.5
σ = 36.3± 0.5


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
XP (8270 entries)
σ/μ = 8.56± 0.11 %
μ = 399.8± 0.4
σ = 34.2± 0.4
WF (8270 entries)
σ/μ = 8.95± 0.13 %
μ = 403.6± 0.5
σ = 36.1± 0.5
0
200
400
600
800
1000
1200

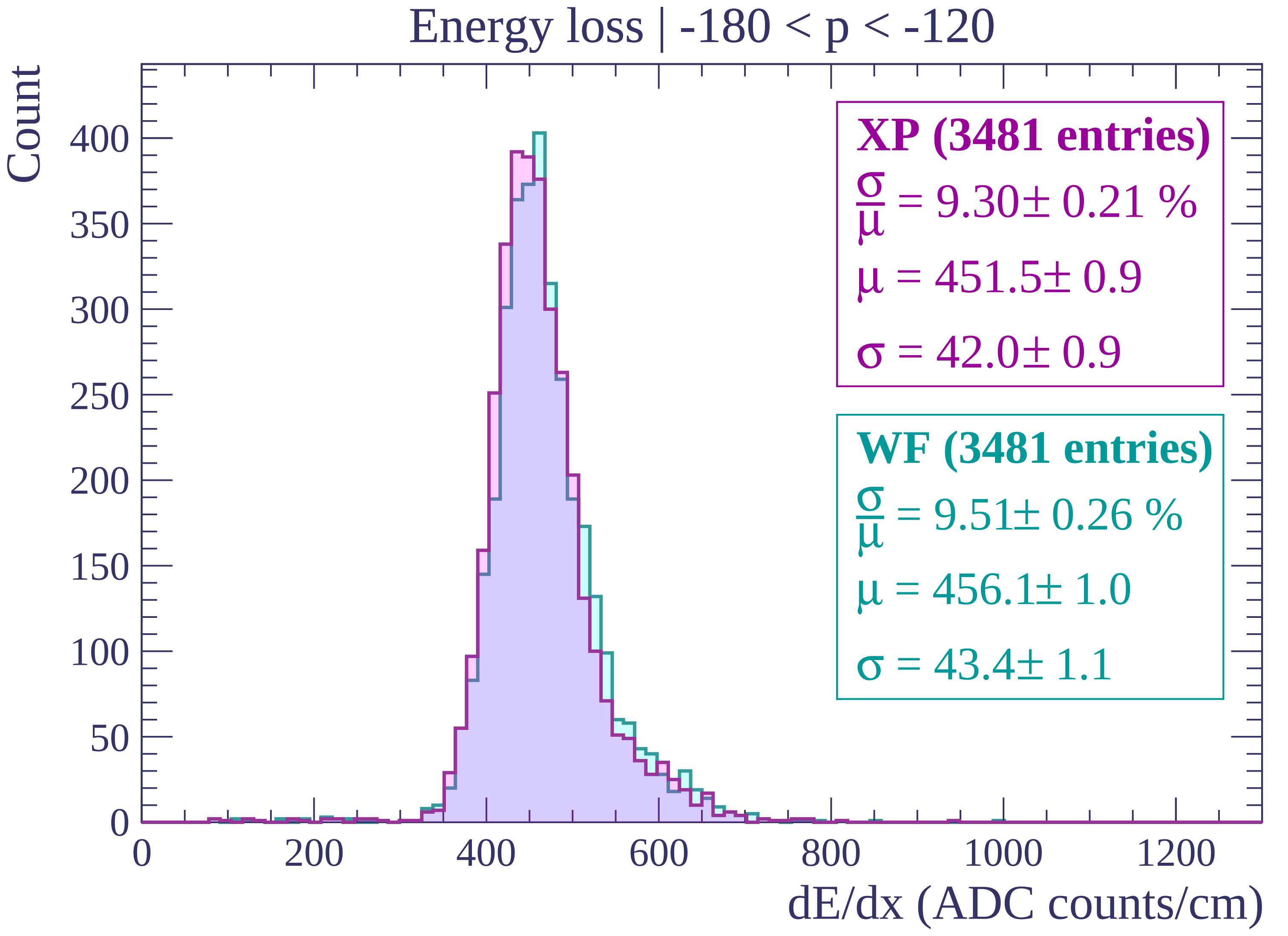


Energy loss | -360 < p < -300
Count
0
200
400
600
800
1000
1200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7683 entries)
σ/μ = 8.62± 0.12 %
μ = 398.3± 0.5
σ = 34.4± 0.4
WF (7683 entries)
σ/μ = 8.88± 0.13 %
μ = 401.8± 0.5
σ = 35.7± 0.5

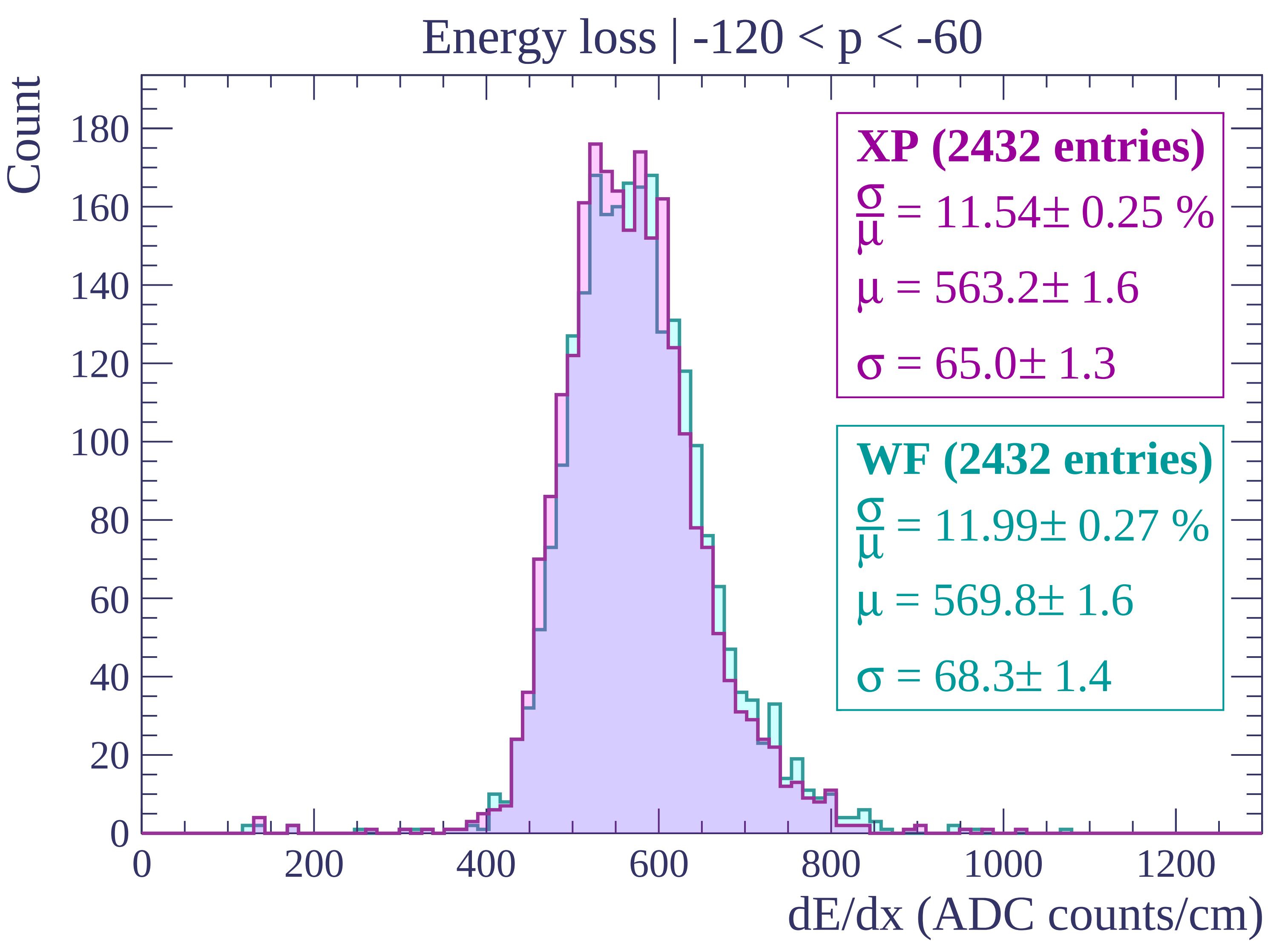

Energy loss | -300 < p < -240
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6796 entries)
σ/μ = 8.41± 0.12 %
μ = 399.8± 0.5
σ = 33.6± 0.4
WF (6796 entries)
σ/μ = 8.78± 0.13 %
μ = 402.8± 0.5
σ = 35.3± 0.5



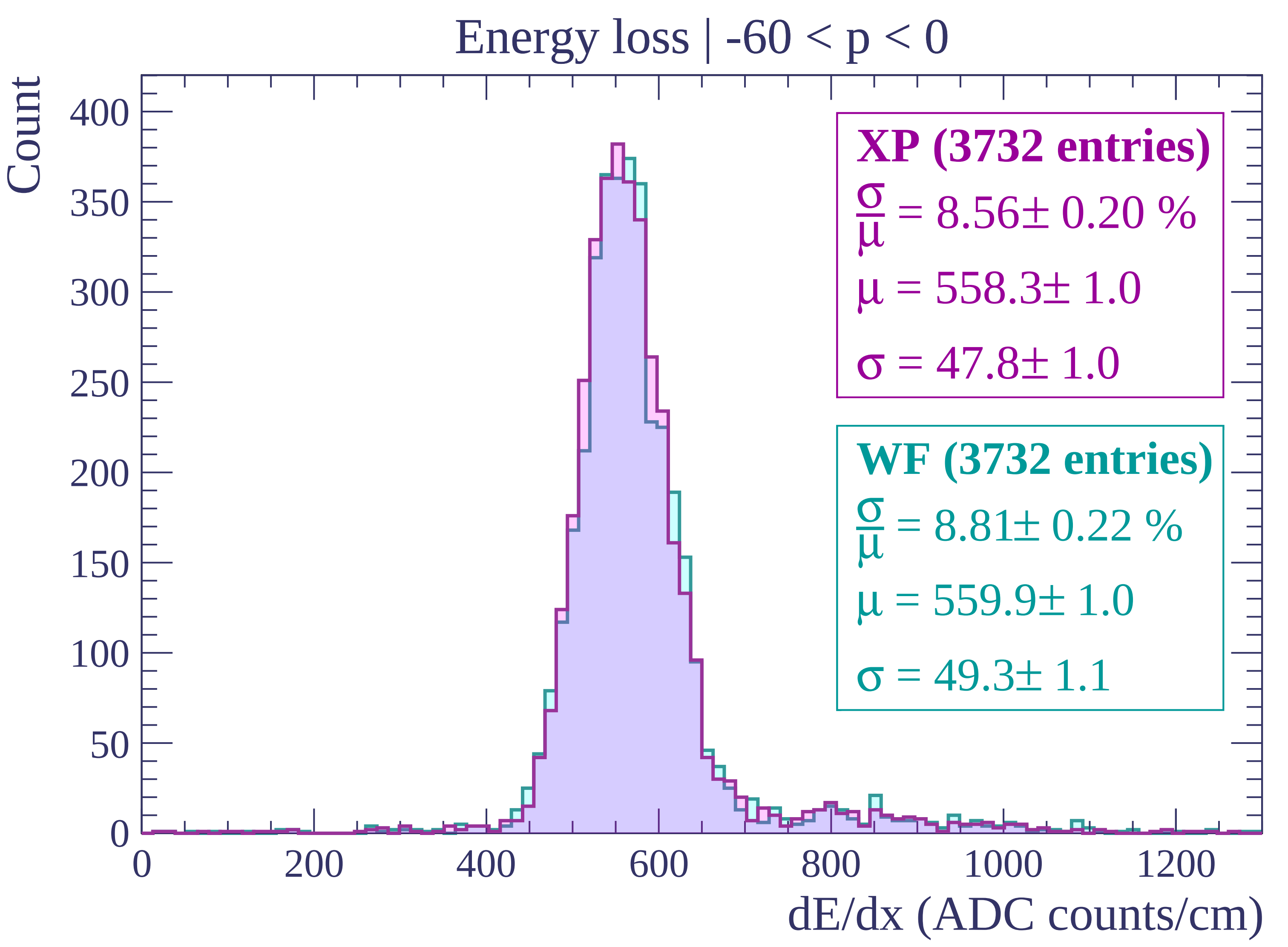
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4865 entries)
$\frac{\sigma}{\mu}$ = 8.52± 0.16 %
μ = 411.9± 0.6
σ = 35.1± 0.6
WF (4865 entries)
$\frac{\sigma}{\mu}$ = 8.93± 0.16 %
μ = 416.4± 0.6
σ = 37.2± 0.6

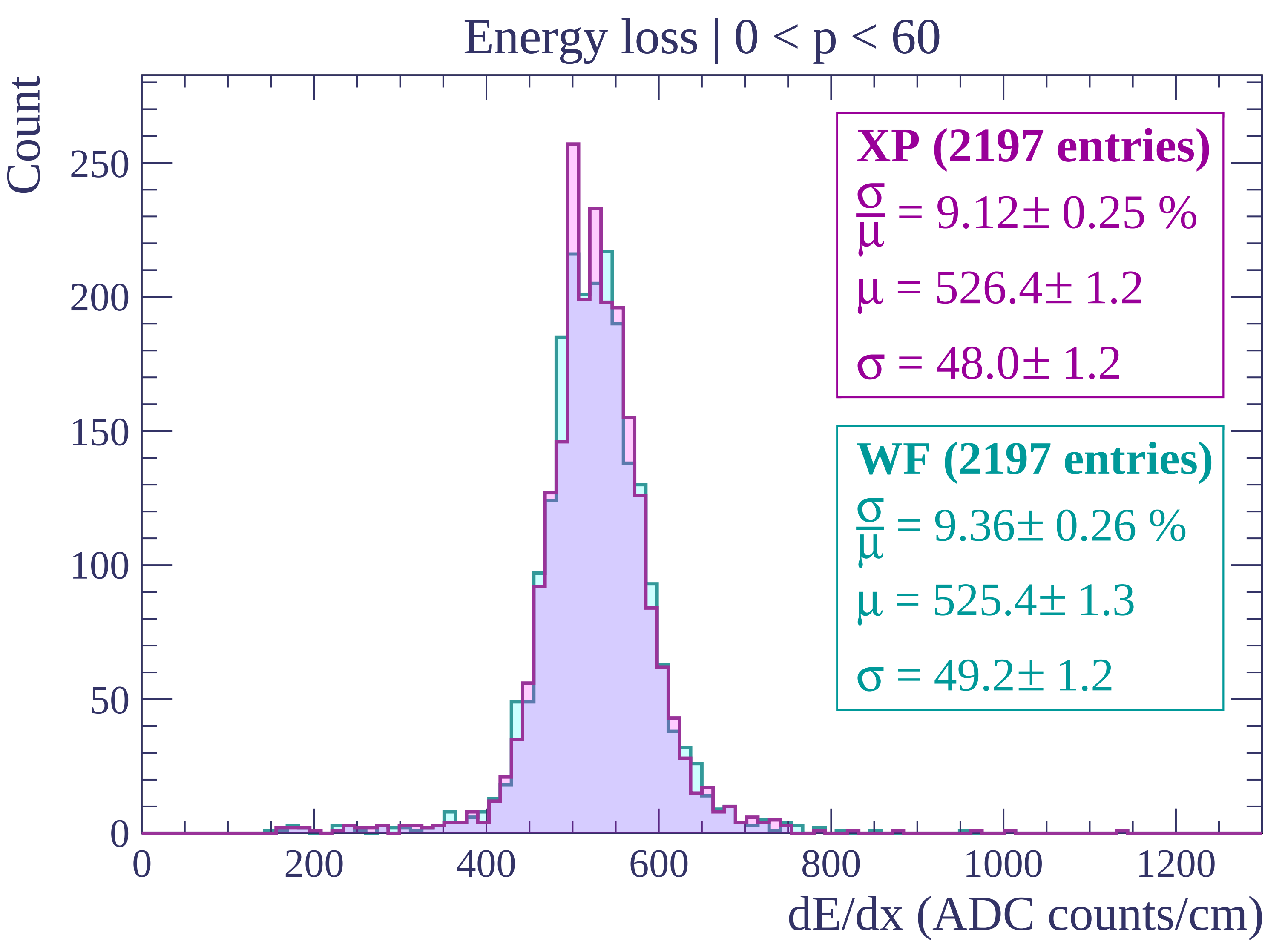


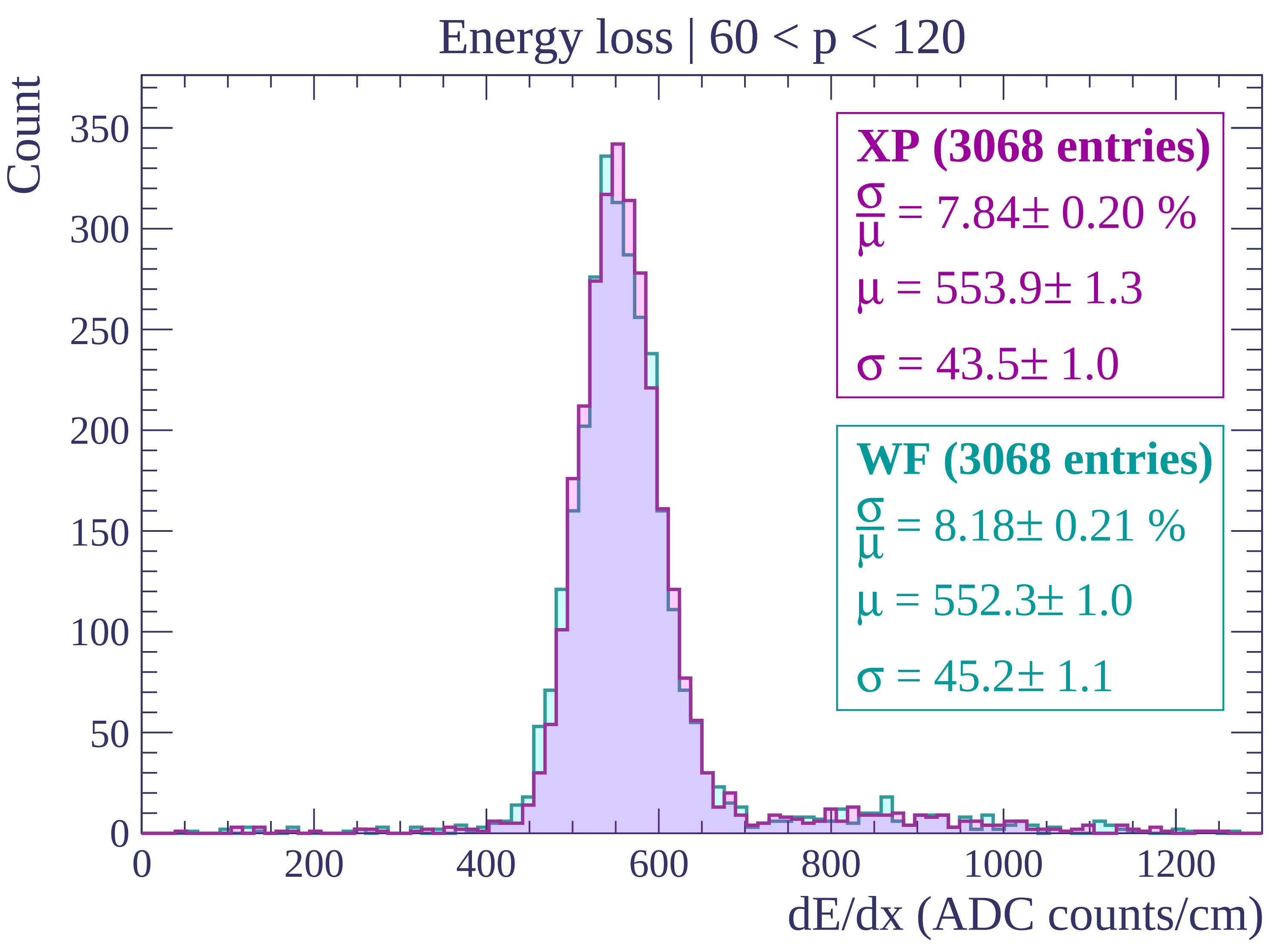
Energy loss | -180 < p < -120
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3481 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.21 %
μ = 451.5± 0.9
σ = 42.0± 0.9
WF (3481 entries)
$\frac{\sigma}{\mu}$ = 9.51± 0.26 %
μ = 456.1± 1.0
σ = 43.4± 1.1



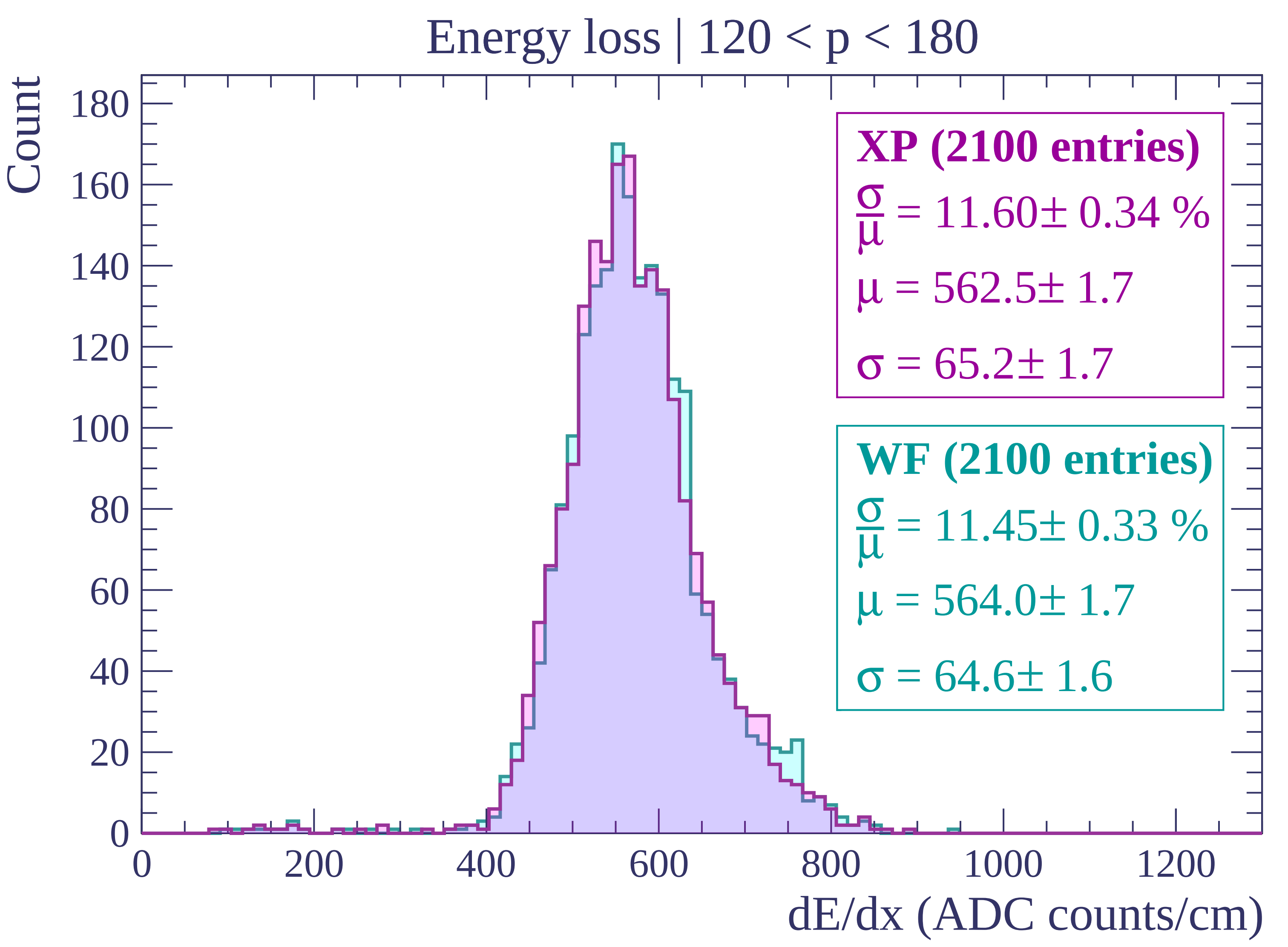
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (2432 entries)
σ/μ = 11.54± 0.25 %
μ = 563.2± 1.6
σ = 65.0± 1.3
WF (2432 entries)
σ/μ = 11.99± 0.27 %
μ = 569.8± 1.6
σ = 68.3± 1.4

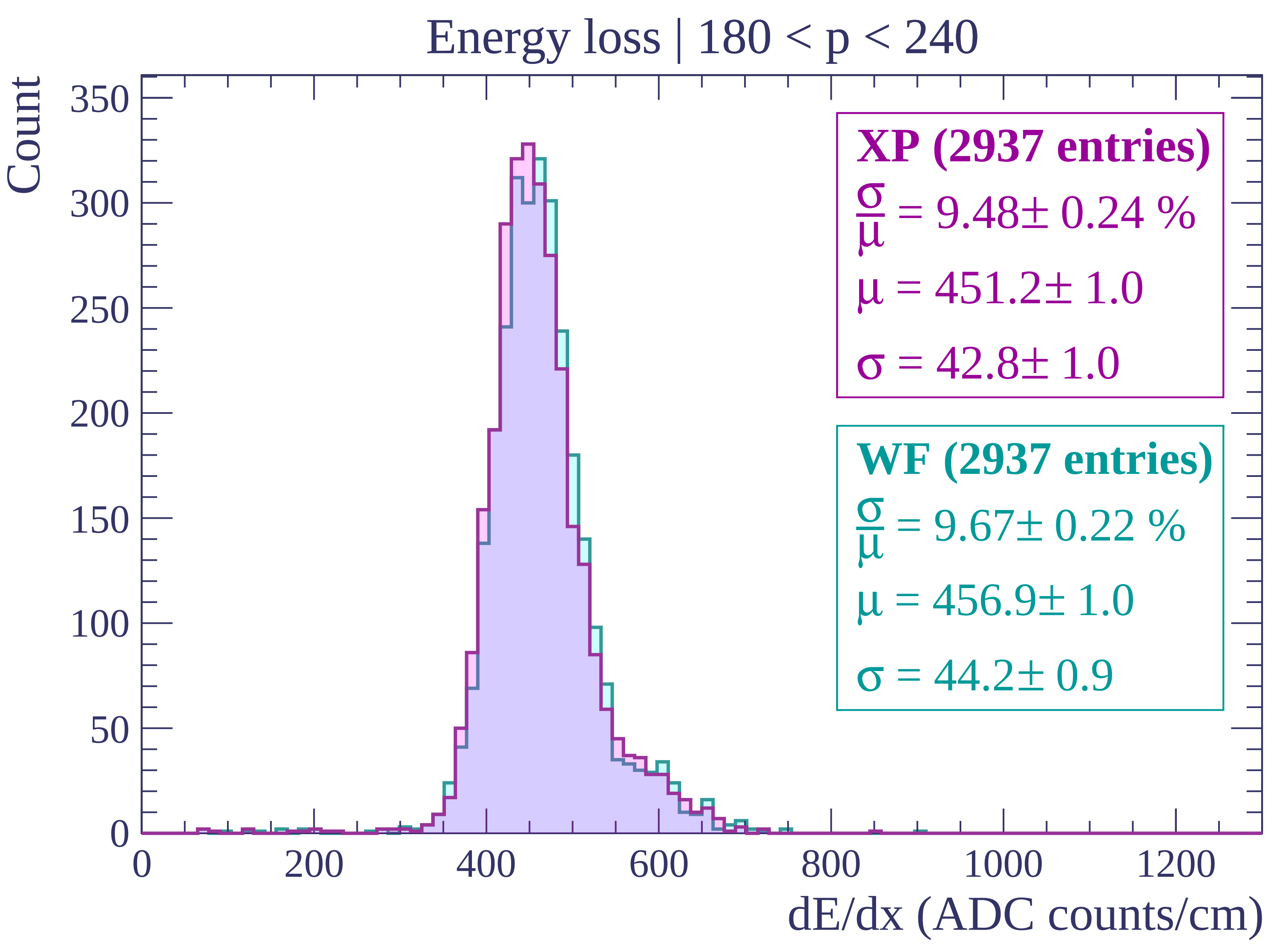

Energy loss | -60 < p < 0
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3732 entries)
σ/μ = 8.56± 0.20 %
μ = 558.3± 1.0
σ = 47.8± 1.0
WF (3732 entries)
σ/μ = 8.81± 0.22 %
μ = 559.9± 1.0
σ = 49.3± 1.1


Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2197 entries)
σ/μ = 9.12± 0.25 %
μ = 526.4± 1.2
σ = 48.0± 1.2
WF (2197 entries)
σ/μ = 9.36± 0.26 %
μ = 525.4± 1.3
σ = 49.2± 1.2

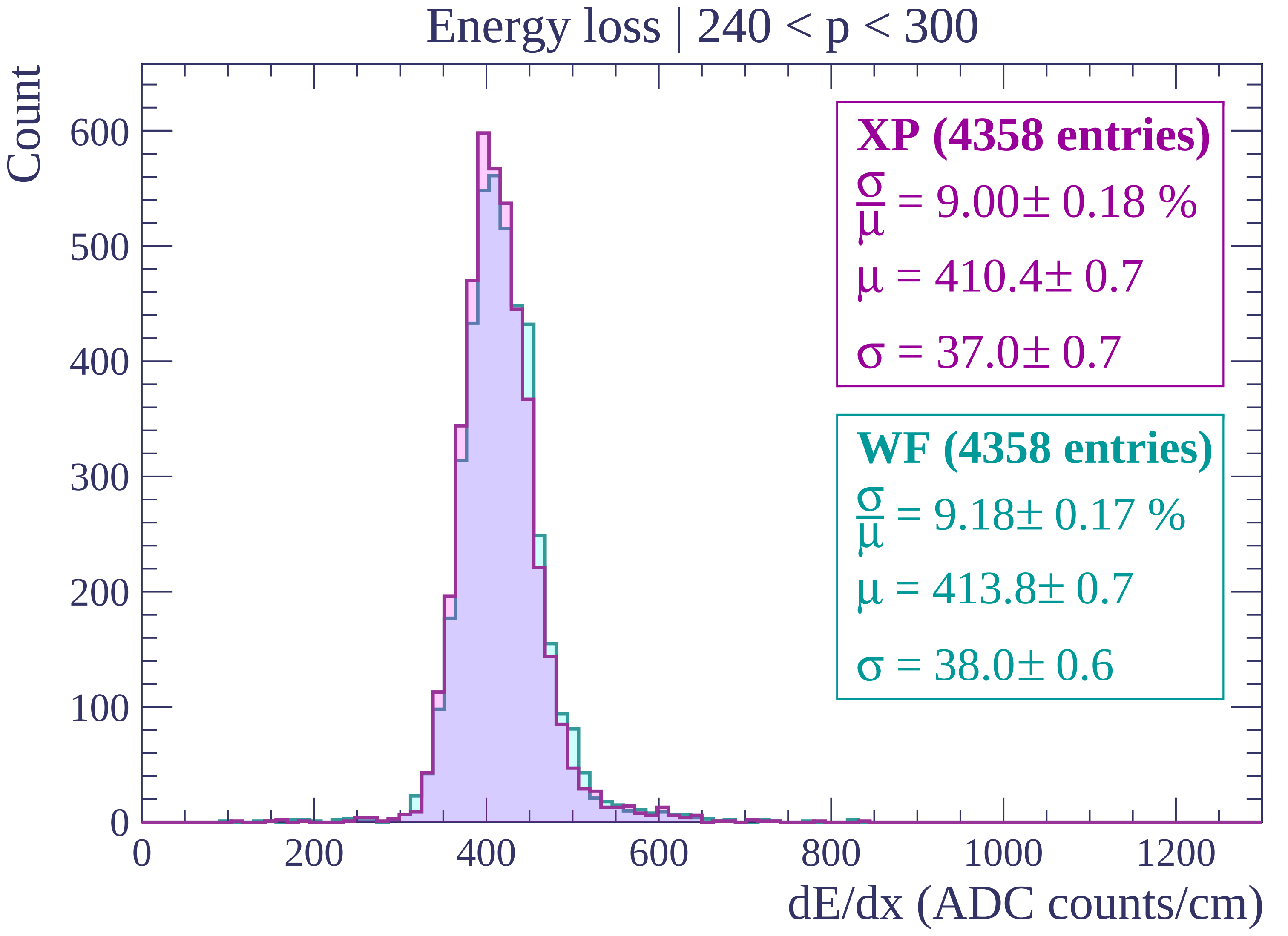


Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3068 entries)
$\frac{\sigma}{\mu}$ = 7.84± 0.20 %
μ = 553.9± 1.3
σ = 43.5± 1.0
WF (3068 entries)
$\frac{\sigma}{\mu}$ = 8.18± 0.21 %
μ = 552.3± 1.0
σ = 45.2± 1.1

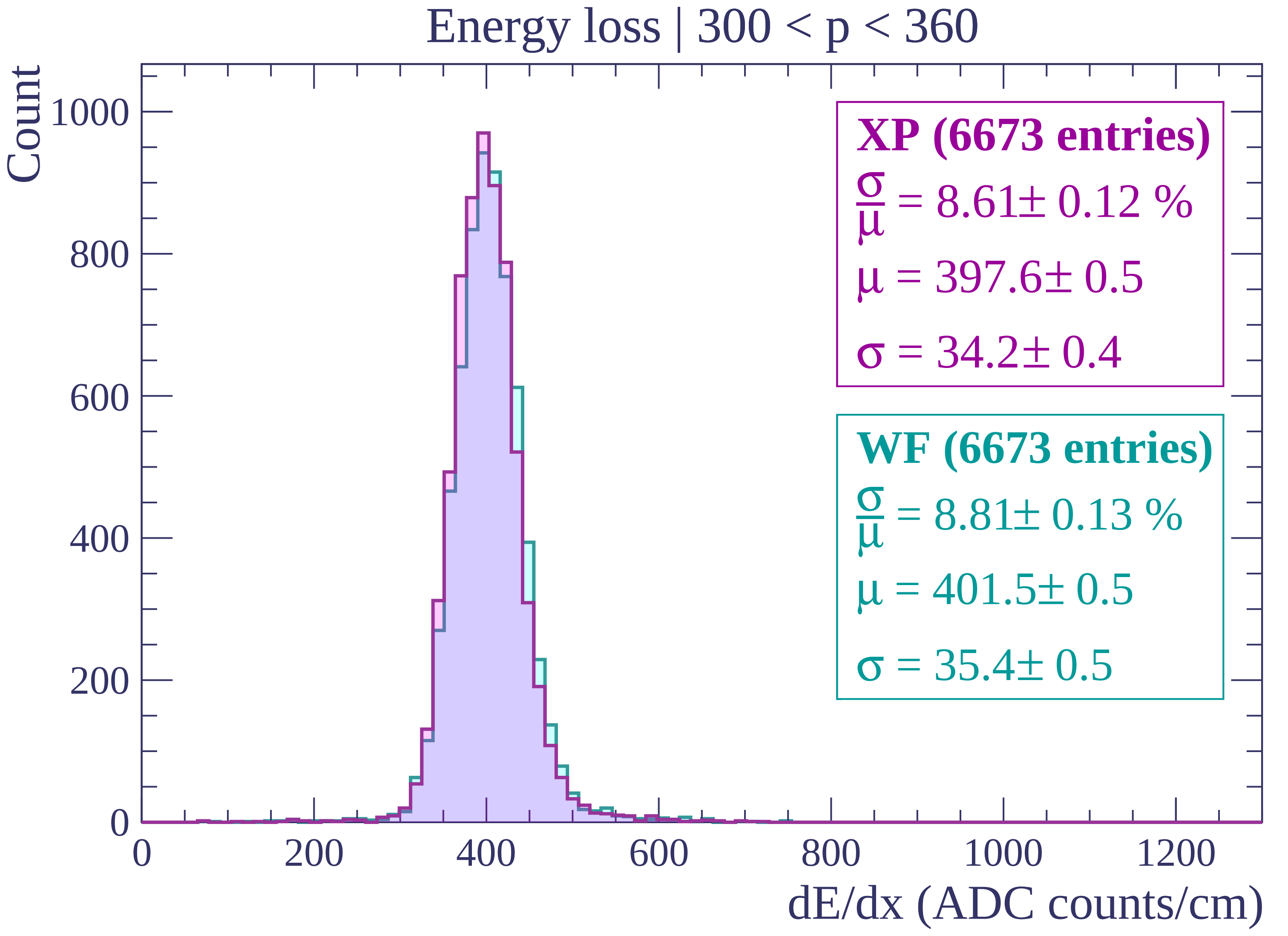

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP (2100 entries)
$\frac{\sigma}{\mu}$ = 11.60± 0.34 %
μ = 562.5± 1.7
σ = 65.2± 1.7
WF (2100 entries)
$\frac{\sigma}{\mu}$ = 11.45± 0.33 %
μ = 564.0± 1.7
σ = 64.6± 1.6



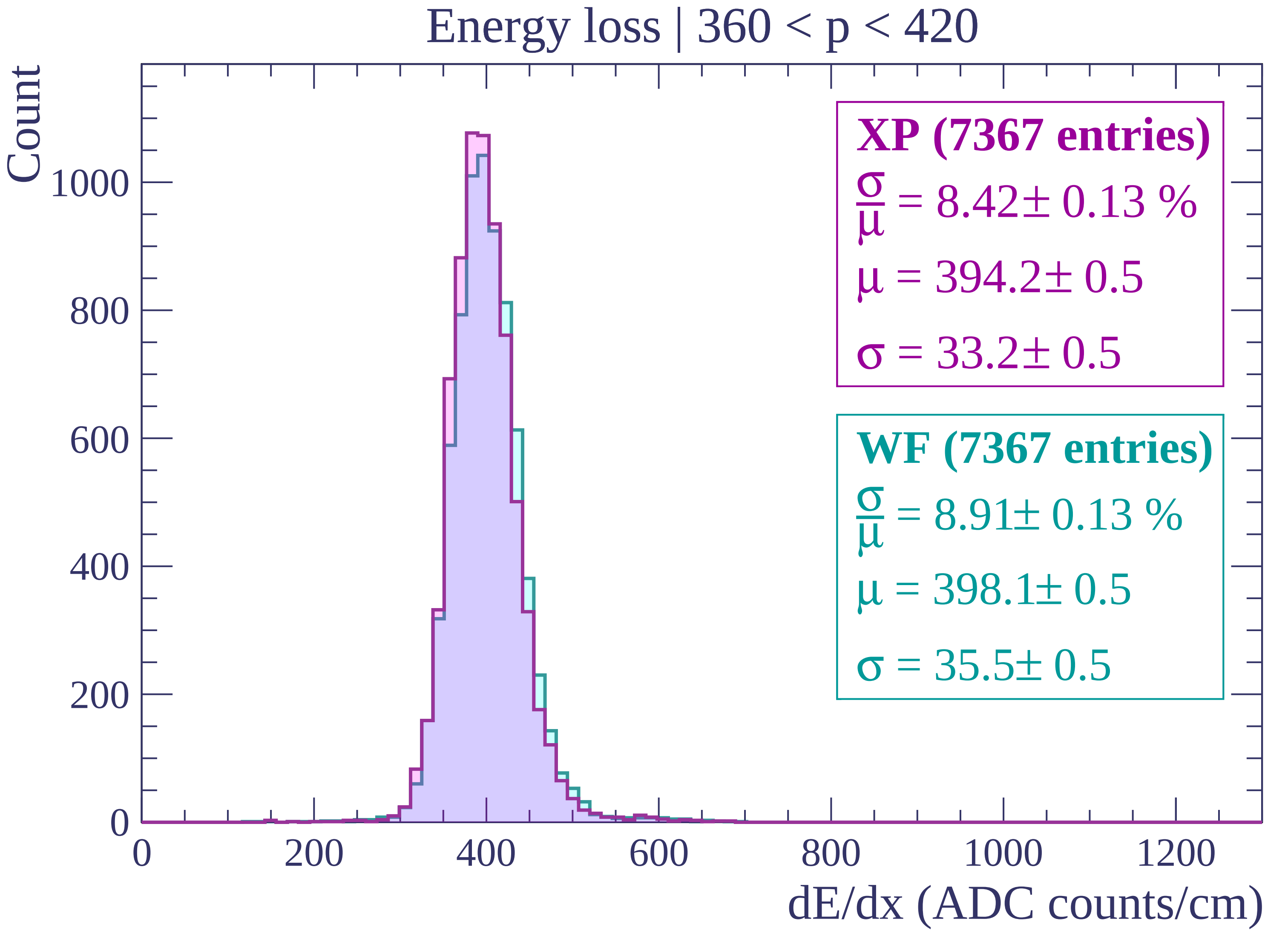
Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2937 entries)
σ/μ = 9.48± 0.24 %
μ = 451.2± 1.0
σ = 42.8± 1.0
WF (2937 entries)
σ/μ = 9.67± 0.22 %
μ = 456.9± 1.0
σ = 44.2± 0.9

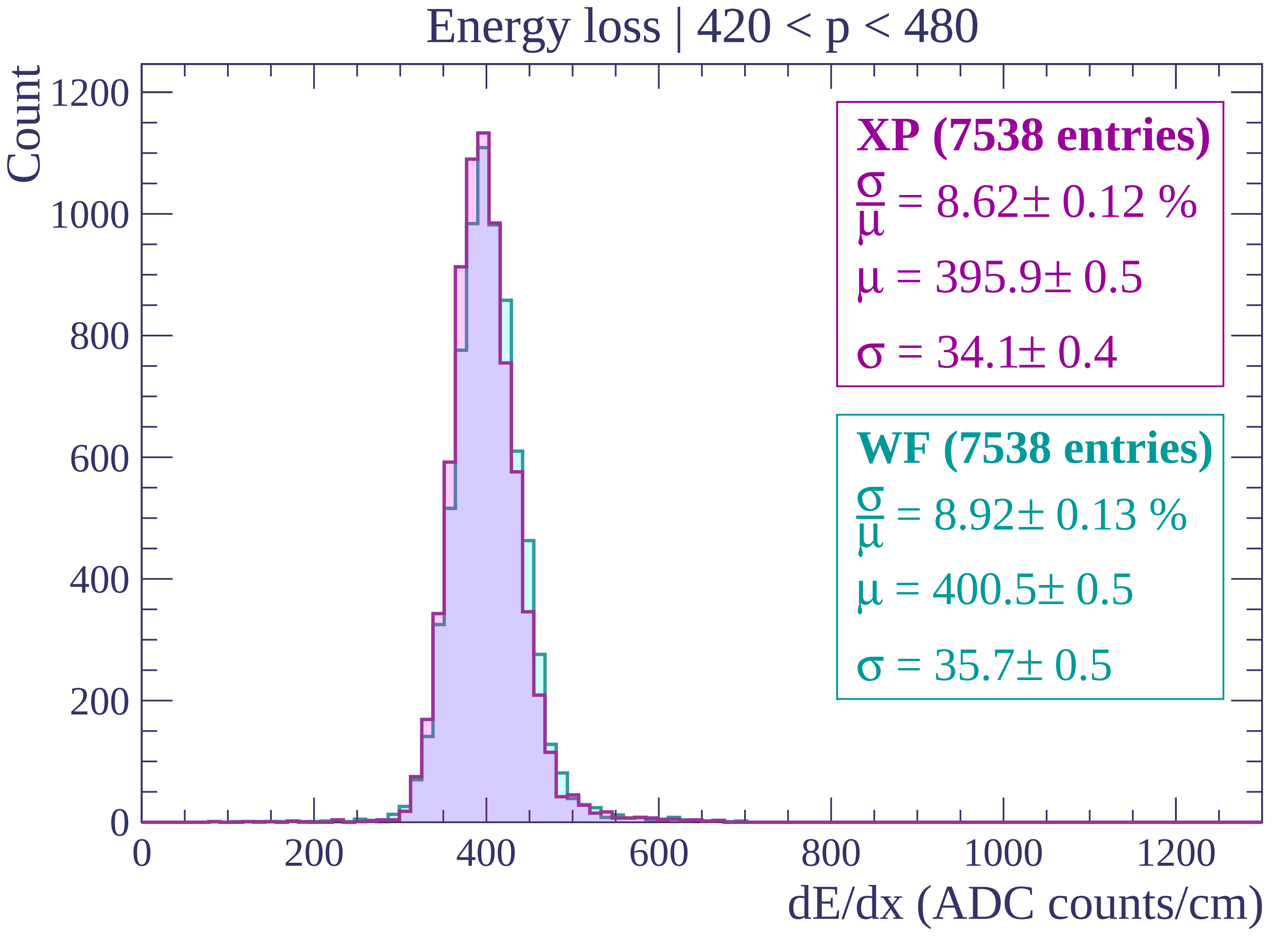


Energy loss | 240 < p < 300
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4358 entries)
σ/μ = 9.00± 0.18 %
μ = 410.4± 0.7
σ = 37.0± 0.7
WF (4358 entries)
σ/μ = 9.18± 0.17 %
μ = 413.8± 0.7
σ = 38.0± 0.6

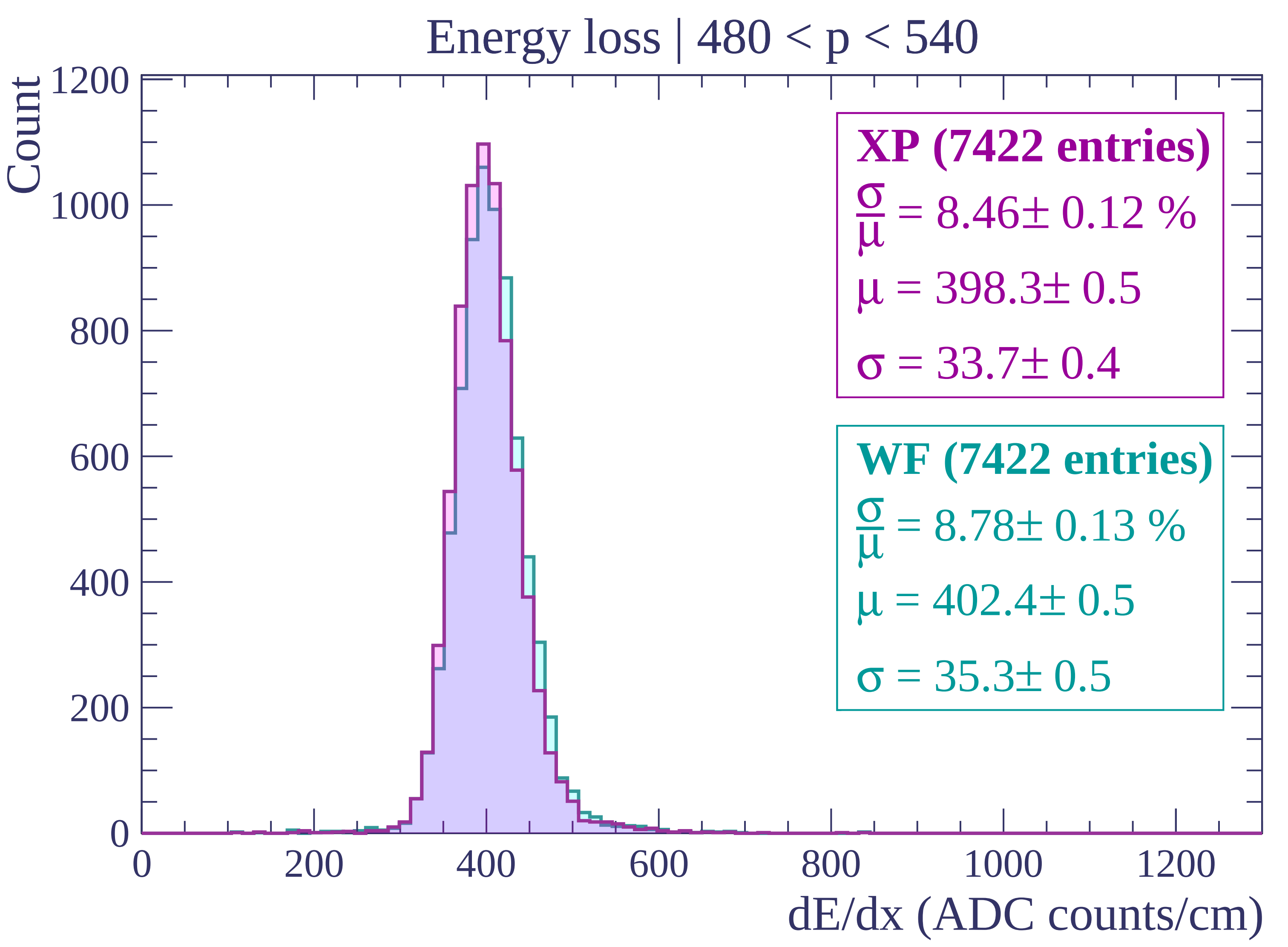


Energy loss | 300 < p < 360
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6673 entries)
$\frac{\sigma}{\mu}$ = 8.61± 0.12 %
μ = 397.6± 0.5
σ = 34.2± 0.4
WF (6673 entries)
$\frac{\sigma}{\mu}$ = 8.81± 0.13 %
μ = 401.5± 0.5
σ = 35.4± 0.5

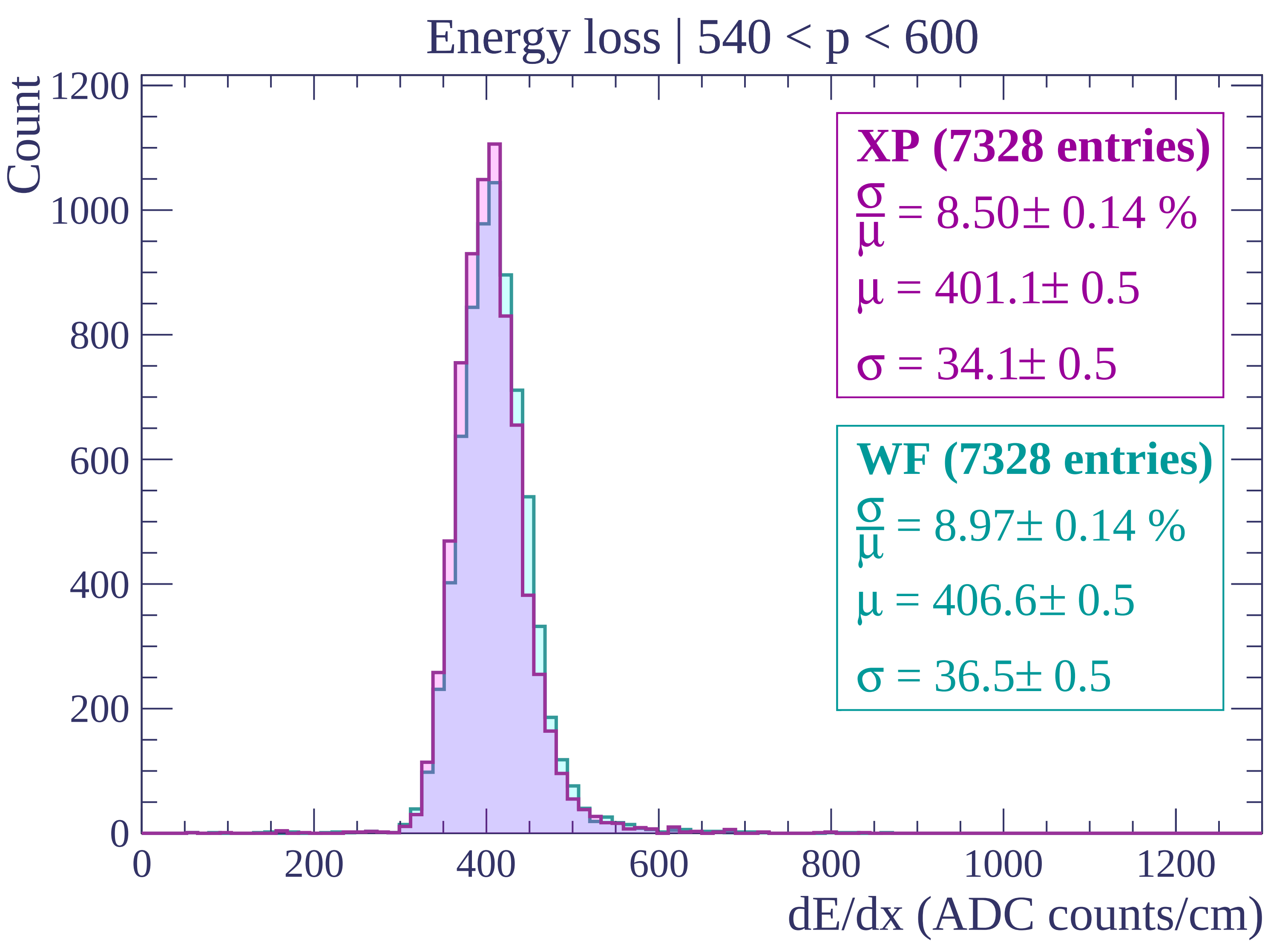

Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
XP (7367 entries)
σ/μ = 8.42± 0.13 %
μ = 394.2± 0.5
σ = 33.2± 0.5
WF (7367 entries)
σ/μ = 8.91± 0.13 %
μ = 398.1± 0.5
σ = 35.5± 0.5
0
200
400
600
800
1000
1200

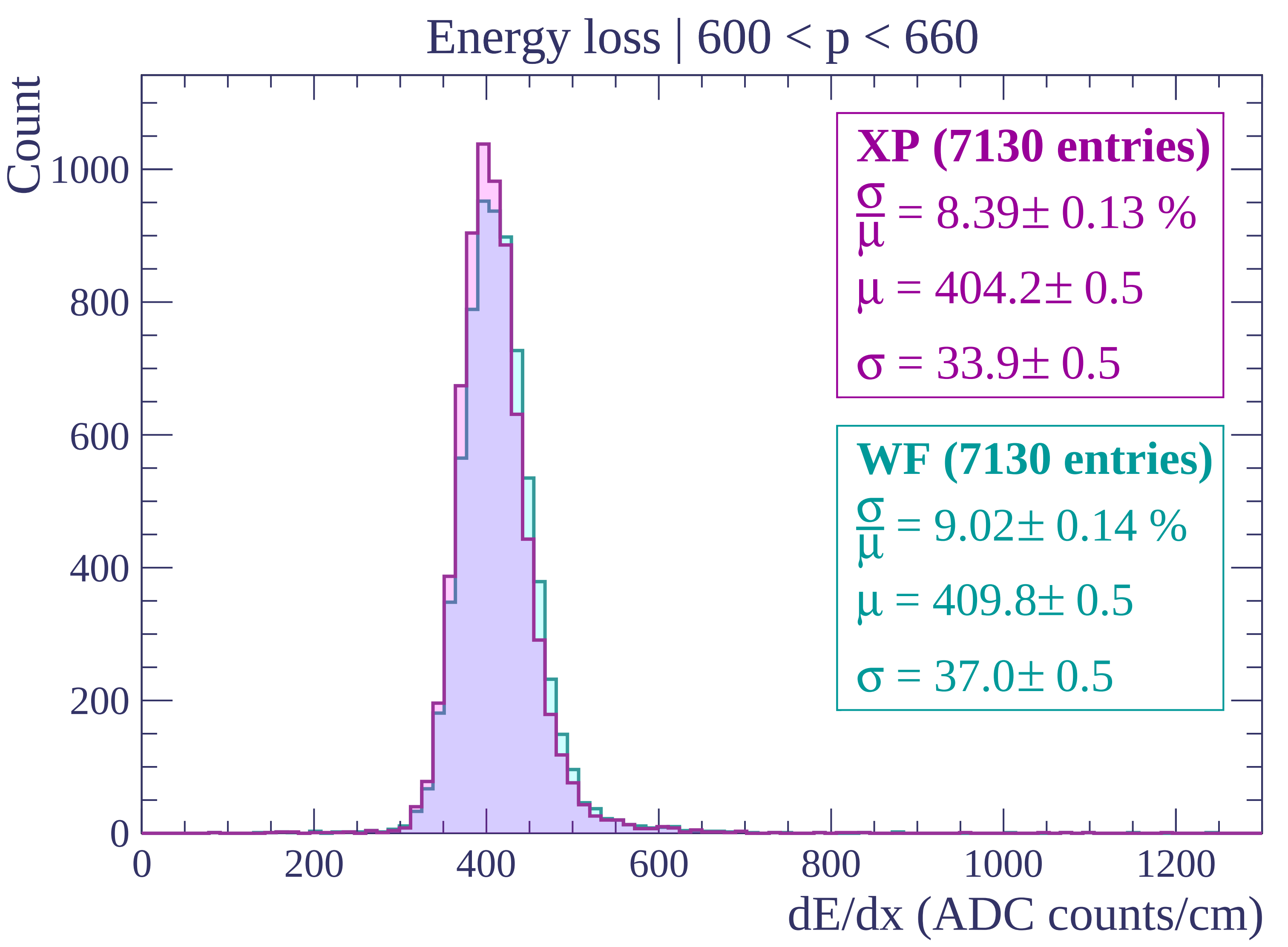


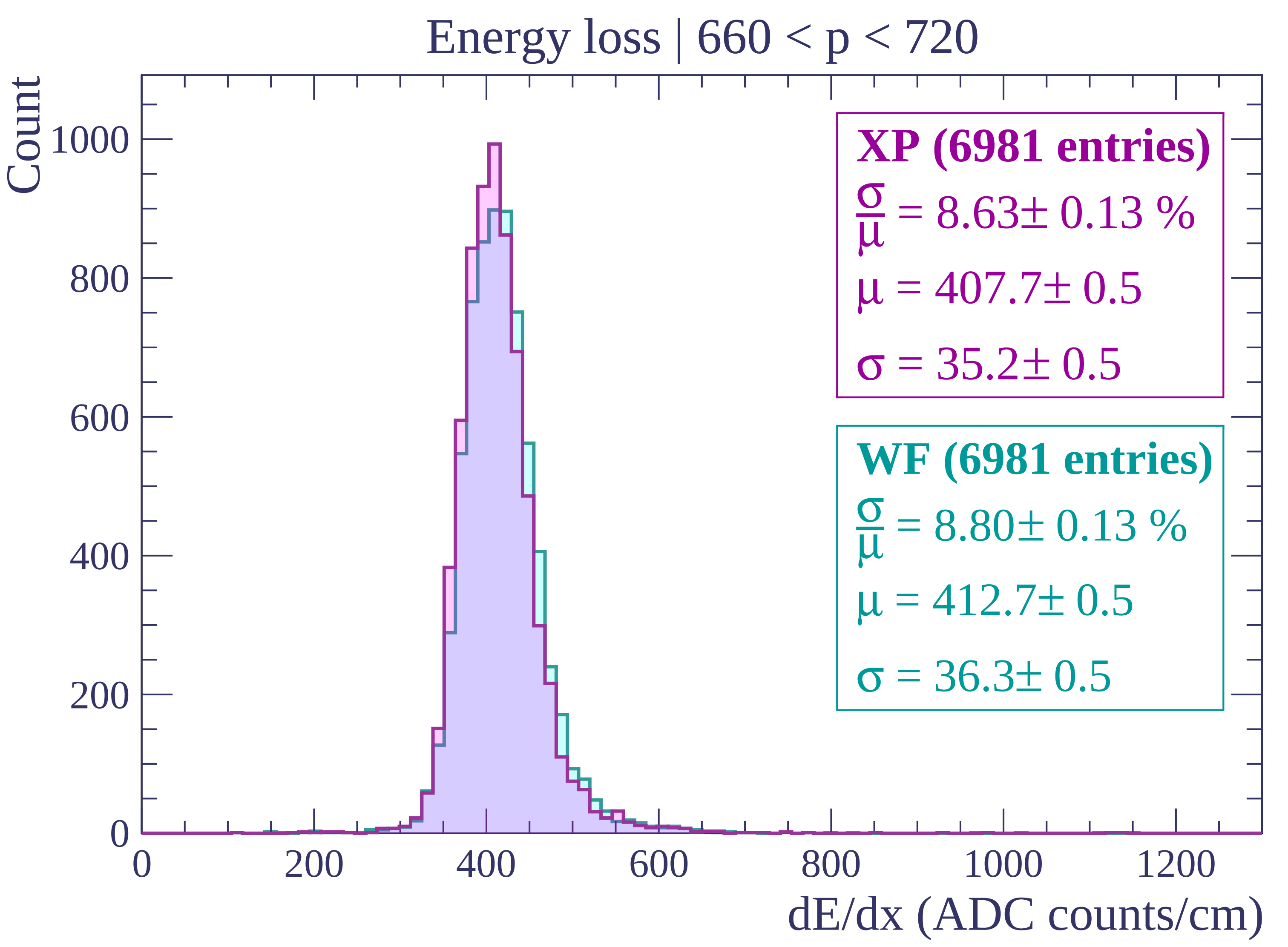
Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7538 entries)
σ/μ = 8.62± 0.12 %
μ = 395.9± 0.5
σ = 34.1± 0.4
WF (7538 entries)
σ/μ = 8.92± 0.13 %
μ = 400.5± 0.5
σ = 35.7± 0.5



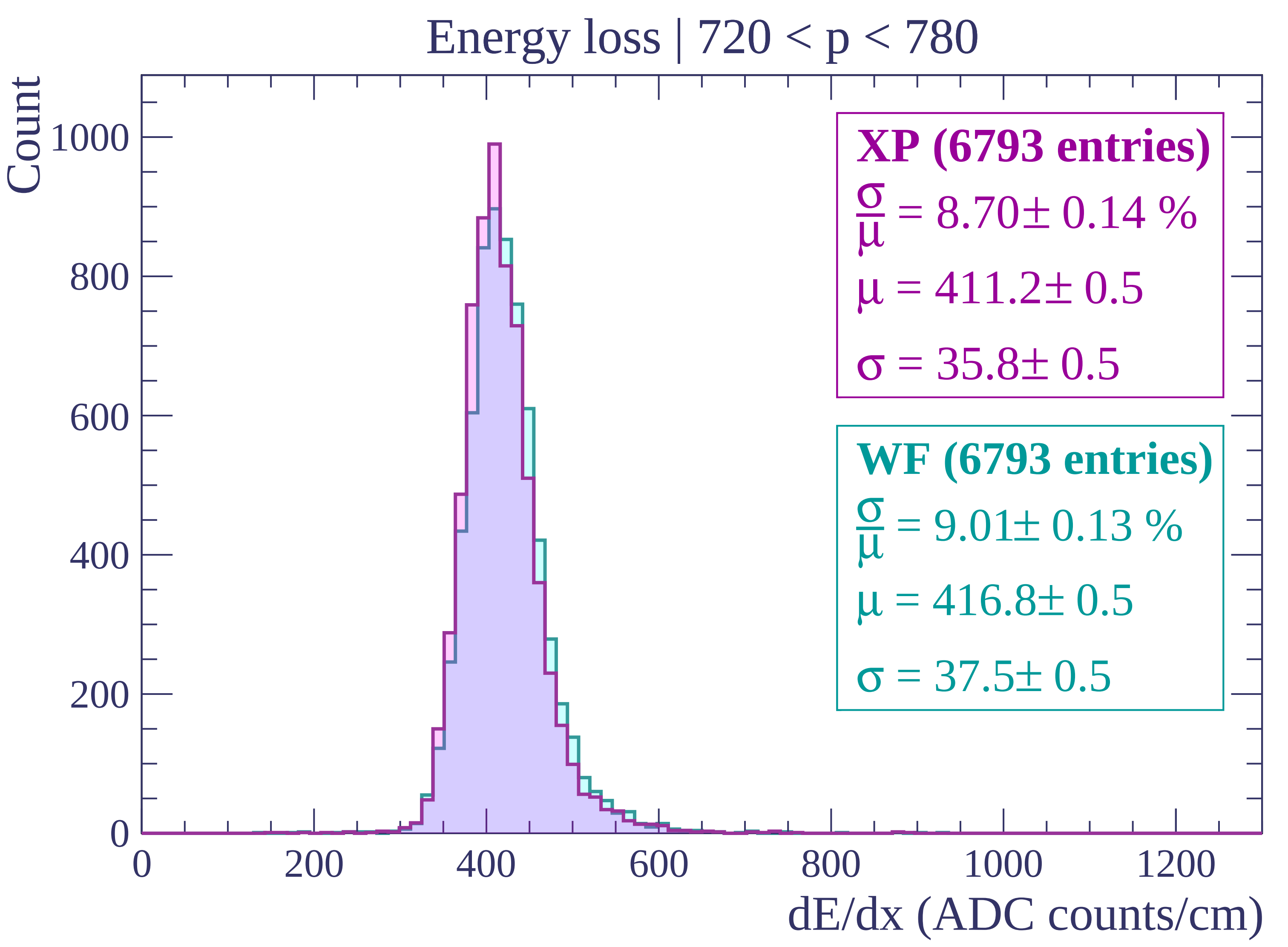
Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
XP (7422 entries)
$\frac{\sigma}{\mu}$ = 8.46± 0.12 %
μ = 398.3± 0.5
σ = 33.7± 0.4
WF (7422 entries)
$\frac{\sigma}{\mu}$ = 8.78± 0.13 %
μ = 402.4± 0.5
σ = 35.3± 0.5


Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7328 entries)
$\frac{\sigma}{\mu}$ = 8.50± 0.14 %
μ = 401.1± 0.5
σ = 34.1± 0.5
WF (7328 entries)
$\frac{\sigma}{\mu}$ = 8.97± 0.14 %
μ = 406.6± 0.5
σ = 36.5± 0.5

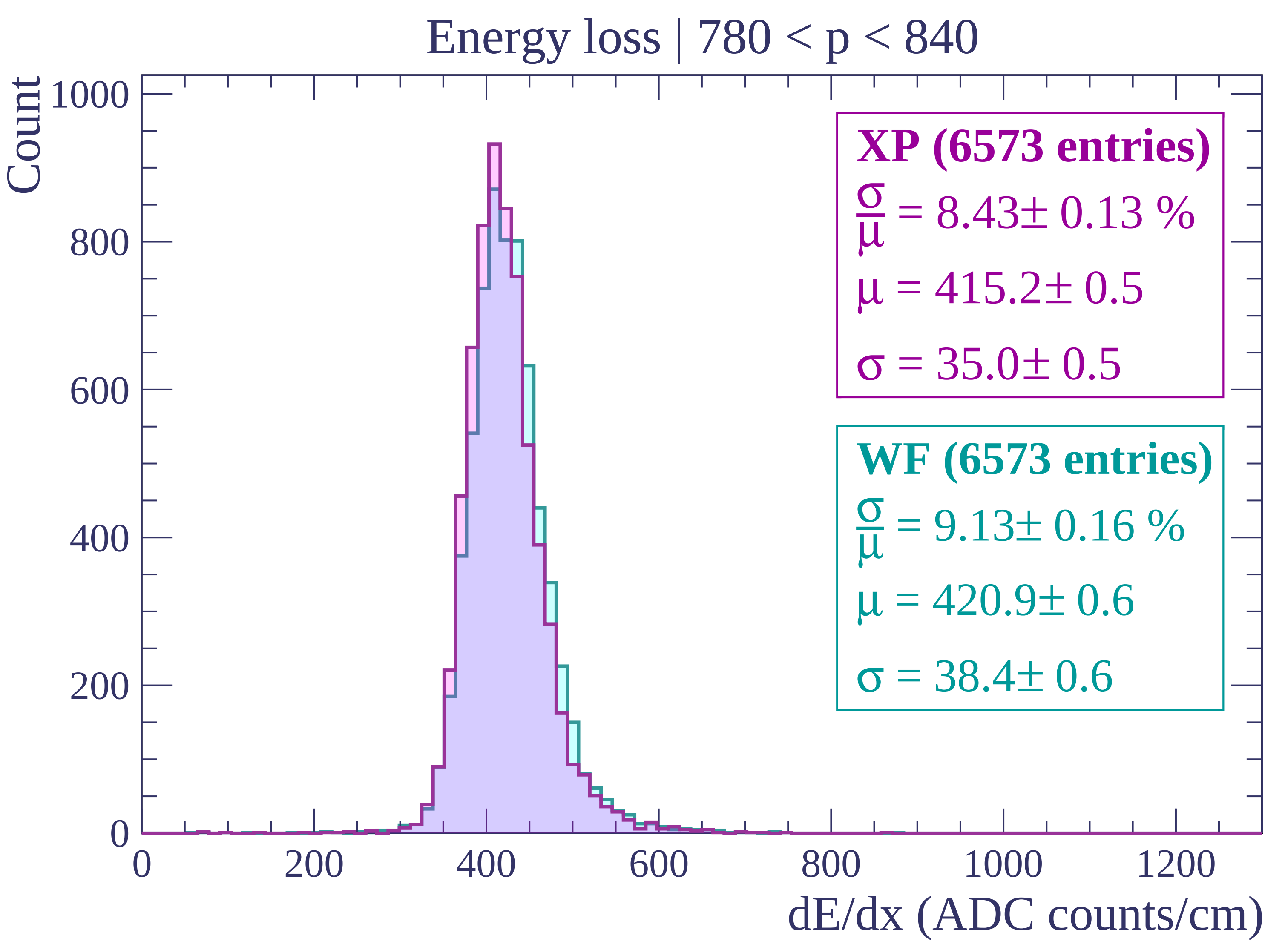

Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7130 entries)
σ/μ = 8.39± 0.13 %
μ = 404.2± 0.5
σ = 33.9± 0.5
WF (7130 entries)
σ/μ = 9.02± 0.14 %
μ = 409.8± 0.5
σ = 37.0± 0.5



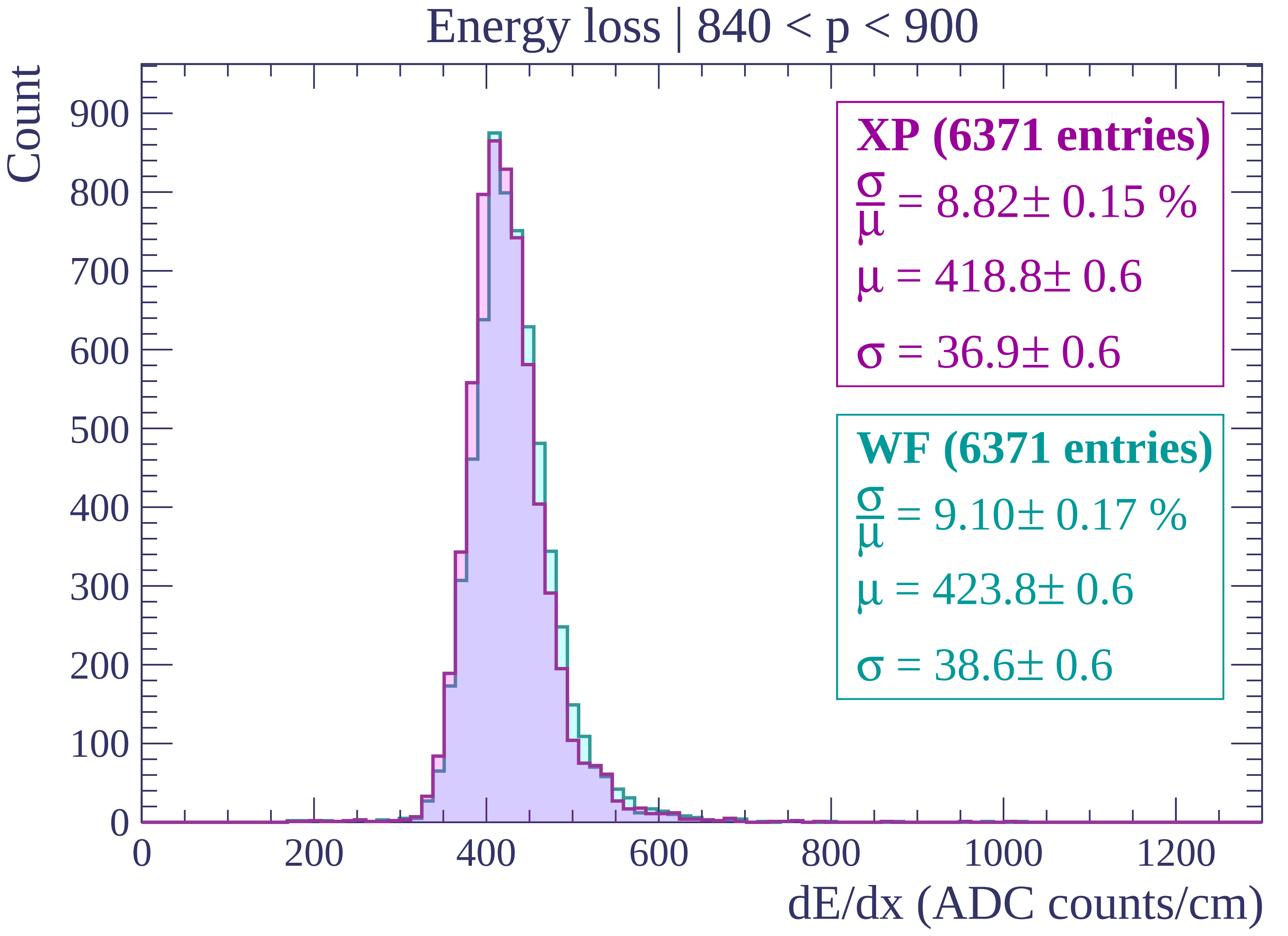
Energy loss | 660 < p < 720
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6981 entries)
$\frac{\sigma}{\mu}$ = 8.63± 0.13 %
μ = 407.7± 0.5
σ = 35.2± 0.5
WF (6981 entries)
$\frac{\sigma}{\mu}$ = 8.80± 0.13 %
μ = 412.7± 0.5
σ = 36.3± 0.5

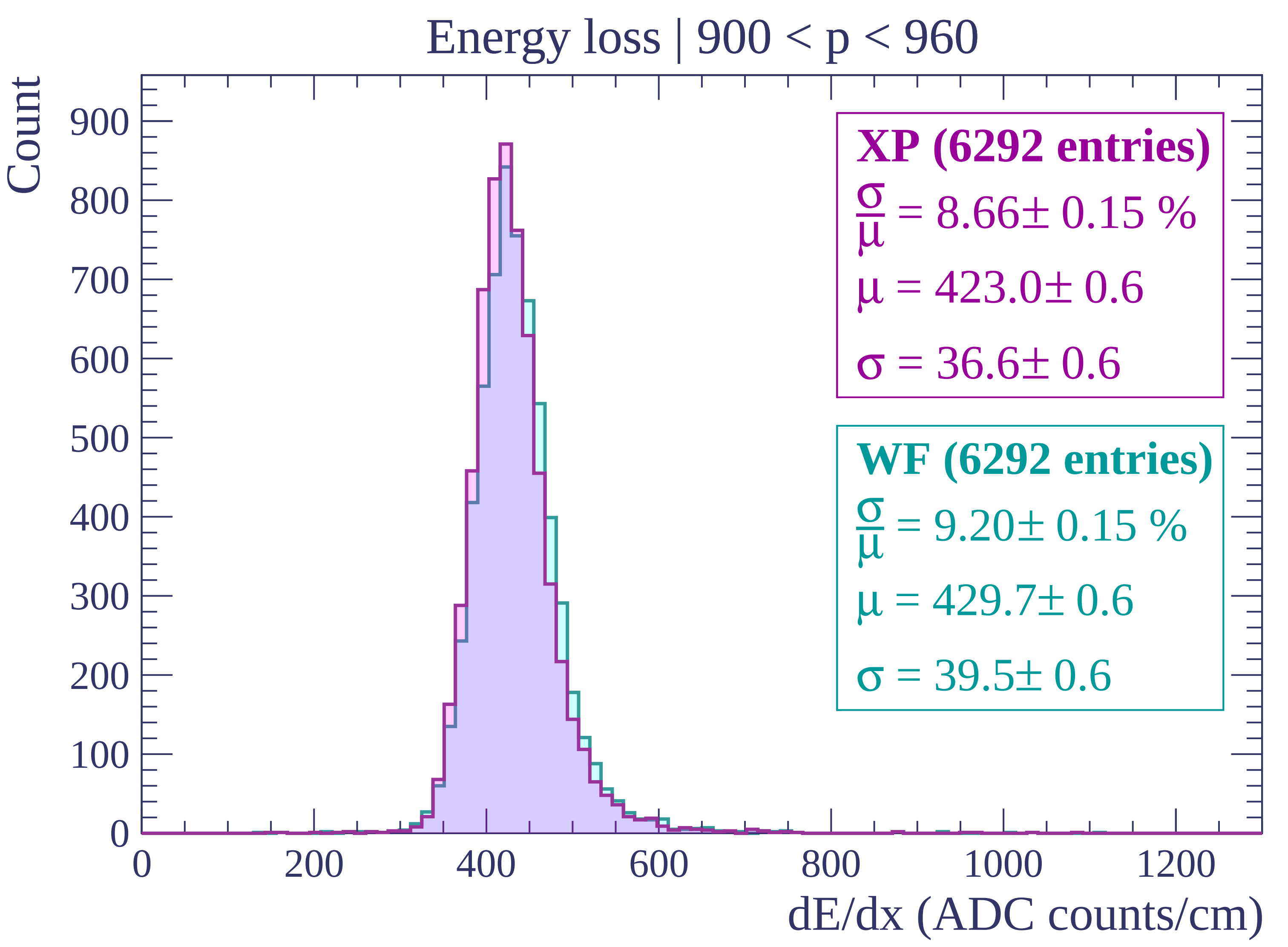


Energy loss | 720 < p < 780
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6793 entries)
σ/μ = 8.70± 0.14 %
μ = 411.2± 0.5
σ = 35.8± 0.5
WF (6793 entries)
σ/μ = 9.01± 0.13 %
μ = 416.8± 0.5
σ = 37.5± 0.5

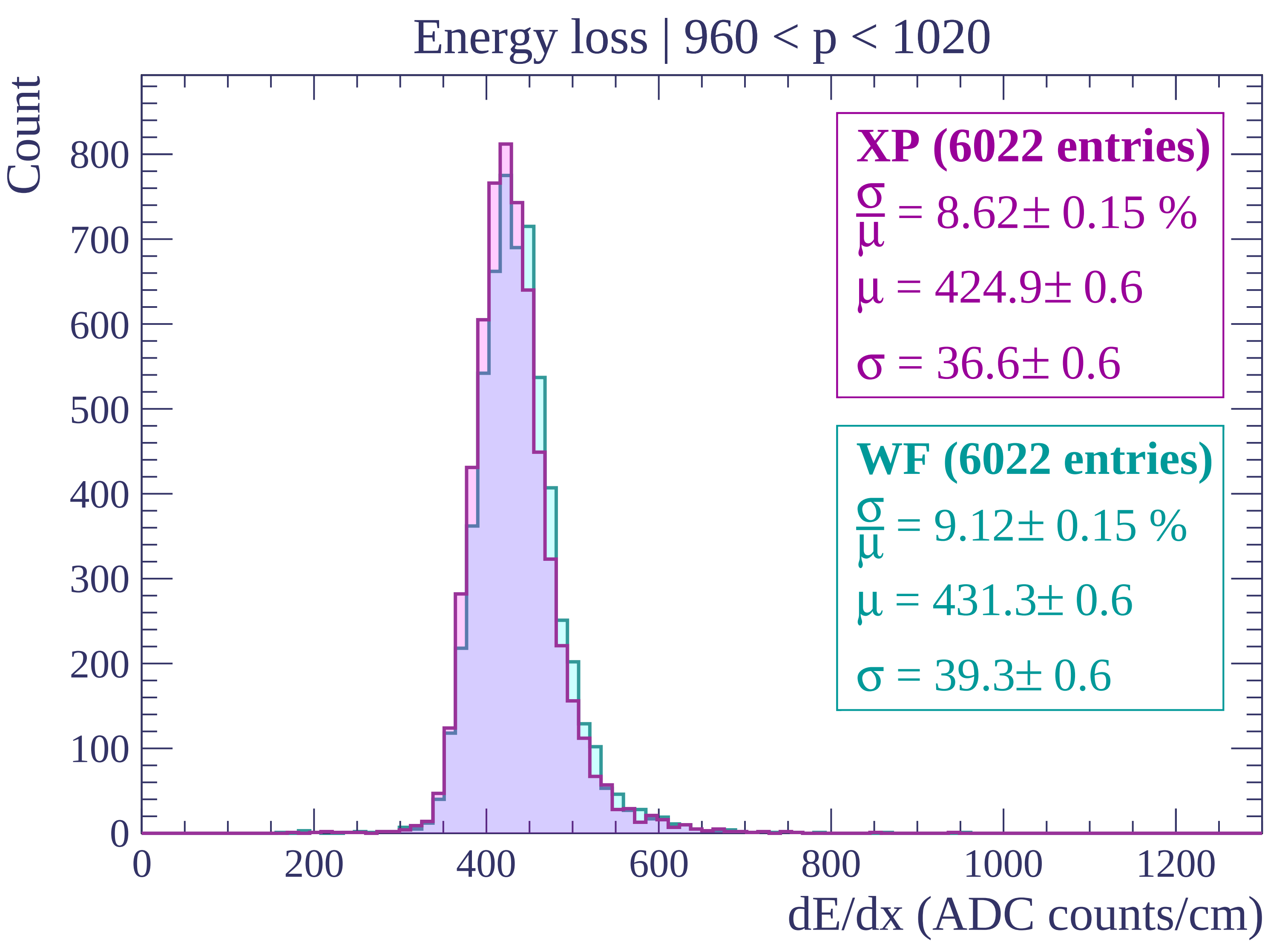


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (6573 entries)
$\frac{\sigma}{\mu}$ = 8.43± 0.13 %
μ = 415.2± 0.5
σ = 35.0± 0.5
WF (6573 entries)
$\frac{\sigma}{\mu}$ = 9.13± 0.16 %
μ = 420.9± 0.6
σ = 38.4± 0.6

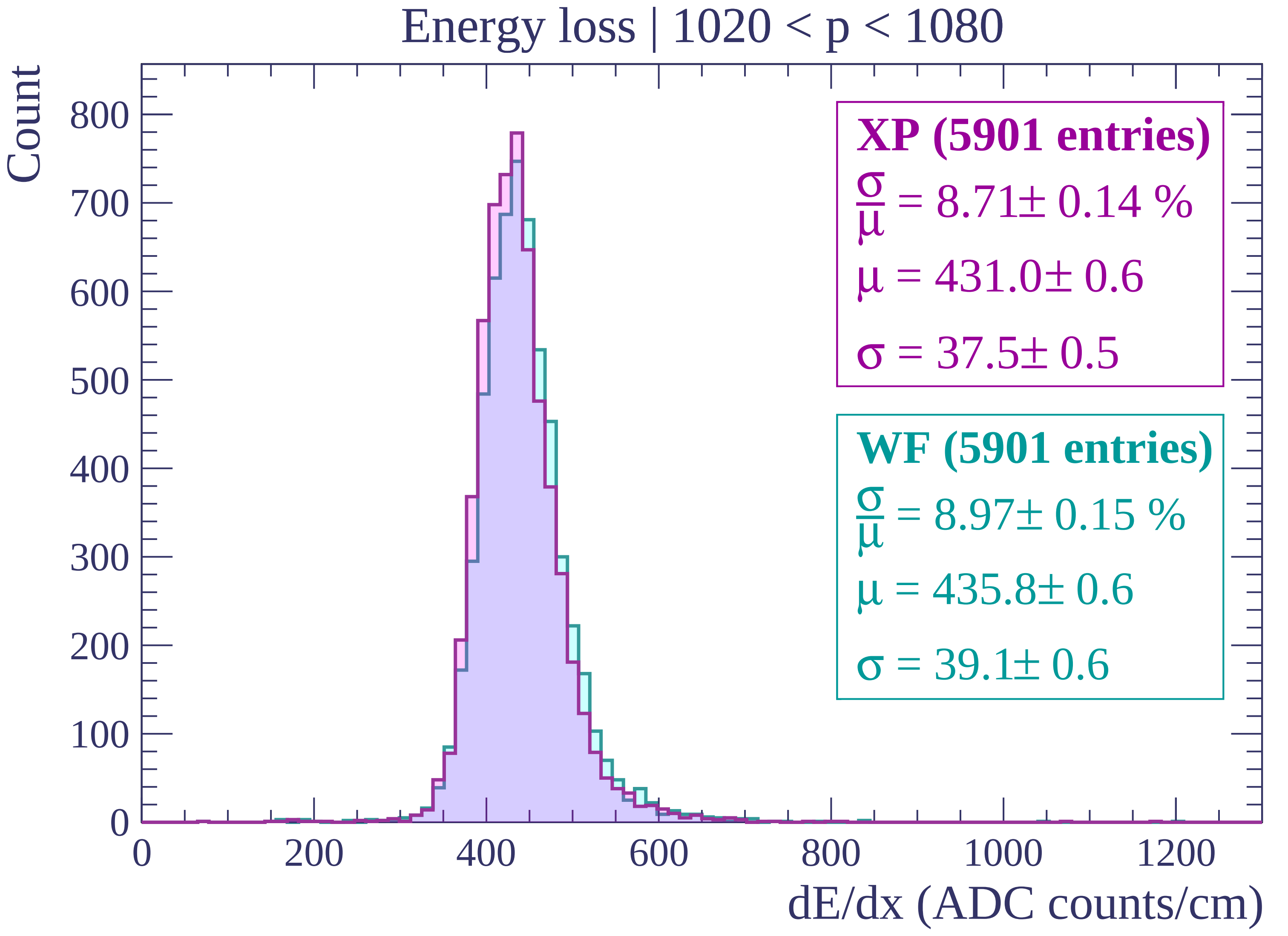

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
900
XP (6371 entries)
σ/μ = 8.82± 0.15 %
μ = 418.8± 0.6
σ = 36.9± 0.6
WF (6371 entries)
σ/μ = 9.10± 0.17 %
μ = 423.8± 0.6
σ = 38.6± 0.6



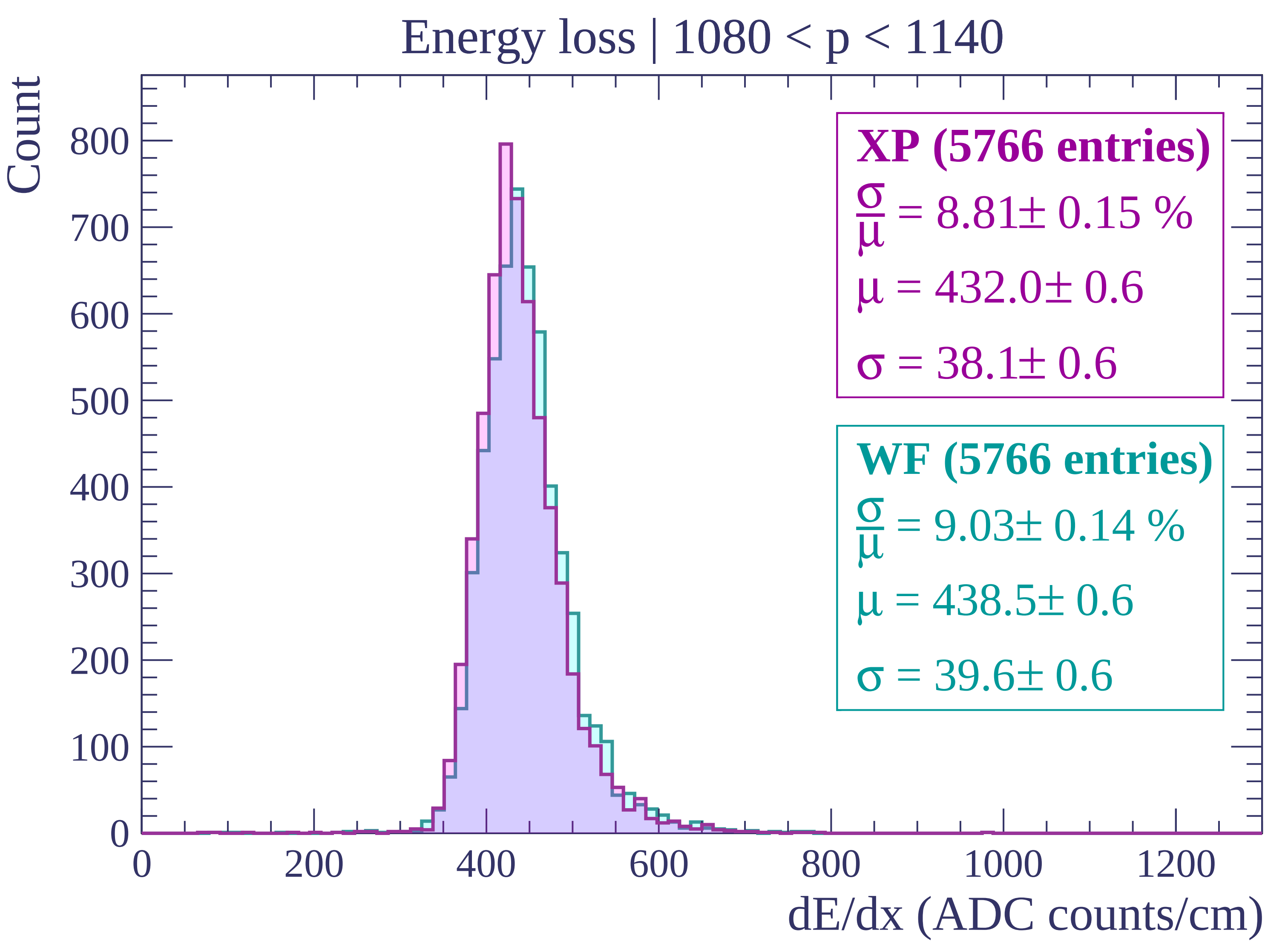
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
XP (6292 entries)
$\frac{\sigma}{\mu}$ = 8.66± 0.15 %
μ = 423.0± 0.6
σ = 36.6± 0.6
WF (6292 entries)
$\frac{\sigma}{\mu}$ = 9.20± 0.15 %
μ = 429.7± 0.6
σ = 39.5± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200



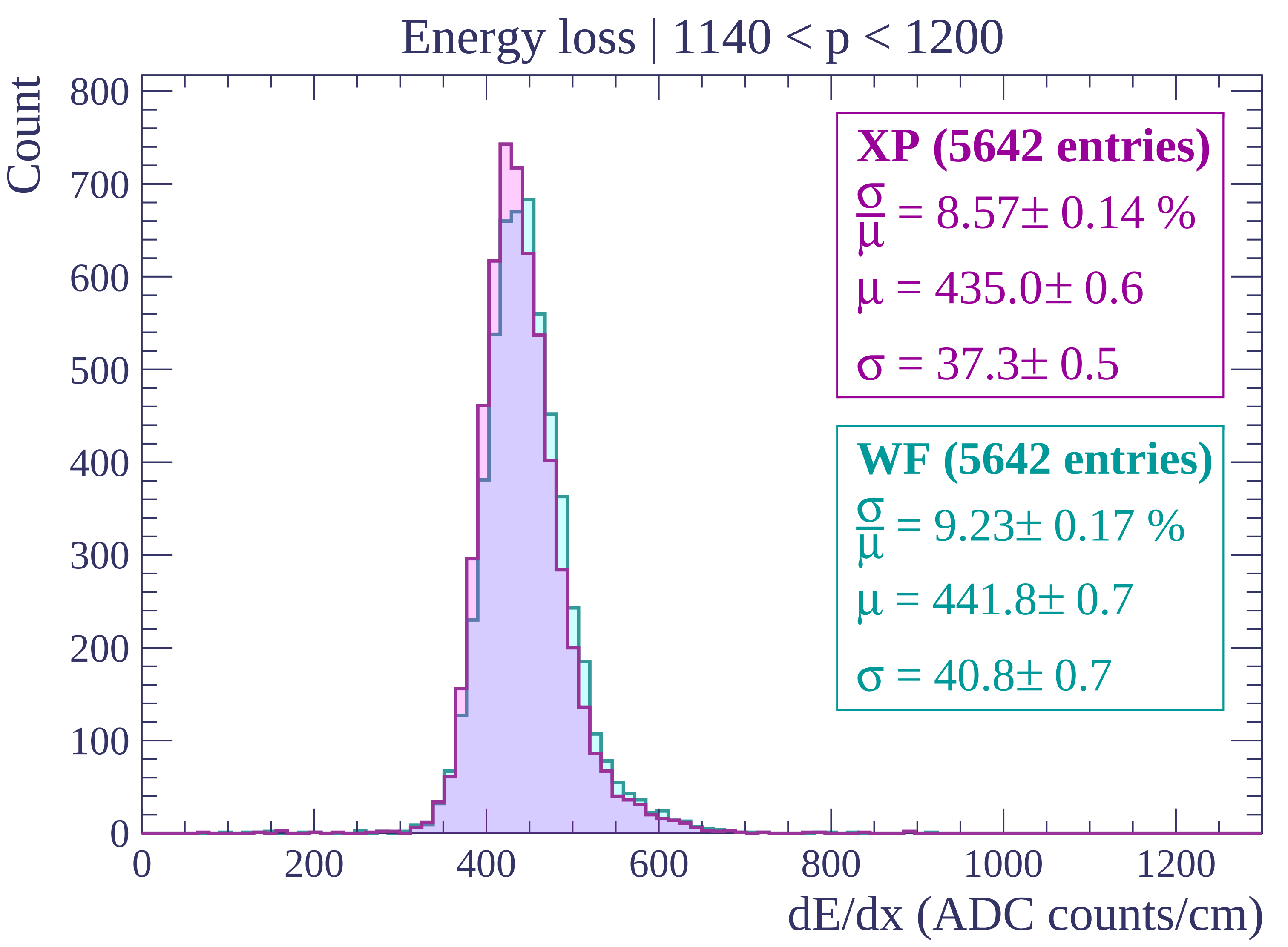
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (6022 entries)
σ/μ = 8.62± 0.15 %
μ = 424.9± 0.6
σ = 36.6± 0.6
WF (6022 entries)
σ/μ = 9.12± 0.15 %
μ = 431.3± 0.6
σ = 39.3± 0.6



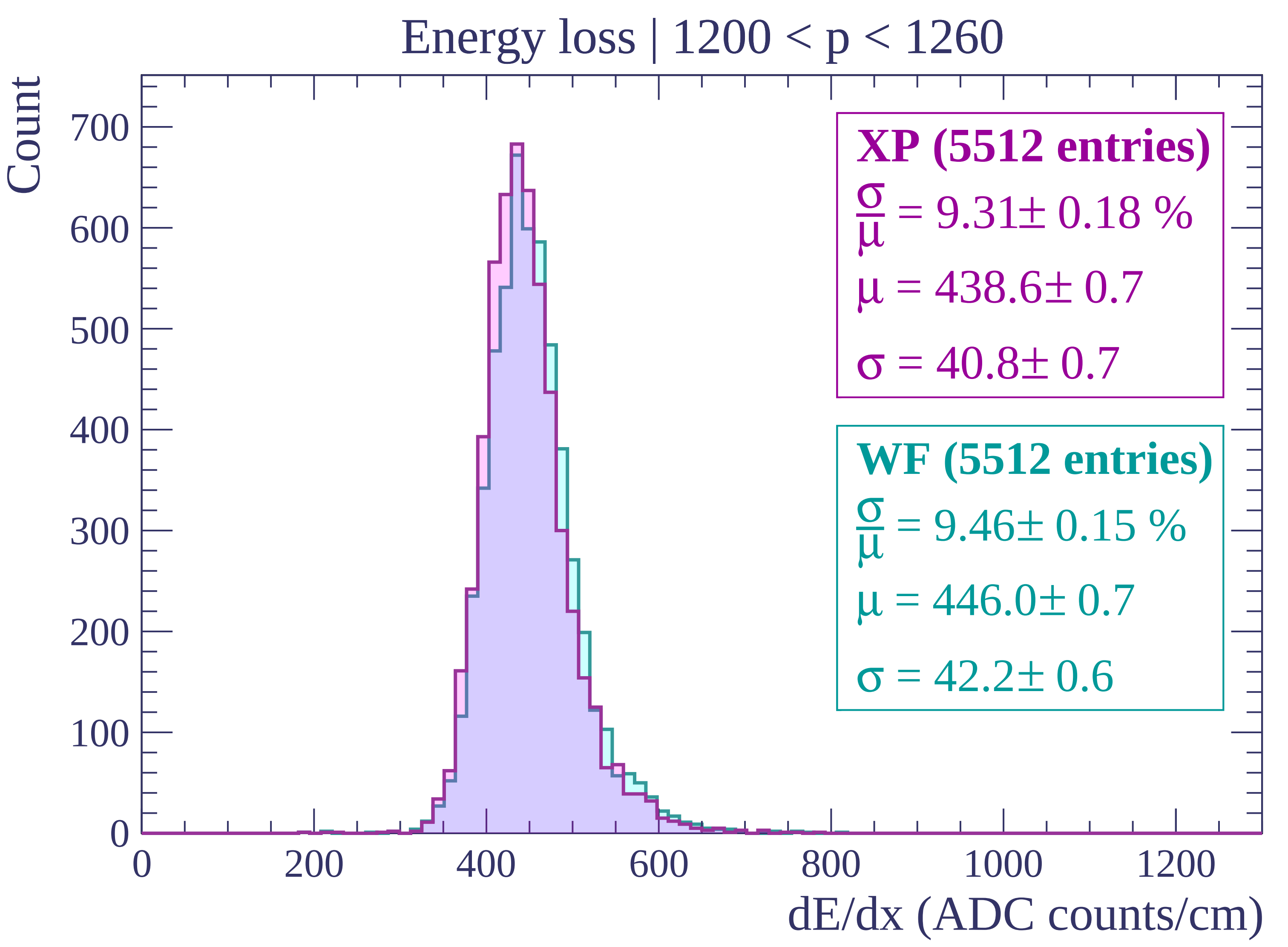
Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5901 entries)
$\frac{\sigma}{\mu}$ = 8.71± 0.14 %
μ = 431.0± 0.6
σ = 37.5± 0.5
WF (5901 entries)
$\frac{\sigma}{\mu}$ = 8.97± 0.15 %
μ = 435.8± 0.6
σ = 39.1± 0.6



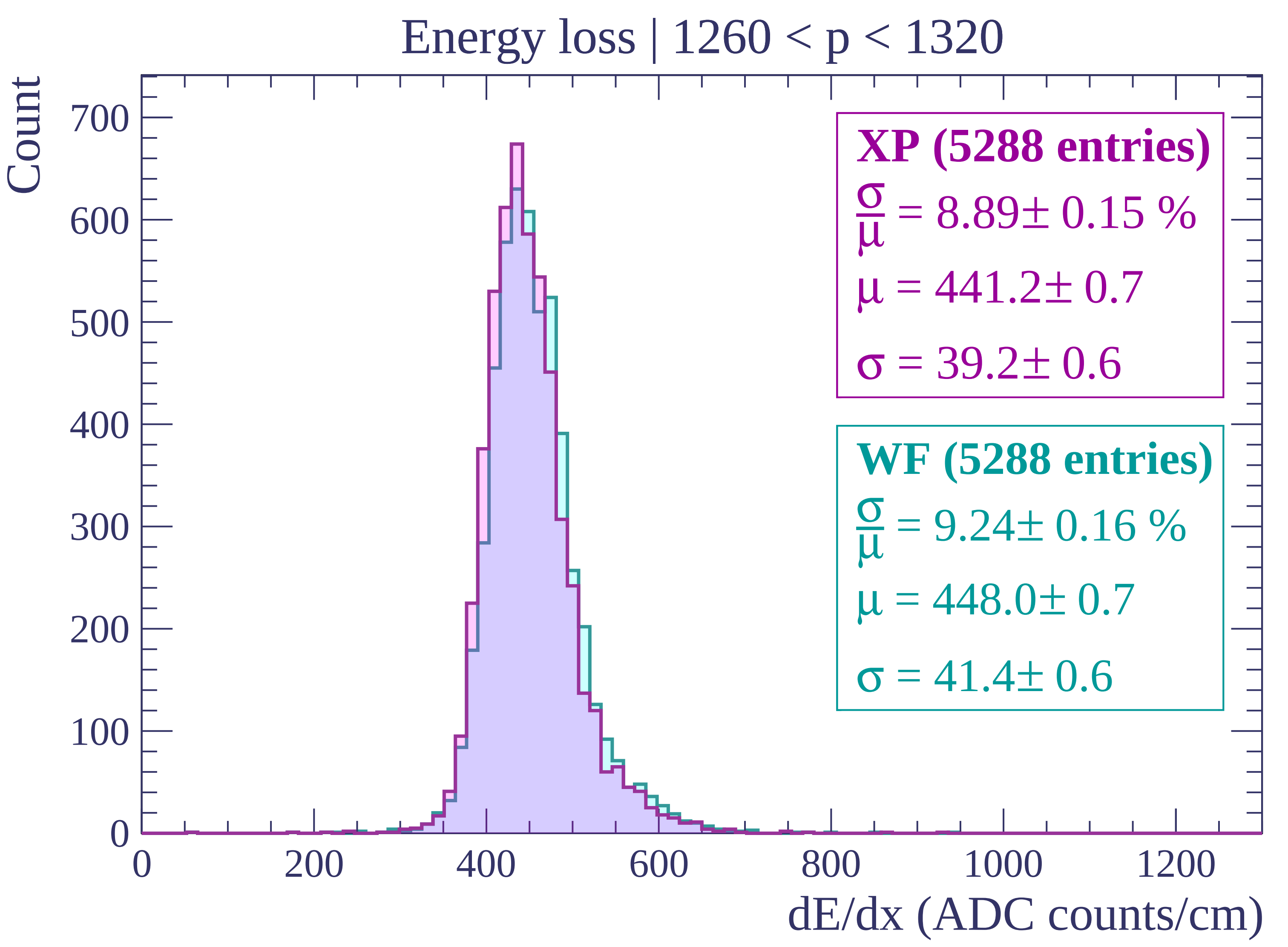
Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5766 entries)
σ/μ = 8.81± 0.15 %
μ = 432.0± 0.6
σ = 38.1± 0.6
WF (5766 entries)
σ/μ = 9.03± 0.14 %
μ = 438.5± 0.6
σ = 39.6± 0.6

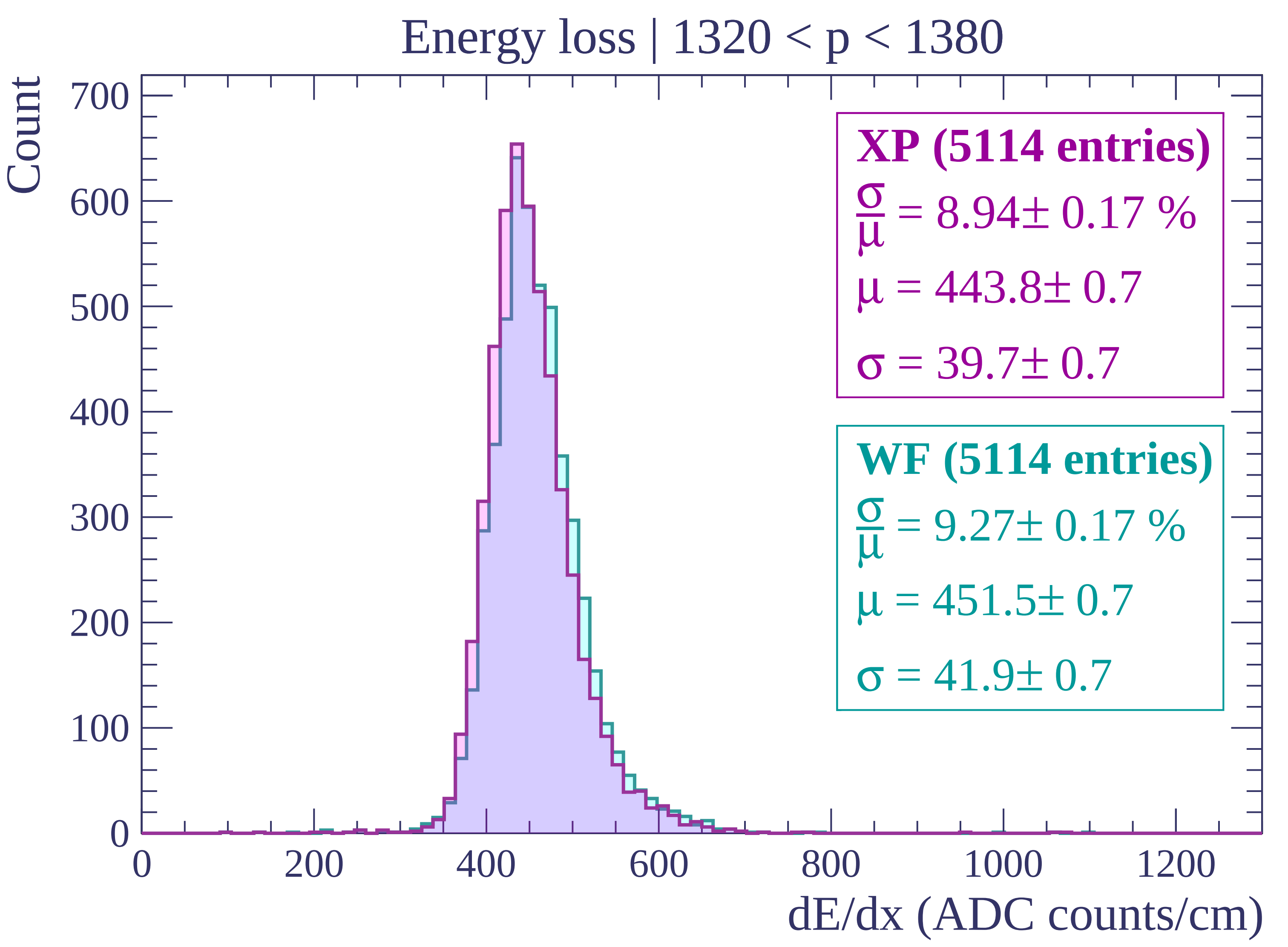


Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (5642 entries)
σ/μ = 8.57± 0.14 %
μ = 435.0± 0.6
σ = 37.3± 0.5
WF (5642 entries)
σ/μ = 9.23± 0.17 %
μ = 441.8± 0.7
σ = 40.8± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

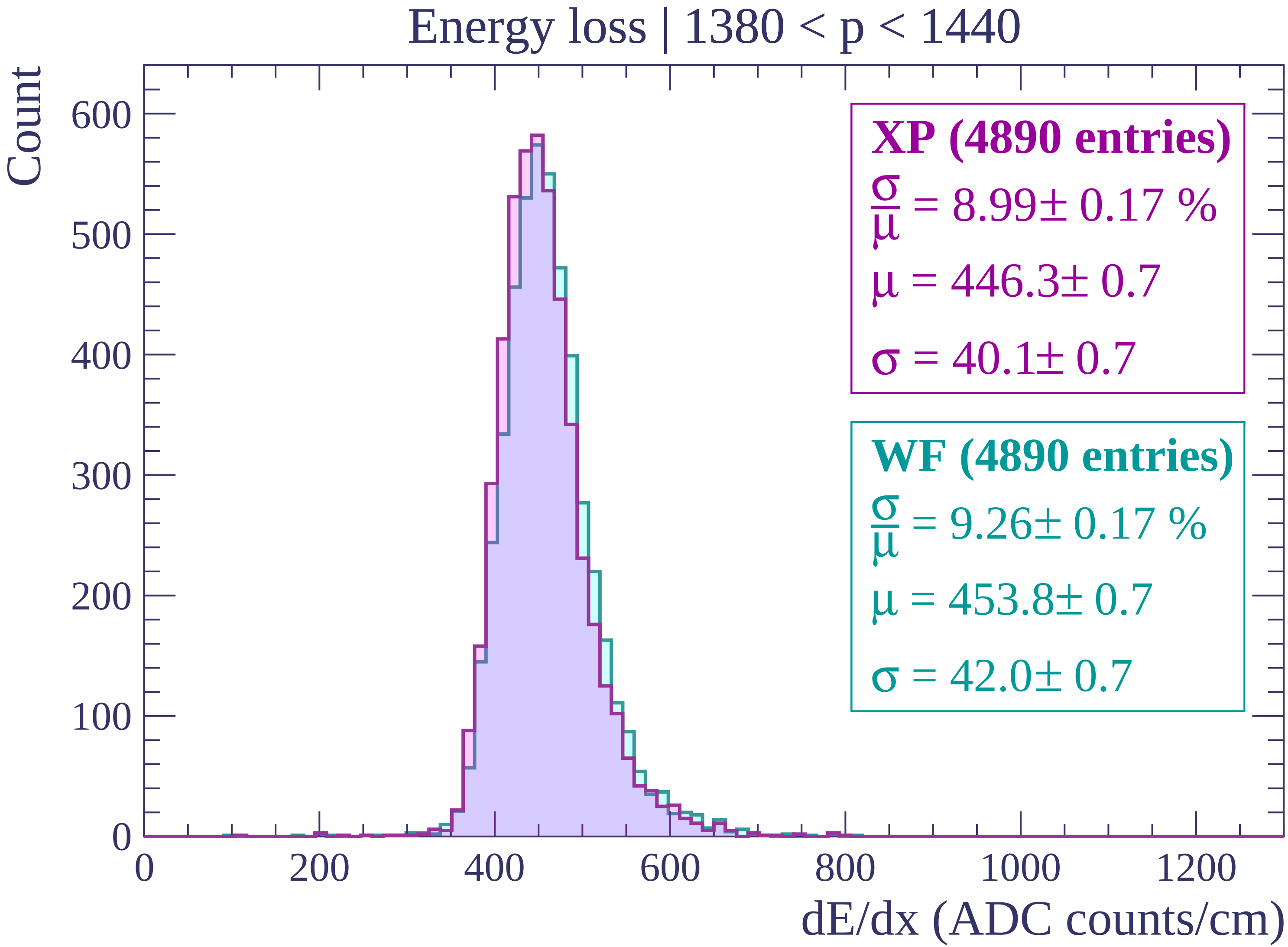


Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
XP (5512 entries)
σ/μ = 9.31± 0.18 %
μ = 438.6± 0.7
σ = 40.8± 0.7
WF (5512 entries)
σ/μ = 9.46± 0.15 %
μ = 446.0± 0.7
σ = 42.2± 0.6
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

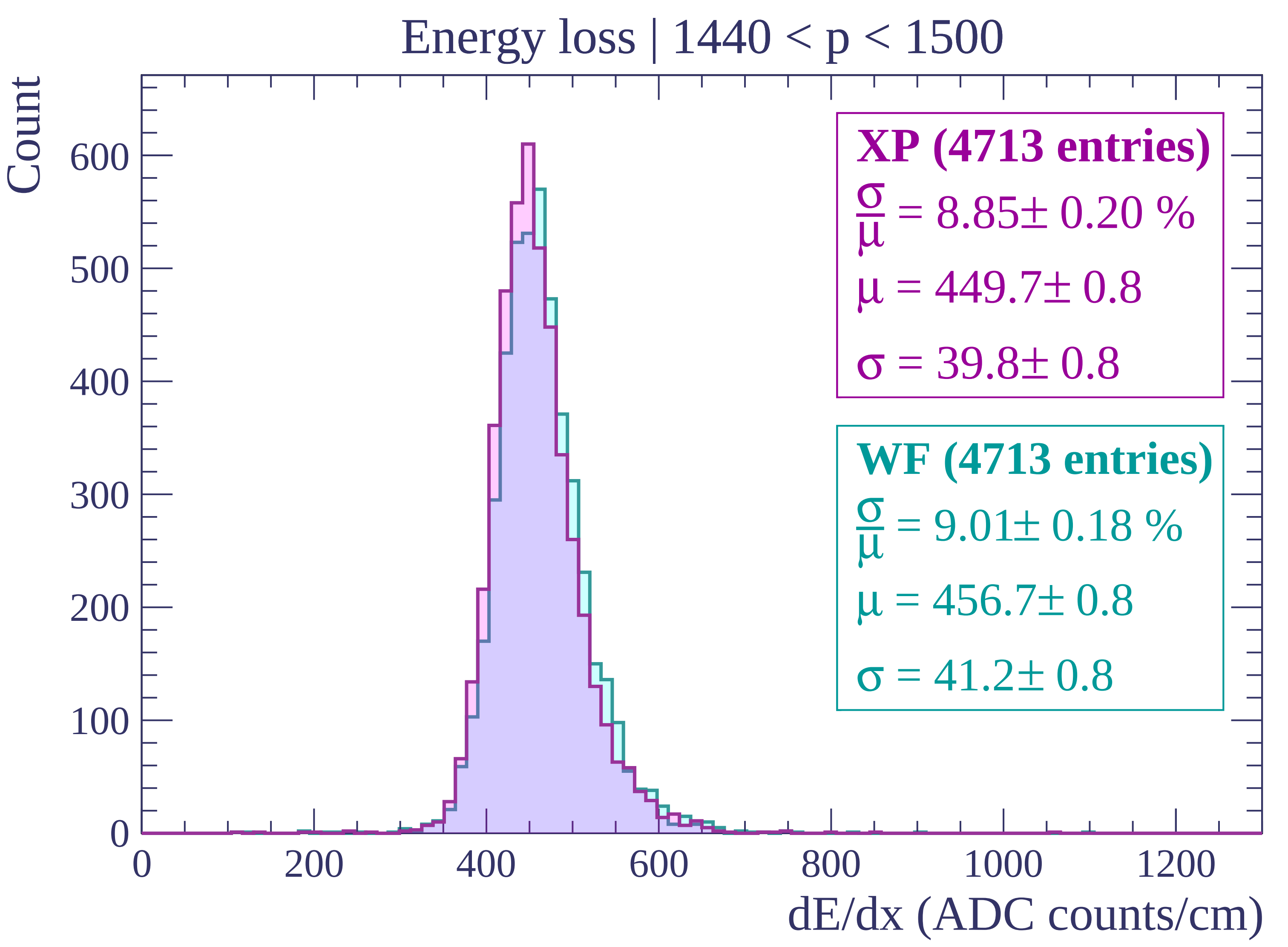

Energy loss | 1260 < p < 1320
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5288 entries)
σ/μ = 8.89± 0.15 %
μ = 441.2± 0.7
σ = 39.2± 0.6
WF (5288 entries)
σ/μ = 9.24± 0.16 %
μ = 448.0± 0.7
σ = 41.4± 0.6



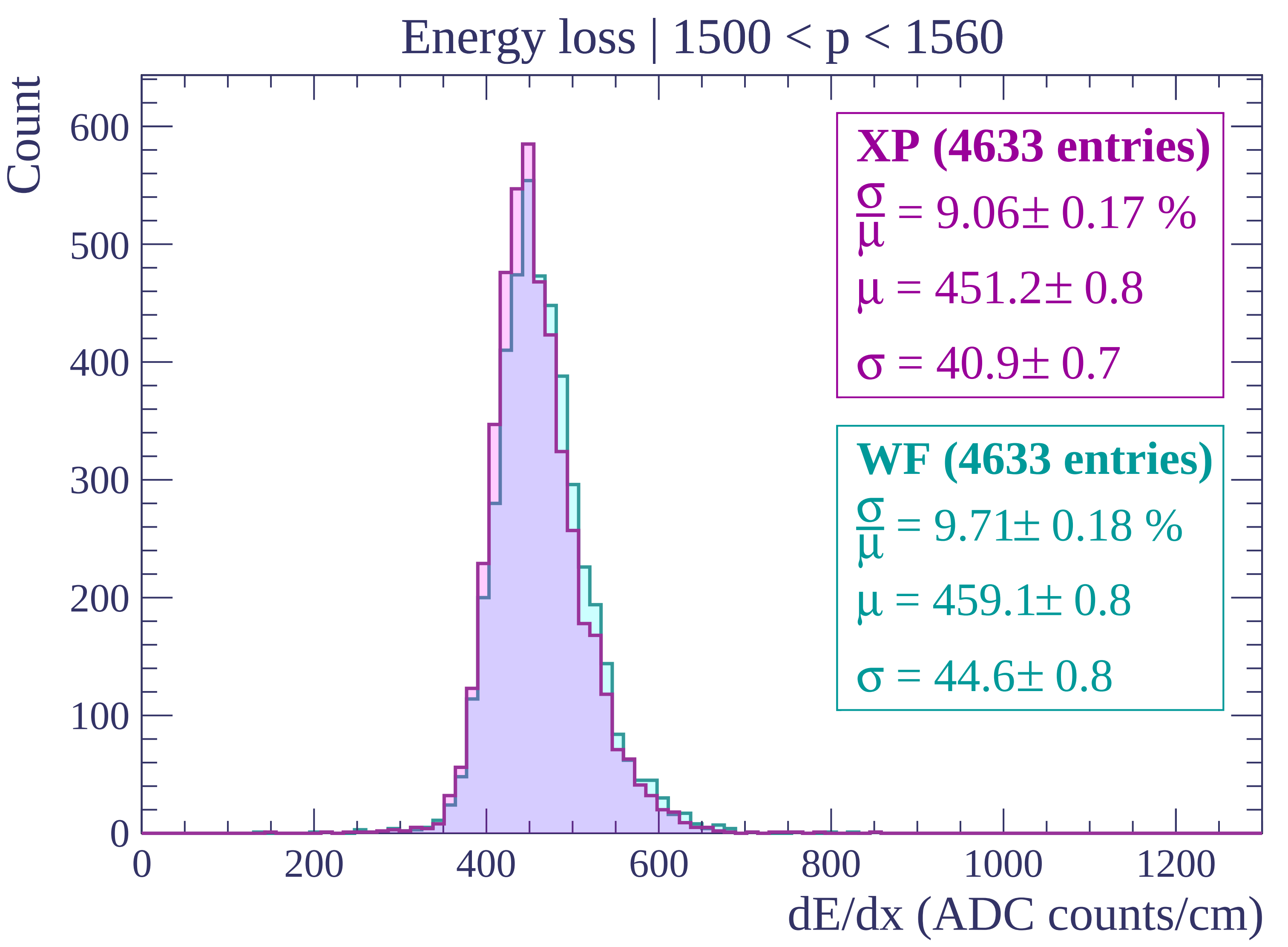
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5114 entries)
$\frac{\sigma}{\mu}$ = 8.94± 0.17 %
μ = 443.8± 0.7
σ = 39.7± 0.7
WF (5114 entries)
$\frac{\sigma}{\mu}$ = 9.27± 0.17 %
μ = 451.5± 0.7
σ = 41.9± 0.7



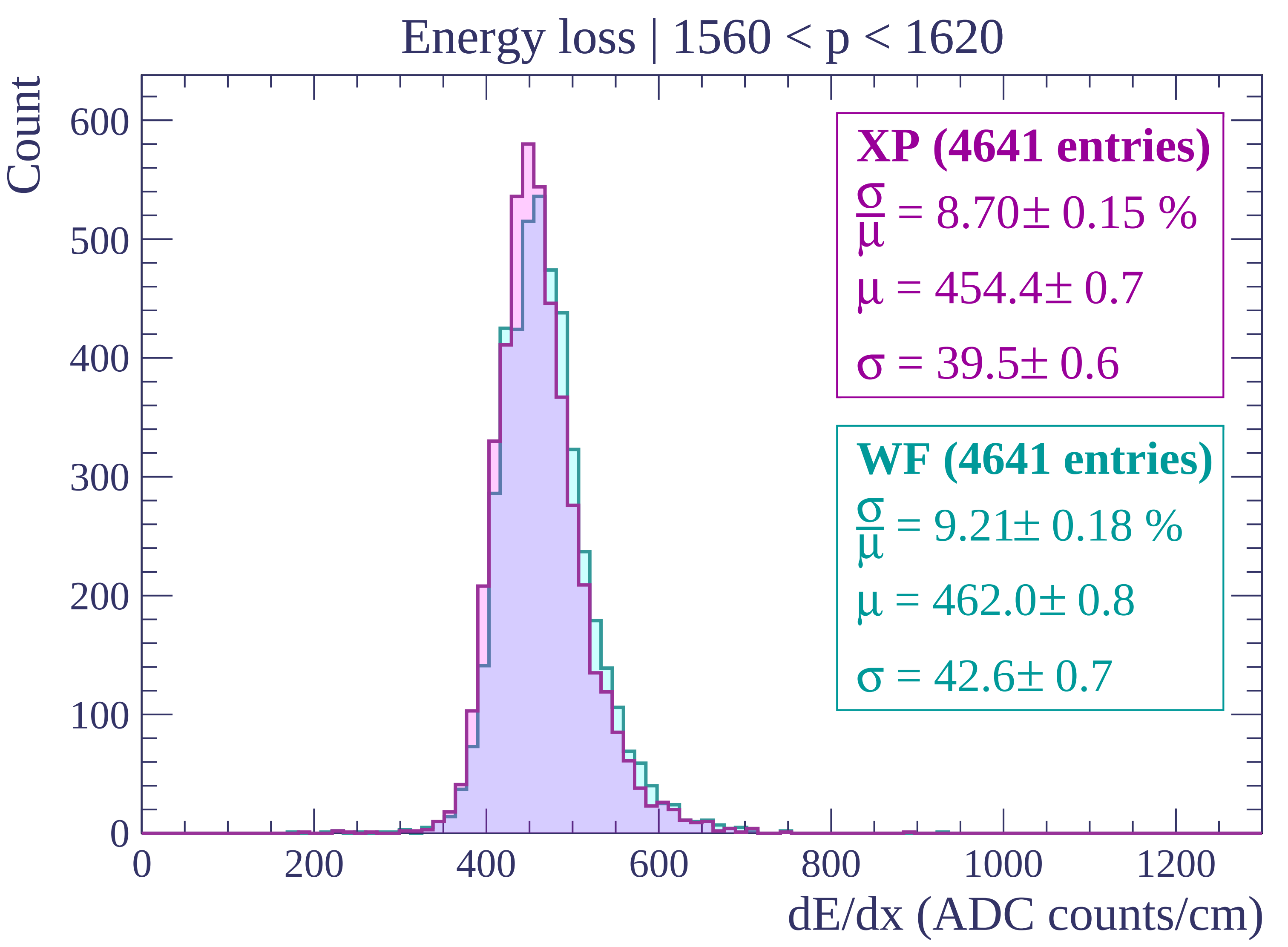
Energy loss | 1380 < p < 1440
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4890 entries)
σ/μ = 8.99± 0.17 %
μ = 446.3± 0.7
σ = 40.1± 0.7
WF (4890 entries)
σ/μ = 9.26± 0.17 %
μ = 453.8± 0.7
σ = 42.0± 0.7



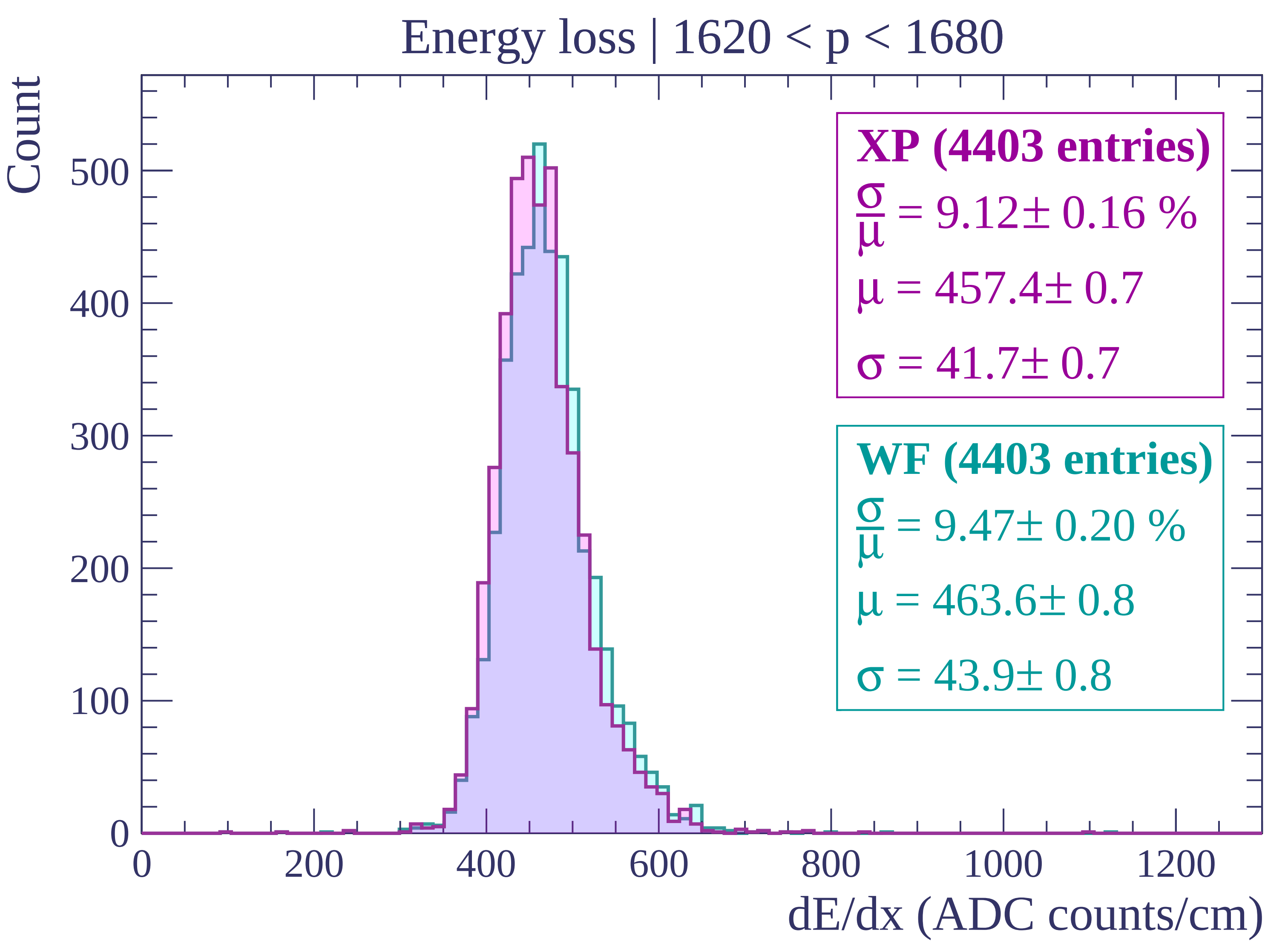
Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4713 entries)
σ/μ = 8.85± 0.20 %
μ = 449.7± 0.8
σ = 39.8± 0.8
WF (4713 entries)
σ/μ = 9.01± 0.18 %
μ = 456.7± 0.8
σ = 41.2± 0.8



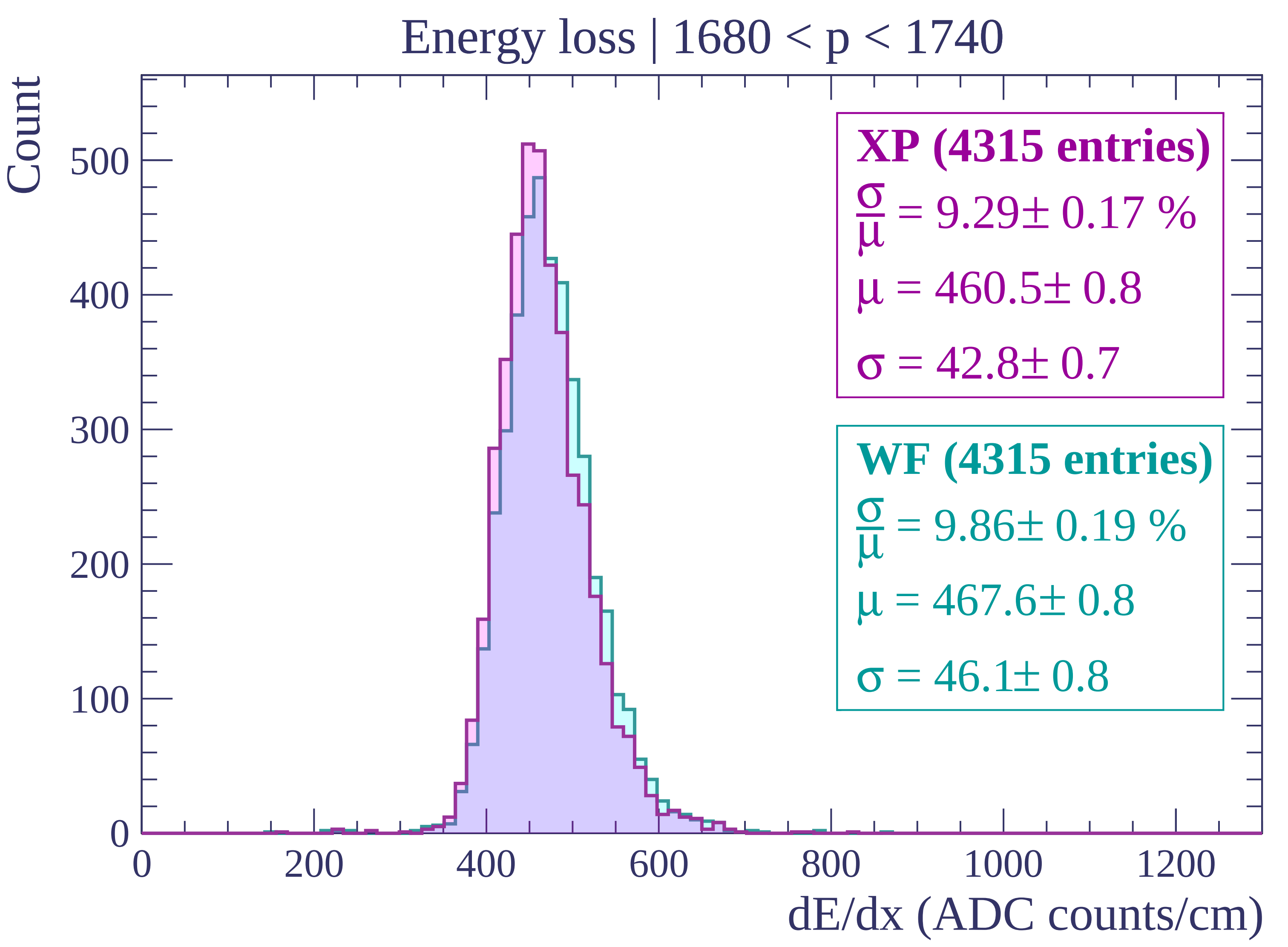
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4633 entries)
σ/μ = 9.06± 0.17 %
μ = 451.2± 0.8
σ = 40.9± 0.7
WF (4633 entries)
σ/μ = 9.71± 0.18 %
μ = 459.1± 0.8
σ = 44.6± 0.8

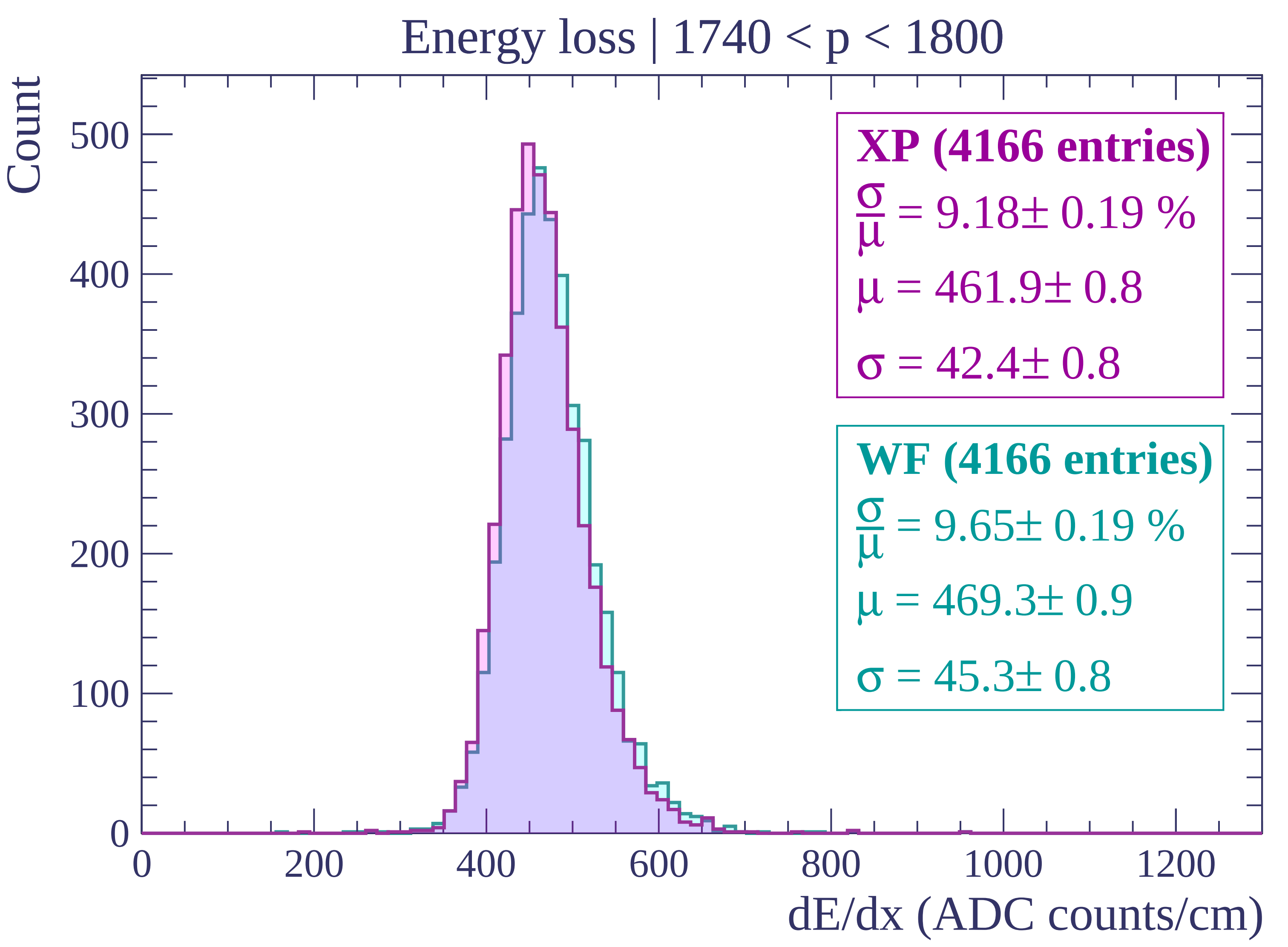


Energy loss | 1560 < p < 1620
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4641 entries)
$\frac{\sigma}{\mu}$ = 8.70± 0.15 %
μ = 454.4± 0.7
σ = 39.5± 0.6
WF (4641 entries)
$\frac{\sigma}{\mu}$ = 9.21± 0.18 %
μ = 462.0± 0.8
σ = 42.6± 0.7

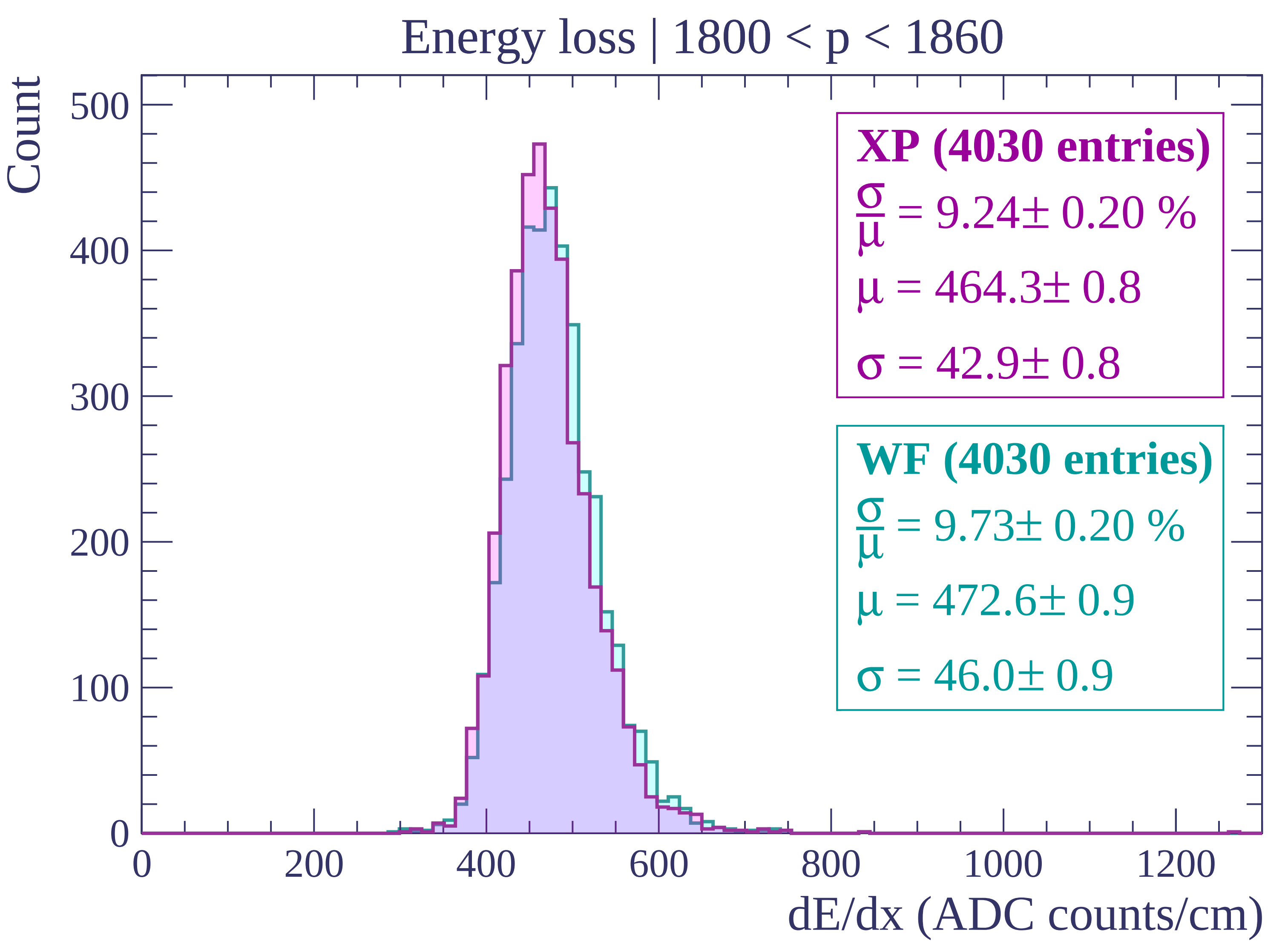


Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4403 entries)
$\frac{\sigma}{\mu}$ = 9.12± 0.16 %
μ = 457.4± 0.7
σ = 41.7± 0.7
WF (4403 entries)
$\frac{\sigma}{\mu}$ = 9.47± 0.20 %
μ = 463.6± 0.8
σ = 43.9± 0.8

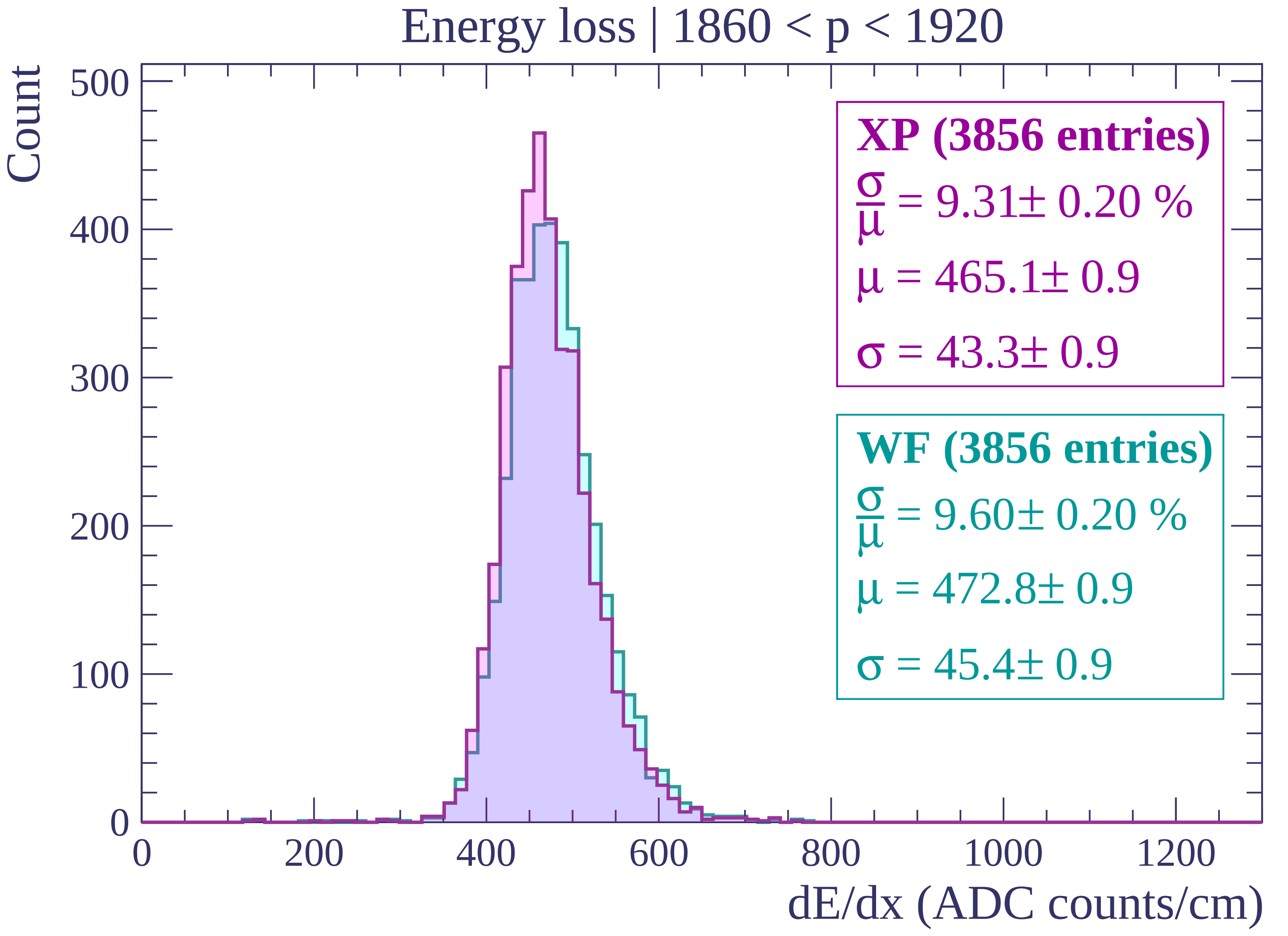

Energy loss | 1680 < p < 1740
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4315 entries)
$\frac{\sigma}{\mu}$ = 9.29± 0.17 %
μ = 460.5± 0.8
σ = 42.8± 0.7
WF (4315 entries)
$\frac{\sigma}{\mu}$ = 9.86± 0.19 %
μ = 467.6± 0.8
σ = 46.1± 0.8



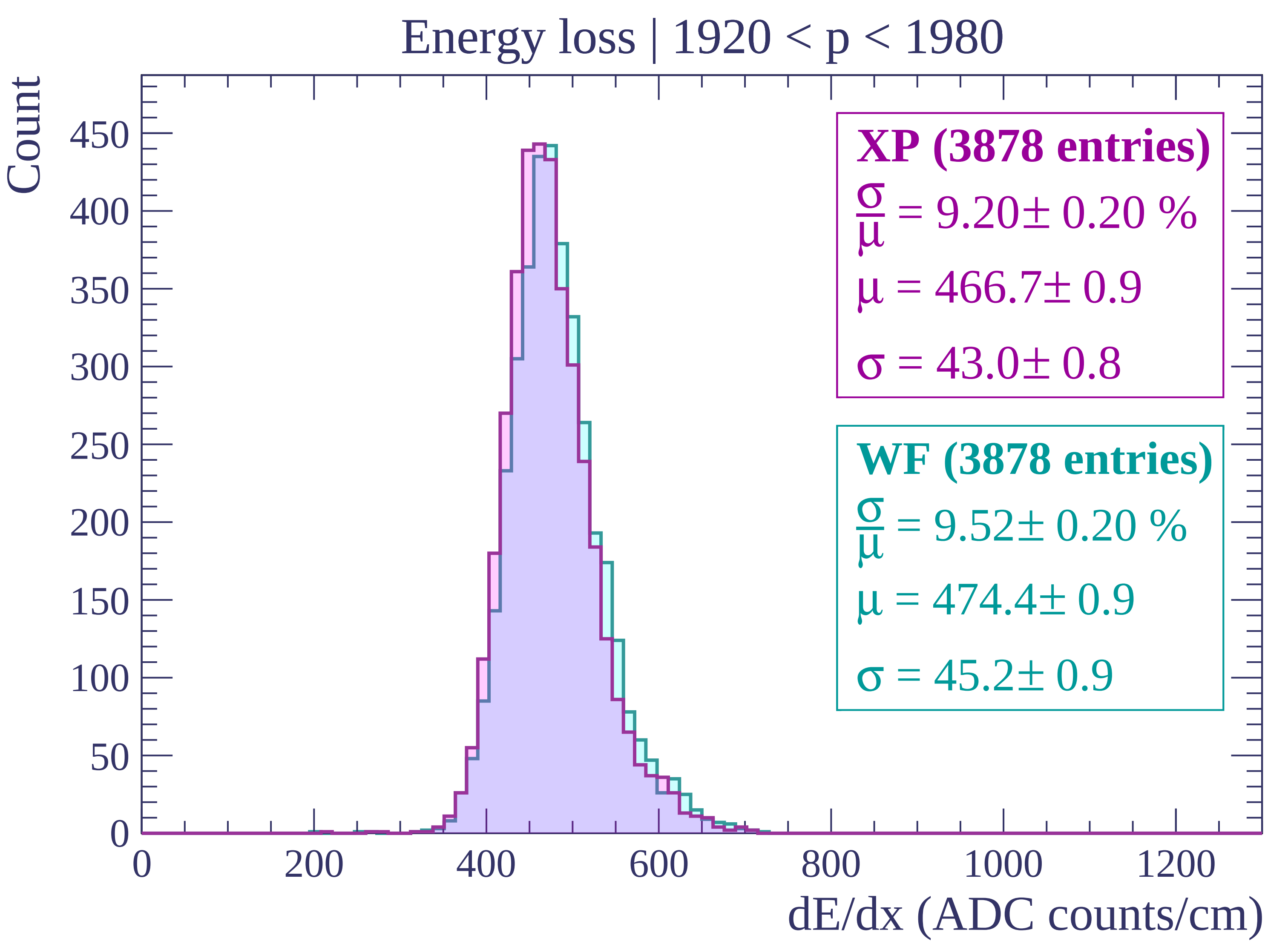
Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4166 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.19 %
μ = 461.9± 0.8
σ = 42.4± 0.8
WF (4166 entries)
$\frac{\sigma}{\mu}$ = 9.65± 0.19 %
μ = 469.3± 0.9
σ = 45.3± 0.8



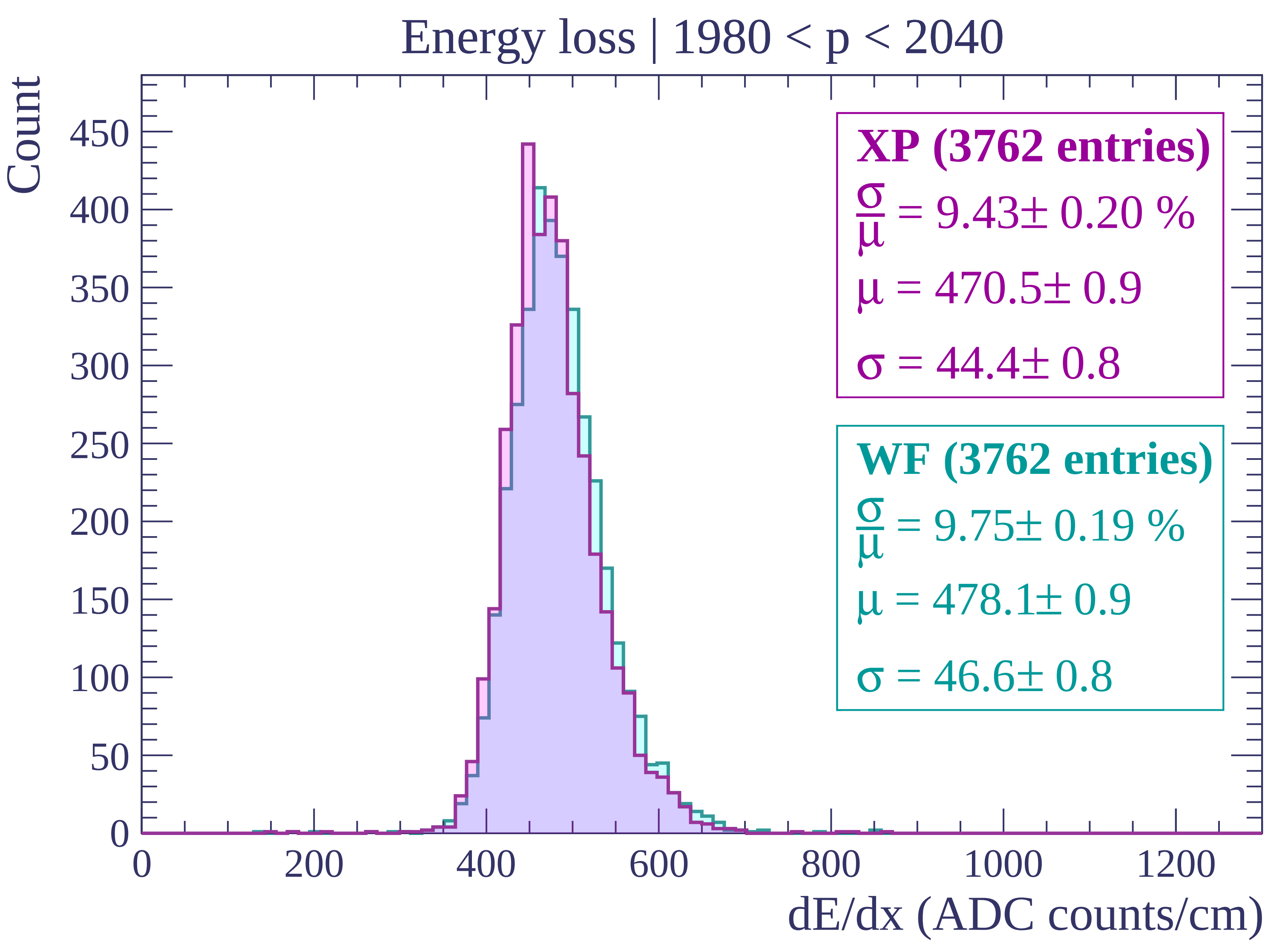
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
100
200
300
400
500
XP (4030 entries)
$\frac{\sigma}{\mu}$ = 9.24± 0.20 %
μ = 464.3± 0.8
σ = 42.9± 0.8
WF (4030 entries)
$\frac{\sigma}{\mu}$ = 9.73± 0.20 %
μ = 472.6± 0.9
σ = 46.0± 0.9



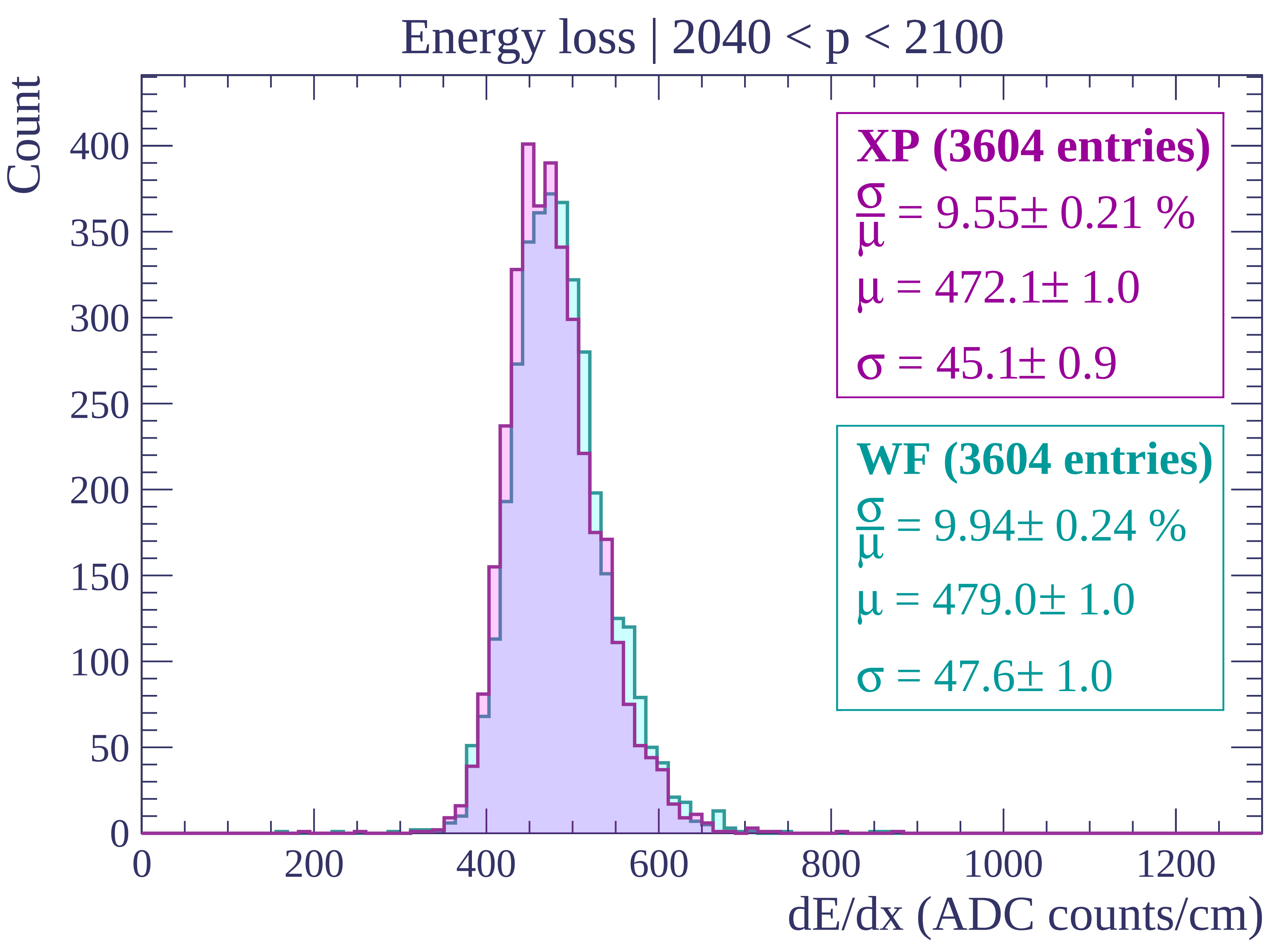
Energy loss | 1860 < p < 1920
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3856 entries)
σ/μ = 9.31± 0.20 %
μ = 465.1± 0.9
σ = 43.3± 0.9
WF (3856 entries)
σ/μ = 9.60± 0.20 %
μ = 472.8± 0.9
σ = 45.4± 0.9

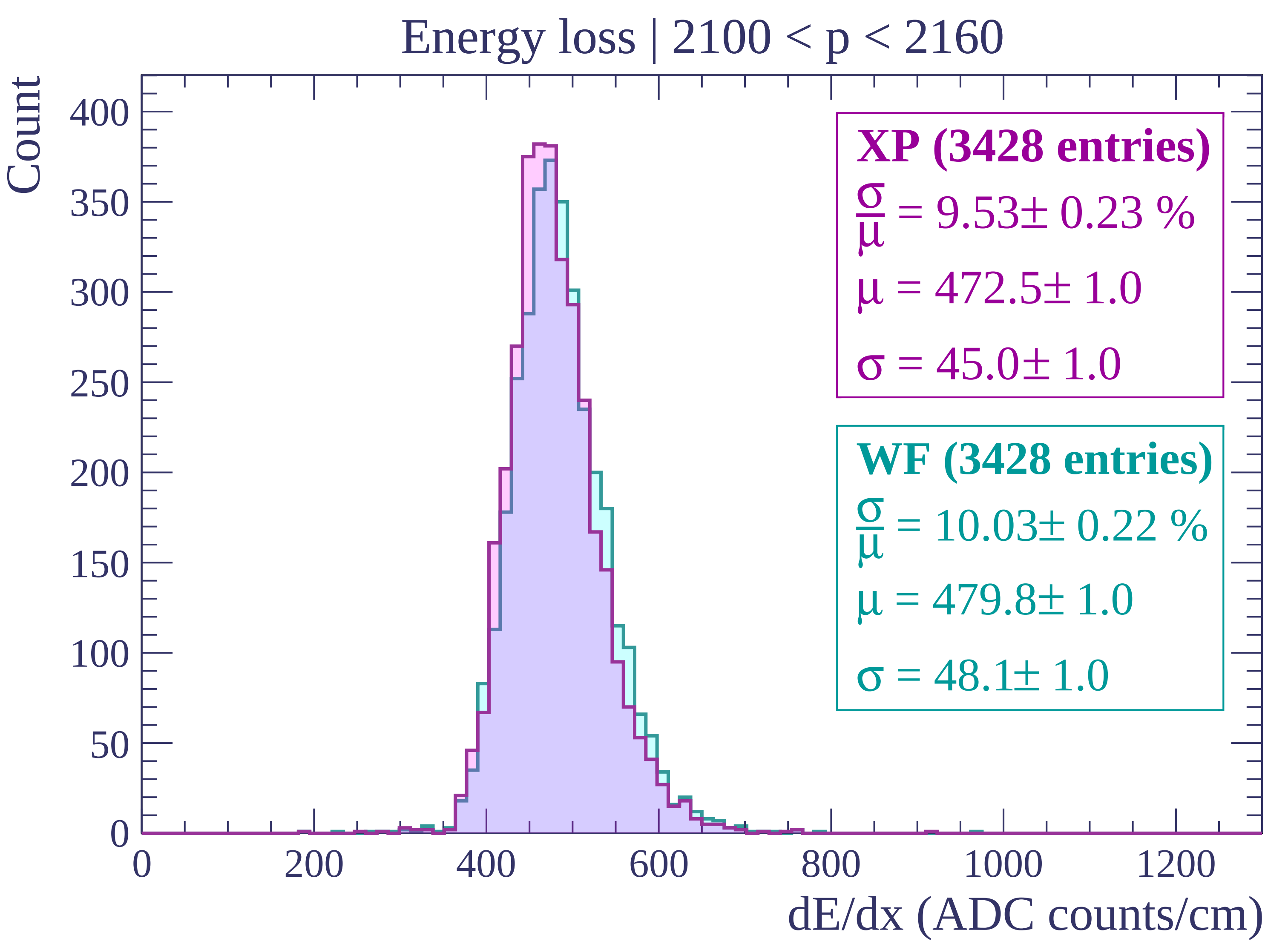


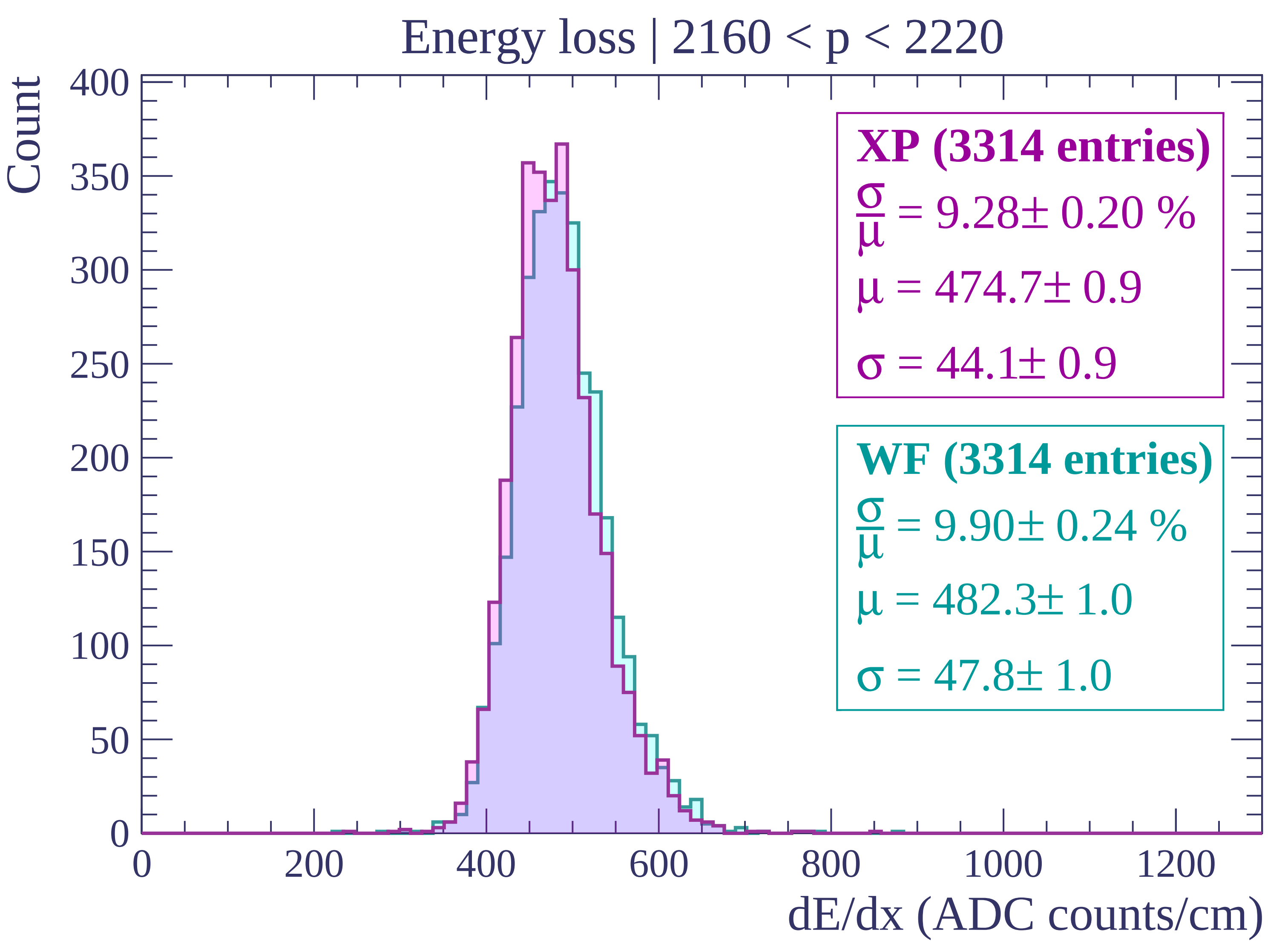
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
XP (3878 entries)
$\frac{\sigma}{\mu}$ = 9.20± 0.20 %
μ = 466.7± 0.9
σ = 43.0± 0.8
WF (3878 entries)
$\frac{\sigma}{\mu}$ = 9.52± 0.20 %
μ = 474.4± 0.9
σ = 45.2± 0.9
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

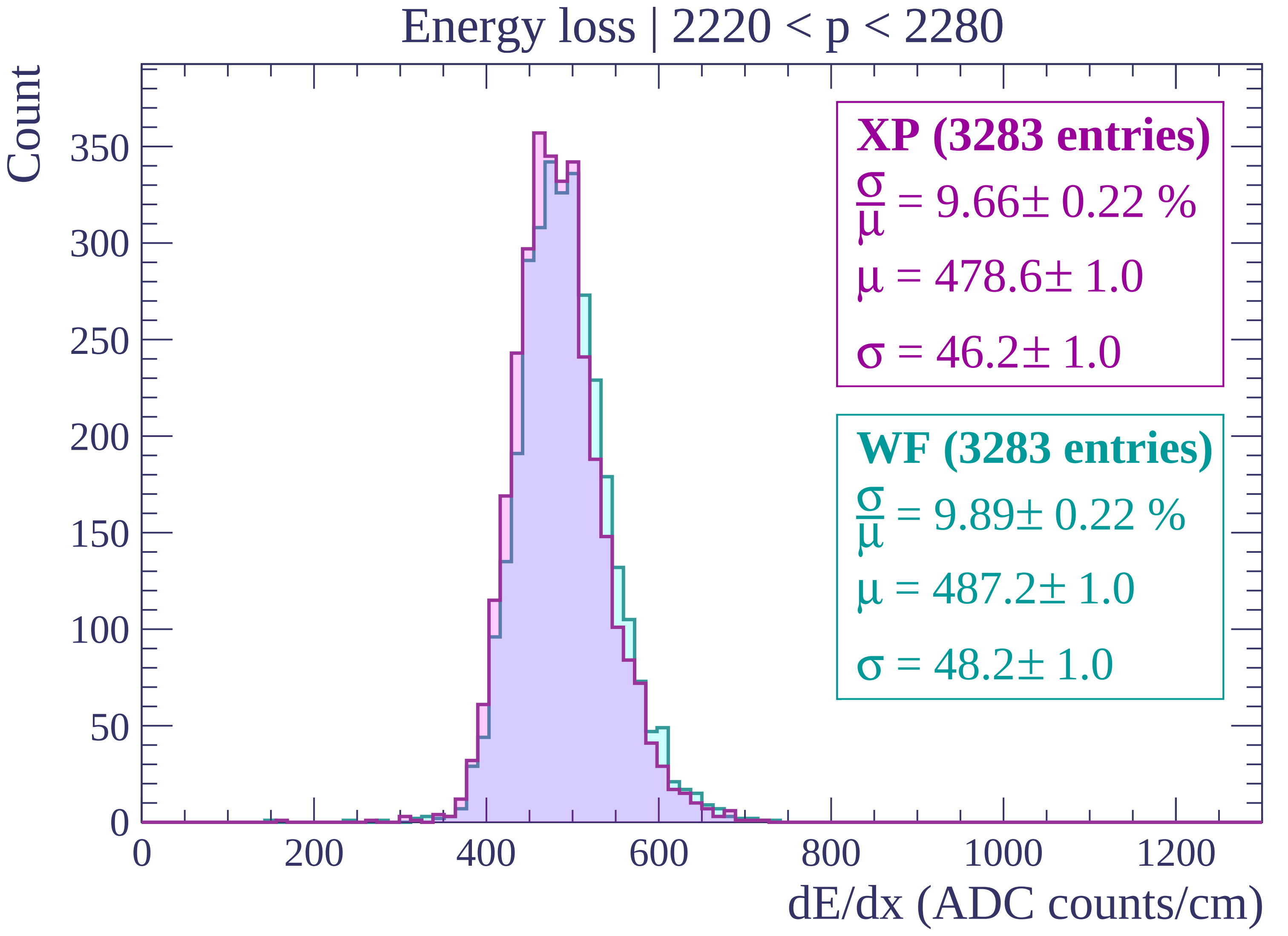


Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
XP (3762 entries)
σ/μ = 9.43± 0.20 %
μ = 470.5± 0.9
σ = 44.4± 0.8
WF (3762 entries)
σ/μ = 9.75± 0.19 %
μ = 478.1± 0.9
σ = 46.6± 0.8
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

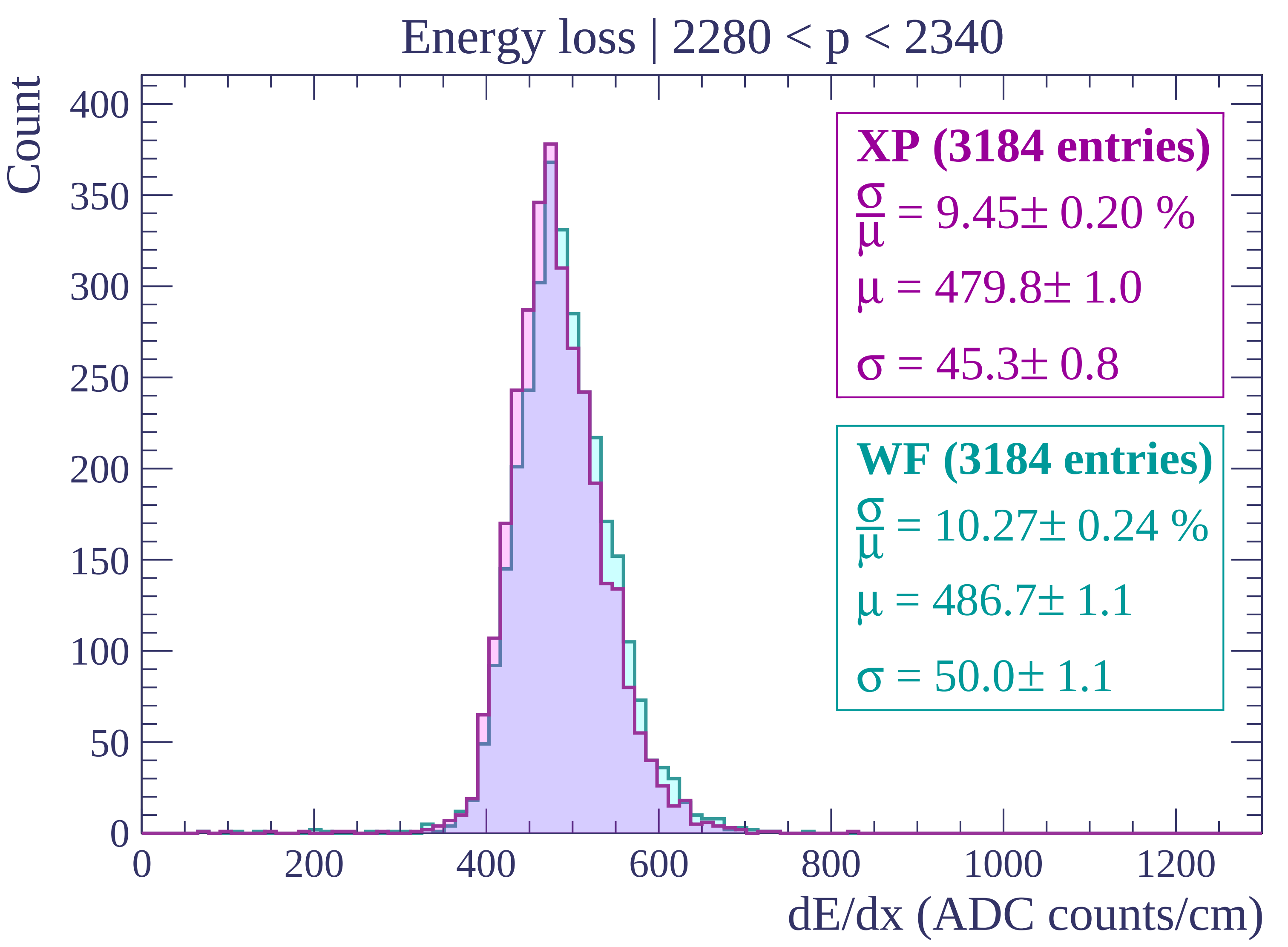

Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3604 entries)
σ/μ = 9.55± 0.21 %
μ = 472.1± 1.0
σ = 45.1± 0.9
WF (3604 entries)
σ/μ = 9.94± 0.24 %
μ = 479.0± 1.0
σ = 47.6± 1.0


Energy loss | 2100 < p < 2160
Count
XP (3428 entries)
σ/μ = 9.53± 0.23 %
μ = 472.5± 1.0
σ = 45.0± 1.0
WF (3428 entries)
σ/μ = 10.03± 0.22 %
μ = 479.8± 1.0
σ = 48.1± 1.0
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

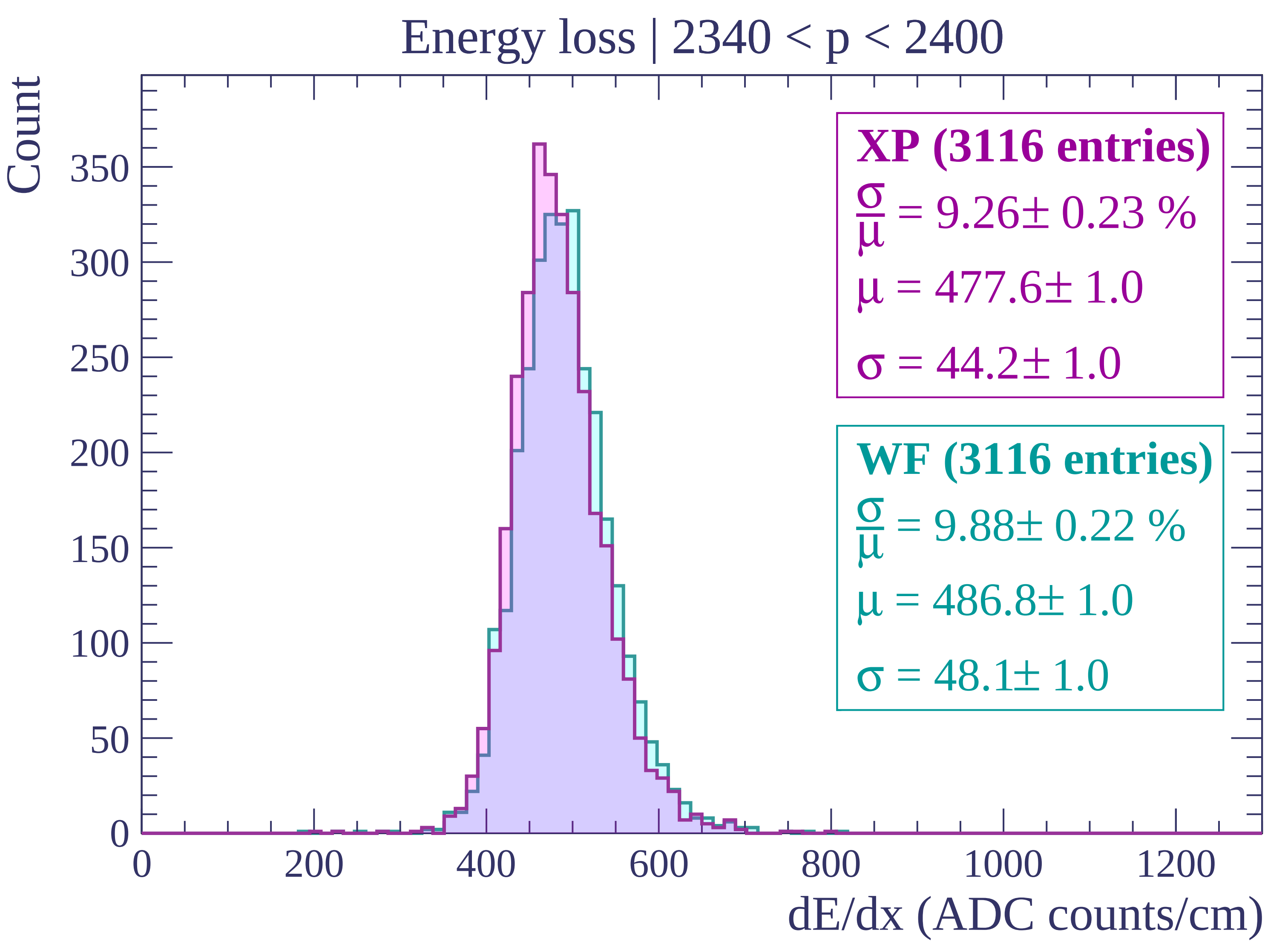

Energy loss | 2160 < p < 2220
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3314 entries)
σ/μ = 9.28± 0.20 %
μ = 474.7± 0.9
σ = 44.1± 0.9
WF (3314 entries)
σ/μ = 9.90± 0.24 %
μ = 482.3± 1.0
σ = 47.8± 1.0



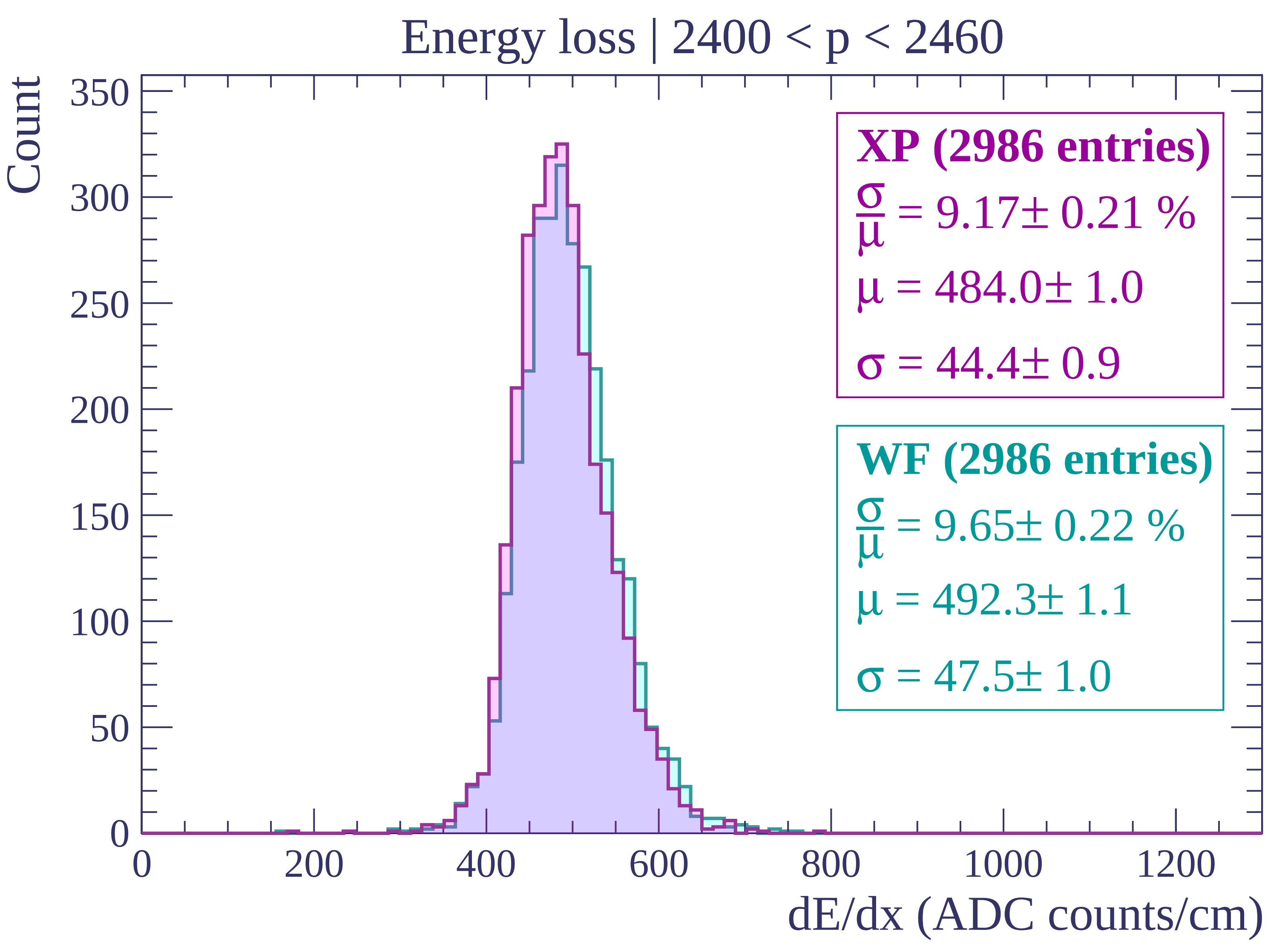
Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3283 entries)
$\frac{\sigma}{\mu}$ = 9.66± 0.22 %
μ = 478.6± 1.0
σ = 46.2± 1.0
WF (3283 entries)
$\frac{\sigma}{\mu}$ = 9.89± 0.22 %
μ = 487.2± 1.0
σ = 48.2± 1.0



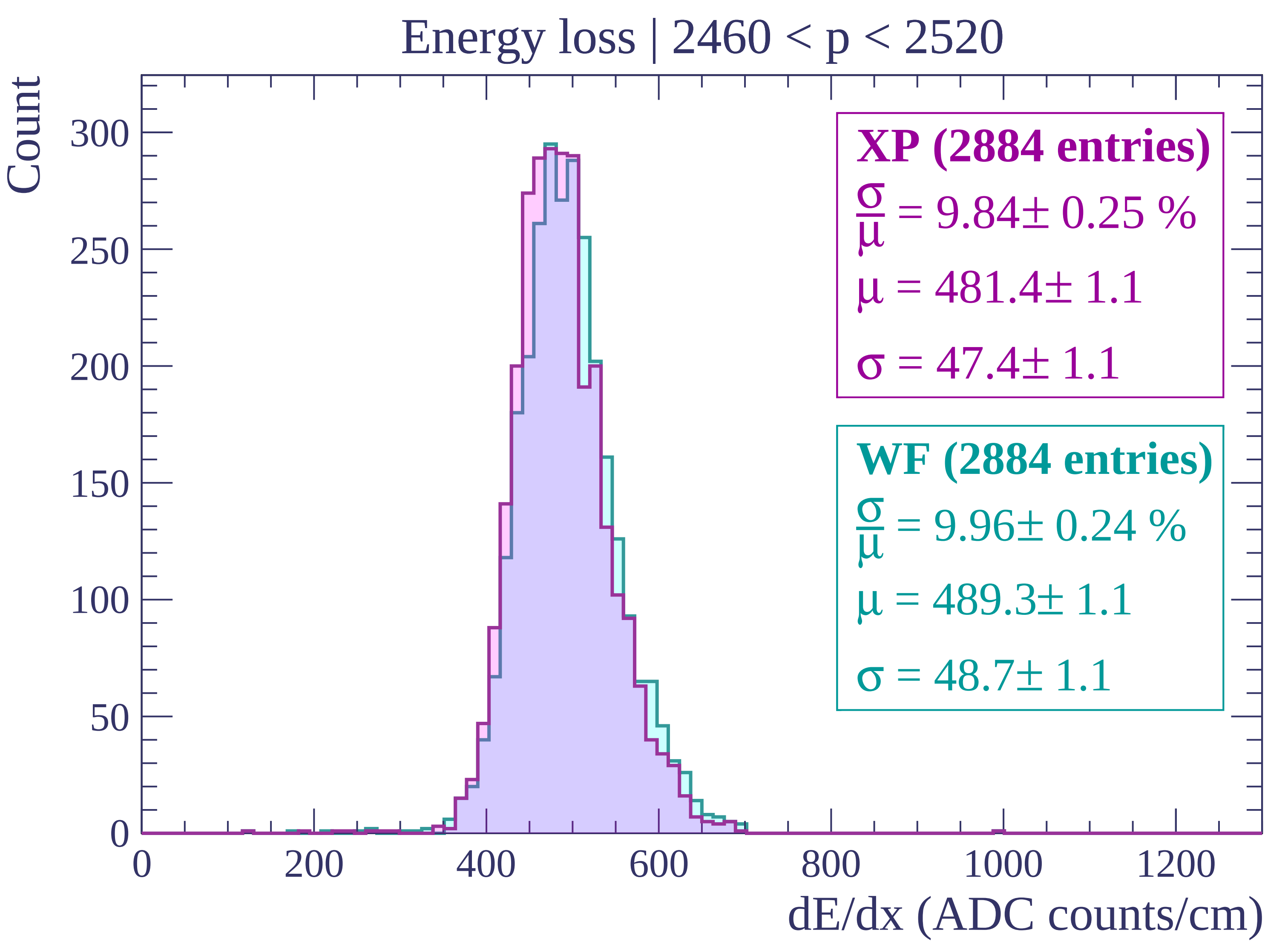
Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3184 entries)
σ/μ = 9.45± 0.20 %
μ = 479.8± 1.0
σ = 45.3± 0.8
WF (3184 entries)
σ/μ = 10.27± 0.24 %
μ = 486.7± 1.1
σ = 50.0± 1.1

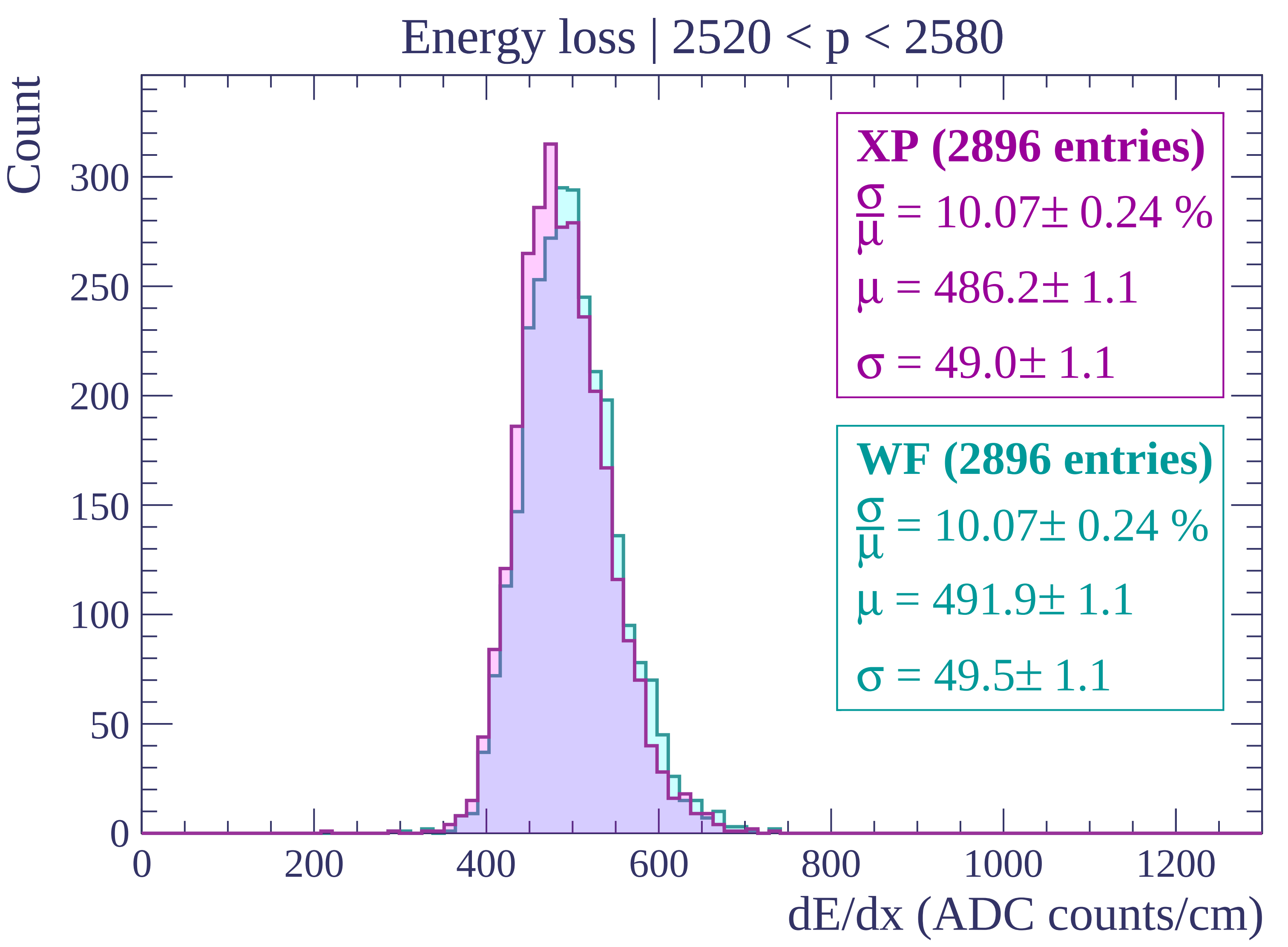


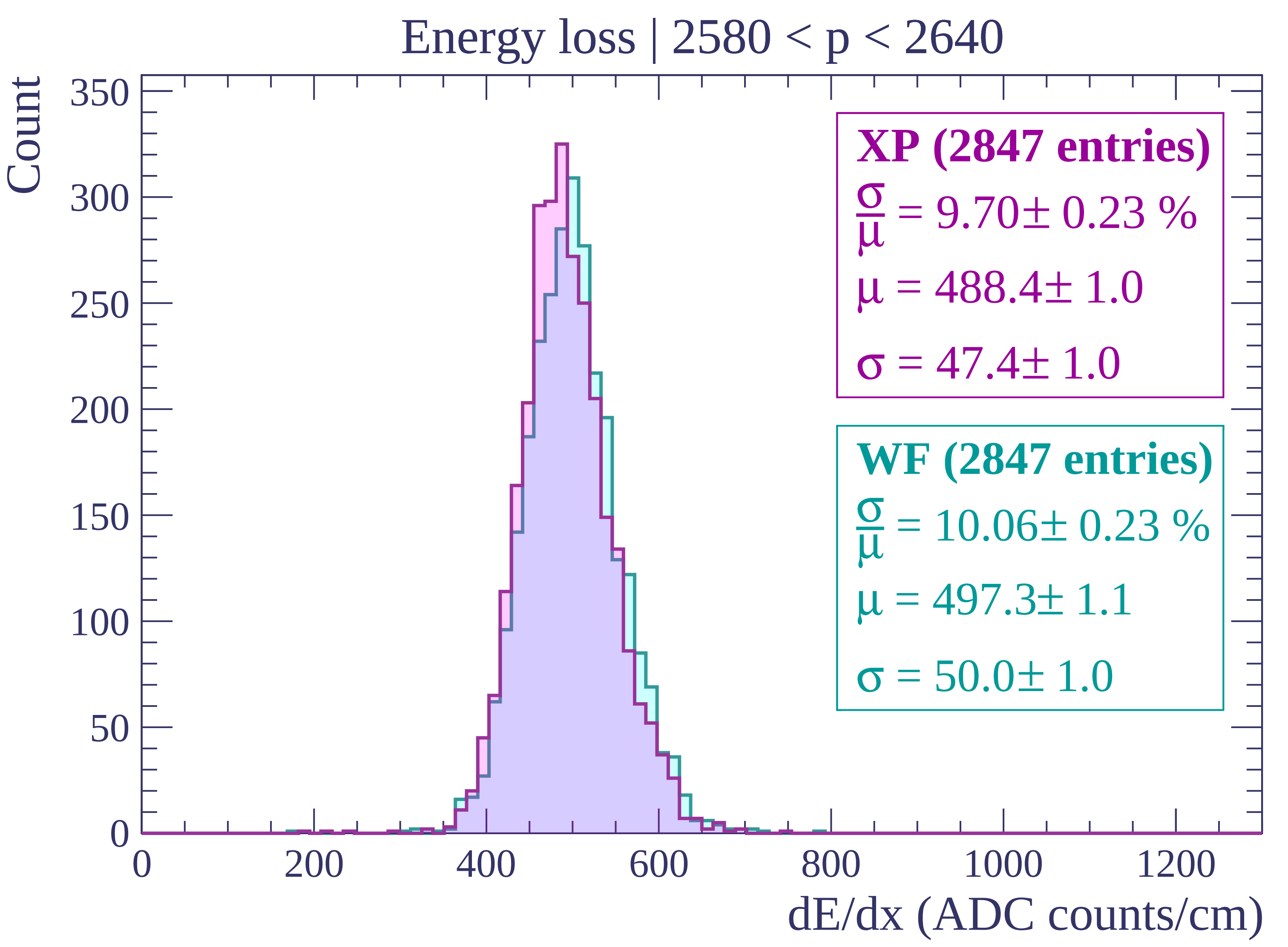
Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3116 entries)
$\frac{\sigma}{\mu}$ = 9.26± 0.23 %
μ = 477.6± 1.0
σ = 44.2± 1.0
WF (3116 entries)
$\frac{\sigma}{\mu}$ = 9.88± 0.22 %
μ = 486.8± 1.0
σ = 48.1± 1.0



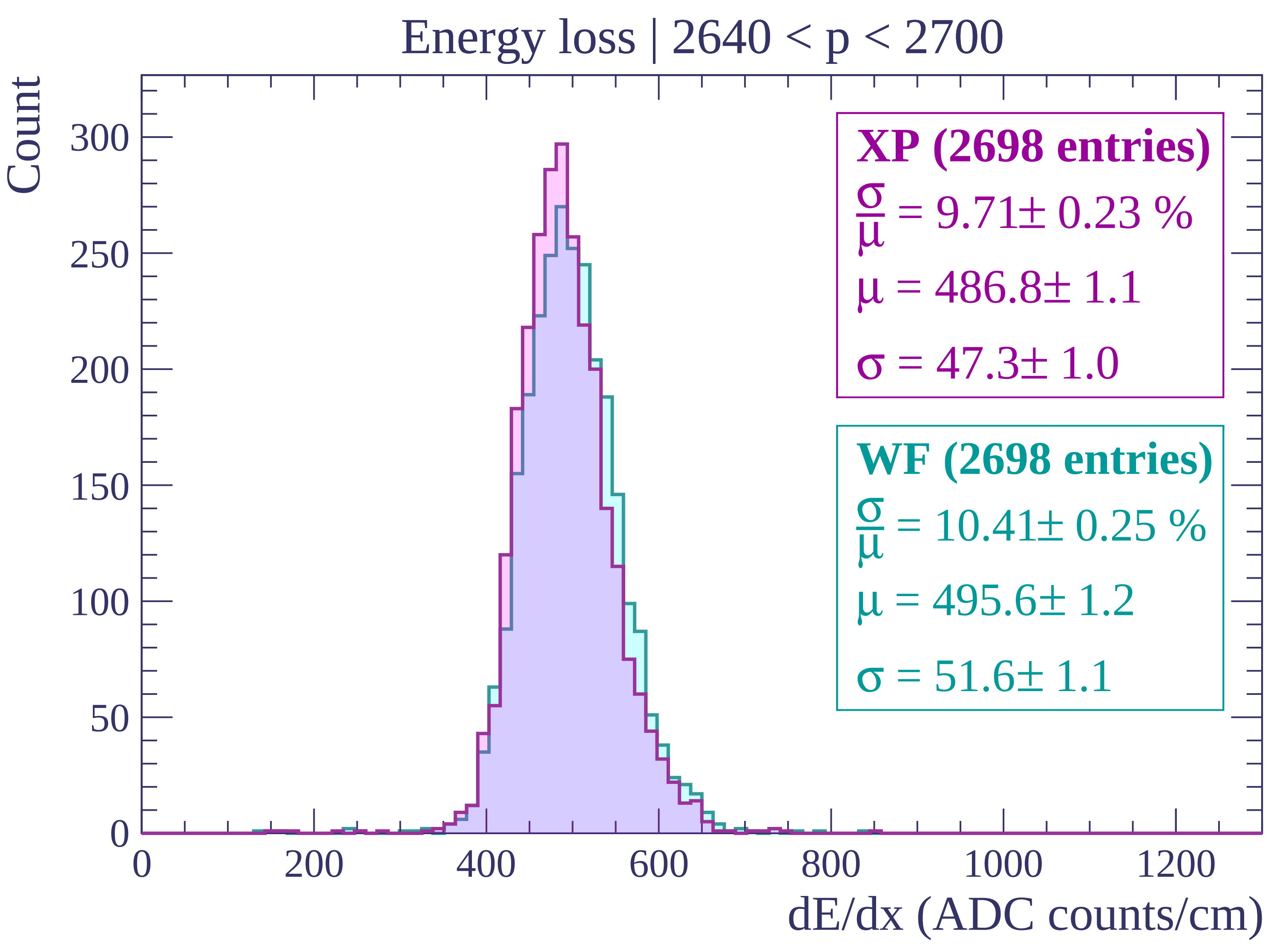
Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2986 entries)
$\frac{\sigma}{\mu}$ = 9.17± 0.21 %
μ = 484.0± 1.0
σ = 44.4± 0.9
WF (2986 entries)
$\frac{\sigma}{\mu}$ = 9.65± 0.22 %
μ = 492.3± 1.1
σ = 47.5± 1.0

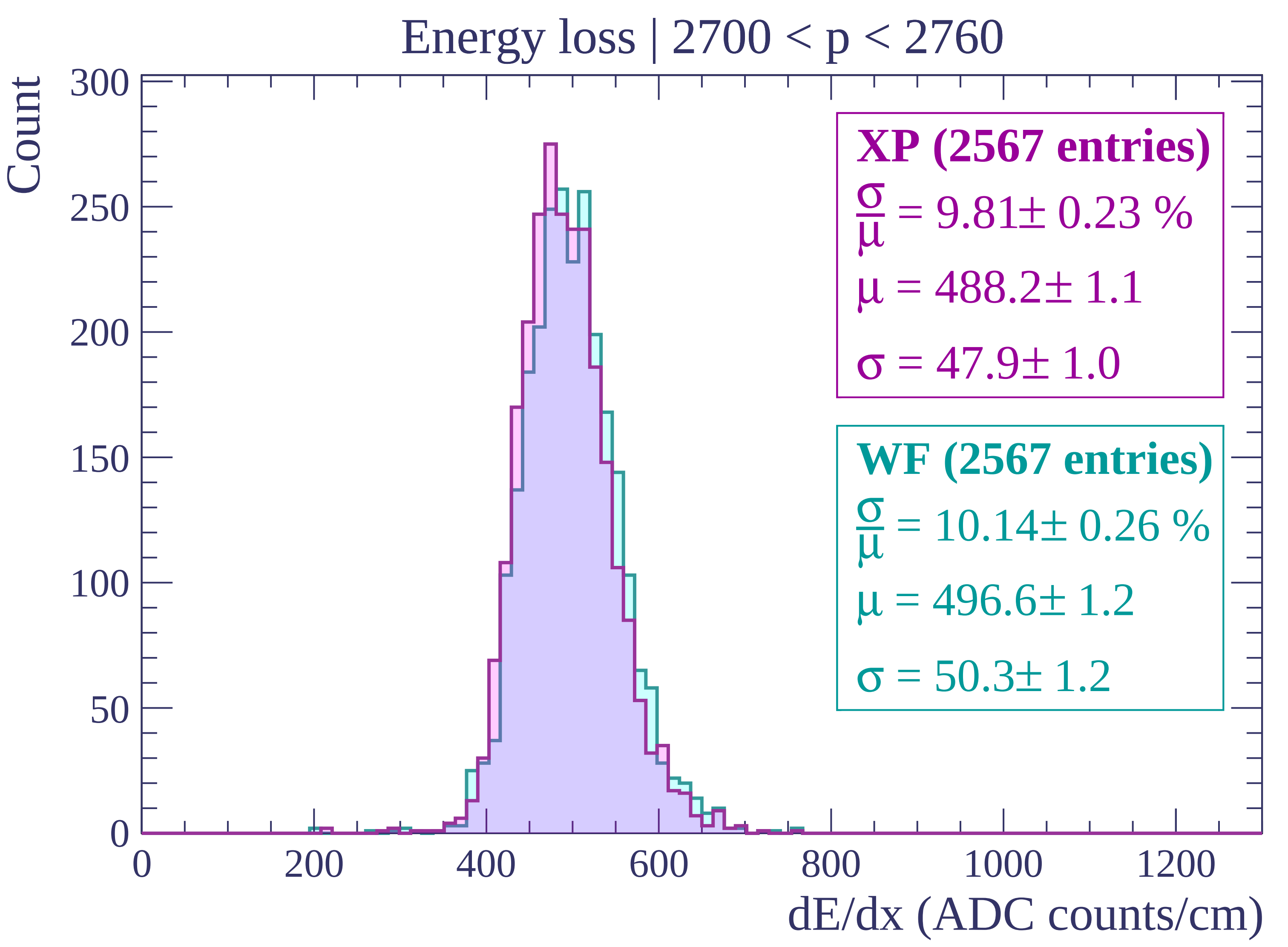

Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
XP (2884 entries)
σ/μ = 9.84± 0.25 %
μ = 481.4± 1.1
σ = 47.4± 1.1
WF (2884 entries)
σ/μ = 9.96± 0.24 %
μ = 489.3± 1.1
σ = 48.7± 1.1
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200


Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2896 entries)
$\frac{\sigma}{\mu}$ = 10.07± 0.24 %
μ = 486.2± 1.1
σ = 49.0± 1.1
WF (2896 entries)
$\frac{\sigma}{\mu}$ = 10.07± 0.24 %
μ = 491.9± 1.1
σ = 49.5± 1.1

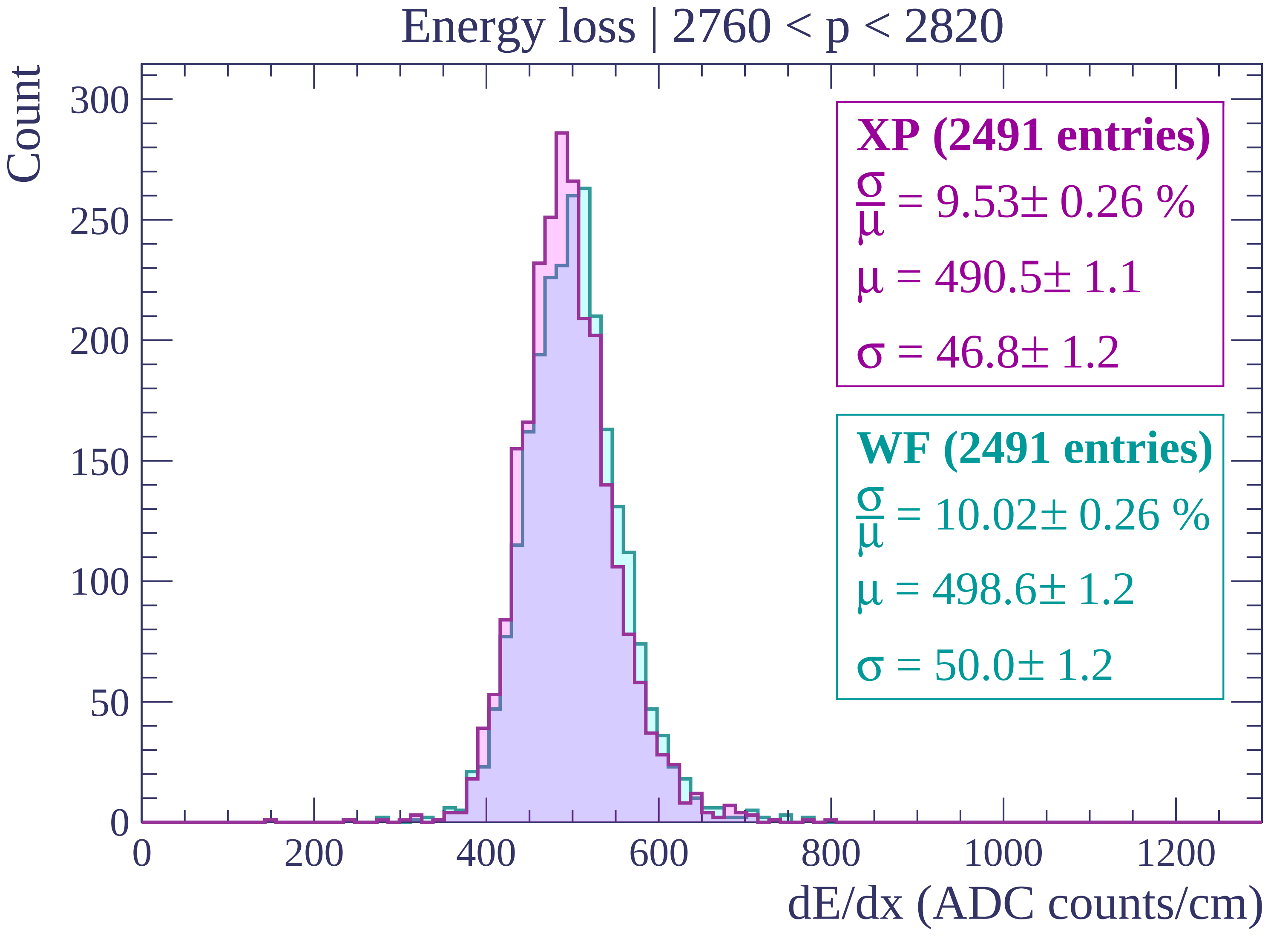


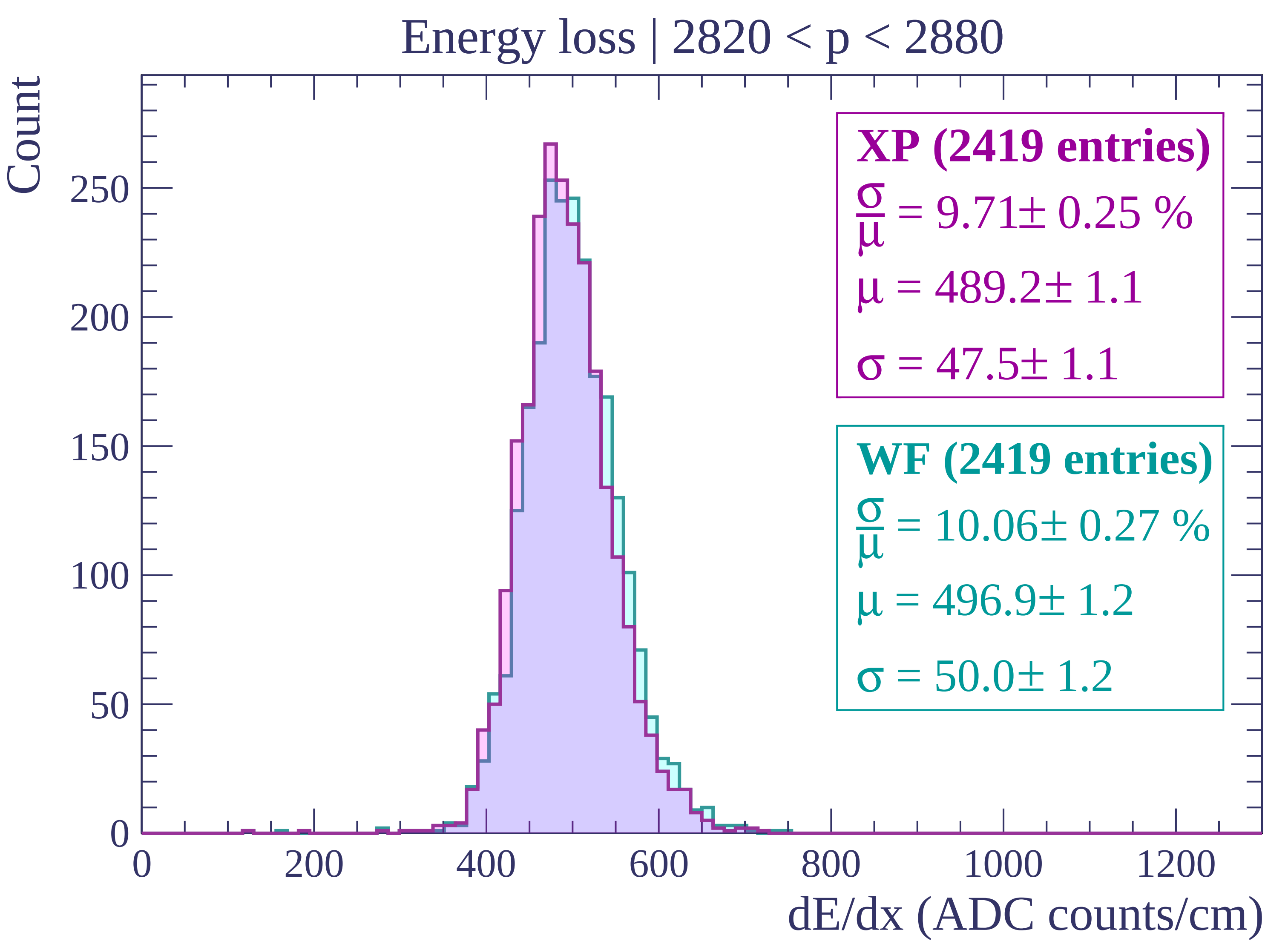
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
XP (2847 entries)
σ/μ = 9.70± 0.23 %
μ = 488.4± 1.0
σ = 47.4± 1.0
WF (2847 entries)
σ/μ = 10.06± 0.23 %
μ = 497.3± 1.1
σ = 50.0± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP (2698 entries)
σ/μ = 9.71± 0.23 %
μ = 486.8± 1.1
σ = 47.3± 1.0
WF (2698 entries)
σ/μ = 10.41± 0.25 %
μ = 495.6± 1.2
σ = 51.6± 1.1

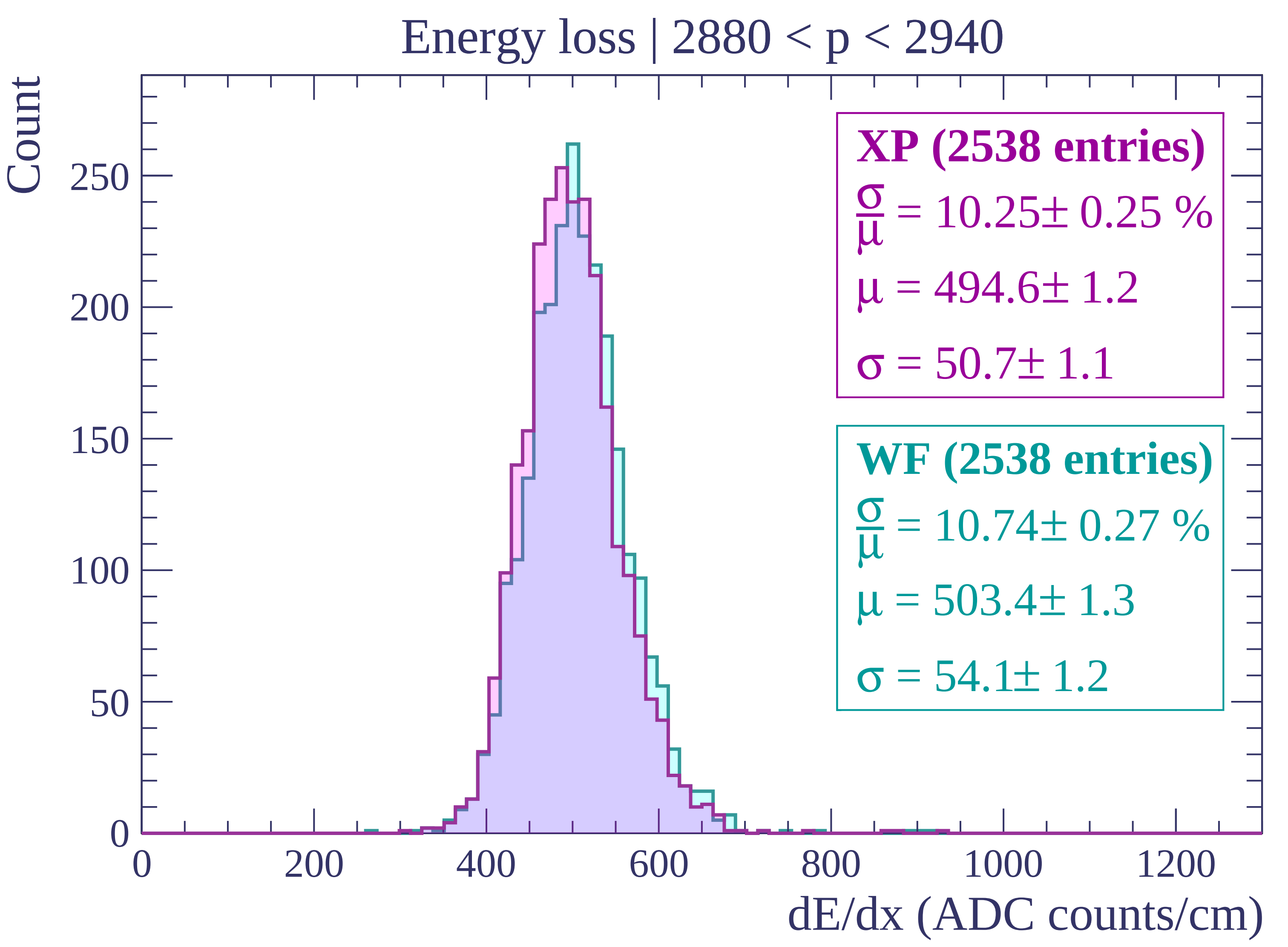


Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2567 entries)
$\frac{\sigma}{\mu}$ = 9.81± 0.23 %
μ = 488.2± 1.1
σ = 47.9± 1.0
WF (2567 entries)
$\frac{\sigma}{\mu}$ = 10.14± 0.26 %
μ = 496.6± 1.2
σ = 50.3± 1.2

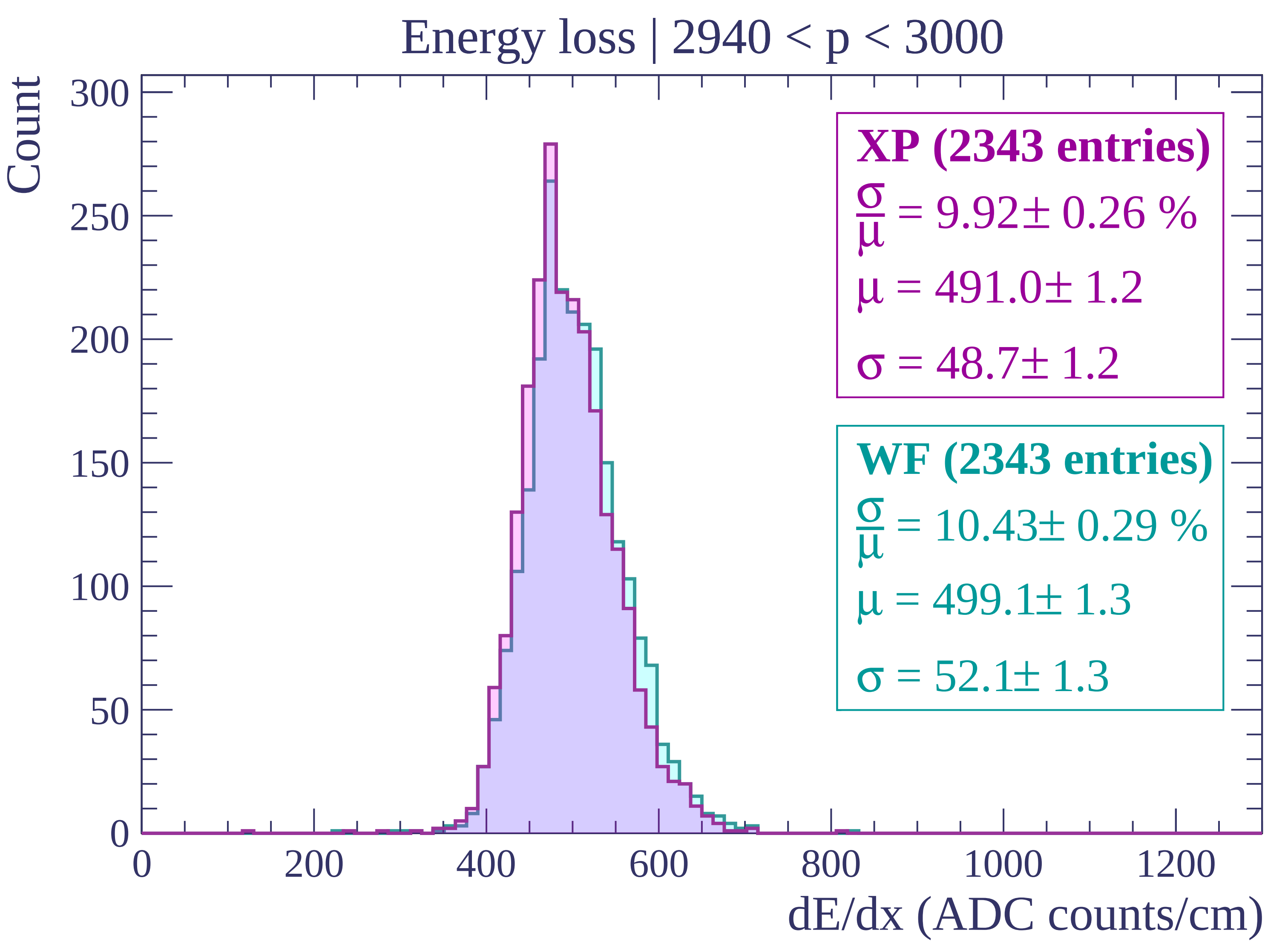

Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
XP (2491 entries)
σ/μ = 9.53± 0.26 %
μ = 490.5± 1.1
σ = 46.8± 1.2
WF (2491 entries)
σ/μ = 10.02± 0.26 %
μ = 498.6± 1.2
σ = 50.0± 1.2

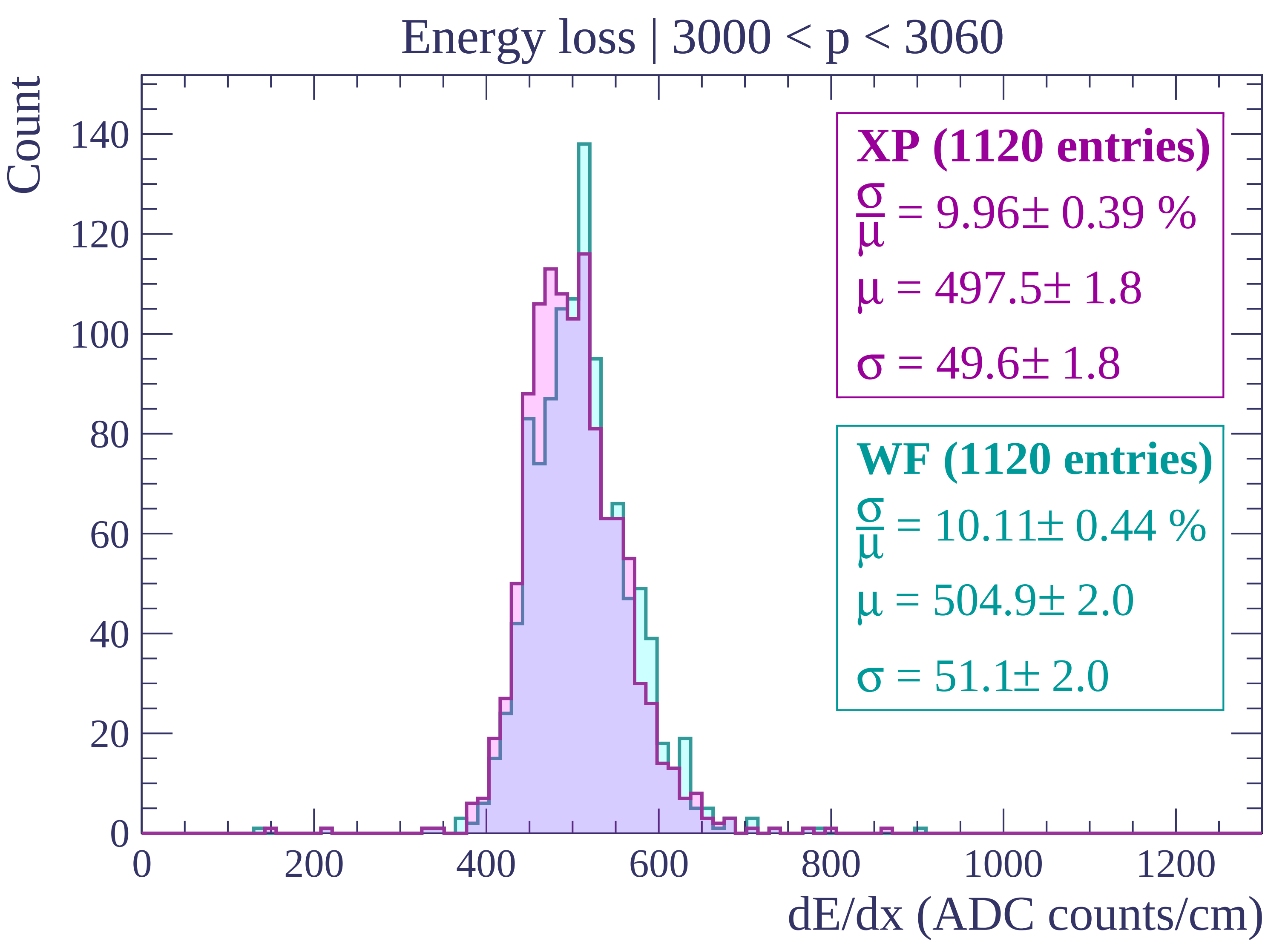


Energy loss | 2820 < p < 2880
Count
XP (2419 entries)
σ/μ = 9.71± 0.25 %
μ = 489.2± 1.1
σ = 47.5± 1.1
WF (2419 entries)
σ/μ = 10.06± 0.27 %
μ = 496.9± 1.2
σ = 50.0± 1.2
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

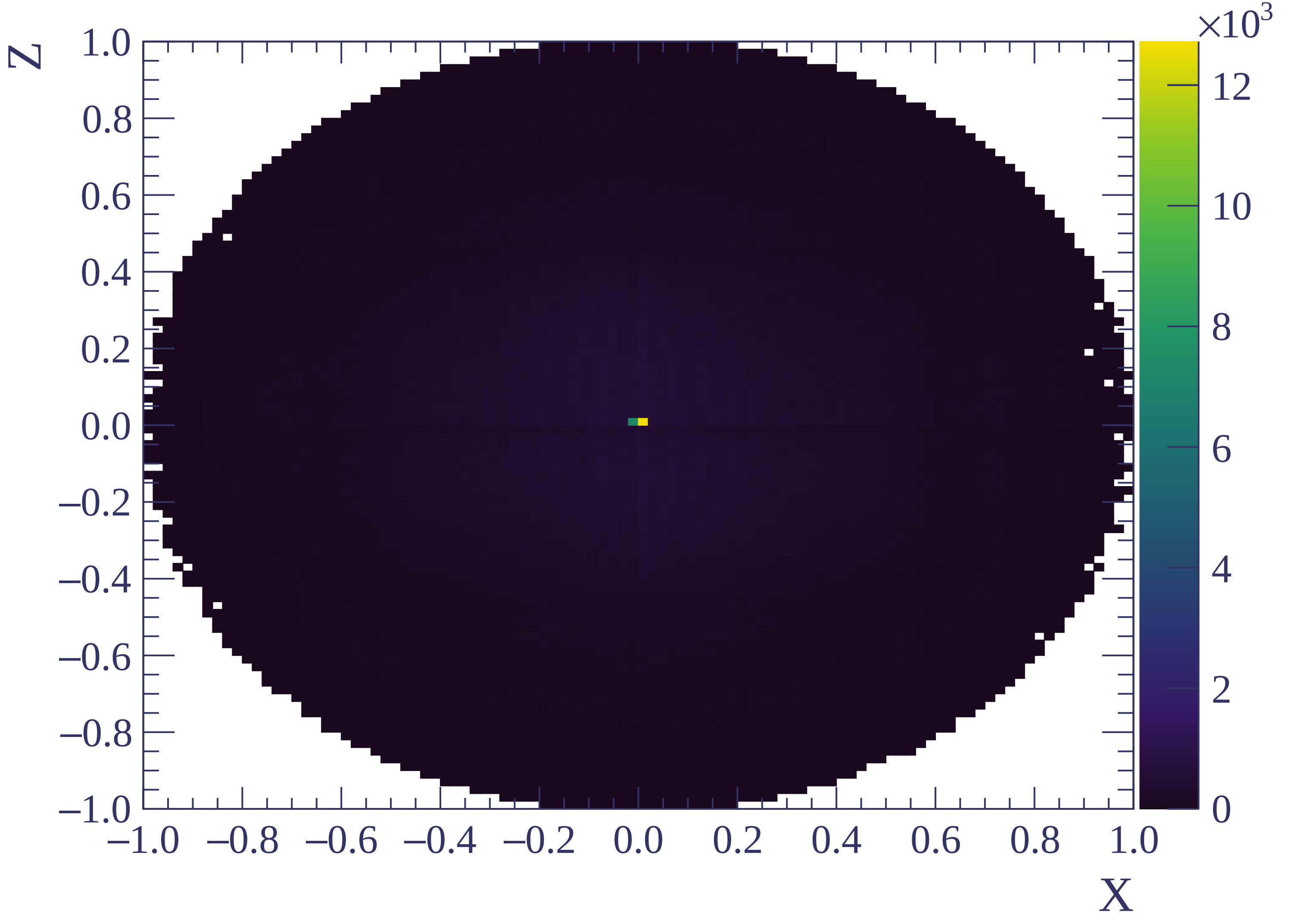


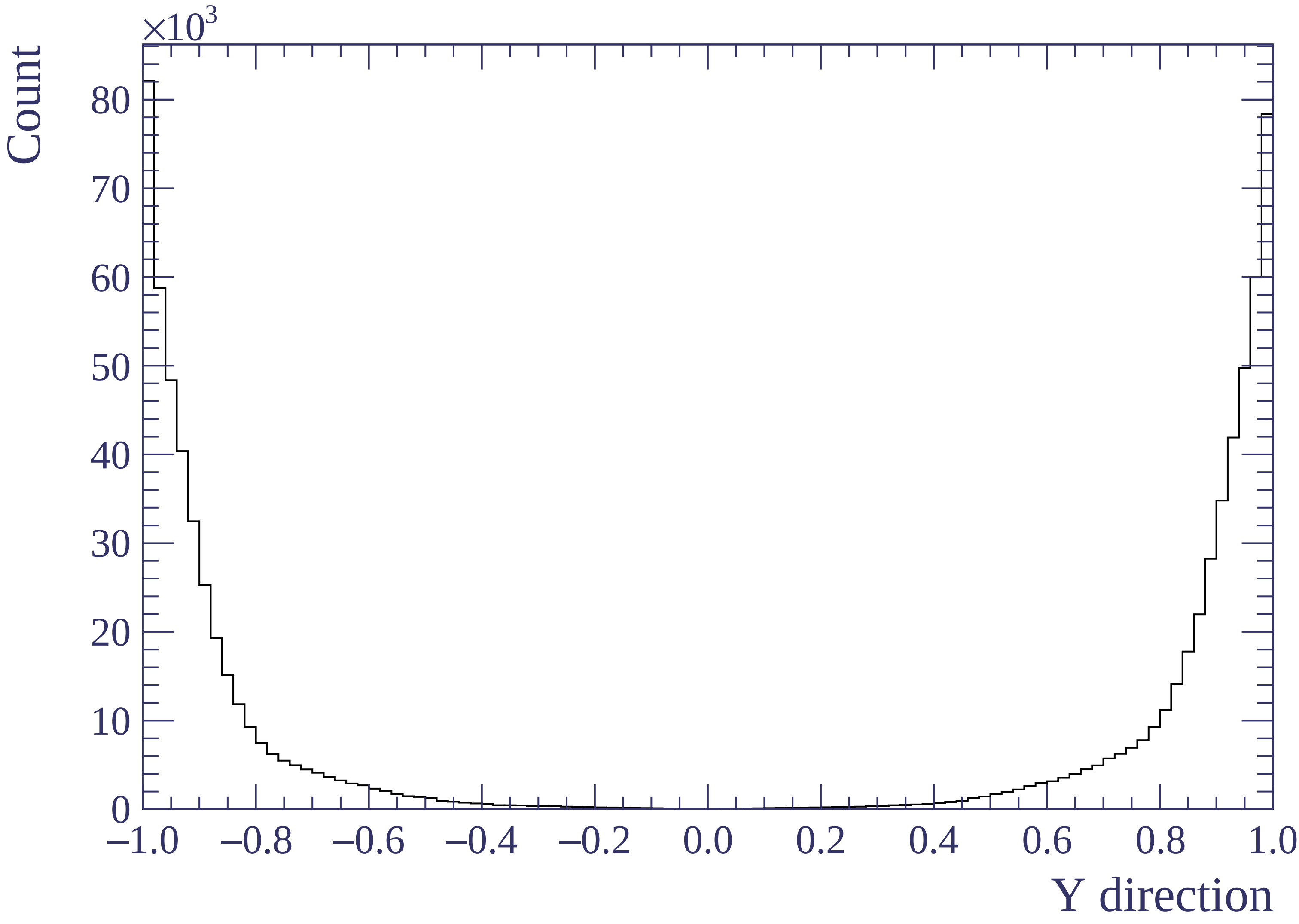
Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2538 entries)
σ/μ = 10.25± 0.25 %
μ = 494.6± 1.2
σ = 50.7± 1.1
WF (2538 entries)
σ/μ = 10.74± 0.27 %
μ = 503.4± 1.3
σ = 54.1± 1.2

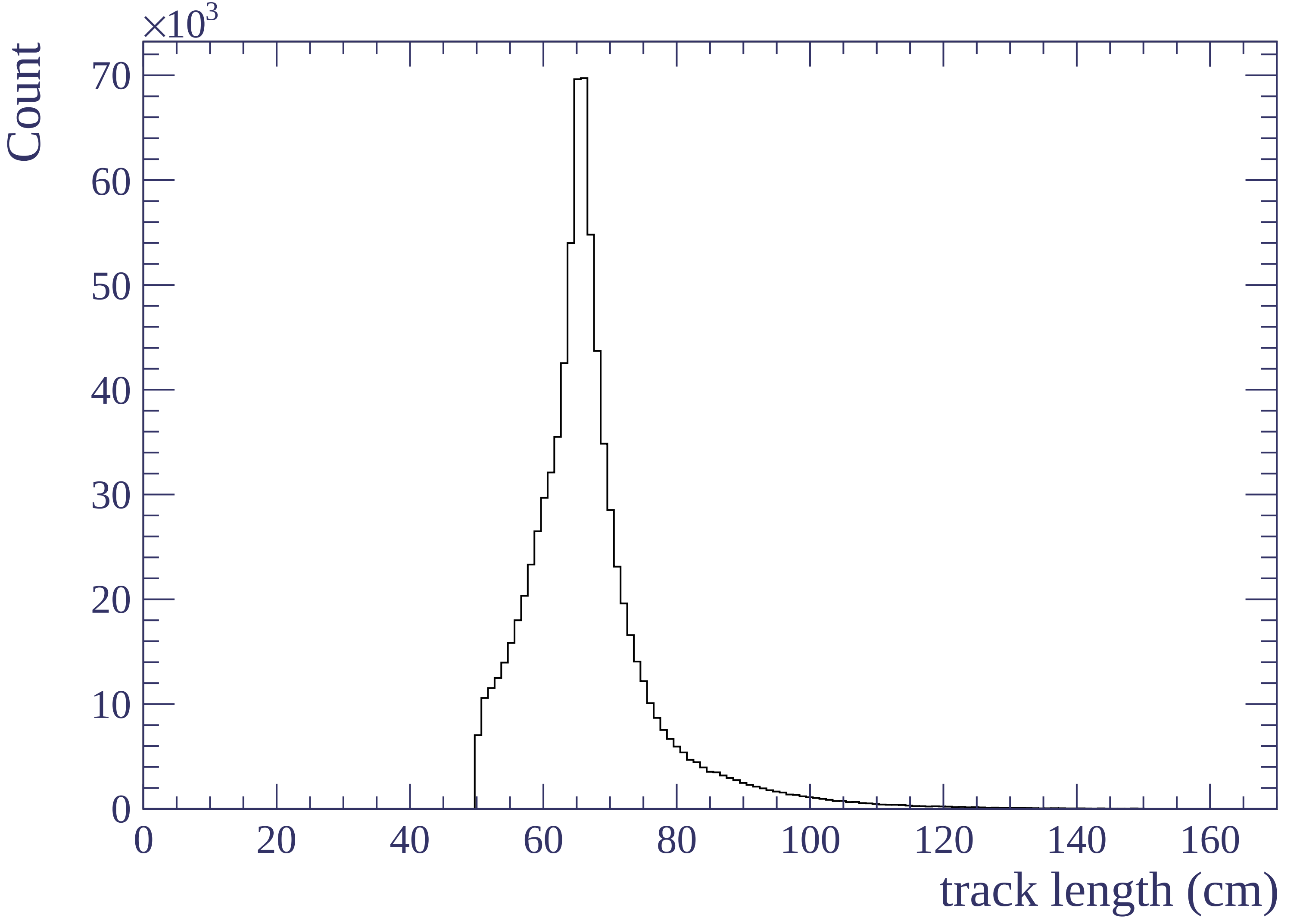

Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2343 entries)
σ/μ = 9.92± 0.26 %
μ = 491.0± 1.2
σ = 48.7± 1.2
WF (2343 entries)
σ/μ = 10.43± 0.29 %
μ = 499.1± 1.3
σ = 52.1± 1.3

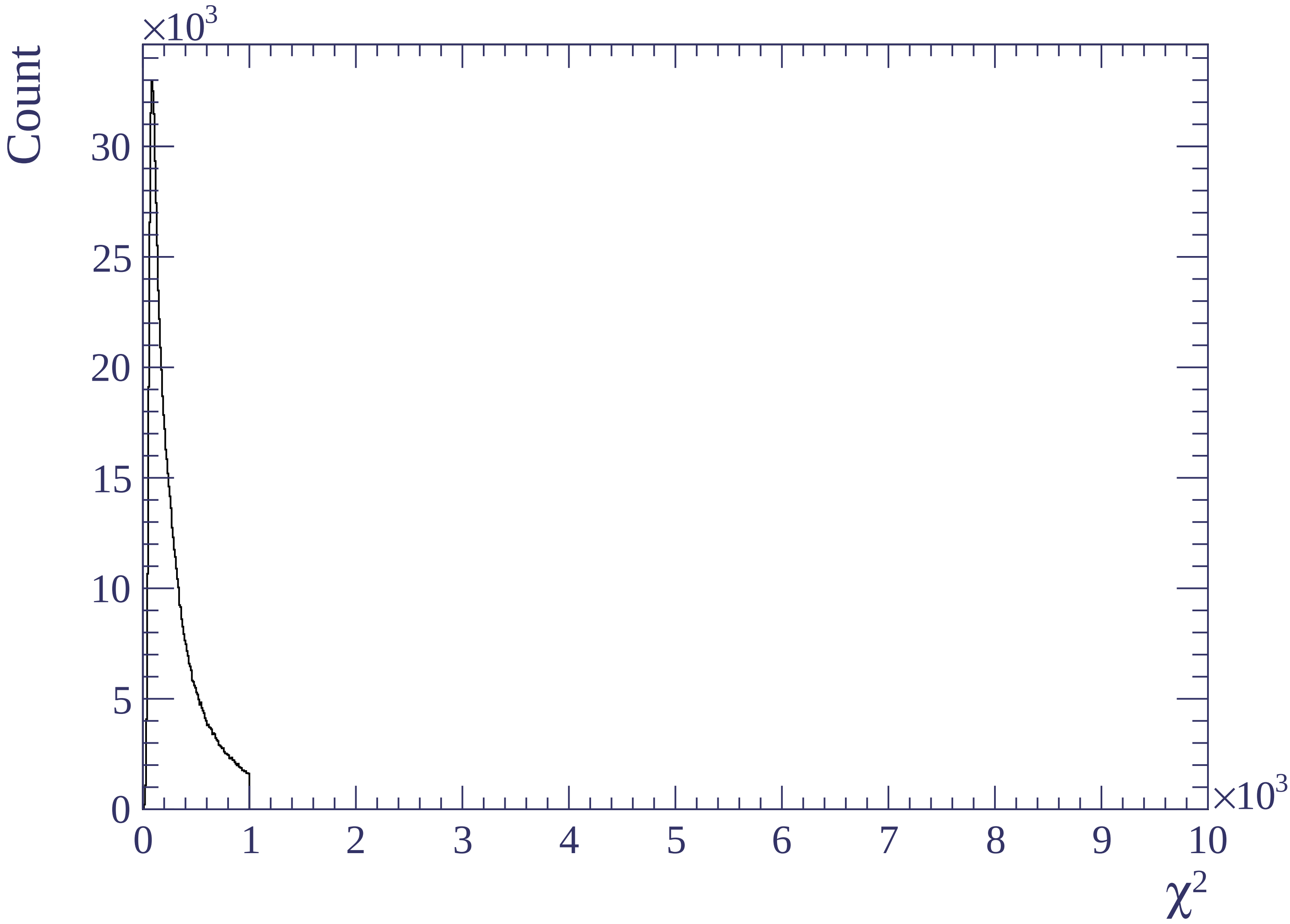


Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1120 entries)
σ/μ = 9.96± 0.39 %
μ = 497.5± 1.8
σ = 49.6± 1.8
WF (1120 entries)
σ/μ = 10.11± 0.44 %
μ = 504.9± 2.0
σ = 51.1± 2.0

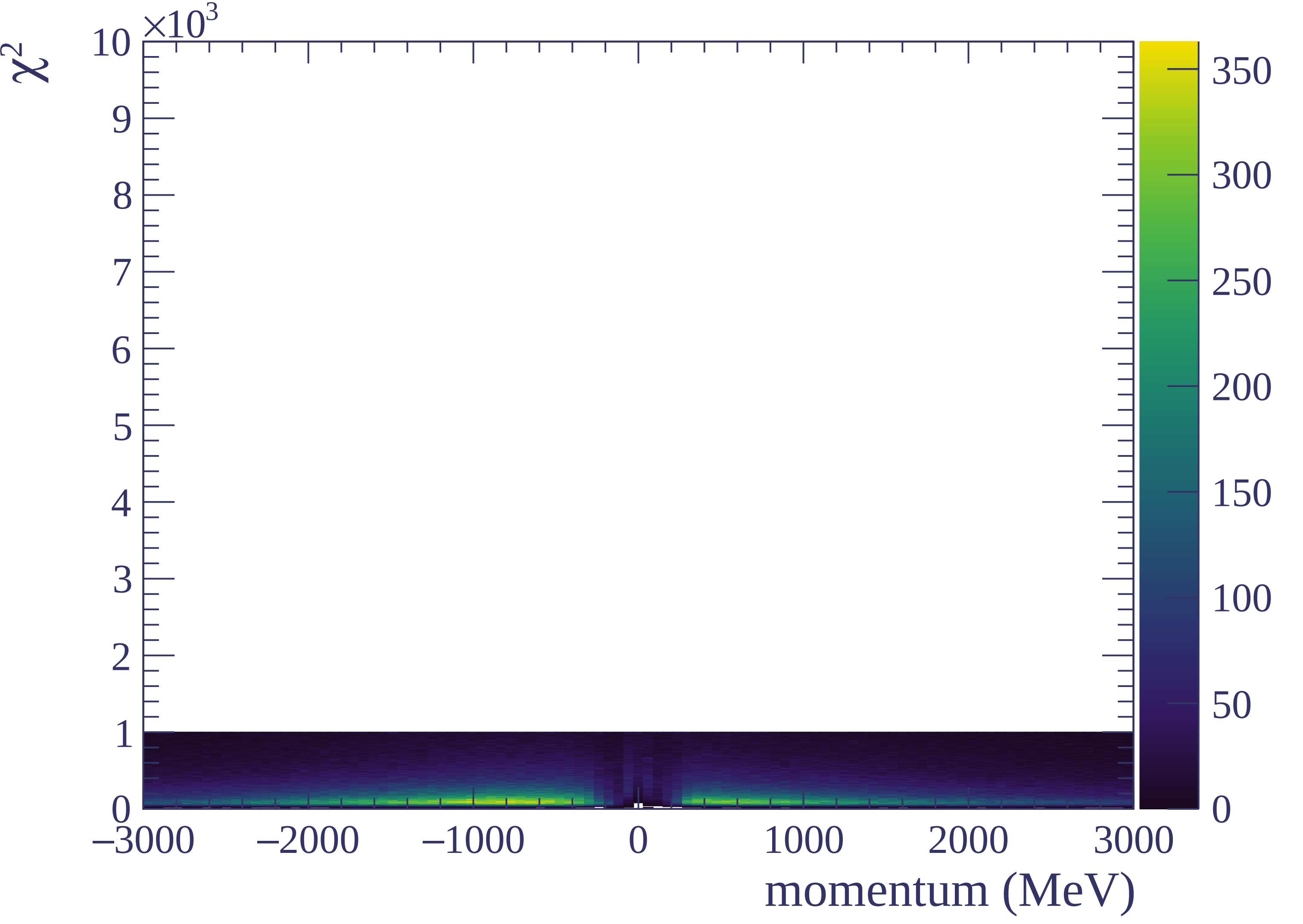

Z
X
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
×10³
12
10
8
6
4
2
0


×10^3
Count
80
70
60
50
40
30
20
10
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

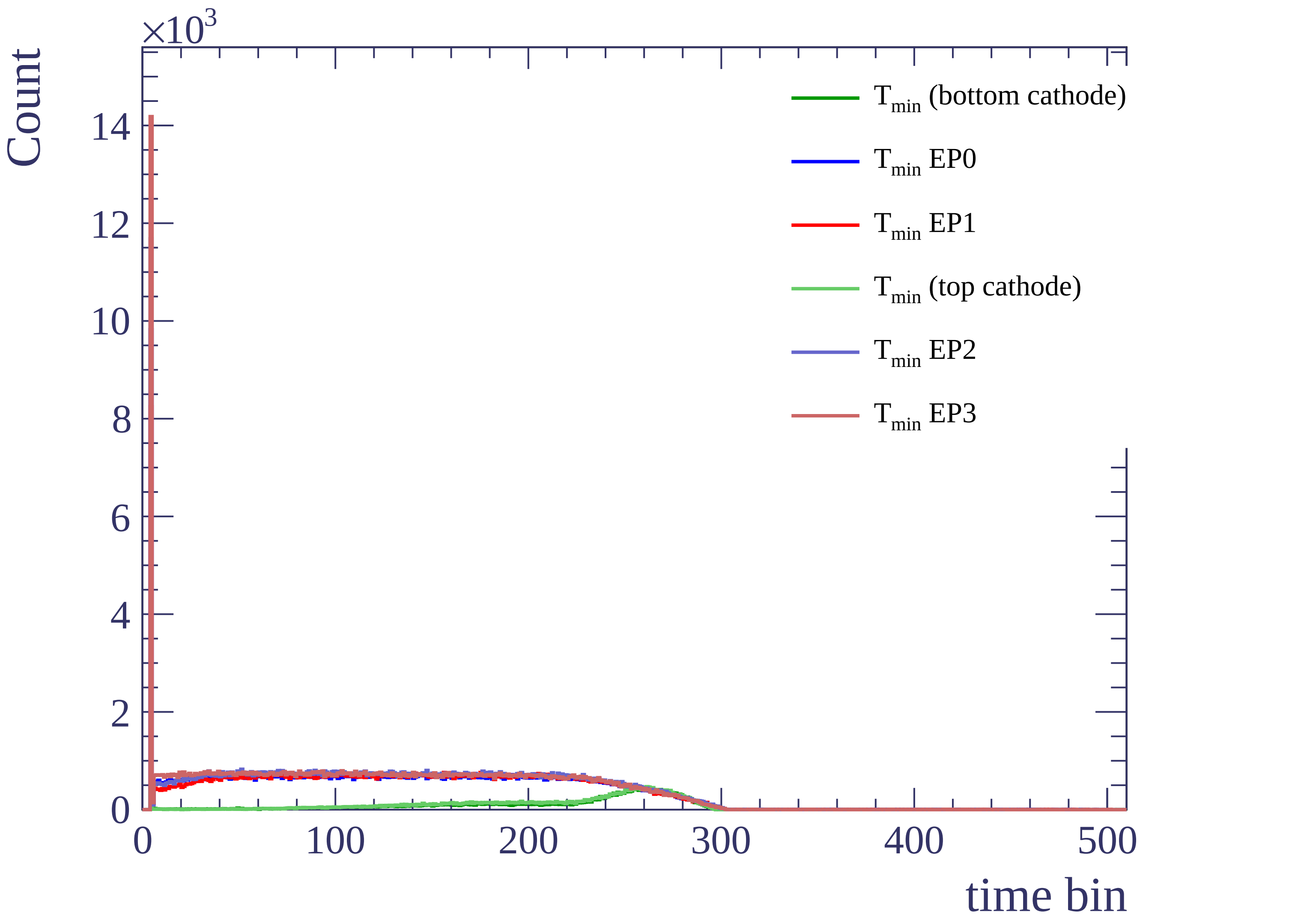

×10^3
Count
0
10
20
30
40
50
60
70
0
20
40
60
80
100
120
140
160
track length (cm)



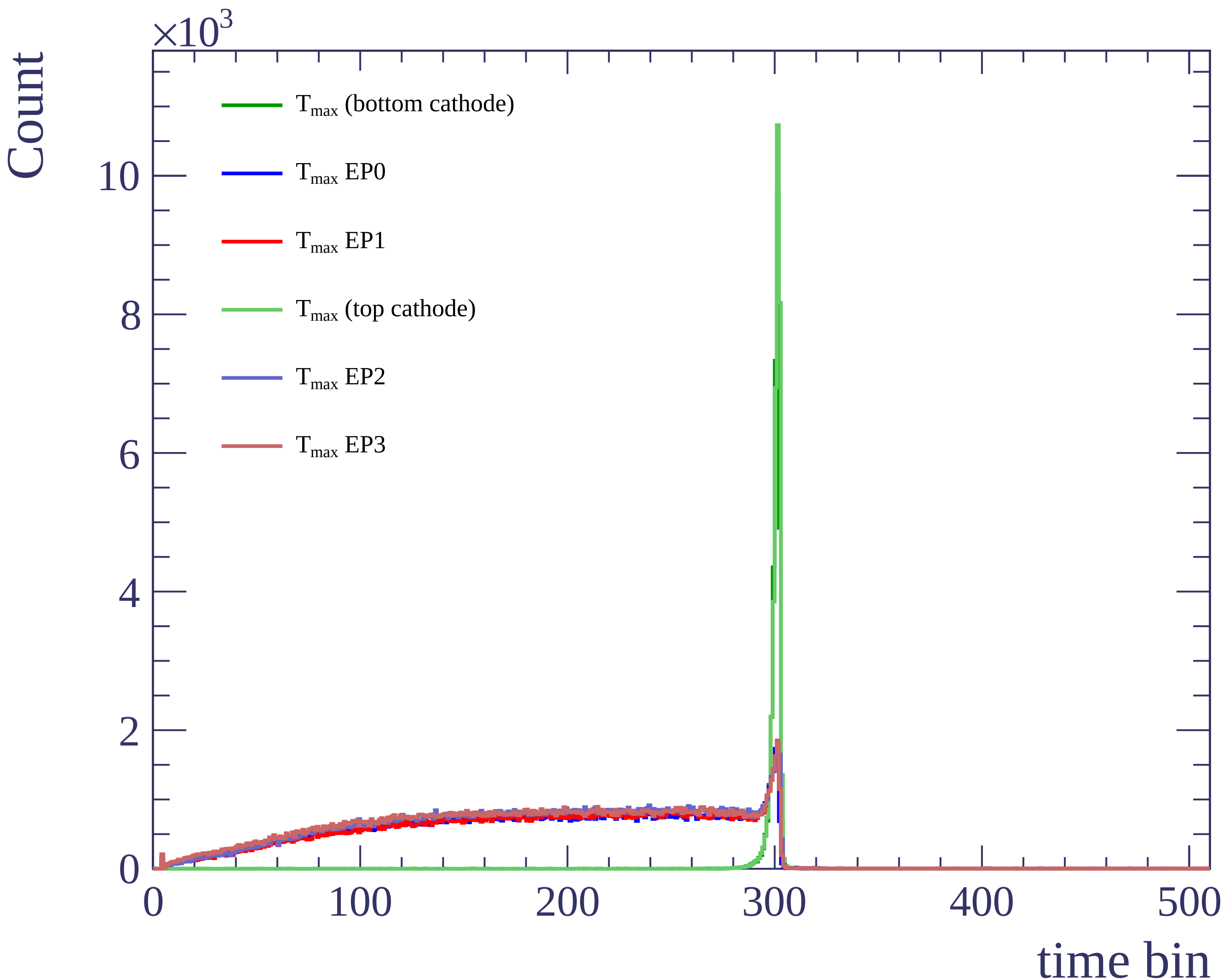
Count
×10^3
30
25
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
350
300
250
200
150
100
50
0



$\times 10^3$
Count
0
2
4
6
8
10
12
14
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



$\times 10^3$
Count
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
2
4
6
8
10
0
100
200
300
400
500
time bin